CAYLEY, Collected Mathematical Papers, Vol. II.

Scan-verified slice: book pages 26–50.

104. On the Theory of Permutants *(Postscript, concluded; book p. 26)*

Postscript. I wish to explain as accurately as I am able, the extent of my obligations to Mr Sylvester in respect of the subject of the present memoir. The term permutant is due to him—intermutant and commutant are merely terms framed between us in analogy with permutant, and the names date from the present year (1851). The theory of commutants is given in my memoir in the Cambridge Philosophical Transactions, [12], and is presupposed in the memoir on Linear Transformations, [13, 14].

It will appear by the last-mentioned memoir that it was by representing the coefficients of a biquadratic function by $a = 1111$, $b = 1112 = 1121 =$ &c., $c = 1122 =$ &c., $d = 1222 =$ &c., $e = 2222$, and forming the commutant

$$\begin{pmatrix} 1111 \\ 2222 \end{pmatrix}$$

that I was led to the function $ae - 4bd + 3c^2$.

The function $ace+2bcd-ad^2-b^2e-c^3$ is mentioned in the memoir on Linear Transformations, as brought into notice by Mr Boole. From the particular mode in which the coefficients $a, b, \ldots$ were represented by symbols such as 1111, &c., I did not perceive that the last-mentioned function could be expressed in the commutant notation.

The notion of a permutant, in its most general sense, is explained by me in my memoir, “Sur les determinants gauches,” Crelle, t. xxxvii. pp. 93—96, [69]; see the paragraph (p. 94) commencing “On obtient ces fonctions, &c.” and which should run as follows: “On obtient ces fonctions (dont je reprends ici la théorie) par les propriétés générales d'un déterminant défini comme suit. En exprimant &c.;” the sentence as printed being “défini. Car en exprimant &c.,” which confuses the sense. [The paragraph is printed correctly 69, p. 411.]

Some time in the present year (1851) Mr Sylvester, in conversation, made to me the very important remark, that as one of a class the above-mentioned function,

$$ace + 2bcd - ad^2 - b^2e - c^3,$$

would be expressed in the commutant notation, viz. by considering

$$00 = a, \quad 01 = 10 = b, \quad 02 = 11 = 20 = c, \quad 12 = 21 = d, \quad 22 = e;$$

and the subject being thereby recalled to my notice, I found shortly afterwards the expression for the function

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd$$

(which cannot be expressed as a commutant) in the form of an intermutant, and I was thence led to see the identity, so to say, of the theory of hyperdeterminants, as given in the memoir on Linear Transformations, with the present theory of intermutants.

It is understood between Mr Sylvester and myself, that the publication of the present memoir is not to affect Mr Sylvester's right to claim the origination, and to be considered as the first publisher of such part as may belong to him of the theory here sketched out.

105. Correction of the Postscript to the Paper on Permutants.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 97—98.]

Mr Sylvester has represented to me that the paragraph relating to his communications conveys an erroneous idea of the nature, purport, and extent of such communications; I have, in fact, in the paragraph in question, singled out what immediately suggested to me the expression of the function

$$6abcd + 3b^2c^2 - 4ac^3 - 4b^3d - a^2d^2$$

as a partial commutant or intermutant, but I agree that a fuller reference ought to have been made to Mr Sylvester's results, and that the paragraph in question would more properly have stood as follows:

"Under these circumstances Mr Sylvester communicated to me a series of formal statements, not only oral but in writing, to the effect that he had discovered a permutation method of obtaining as many invariants—viz. commutantive invariants—by direct inspection from a function of any degree of any number of letters as the index of the degree contains even factors; one of these commutantive invariants being in fact the function

$$ace + 2bcd - ad^2 - b^2e - c^3,$$

expressible, according to Mr Sylvester's notation of my memoir in the *Camb. Phil. Trans.*, supposing $00 = a$, $01 = 10 = b$, $02 = 11 = 20 = c$, $12 = 21 = d$, $22 = e$, by

$$\begin{pmatrix} 00 & 11 & 22 \\ a & b & c \\ b & c & d \\ c & d & e \end{pmatrix},$$

and so on."

Mr Sylvester and I shall, I have no doubt, be able to agree to a joint statement of any further correction or explanation which may be required.

106. On the Singularities of Surfaces.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 166—171.]

In the following paper, for symmetry of nomenclature and in order to avoid ambiguities, I shall, with reference to plane curves and in various phrases and compound words, use the term "node" as synonymous with double point, and the term "spinode" as synonymous with cusp.

I shall, besides, have occasion to consider the several singularities which I call the "flecnode," the "oscnode," the "fleflecnode," and the "tacnode:" the flecnode is a double point which is a point of inflexion on one of the branches through it; the oscnode is a double point which is a point of osculation on one of the branches through it; the fleflecnode is a double point which is a point of inflexion on each of the branches through it; and the tacnode is a double point where two branches touch.

And it may be proper to remark here, that a tacnode may be considered as a point resulting from the coincidence and amalgamation of two double points (and therefore equivalent to twelve points of inflexion) or, in a different point of view, as a point uniting the characters of a spinode and a flecnode.

I wish to call to mind here the definition of conjugate tangent lines of a surface, viz. that a tangent to the curve of contact of the surface with any circumscribed developable and the corresponding generating line of the developable, are conjugate tangents of the surface.

Suppose, now, that an absolutely arbitrary surface of any order be intersected by a plane: the curve of intersection has not in general any singularities other than such as occur in a perfectly arbitrary curve of the same order; but as a plane can be made to satisfy one, two, or three conditions, the curve may be made to acquire singularities which do not occur in such absolutely arbitrary curve.

Let a single condition only be imposed on the plane. We may suppose that the curve of intersection has a node; the plane is then a tangent plane and the node is the point of contact—of course any point on the surface may be taken for the node. We may if we please use the term "nodes of a surface," "node-planes of a surface," as synonymous with the points and tangent planes of a surface. And it will be convenient also to use the word node-tangents to denote the tangents to the curve of intersection at the node; it may be noticed here that the node-tangents are conjugate tangents of the surface.

Next let two conditions be imposed upon the plane: there are three distinct cases to be considered.

First, the curve of intersection may have a flecnode. The plane (which is of course still a tangent plane at the flecnode) may be termed a flecnode-plane; the flecnodes are singular points on the surface lying on a curve which may be termed the "flecnode-curve,"[1] and the flecnode-planes give rise to a developable which may be termed the flecnode-develope. The "flecnode-tangents" are the tangents to the curve of intersection at the flecnode; the tangent to the inflected branch may be termed the "singular flecnode-tangent," and the tangent to the other branch the "ordinary flecnode-tangent."

Secondly, the curve of intersection may have a spinode. The plane (which is of course still a tangent plane at the spinode) may be termed a spinode-plane; the spinodes are singular points on the surface lying on a curve which may be termed the "spinode-curve."[2] And the

[1]The flecnode-curve, defined as the locus of the points through which can be drawn a line meeting the surface in four consecutive points, was, so far as I am aware, first noticed in Mr Salmon's paper "On the Triple Tangent Planes of a Surface of the Third Order" (Journal, t. iv. [1849], pp. 252—260), where Mr Salmon, among other things, shows that the order of the surface being n, the curve in question is the intersection of the surface with a surface of the order $11n - 24$.

[2]The notion of a spinode, considered as the point where the indicatrix is a parabola (on which account the spinode has been termed a parabolic point) may be found in Dupin's *Developpements de Geometrie*: the most important step in the theory of these points is contained in Hesse's memoir "Ueber die Wendepuncte der Curven dritter Ordnung" (Crelle, t. xxviii. [1848], pp. 97—107), where it is shown that the spinode-curve is the curve of

spinode-planes give rise to a developable which may be termed the "spinode-develope." Also the "spinode-tangent" is the tangent to the curve of intersection at the spinode.

Thirdly, the curve of intersection may have two nodes, or what may be termed a "node-couple." The plane (which is a tangent plane at each of the nodes and therefore a double tangent plane) may be also termed a "node-couple-plane." The node-couples are pairs of singular points on the surface lying in a curve which may be termed the "node-couple-curve," and the node-couple-planes give rise to a developable which may be termed the "node-couple-develope." The tangents to the curve of intersection at the two nodes of a node-couple might, if the term were required, be termed the "node-couple-tangents." Also one of the nodes of a node-couple may be termed a "node-with-node," and the tangents to the curve of intersection at such point will be the "node-with-node-tangents."

It is hardly necessary to remark that the flecnode-curve is not the edge of regression of the flecnode-develope, and the like remark applies *mutatis mutandis* to the spinode-curve and the node-couple curve.

Finally, let three conditions be imposed upon the plane: there are six distinct cases to be considered, in each of which we have no longer curves and developes, but only singular points and singular tangent planes determinate in number.

First, the curve of intersection may have an oscnode. The plane (which continues a tangent plane at the oscnode) is an "oscnode-plane." The "oscnode-tangents" are the tangents to the curve of intersection at the oscnode; the tangent to the osculating branch is the "singular oscnode-tangent;" and the tangent to the other branch the "ordinary oscnode-tangent."

Secondly, the curve of intersection may have a fleflecnode. The plane (which continues a tangent plane at the fleflecnode) is a "fleflecnode-plane." The "fleflecnode-tangents" are the tangents to the curve of intersection at the fleflecnode.

Thirdly, the curve of intersection may have a tacnode. The plane (which continues a tangent plane at the tacnode) is a "tacnode-plane." The "tacnode-tangent" is the tangent to the curve of intersection at the tacnode.

Fourthly, the curve of intersection may have a node and a flecnode, or what may be termed a node-and-flecnode. The plane (which is a tangent plane at the node and also at the flecnode, where it is obviously a flecnode-plane) is a "node-and-flecnode-plane." The "node-and-flecnode-tangents," if the term were required, would be the tangents to the curve of intersection at the node and at the flecnode of the node-and-flecnode. The node of the node-and-flecnode may be distinguished as the node-with-flecnode, and the flecnode as the flecnode-with-node, and we have thus the terms "node-with-flecnode-tangents," "flecnode-with-node-tangents," "singular flecnode-with-node-tangent," and "ordinary flecnode-with-node-tangent."

Fifthly, the curve of intersection may have a node and also a spinode, or what may be termed a "node-and-spinode." The plane (which is a tangent plane at the node, and is also a tangent plane at the spinode, where it is obviously a spinode-plane) is a "node-and-spinode-plane." The node-and-spinode-tangents, if the term were required, would be the tangents at the node and the tangent at the spinode of the node-and-spinode to the curve of intersection. The node of the node-and-spinode may be distinguished as the "node-with-spinode," and the spinode as the "spinode-with-node," and we have thus the terms "node-with-spinode-tangent," "spinode-with-node-tangent."

Sixthly, the curve of intersection may have three nodes, or what may be termed a "node-triplet." The plane (which is a triple tangent plane touching the surface at each of the nodes) is a "node-triplet-plane." The "node-triplet-tangents," if the term were required, would be the tangents to the curve of intersection at the nodes of the node-triplet. Each node of the node-triplet may be distinguished as a "node-with-node-couple," and the tangents to the curve of

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intersection of the surface supposed as before of the order n, with a certain surface of the order $4(n-2)$. See also Mr Salmon's memoir "On the Condition that a Plane should touch a surface along a Curve Line" (Journal, t. iii. [1848], pp. 44—46).

intersection at such nodes are "node-with-node-couple-tangents." The terms "node-couple-with-node," "node-couple-with-node-tangent," might be made use of if necessary.

It should be remarked that the oscnodes lie on the flecnode-curve, as do also the fleflecnodes; these latter points are real double points of the flecnode-curve. The tacnodes are points of intersection and (what will appear in the sequel) points of contact of the flecnode-curve, the spinode-curve, and the node-couple-curve. The spinode-with-nodes are points of intersection of the spinode-curve and node-couple-curve, and the flecnode-with-nodes are points of intersection of the flecnode-curve and node-couple-curve; the node-with-node-couples are real double points (entering in triplets) of the node-couple-curve.

Consider for a moment an arbitrary curve on the surface; the locus of the node-tangents at the different points of this curve is in general a skew surface, which may however, in cases to be presently considered, degenerate in different ways.

Reverting now to the flecnode-curve, it may be shown that the singular flecnode-tangent coincides with the tangent of the flecnode-curve. For consider on a surface two consecutive points such that the line joining them meets the surface in two points consecutive to the first-mentioned two points. The line meets the surface in four consecutive points, it is therefore a singular flecnode-tangent; each of the first-mentioned two points must be on the flecnode-curve, or the singular flecnode-tangent touches the flecnode-curve. The two flecnode-tangents are by a preceding observation conjugate tangents. It follows that the skew surface, locus of the flecnode-tangents, breaks up into two surfaces, each of which is a developable, viz. the locus of the singular flecnode-tangents is the developable having the flecnode-curve for its edge of regression, and the locus of the ordinary flecnode-tangents is the flecnode-develope. Of course at the tacnode, the tacnode-tangent touches the flecnode-curve.

Passing next to the spinode-curve, the spinode-plane and the tangent-plane at a consecutive point along the spinode-tangent are identical, or their line of intersection is indeterminate.[3] The spinode-tangent is therefore the conjugate tangent to any other tangent line at the spinode, and therefore to the tangent to the spinode-curve. It follows that the surface locus of the spinode-tangents degenerates into a developable surface twice repeated, viz. the spinode-develope.

Consider the tacnode as two coincident nodes; each of these nodes, by virtue of its constituting, in conjunction with the other, a tacnode, is on the spinode-curve; or, in other words, the tacnode-tangent touches the spinode-curve, and the same reasoning proves that it touches the node-couple-curve. It has already been seen that the tacnode-tangent touches the flecnode-curve; consequently the tacnode is a point, not of simple intersection only, but of contact, of the flecnode-curve, the spinode-curve, and the node-couple-curve.

In virtue of the principle of the spinode-plane being identical with the tangent plane at a consecutive point along the spinode tangent, it appears that the tacnode-plane is a stationary plane, as well of the flecnode-develope as of the spinode-develope, and it would at first sight appear that it must be also a stationary tangent plane of the node-couple-develope. But this is not so; the node-with-node-planes envelope, not the node-couple-develope, but the node-couple-develope twice repeated: the tacnode-plane is in a sense a stationary plane on such duplicate developable, but not in any manner on the single developable. The tacnode-plane is an ordinary tangent plane of the node-couple-develope.

Consider now a spinode-with-node, which we have seen is a point of intersection of the spinode-curve and node-couple-curve. The tangent plane at a consecutive point along the spinode-with-node-tangent, is identical with the spinode-with-node-plane; the curve of intersection of the tangent plane at such consecutive point has therefore a node at the node-with-spinode, or the tangent plane in question is a node-couple-plane, and the point of contact is a point on the node-couple-curve. Consequently the spinode-with-node-tangent touches the node-couple-curve, and thence also the spinode-with-node-plane is a stationary tangent plane of the

[3]It must not be inferred that the tangent plane at such consecutive point is a spinode-plane; this is obviously not the case.

node-couple-develope.

It should be remarked that no circumscribed developable can have a stationary tangent plane except the tangent planes at the points where the curve of contact meets the spinode-curve, and any one of these planes is only a stationary plane when the curve of contact touches the spinode-tangent; and that the node-couple-curve and the flecnode-curve do not intersect the spinode-curve except in the points which have been discussed.

Recapitulating, the node-couple-curve and the spinode-curve touch at the tacnodes, and intersect at the spinode-with-nodes: moreover, the tacnode-planes are stationary planes of the spinode-develope, and the spinode-with-node-planes are stationary planes of the node-couple-develope. Besides this, the two curves are touched at the tacnodes by the flecnode-curve, and the tacnode-planes are stationary planes of the flecnode-develope.

107. On the Theory of Skew Surfaces.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 171—173.]

A SURFACE of the n^{th} order is a surface which is met by an indeterminate line in n points. It follows immediately that a surface of the n^{th} order is met by an indeterminate plane in a curve of the n^{th} order.

Consider a skew surface or the surface generated by a singly infinite series of lines, and let the surface be of the n^{th} order. Any plane through a generating line meets the surface in the line itself and in a curve of the $(n-1)^{\text{th}}$ order. The generating line meets this curve in $(n-1)$ points. Of these points one, viz. that adjacent to the intersection of the plane with the consecutive generating line, is a unique point; the other $(n-2)$ points form a system. Each of the $(n-1)$ points are *sub modo* points of contact of the plane with the surface, but the proper point of contact is the unique point adjacent to the intersection of the plane with the consecutive generating line. Thus every plane through a generating line is an ordinary tangent plane, the point of contact being a point on the generating line.

It is not necessary for the present purpose, but I may stop for a moment to refer to the known theorems that the anharmonic ratio of any four tangent planes through the same generating line is equal to the anharmonic ratio of their points of contact, and that the locus of the normals to the surface along a generating line is a hyperbolic paraboloid.

Returning to the $(n-2)$ points in which, together with the point of contact, a generating line meets the curve of intersection of the surface and a plane through the generating line, these are fixed points independent of the particular plane, and are the points in which the generating line is intersected by other generating lines.

There is therefore on the surface a double curve intersected in $(n-2)$ points by each generating line of the surface—a property which, though insufficient to determine the order of this double curve, shows that the order cannot be less than $(n-2)$.

(Thus for $n=4$, the above reasoning shows that the double-curve must be at least of the second order: assuming for a moment that it is in any case precisely of this order, it obviously cannot be a plane curve, and must therefore be two non-intersecting lines. This suggests at any rate the existence of a class of skew surfaces of the fourth order generated by a line which always passes through two fixed lines and by some other condition not yet ascertained; and it would appear that surfaces of the second order constitute a degenerate species belonging to the class in question.)

In particular cases a generating line will be intersected by the consecutive generating line. Such a generating line touches the double curve.

Consider now a point not on the surface; the planes determined by this point and the generating lines of the surface are the tangent planes through the point; the intersections of consecutive tangent planes are the tangent lines through the point; and the cone generated by these tangent lines or enveloped by the tangent planes is the tangent cone corresponding to the point. This cone is of the n^{th} class. For considering a line through the point, this line meets the surface in n points, i.e. it meets n generating lines of the surface; and the planes through the line and these n generating lines, are of course tangent planes to the cone: that is, n tangent planes can be drawn to the cone through a given line passing through the vertex.

The cone has not in general any lines of inflexion, or, what is the same thing, stationary tangent planes. For a stationary tangent plane would imply the intersection of two consecutive generating lines of the surface. And since the number of generating lines intersected by a consecutive generating line, and therefore the number of planes through two consecutive generating lines, is finite, no such plane passes through an indeterminate point.

The tangent cone will have in general a certain number of double tangent planes; let this number be χ. We have therefore a cone of the class n, number of double tangent planes χ, number of stationary tangent planes 0. Hence, if m be the order of the cone, α the number of its double lines, and β the number of its cuspidal or stationary lines,

$$\begin{aligned} m &= n(n-1) - 2\chi, \\ \beta &= 3n(n-2) - 6\chi, \\ \alpha &= \tfrac{1}{2}n(n-2)(n^2-9) - 2\chi(n^2-n-6) + 2\chi(\chi-1). \end{aligned}$$

This is the proper tangent cone, but the cone through the double curve is *sub modo* a tangent cone, and enters as a square factor into the equation of the general tangent cone of the order $n(n-1)$. Hence, if X be the order of the double curve, and therefore of the cone through this curve,

$$m + 2X = n(n-1),$$

and therefore $X = \chi$; that is, the number of double tangent planes to the tangent cone is equal to the order of the double curve.

It does not appear that there is anything to determine χ; and if this is so, skew surfaces of the n^{th} order may be considered as forming different families according to the order of the double curve upon them. To complete the theory, it should be added that a plane intersects the surface in a curve of the n^{th} order having χ double points but no cusps.

108. On certain Multiple Integrals connected with the Theory of Attractions.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 174—178.]

It is easy to deduce from Mr Boole's formula, given in my paper "On a Multiple Integral connected with the theory of Attractions," Journal, t. ii. [1847], pp. 219—223, [44], the equation

$$\int \frac{d\xi\, d\eta \ldots}{\left[(\xi-\alpha)^2+(\eta-\beta)^2+\ldots+v^2\right]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n-q)\,\Gamma(q+1)} \int_0^{\epsilon} \frac{s^{q-1}(e-s)^q\, ds}{(1+s)^{\frac{1}{2}n+q}},$$

where n is the number of variables of the multiple integral, and the condition of the integration is

$$(\xi-\alpha)^2+(\eta-\beta)^2+\ldots < f^2;$$

also where

$$\frac{(\xi-\alpha)^2+(\eta-\beta)^2+\ldots}{f^2} - \frac{v^2}{ef^2} = 1,$$

and e is the positive root of

$$\xi^2+\eta^2+\ldots = v^2.$$

Suppose $f = f_1 = \ldots = f_n$, and write $(\alpha-\alpha_1)^2+\ldots = k^2$; we obtain

$$\int \frac{d\xi\, d\eta \ldots}{\left[(\xi-\alpha)^2+\ldots+v^2\right]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n-q)\,\Gamma(q+1)} \int_0^{\epsilon} \frac{s^{q-1}(\epsilon-s)^q\, ds}{(1+s)^{\frac{1}{2}n+q}},$$

the limiting condition for the multiple integral being

$$(\xi-\alpha_1)^2+(\eta-\beta_1)^2+\ldots < f_1^2,$$

and the function e, and limit ϵ, being given by

$$e^2+\xi^2+\ldots = v^2, \quad \epsilon \text{ denoting, as before, the positive root.}$$

Observing that the quantity under the integral sign on the second side vanishes for $s=\epsilon$, there is no difficulty in deducing, by a differentiation with respect to ξ_1, the formula

$$\int \frac{dS}{\left[(\xi-\alpha)^2+\ldots+v^2\right]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n-q)\,\Gamma(q)} \int_0^{\epsilon} \frac{s^{q-1}(\epsilon-s)^{q-1}\, ds}{(1+s)^{\frac{1}{2}n+q-1}},$$

where dS is the element of the surface $(\xi-\alpha_1)^2+\ldots = f_1^2$ and the integration is extended over the entire surface.

A slight change of form is convenient. We have

$$s^{q-1}(\epsilon-s)^{q-1}\, ds = \frac{1}{s(1+s)} \cdot \frac{s^q(\epsilon-s)^q}{\epsilon}\, d\epsilon,$$

if we suppose

$$\epsilon = \frac{\xi_1^2+\eta_1^2+\ldots-f_1^2}{(\xi-\alpha_1)^2+(\eta-\beta_1)^2+\ldots}.$$

The formulae then become

$$\int \frac{dS}{\left[(\xi-\alpha)^2+\ldots+v^2\right]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n-q)\,\Gamma(q)} \int_0^{\epsilon} \frac{(\xi_1^2+\ldots-f_1^2)^{q-1}\, d\epsilon}{\epsilon(1+\epsilon)\left[(\xi-\alpha_1)^2+\ldots\right]^{\frac{1}{2}n-q-1}},$$

in which ϵ is the positive root of the equation

$$\rho^2 e^2 + \lambda e - \mu = 0.$$

I propose to transform these formulae by means of the theory of images; it will be convenient to investigate some preliminary formulae.

Suppose $l^2 = \alpha^2 + \beta^2 + \ldots$, $l_1^2 = \alpha_1^2 + \beta_1^2 + \ldots$; also consider the new constants $a, b, \ldots, a_1, b_1, \ldots, u, \mu$, determined by the equations

$$\frac{S}{u} = \frac{\alpha}{a} = \frac{\beta}{b} = \ldots = \frac{l^2}{\mu} = \frac{l_1^2}{S},$$

where S is arbitrary.

Then, putting

$$x = \frac{S\xi}{\xi^2 + \eta^2 + \ldots}, \quad y = \frac{S\eta}{\xi^2 + \eta^2 + \ldots}, \quad \ldots$$

it is easy to see that

$$(x-a)^2 + (y-b)^2 + \ldots = \frac{S^2}{(\xi^2 + \eta^2 + \ldots)^2} \left\{ (\xi - \alpha)^2 + (\eta - \beta)^2 + \ldots \right\},$$

and

$$\frac{a}{l^2} = \frac{\alpha}{l^2 + uS}, \quad \frac{S^2 \alpha}{l^2 + uS} = a, \quad \frac{S^2 \alpha_1}{l_1^2 + \mu S} = a_1.$$

Proceeding to express the single integrals in terms of the new constants, we have in the first place $k^2 = h^2 \cdot l^2$, where

$$h^2 = \frac{(l^2 + uS)(l_1^2 + \mu S)}{S^2} = \frac{l^2 l_1^2 + S(\mu l^2 + u l_1^2) + u\mu S^2}{S^2};$$

or if we write

$$\alpha\alpha_1 + \beta\beta_1 + \ldots = l l_1 \cos\omega,$$

we have

$$h^2 = l^2 + l_1^2 - 2 l l_1 \cos\omega.$$

Hence also $k^{-2} = h^{-2} \cdot l^{-2}$, where

$$\frac{l^2 + uS}{S^2 \alpha} = \frac{1}{a}, \quad \text{whence } \frac{(l^2 + uS)(l_1^2 + \mu S)}{S^2 l l_1 \cos\omega} = \frac{1}{\cos\omega};$$

$$\frac{(k^2 - \lambda^2)(l^2 + uS)(l_1^2 + \mu S)}{S^2 l l_1} = \frac{1}{\rho^2},$$

where $\rho^2 = l^2 + l_1^2 - 2 l l_1 \cos\omega$, that is $\rho^2 = (a - a_1)^2 + (b - b_1)^2 + \ldots$; consequently

$$\frac{1}{k^2} \cdot \frac{l^2 + uS}{l_1^2 + \mu S} = \frac{S^2}{\Pi},$$

where Π is given by

$$\Pi = \rho^2 e^2 + (\mu - \lambda^2)\, e - u\mu,$$

and it is clear that e will be the positive root of

$$\Pi = \rho^2 e^2 + (\mu - \lambda^2)\, e - u\mu = 0.$$

It may be noticed that, in the particular case of $u=0$, the roots of this equation are 0, and $\dfrac{\mu}{\lambda^2-\mu}$. Consequently if $\rho^2-f_1^2$ and l^2-f^2 are of opposite signs, we have $e=0$; but if $\rho^2-f_1^2$ and l^2-f^2 are of the same sign, $e=\dfrac{\mu}{\lambda^2-\mu}$.

In order to transform the double integrals, considering the new variables $x, y, \ldots$, I write $x^2+y^2+\ldots=r^2$ and

$$\xi=\frac{Sx}{x^2+y^2+\ldots}, \quad \eta=\frac{Sy}{x^2+y^2+\ldots}, \quad \ldots$$

whence also, if $\xi^2+\eta^2+\ldots=\rho^2$ (which gives $r\rho=S^2$), we have

$$\frac{d\xi}{\xi}=\frac{dx}{x}-\frac{2x\,dx+2y\,dy+\ldots}{x^2+y^2+\ldots}, \quad \text{\&c.};$$

also it is immediately seen that

$$(\xi-\alpha)^2+\ldots+v^2=\frac{S^2}{(x^2+y^2+\ldots)^2}\left\{(x-a)^2+(y-b)^2+\ldots+u^2\right\},$$

$$(\xi-\alpha_1)^2+\ldots-f_1^2=\frac{S^2}{(x^2+y^2+\ldots)^2}\left\{(x-a_1)^2+(y-b_1)^2+\ldots-f_1'^2\right\};$$

and from the latter equation it follows that the limiting condition for the first integral is

$$(x-a_1)^2+(y-b_1)^2+\ldots>f_1'^2$$

(there is no difficulty in seeing that the sign $<$ in the former limiting condition gives rise here to the sign $>$), and that the second integral has to be extended over the surface $(x-a_1)^2+(y-b_1)^2+\ldots=f_1'^2$. Also if dS' represent the element of this surface, we may obtain

$$d\xi\,d\eta\ldots=\frac{S^n}{r^{2n}}\,dx\,dy\ldots, \qquad dS=\frac{S^{n-1}}{r^{2(n-1)}}\,dS';$$

and, combining the above formulae, we obtain

$$\int\frac{dx\,dy\ldots}{(x^2+y^2+\ldots)^{\frac{1}{2}n+q}\left\{(x-a)^2+(y-b)^2+\ldots+u^2\right\}^{\frac{1}{2}n-q}}$$
$$=\frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n-q)\,\Gamma(q+1)(l^2+u^2)^{\frac{1}{2}n-q}}\int_0^{\epsilon}\frac{s^{q-1}(e-s)^q\,ds}{s(1+s)^{\frac{1}{2}n+q}},$$

the limiting condition of the multiple integral being

$$(x-a_1)^2+(y-b_1)^2+\ldots>f_1'^2;$$

and

$$\int\frac{dS'}{(x^2+y^2+\ldots)^{\frac{1}{2}n+q-1}\left\{(x-a)^2+(y-b)^2+\ldots+u^2\right\}^{\frac{1}{2}n-q}}$$
$$=\frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n-q)\,\Gamma(q)\,(l^2+u^2)^{\frac{1}{2}n-q}(l_1^2-f_1^2)}\int_0^{\epsilon}\frac{s^{q-1}(e-s)^{q-1}\,ds}{(1+s)^{\frac{1}{2}n+q-1}},$$

where dS' is the element of the surface $(x-a_1)^2+(y-b_1)^2+\ldots=f_1'^2$, and the integration extends over the entire surface.

In these formulae, l, l_1, ρ, Π denote as follows:

$$\begin{aligned} l^2 &= a^2 + b^2 + \ldots, \\ l_1^2 &= a_1^2 + b_1^2 + \ldots, \\ \rho^2 &= (a - a_1)^2 + (b - b_1)^2 + \ldots, \\ \Pi &= \rho^2 e^2 + (\mu - \lambda^2)\, e - u\mu, \end{aligned}$$

and e is the positive root of the equation $\Pi = 0$.

The only obviously integrable case is that for which in the second formula $q = \frac{1}{2}$; this gives

$$\int \frac{dS'}{(x^2+y^2+\ldots)^{\frac{1}{2}n} \left\{(x-a)^2+(y-b)^2+\ldots+u^2\right\}^{\frac{1}{2}n-1}} = \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma(\frac{1}{2}n)(l^2+u^2)^{\frac{1}{2}n-1}(l_1^2-f_1^2)(1+e)^{\frac{1}{2}n-1}}.$$

In the case of $u = 0$, we have, as before, when $\rho^2 - f_1^2$ and $l^2 - f^2$ are of opposite signs, $e = 0$, and therefore $1 + e = 1$; but when $\rho^2 - f_1^2$ and $l^2 - f^2$ are of the same sign, the value before found for e gives

$$1 + e = \frac{\rho^2}{\rho^2 - f_1^2} \left\{ l_1^2 + (\rho^2 - f_1^2)(l^2 - f^2) \right\} / l_1^2.$$

Consider the image of the origin with respect to the sphere $(x - a_1)^2 + (y - b_1)^2 + \ldots = f_1^2$; the coordinates of this image are

$$\frac{f_1^2 a_1}{l_1^2}, \quad \frac{f_1^2 b_1}{l_1^2}, \quad \ldots,$$

and consequently, if $\hbar$ be the distance of this image from the point $(a, b, \ldots)$, we have

$$\hbar^2 = \frac{1}{l_1^2} \left\{ l_1^2\, l^2 + (\rho^2 - f_1^2)(l_1^2 - f_1^2) \right\};$$

whence, by a simple reduction, or the values of the integral are: when $\rho^2 - f_1^2$ and $l^2 - f^2$ have opposite signs,

$$I = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n)} \cdot \frac{f_1^{n-1}}{l_1^{n-1}};$$

when $\rho^2 - f_1^2$ and $l^2 - f^2$ have the same sign,

$$I = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n)} \cdot \frac{f_1^{n-1}}{\hbar^{n-2}(\rho^2 - f_1^2)/l_1^{n-1}},$$

where $\hbar$ is the distance from the point $(a, b, \ldots)$ of the image of the origin with respect to the sphere $(x - a_1)^2 + \ldots = f_1^2$.

Considered in reference to the theory of images, the first formula gives the potential of a multiple layer (the density being inversely as the $(n-1)^{\text{th}}$ power of the distance) spread over the surface of an $(n-1)$-fold space, due to an attracting mass (the law of attraction being the inverse $(n-1)^{\text{th}}$ power of the distance) situated at a given point; and the second formula gives the potential of the attracting mass itself. It is easy to see that the formula agrees with the ordinary theory of electrical images.

Stone Buildings, August 6, 1850.

109. On the Rationalisation of certain Algebraical Equations.

[From the Cambridge and Dublin Mathematical Journal, vol. viii. (1853), pp. 97—101.]

Suppose

$$x + y = 0, \quad x^2 = a, \quad y^2 = b;$$

then if we multiply the first equation by 1, xy, and reduce by the two others, we have

$$x + y = 0, \qquad bx + ay = 0,$$

from which, eliminating x, y,

$$\begin{vmatrix} 1 & 1 \\ b & a \end{vmatrix} = 0;$$

which is the equation between a and b; or, considering x, y as quadratic radicals, the rational equation between x, y. So if the original equation be multiplied by x, y, we have

$$a + xy = 0, \qquad b + xy = 0;$$

or, eliminating 1, xy,

$$\begin{vmatrix} a & 1 \\ b & 1 \end{vmatrix} = 0,$$

which may be in like manner considered as the rational equation between x, y.

The preceding results are of course self-evident, but by applying the same process to the equations

$$x + y + z = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c,$$

we have results of some elegance. Multiply the equation first by 1, yz, zx, xy, reduce and eliminate the quantities x, y, z, xyz; we have the rational equation

$$\begin{vmatrix} 1 & 1 & 1 & . \\ . & c & b & 1 \\ c & . & a & 1 \\ b & a & . & 1 \end{vmatrix} = 0;$$

and again, multiply the equation by x, y, z, xyz, reduce and eliminate the quantities 1, yz, zx, xy, the result is

$$\begin{vmatrix} a & b & c & . \\ a & . & 1 & 1 \\ b & 1 & . & 1 \\ c & 1 & 1 & . \end{vmatrix} = 0,$$

which is of course equivalent to the preceding one (the two determinants are in fact identical in value), but the form is essentially different. The former of the two forms is that given in my paper "On a theorem in the Geometry of Position" (Journal, vol. II. [1841] p. 270 [1]): it was only very recently that I perceived that a similar process led to the latter of the two forms.

Similarly, if we have the equations

$$x + y + z + w = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c, \quad w^2 = d,$$

then multiplying by 1, yz, zx, xy, xw, yw, zw, $xyzw$, reducing and eliminating the quantities in the outside row,

$$\big(x,\ y,\ z,\ w,\ yzw,\ zwx,\ wxy,\ xyz \big),$$

we have the result

$$\begin{vmatrix} 1 & 1 & 1 & 1 & . & . & . & . \\ . & c & b & . & . & . & 1 & 1 \\ c & . & a & . & . & 1 & . & 1 \\ b & a & . & . & 1 & . & . & 1 \\ d & . & . & a & . & 1 & 1 & . \\ . & d & . & b & 1 & . & 1 & . \\ . & . & d & c & 1 & 1 & . & . \\ . & . & . & . & a & b & c & d \end{vmatrix} = 0;$$

so if we multiply the equations by x, y, z, w, yzw, zwx, wxy, xyz, reduce and eliminate the quantities in the outside row

$$\big(1,\ yz,\ zx,\ xy,\ xw,\ yw,\ zw,\ xyzw\big),$$

we have the result

$$\begin{vmatrix} a & . & . & 1 & 1 & 1 & . & . \\ b & 1 & . & . & 1 & . & 1 & . \\ c & 1 & 1 & . & . & . & . & 1 \\ d & . & . & . & 1 & 1 & 1 & . \\ . & d & . & . & . & c & b & 1 \\ . & . & d & . & c & . & a & 1 \\ . & . & . & d & b & a & . & 1 \\ . & a & b & c & . & . & . & 1 \end{vmatrix} = 0,$$

which however is not essentially distinct from the form before obtained, but may be derived from it by an interchange of lines and columns.

And in general for any even number of quadratic radicals the two forms are not essentially distinct, but may be derived from each other by interchanging lines and columns, while for an odd number of quadratic radicals the two forms cannot be so derived from each other, but are essentially distinct.

I was indebted to Mr Sylvester for the remark that the above process applies to radicals of a higher order than the second. To take the simplest case, suppose

$$x + y = 0, \quad x^3 = a, \quad y^3 = b;$$

and multiply first by 1, x^2y, xy^2; this gives

$$x + y = 0, \qquad x^2y + ay = 0, \qquad bx + xy^2 = 0;$$

or, eliminating,

$$\begin{vmatrix} 1 & 1 & . \\ . & a & 1 \\ b & . & 1 \end{vmatrix} = 0\,;$$

next multiply by x, y, x^2y^2; this gives

$$x^2 + xy = 0, \qquad xy + y^2 = 0, \qquad bx^2 + ay^2 = 0;$$

or, eliminating,

$$\begin{vmatrix} 1 & . & 1 \\ . & 1 & 1 \\ b & a & . \end{vmatrix} = 0\,;$$

and lastly, multiply by x^2, y^2, xy; this gives

$$a + x^2y = 0, \qquad b + xy^2 = 0, \qquad x^2y + xy^2 = 0;$$

or, eliminating,

$$\begin{vmatrix} a & 1 & . \\ b & . & 1 \\ . & 1 & 1 \end{vmatrix} = 0\,;$$

where it is to be remarked that the second and third forms are not essentially distinct, since the one may be derived from the other by the interchange of lines and columns.

Applying the preceding process to the system

$$x+y+z=0, \quad x^3=a, \quad y^3=b, \quad z^3=c,$$

multiply first by 1, xyz, $x^2y^2z^2$, x^2z, y^2x, z^2y, x^2y, y^2z, z^2x, reduce and eliminate the quantities in the outside row

$$\big(x,\ y,\ z,\ y^2z^2,\ z^2x^2,\ x^2y^2,\ x^2yz,\ y^2zx,\ z^2xy\big);$$

the result is

$$\begin{vmatrix} 1 & 1 & 1 & . & . & . & . & . & . \\ . & . & . & 1 & 1 & 1 & . & . & . \\ . & . & . & . & . & . & a & b & c \\ . & . & a & 1 & . & . & . & . & 1 \\ b & . & . & . & 1 & . & . & . & 1 \\ . & c & . & . & . & 1 & 1 & . & . \\ . & a & . & 1 & . & . & . & . & 1 \\ . & . & b & . & 1 & . & 1 & . & . \\ c & . & . & . & . & 1 & . & 1 & . \end{vmatrix} = 0\,;$$

next multiply by x, y, z, y^2z^2, z^2x^2, x^2y^2, x^2yz, y^2zx, z^2xy, reduce and eliminate the quantities in the outside row

$$\big(x^2,\ y^2,\ z^2,\ yz,\ zx,\ xy,\ xy^2z^2,\ yz^2x^2,\ zx^2y^2\big),$$

the result is

$$\begin{vmatrix} 1 & . & . & . & 1 & 1 & . & . & . \\ . & 1 & . & 1 & . & 1 & . & . & . \\ . & . & 1 & 1 & 1 & . & . & . & . \\ . & c & b & . & . & . & 1 & . & . \\ c & . & a & . & . & . & . & . & 1 \\ b & a & . & . & . & . & . & 1 & . \\ . & . & a & . & . & . & . & 1 & 1 \\ . & b & . & . & . & . & 1 & . & 1 \\ c & . & . & . & . & . & 1 & 1 & . \end{vmatrix} = 0\,;$$

lastly, multiply by x^2, y^2, z^2, yz, zx, xy, xy^2z^2, yz^2x^2, zx^2y^2, reduce and eliminate the quantities in the outside row

$$\big(1,\ xyz,\ x^2y^2z^2,\ x^2z,\ y^2x,\ z^2y,\ x^2y,\ y^2z,\ z^2x\big),$$

the result is

$$\begin{vmatrix} a & . & . & . & 1 & . & . & 1 & . \\ b & . & . & 1 & . & 1 & . & . & . \\ c & . & 1 & . & . & . & 1 & . & . \\ . & 1 & 1 & . & . & 1 & . & . & . \\ . & 1 & . & 1 & . & . & 1 & . & . \\ . & 1 & . & . & 1 & . & . & . & 1 \\ . & . & 1 & c & . & b & . & . & . \\ . & . & 1 & . & a & . & . & . & c \\ . & . & 1 & . & b & . & a & . & . \end{vmatrix} = 0\,;$$

where, as in the case of two cubic radicals, two forms, viz. the first and third forms of the rational equation, are not essentially distinct, but may be derived from each other by interchanging lines and columns.

And in general, whatever be the number of cubic radicals, two of the three forms are not essentially distinct, but may be derived from each other by interchanging lines and columns.

110. Note on the Transformation of a Trigonometrical Expression.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 61—62.]

The differential equation

$$\frac{dx}{(a+x)\sqrt{c+x}} + \frac{dy}{(a+y)\sqrt{c+y}} + \frac{dz}{(a+z)\sqrt{c+z}} = 0,$$

integrated so as to be satisfied when the variables are simultaneously infinite, gives by direct integration

$$\tan^{-1}\sqrt{\frac{a-c}{c+x}} + \tan^{-1}\sqrt{\frac{a-c}{c+y}} + \tan^{-1}\sqrt{\frac{a-c}{c+z}} = 0,$$

and, by Abel's theorem,

$$\begin{vmatrix} 1 & x & (a+x)\sqrt{c+x} \\ 1 & y & (a+y)\sqrt{c+y} \\ 1 & z & (a+z)\sqrt{c+z} \end{vmatrix} = 0.$$

To show *à posteriori* the equivalence of these two equations, I represent the determinant by the symbol Ω, and expressing it in the form

$$\Omega = \begin{vmatrix} 1 & a+x & (a+x)\sqrt{c+x} \\ 1 & a+y & (a+y)\sqrt{c+y} \\ 1 & a+z & (a+z)\sqrt{c+z} \end{vmatrix},$$

I write for the moment $\xi = \sqrt{\dfrac{a-c}{c+x}}$, etc., and we get

$$\Omega = \begin{vmatrix} 1 & (a-c)(1+\xi^2) & (a-c)^{3/2}(\xi+\xi^3) \\ 1 & (a-c)(1+\eta^2) & (a-c)^{3/2}(\eta+\eta^3) \\ 1 & (a-c)(1+\zeta^2) & (a-c)^{3/2}(\zeta+\zeta^3) \end{vmatrix}.$$

Pulling out the common factors and simplifying,

$$\Omega = (a-c)^{5/2}\,(\xi-\eta)(\eta-\zeta)(\zeta-\xi)\,(\xi\eta+\eta\zeta+\zeta\xi-1);$$

or, replacing ξ, η, ζ by their values, we have identically

$$\begin{vmatrix} 1 & x & (a+x)\sqrt{c+x} \\ 1 & y & (a+y)\sqrt{c+y} \\ 1 & z & (a+z)\sqrt{c+z} \end{vmatrix} = (a-c)^{5/2}\,\frac{(\xi-\eta)(\eta-\zeta)(\zeta-\xi)(\xi\eta+\eta\zeta+\zeta\xi-1)}{1};$$

and the equation

$$\xi\eta+\eta\zeta+\zeta\xi-1=0, \quad \text{i.e.} \quad \sqrt{\tfrac{a-c}{c+x}}\sqrt{\tfrac{a-c}{c+y}} + \sqrt{\tfrac{a-c}{c+y}}\sqrt{\tfrac{a-c}{c+z}} + \sqrt{\tfrac{a-c}{c+z}}\sqrt{\tfrac{a-c}{c+x}} - 1 = 0,$$

is of course equivalent to the trigonometrical equation

$$\tan^{-1}\sqrt{\frac{a-c}{c+x}} + \tan^{-1}\sqrt{\frac{a-c}{c+y}} + \tan^{-1}\sqrt{\frac{a-c}{c+z}} = 0,$$

which shows the equivalence of the two equations in question.

111. On a Theorem of M. Lejeune-Dirichlet's.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 163—165.]

The following formula,

$$\Sigma\, q^{ax^2+2bxy+cy^2} \cdot \Sigma\, q^{a'x'^2+2b'x'y'+c'y'^2} \cdots \;=\; \frac{1}{S}\,\Sigma\,\epsilon\left(\frac{D}{n}\right)\left(\frac{D}{n'}\right)\cdots q^{nn'\cdots/P},$$

is given in Lejeune-Dirichlet's well-known memoir "Recherches sur diverses applications &c." (*Crelle*, t. xxi. [1840] p. 8). The notation is as follows: On the left-hand side (a, b, c), (a', b', c'), ... are a system of properly primitive forms to the negative determinant $-D$ (i.e. a system of positive forms); x, y are positive or negative integers including zero, such that in the sum $\sum q^{ax^2+2bxy+cy^2}$, $ax^2 + 2bxy + cy^2$ is prime to $2D$, and similarly in the other sums; q is indeterminate and the summations extend to the values first mentioned, of x and y. On the right-hand side we have to consider the form of D, viz. we have $D = PS^2$ or else $D = 2PS^2$, where S^2 is the greatest square factor in D and where P is odd: this obviously defines P, and the values of S, ϵ, which are always ± 1 (or, as I prefer to express it, are always $\pm$) are given as follows, viz.

$D = PS^2$,	$P \equiv 1 \pmod 4$,	$S, \epsilon = +,\ +$,
$D = PS^2$,	$P \equiv 3 \pmod 4$,	$S, \epsilon = -,\ +$,
$D = 2PS^2$,	$P \equiv 1 \pmod 4$,	$S, \epsilon = +,\ -$,
$D = 2PS^2$,	$P \equiv 3 \pmod 4$,	$S, \epsilon = -,\ -$;

n, n' are any positive numbers prime to $2D$, $\left(\frac{D}{n}\right)$ is Legendre's symbol as generalized by Jacobi, viz. in general if p be a positive or negative prime not a factor of n, then $\left(\frac{n}{p}\right) = +1$ or -1 according as n is or is not a quadratic residue of p (or, what is the same thing, p being positive, $\left(\frac{n}{p}\right) \equiv n^{\frac{1}{2}(p-1)} \pmod p$), and for $P = pp'p'' \ldots$,

$$\left(\tfrac{n}{P}\right) = \left(\tfrac{n}{p}\right)\left(\tfrac{n}{p'}\right)\left(\tfrac{n}{p''}\right)\cdots$$

and the summation extends to all the values of n, n' of the form above mentioned. In the particular case $D = -1$, it is necessary that the second side should be doubled.

The method of reducing the equation is indicated in the memoir. The following are a few particular cases.

$$D = -1, \qquad \Sigma\, q^{x^2+y^2} \;=\; 4\Sigma\left(\tfrac{-1}{n}\right) q^n,$$

or $$2\left(1+2q+2q^4+2q^9+\ldots\right)\left(q+q^4+q^9+q^{16}+\ldots\right) \;=\; \frac{q}{1-q} - \frac{q^3}{1-q^3} + \frac{q^5}{1-q^5} - \frac{q^7}{1-q^7} + \&\text{c.}$$

$$D = -2, \qquad \Sigma\, q^{x^2+2y^2} \;=\; 2\,\Sigma\left(\tfrac{-2}{n}\right) q^n,$$

or $$\left(1 + 2q^2 + 2q^8 + 2q^{18}\ldots\right)\left(q + q^3 + q^9 + q^{17}\ldots\right) \;=\; \frac{q}{1-q^2} + \frac{q^3}{1-q^6} - \frac{q^5}{1-q^{10}} - \&\text{c.}$$

an example given in the memoir.

For $D = -3$,

$$(q+q^4+q^9+q^{16}+q^{25}\ldots)(1+2q^2+2q^6+2q^{12}\ldots)+2(q^2+q^3+q^6+q^{12}+\ldots)(q^4+q^9+q^{16}+q^{25}\ldots)$$

$$= \frac{q}{1-q^2} - \frac{q+q^2}{1-q^3} + \frac{q^3-q^4}{1-q^6} + \frac{q^7-q^8}{1-q^9} + \&\text{c.}$$

I am not aware that the above theorem is quoted or referred to in any subsequent memoir on Elliptic Functions, or on the class of series to which it relates; and the theorem is so distinct in its origin and form from all other theorems relating to the same class of series, and, independently of the researches in which it originates, so remarkable as a result, that I have thought it desirable to give a detached statement of it in this paper.

112. Demonstration of a Theorem relating to the Products of Sums of Squares.

[From the Philosophical Magazine, vol. IV. (1852), pp. 515—519.]

Mr Kirkman, in his paper "On Pluquaternions and Homoid Products of Sums of n Squares" (*Phil. Mag.* vol. xxxiii. [1848] pp. 447—459 and 494—509), quotes from a note of mine the following passage: "The complete test of the possibility of the product of 2^m squares by 2^m squares reducing itself to a sum of 2^m squares is the following—forming the complete systems of triplets for (2^m-1) things, if eab, ecd, fac, fdb be any four of them, we must have, paying attention to the signs alone,

$$(\pm eab)(\pm ecd) \;=\; (\pm fac)(\pm fdb)\,;$$

i.e. if the first two are of the same sign, the last two must be so also, and *vice versa*; I believe that, for a system of seven, two conditions of this kind being satisfied would imply the satisfaction of all the others: it remains to be shown that the complete system of conditions cannot be satisfied for fifteen things." I propose to explain the meaning of the theorem, and to establish the truth of it, without in any way assuming the existence of imaginary units.

The identity to be established is

$$(w^2+a^2+b^2+\ldots)(w_1^2+a_1^2+b_1^2+\ldots) \;=\; w_{\prime\prime}^2+a_{\prime\prime}^2+b_{\prime\prime}^2+\ldots,$$

where the 2^m quantities $w, a, b, c, \ldots$ and the 2^m quantities $w_1, a_1, b_1, c_1, \ldots$ are given quantities in terms of which the 2^m quantities $w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$ have to be determined.

Without attaching any meaning whatever to the symbols $a_0, b_0, c_0, \ldots$ I write down the expressions

$$ww_1+aa_1+bb_1+cc_1+\ldots, \qquad w_1a_0+a_1b_0+b_1c_0+\ldots,$$

and I multiply as if $a_0, b_0, c_0, \ldots$ really existed, taking care to multiply without making any transposition in the order *inter se* of two symbols a_0, b_0 combined in the way of multiplication. This gives a quasi-product

$$\begin{aligned} ww_1 \;&+\; (aw_1+a_1w)\,a_0+(bw_1+b_1w)\,b_0+\ldots \\ &+\; aa_1a_0^2+bb_1b_0^2+\ldots \\ &+\; ab_1a_0b_0+a_1b\,b_0a_0+\ldots \end{aligned}$$

Suppose, now, that a quasi-equation, such as

$$a_0b_0c_0 \;=\; +,$$

means that in the expression of the quasi-product

$$a_0b_0,\; b_0c_0,\; c_0a_0,\; c_0b_0,\; a_0c_0,\; b_0a_0$$

are to be replaced by

$$c_0,\; a_0,\; b_0,\; -a_0,\; -b_0,\; -c_0;$$

and that a quasi-equation, such as $a_0b_0c_0=-$, means that in the expression of the quasi-product

$$a_0b_0,\; b_0c_0,\; c_0a_0,\; b_0a_0,\; c_0b_0,\; a_0c_0$$

are to be replaced by

$$-c_0,\; -a_0,\; -b_0,\; +a_0,\; +b_0,\; +c_0.$$

It is in the first place clear that the quasi-equation $a_0b_0c_0 = +$ may be written in any one of the six forms

$$a_0b_0c_0 = b_0c_0a_0 = c_0a_0b_0 = -a_0c_0b_0 = -c_0b_0a_0 = -b_0a_0c_0\,,$$

and so for the quasi-equation $a_0b_0c_0 = -$. This being premised, if we form a system of quasi-equations, such as

$$a_0b_0c_0,\ e_0a_0b_0,\ \ldots,$$

where the system of triplets contains each duad once, and once only, and the arbitrary signs are chosen at pleasure; if, moreover, in the expression of the quasi-product we replace $a_0^2, b_0^2, \ldots$ each by -1, it is clear that the quasi-product will assume the form

$$w_{\prime\prime}w_1 + a_{\prime\prime}a_0 + b_{\prime\prime}b_0 + c_{\prime\prime}c_0 + \ldots,$$

$w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$ being determinate functions of $w, a, b, c, \ldots$; $w_1, a_1, b_1, c_1, \ldots$, homogeneous of the first order in the quantities of each set; the value of $w_{\prime\prime}$ being obviously in every case

$$w_{\prime\prime} = ww_1 - aa_1 - bb_1 - cc_1 - \ldots,$$

and $a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$ containing in every case the terms $aw_1 + a_1w$, $bw_1 + b_1w$, $cw_1 + c_1w$, $\ldots$ but the form of the remaining terms depending as well on the triplets entering into the system of quasi-equations as on the values given to the signs $\pm$.

[The demonstration continues on book p. 51 et seq., outside the present pp. 26–50 slice.]

CAYLEY, Collected Mathematical Papers, Vol. II.

Scan-verified retype: source PDF pp. 51–75 (book pp. 35–59).

Papers 108–114.

108.

ON CERTAIN MULTIPLE INTEGRALS CONNECTED WITH THE THEORY OF ATTRACTIONS.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 174–178.]

It is easy to deduce from Mr Boole's formula, given in my paper "On a Multiple Integral connected with the theory of Attractions," *Journal*, t. ii. [1847], pp. 219–223, [44], the equation

$$\int \frac{d\xi\, d\eta \cdots}{\left[(\xi-\alpha)^2+(\eta-\beta)^2+\cdots+v^2\right]^{\frac{1}{2}n-q}} \;=\; \theta_1{}^q\,\Gamma\big(\tfrac{1}{2}n-q\big)\,\Gamma(q+1)\int_e^\infty \frac{s^{n-q-1}\,(\theta_1{}^2-\sigma)^q\,ds}{\sqrt{\big(s+\frac{f^2}{\theta_1{}^2}\big)\big(s+\frac{g^2}{\theta_1{}^2}\big)\cdots}}\,,$$

where n is the number of variables of the multiple integral, and the condition of the integration is

$$\frac{(\xi-\alpha)^2}{f^2}+\frac{(\eta-\beta)^2}{g^2}+\cdots \;\leqq\; 1\,;$$

also where

$$\sigma \;=\; \frac{(\alpha-\alpha_1)^2}{s+\frac{f^2}{\theta_1{}^2}} \;+\; \frac{(\beta-\beta_1)^2}{s+\frac{g^2}{\theta_1{}^2}} \;+\;\cdots\;+\;\frac{v^2}{s}\,,$$

and e is the positive root of

$$\theta_1{}^2 \;=\; \frac{(\alpha-\alpha_1)^2}{e+\frac{f^2}{\theta_1{}^2}} \;+\; \frac{(\beta-\beta_1)^2}{e+\frac{g^2}{\theta_1{}^2}} \;+\;\cdots\;+\;\frac{v^2}{e}\,.$$

Suppose $f=g=\cdots=\theta_1$, and write $(\alpha-\alpha_1)^2+\cdots=k^2$, we obtain

$$\int \frac{d\xi\, d\eta\cdots}{\left[(\xi-\alpha)^2+\cdots+v^2\right]^{\frac{1}{2}n-q}} \;=\; \frac{\pi^{\frac{1}{2}n}}{\Gamma\big(\frac{1}{2}n-q\big)\,\Gamma(q+1)}\int_e^\infty \frac{s^{q-1}\,(\theta_1{}^2-\sigma)^q\,ds}{(1+s)^{\frac{1}{2}n}}\,,$$

the limiting condition for the multiple integral being

$$(\xi-\alpha_1)^2+\cdots \;\leqq\; \theta_1{}^2,$$

and the function σ, and limit e, being given by

$$\sigma \;=\; \frac{(\alpha-\alpha_1)^2}{1+s}+\frac{(\beta-\beta_1)^2}{1+s}+\cdots+\frac{v^2}{s}\,,\qquad \theta_1{}^2 \;=\; \frac{(\alpha-\alpha_1)^2}{1+e}+\frac{(\beta-\beta_1)^2}{1+e}+\cdots+\frac{v^2}{e}\,,$$

e denoting, as before, the positive root. Observing that the quantity under the integral sign on the second side vanishes for $s=e$, there is no difficulty in deducing, by a differentiation with respect to θ_1, the formula

$$\int \frac{d\Sigma}{\left[(\xi-\alpha)^2+\cdots+v^2\right]^{\frac{1}{2}n-q+1}\,\theta_1} \;=\; \frac{2\theta_1{}^{q+1}\pi^{\frac{1}{2}n}}{\Gamma\big(\frac{1}{2}n-q\big)\,\Gamma(q)}\int_e^\infty \frac{s^{q-1}\,(\theta_1{}^2-\sigma)^{q-1}\,ds}{(1+s)^{\frac{1}{2}n}}\,,$$

where $d\Sigma$ is the element of the surface ${\theta_1}^2 = (\xi-\alpha_1)^2+\cdots+{\alpha_1}^2$ and the integration is extended over the entire surface.

A slight change of form is convenient. We have

$$ {\theta_1}^2 - \sigma \;=\; {\theta_1}^2 - \frac{k^2}{1+s} - \frac{v^2}{s} \;=\; \frac{1}{s(1+s)}\left({\theta_1}^2 s^2 + \chi s - v^2\right), $$

if we suppose

$$ \chi \;=\; {\theta_1}^2 - k^2 - v^2. $$

The formulæ then become

$$ \int \frac{d\xi\,\cdots}{\left[(\xi-\alpha)^2+\cdots+v^2\right]^{\frac12 n-q}} \;=\; \frac{\pi^{\frac12 n}}{\Gamma\left(\frac12 n-q\right)\Gamma(q+1)} \int_e^\infty \frac{({\theta_1}^2 s^2+\chi s-v^2)^q\,ds}{s(1+s)^{\frac12 n+q}}\,, $$

$$ \int \frac{d\Sigma}{\left[(\xi-\alpha)^2+\cdots+v^2\right]^{\frac12 n-q+1}\theta_1} \;=\; \frac{2\pi^{\frac12 n}\theta_1}{\Gamma\left(\frac12 n-q\right)\Gamma(q)} \int_e^\infty \frac{({\theta_1}^2 s^2+\chi s-v^2)^{q-1}\,ds}{(1+s)^{\frac12 n+q-1}}\,, $$

in which e is the positive root of the equation

$$ {\theta_1}^2 s^2 + \chi s - v^2 \;=\; 0. $$

I propose to transform these formulæ by means of the theory of images; it will be convenient to investigate some preliminary formulæ. Suppose $\lambda^2 = \alpha^2+\beta^2+\cdots$, ${\lambda_1}^2 = {\alpha_1}^2+{\beta_1}^2+\cdots$; also consider the new constants $a, b, \ldots,\ a_1, b_1, \ldots,\ u, f_1$, determined by the equations

$$ \frac{\partial^2 u}{\partial\lambda^2+v^2} = \alpha, \qquad \frac{\partial^2 u_1}{{\lambda_1}^2-{f_1}^2} = a_1, \qquad \ldots \qquad \frac{\partial^2 v}{\lambda^2+v^2} = u, \quad \frac{\partial^2 f_1}{{\lambda_1}^2-{f_1}^2} = f_1\,, $$

where S is arbitrary. Then, putting

$$ l^2 = a^2+b^2+\cdots\,, \qquad {l_1}^2 = {a_1}^2+{b_1}^2+\cdots\,, $$

it is easy to see that

$$ (\lambda^2+v^2)(l^2+u^2) \;=\; S^2, \qquad ({\lambda_1}^2-{f_1}^2)({l_1}^2-{f_1}^2) \;=\; S^2{\alpha_1}^2/\alpha^2, $$

and

$$ \frac{\partial^2 u}{l^2+u^2} \;=\; \lambda, \qquad \frac{\partial^2 f_1}{{l_1}^2-{f_1}^2} \;=\; \lambda_1\,, $$

and so on.

Proceeding to express the single integrals in terms of the new constants, we have in the first place $k^2 = \theta^2 k'^2$, where

$$ k'^2 \;=\; \frac{p^2}{(l^2+u^2)} + \frac{{l_1}^2}{({l_1}^2-{f_1}^2)^2} + \cdots\,; $$

or if we write

$$ aa_1+bb_1+\cdots \;=\; l\,l_1\,\cos\omega, $$

we have

$$ k'^2 \;=\; \frac{l^2}{l^2+u^2} + \frac{{l_1}^2}{{l_1}^2-{f_1}^2} - \frac{2\,l\,l_1\cos\omega}{(l^2+u^2)^{\frac12}({l_1}^2-{f_1}^2)^{\frac12}}\,. $$

Hence also $\chi = \theta^2\chi'$, where

$$ \chi' \;=\; \frac{{f_1}^2}{({l_1}^2-{f_1}^2)^2} - k'^2 - \frac{u^2}{(l^2+u^2)^2}\,, $$

whence

$$-\chi' \;=\; \frac{1}{l^2+u^2} + \frac{1}{{l_1}^2-{f_1}^2} + \frac{2\,l\,l_1\cos\omega}{(l^2+u^2)^{\frac{1}{2}}({l_1}^2-{f_1}^2)^{\frac{1}{2}}} \;=\; \frac{1}{(l^2+u^2)({l_1}^2-{f_1}^2)}\bigl[\,p^2+u^2-{f_1}^2\,\bigr],$$

where $p^2 = l^2 + {l_1}^2 - 2\,l\,l_1\cos\omega$, that is

$$p^2 \;=\; (a-a_1)^2 + (b-b_1)^2 + \cdots\,;$$

consequently ${\theta_1}^2 s^2 + \chi s - v^2 = -\theta^2 S^2\,\Pi$, where Π is given by

$$\Pi \;=\; \frac{{f_1}^2}{({l_1}^2-{f_1}^2)^2}\,s^2 - \frac{(p^2+u^2-{f_1}^2)}{(l^2+u^2)({l_1}^2-{f_1}^2)}\,s - \frac{u^2}{(l^2+u^2)^2}\,,$$

and it is clear that e will be the positive root of

$$0 \;=\; \frac{{f_1}^2}{({l_1}^2-{f_1}^2)^2}\,e^2 - \frac{(p^2+u^2-{f_1}^2)}{(l^2+u^2)({l_1}^2-{f_1}^2)}\,e - \frac{u^2}{(l^2+u^2)^2}\,.$$

It may be noticed that, in the particular case of $u=0$, the roots of this equation are 0 and $\dfrac{(p^2-{f_1}^2)({l_1}^2-{f_1}^2)}{{l_1}^2{f_1}^2}$. Consequently if $p^2-{f_1}^2$ and ${l_1}^2-{f_1}^2$ are of opposite signs, we have $e=0$; but if $p^2-{f_1}^2$ and ${l_1}^2-{f_1}^2$ are of the same sign, $e = \dfrac{(p^2-{f_1}^2)({l_1}^2-{f_1}^2)}{{l_1}^2{f_1}^2}$.

In order to transform the double integrals, considering the new variables $x, y, \ldots$, I write $x^2+y^2+\cdots = r^2$ and

$$\xi \;=\; \frac{Sx}{r^2}, \quad \ldots$$

whence also, if $\xi^2+\eta^2+\cdots = \rho^2$ (which gives $r\rho = S$), we have

$$\alpha \;=\; \frac{d\xi}{d\rho^2}\,, \quad \ldots\,;$$

also it is immediately seen that

$$(\xi-\alpha)^2 + \cdots + v^2 \;=\; \frac{S^2}{(l^2+u^2)\,r^2}\bigl[(x-a)^2+\cdots+u^2\bigr],$$

$$(\xi-\alpha_1)^2 + \cdots - {\theta_1}^2 \;=\; \frac{S^2}{({l_1}^2-{f_1}^2)\,r^2}\bigl[(x-a_1)^2+\cdots-{f_1}^2\bigr];$$

and from the latter equation it follows that the limiting condition for the first integral is $(x-a_1)^2+\cdots \leqq {f_1}^2$ (there is no difficulty in seeing that the sign $<$ in the former limiting condition gives rise here to the sign $>$), and that the second integral has to be extended over the surface $(x-a_1)^2+\cdots = {f_1}^2$. Also if dS represent the element of this surface, we may obtain

$$d\xi\,d\eta\,\cdots \;=\; \frac{S^n}{r^{2n}}\,dx\,dy\,\cdots\,, \qquad d\Sigma \;=\; \frac{S^{n-1}}{r^{2n-2}}\,dS\,;$$

and, combining the above formulæ, we obtain

$$\int\frac{dx\,dy\,\cdots}{(x^2+y^2+\cdots)^{\frac{1}{2}n+q}\bigl[(x-a)^2+(y-b)^2+\cdots+u^2\bigr]^{\frac{1}{2}n-q}} \;=\; \frac{\pi^{\frac{1}{2}n}}{\Gamma\bigl(\frac{1}{2}n-q\bigr)\,\Gamma(q+1)\,(l^2+u^2)^{\frac{1}{2}n-q}}\int_e^\infty\frac{\Pi^q\,ds}{s\,(1+s)^{\frac{1}{2}}}$$

the limiting condition of the multiple integral being

$$(x-a_1)^2+(y-b_1)^2+\cdots \;\leqq\; {f_1}^2\,;$$

and

$$\int \frac{dS}{(x^2+y^2+\cdots)^{\frac{1}{2}n+q-1}\left[(x-a)^2+(y-b)^2+\cdots+u^2\right]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma\left(\frac{1}{2}n-q\right)\Gamma(q)\,(l^2+u^2)^{\frac{1}{2}n-q}\,({l_1}^2-{f_1}^2)} \int_e^\infty$$

where dS is the element of the surface $(x-a_1)^2+(y-b_1)^2+\cdots={f_1}^2$, and the integration extends over the entire surface. In these formulæ, l, l_1, p, Π denote as follows:

$$l^2=a^2+b^2+\cdots, \quad {l_1}^2={a_1}^2+{b_1}^2+\cdots, \quad p^2=(a-a_1)^2+(b-b_1)^2+\cdots,$$

$$\Pi = \frac{{f_1}^2}{({l_1}^2-{f_1}^2)^2}\,s^2 - \frac{(p^2+u^2-{f_1}^2)}{(l^2+u^2)\,({l_1}^2-{f_1}^2)}\,s - \frac{u^2}{(l^2+u^2)^2},$$

and e is the positive root of the equation $\Pi=0$.

The only obviously integrable case is that for which in the second formula $q=1$; this gives

$$\int \frac{dS}{(x^2+y^2+\cdots)^{\frac{1}{2}n}\left[(x-a)^2+(y-b)^2+\cdots+u^2\right]^{\frac{1}{2}n-1}} = \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma\left(\frac{1}{2}n\right)(l^2+u^2)^{\frac{1}{2}n-1}\,({l_1}^2-{f_1}^2)\,(1+e)^{\frac{1}{2}n-1}}.$$

In the case of $u=0$, we have, as before, when $p^2-{f_1}^2$ and ${l_1}^2-{f_1}^2$ are of opposite signs, $e=0$, and therefore $1+e=1$; but when $p^2-{f_1}^2$ and ${l_1}^2-{f_1}^2$ are of the same sign, the value before found for e gives

$$1+e = \frac{1}{{l_1}^2{f_1}^2}\left[{l_1}^2{f_1}^2+(p^2-{f_1}^2)({l_1}^2-{f_1}^2)\right].$$

Consider the image of the origin with respect to the sphere $(x-a_1)^2+(y-b_1)^2+\cdots={f_1}^2$; the coordinates of this image are

$$\frac{a_1}{{l_1}^2}({l_1}^2-{f_1}^2), \qquad \frac{b_1}{{l_1}^2}({l_1}^2-{f_1}^2), \quad \ldots,$$

and consequently, if μ be the distance of this image from the point $(a, b, \ldots)$, we have

$$\mu^2 = \left[a-\tfrac{a_1}{{l_1}^2}({l_1}^2-{f_1}^2)\right]^2+\cdots = \frac{1}{{l_1}^2}\left[l^2{f_1}^2+(p^2-{f_1}^2)({l_1}^2-{f_1}^2)\right];$$

whence, by a simple reduction,

$$1+e = \frac{{l_1}^2\mu^2}{{l_1}^2{f_1}^2},$$

or the values of the integral are

$$p^2-{f_1}^2 \text{ and } {l_1}^2-{f_1}^2 \text{ opposite signs}, \quad I = \frac{2\pi^{\frac{1}{2}n}}{\Gamma\left(\frac{1}{2}n\right)}\cdot\frac{f_1}{l^{n-2}({l_1}^2-{f_1}^2)},$$

$$p^2-{f_1}^2 \text{ and } {l_1}^2-{f_1}^2 \text{ the same sign}, \quad I = \frac{2\pi^{\frac{1}{2}n}}{\Gamma\left(\frac{1}{2}n\right)}\cdot\frac{{f_1}^{n-1}}{l^{n-2}\,\mu^{n-2}({l_1}^2-{f_1}^2)},$$

where μ is the distance from the point $(a, b, \ldots)$ of the image of the origin with respect to the sphere $(x-a_1)^2+\cdots-{f_1}^2=0$.

Stone Buildings, August 6, 1850.

109.

ON THE RATIONALISATION OF CERTAIN ALGEBRAICAL EQUATIONS.

[From the Cambridge and Dublin Mathematical Journal, vol. viii. (1853), pp. 97–101.]

SUPPOSE

$$x + y = 0, \qquad x^2 = a, \qquad y^2 = b\,;$$

then if we multiply the first equation by 1, xy, and reduce by the two others, we have

$$x + y = 0, \qquad bx + ay = 0,$$

from which, eliminating x, y,

$$\begin{vmatrix} 1 & 1 \\ b & a \end{vmatrix} = 0\,;$$

which is the equation between a and b; or, considering x, y as quadratic radicals, the rational equation between x, y. So if the original equation be multiplied by x, y, we have

$$a + xy = 0, \qquad b + xy = 0\,;$$

or, eliminating 1, xy,

$$\begin{vmatrix} a & 1 \\ b & 1 \end{vmatrix} = 0,$$

which may be in like manner considered as the rational equation between x, y.

The preceding results are of course self-evident, but by applying the same process to the equations

$$x + y + z = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c,$$

we have results of some elegance. Multiply the equation first by 1, yz, zx, xy, reduce and eliminate the quantities x, y, z, xyz, we have the rational equation

$$\begin{vmatrix} 1 & 1 & 1 & \cdot \\ \cdot & c & b & \cdot \\ c & \cdot & a & \cdot \\ b & a & \cdot & \cdot \end{vmatrix} = 0\,;$$

and again, multiply the equation by x, y, z, xyz, reduce and eliminate the quantities 1, yz, zx, xy, the result is

$$\begin{vmatrix} a & b & c & \cdot \\ a & \cdot & 1 & 1 \\ b & 1 & \cdot & 1 \\ c & 1 & 1 & \cdot \end{vmatrix} = 0,$$

which is of course equivalent to the preceding one (the two determinants are in fact identical in value), but the form is essentially different. The former of the two forms is that given in my paper "On a theorem in the Geometry of Position" (*Journal*, vol. ii. [1841] p. 270 [1]): it was only very recently that I perceived that a similar process led to the latter of the two forms.

Similarly, if we have the equations

$$x + y + z + w = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c, \quad w^2 = d,$$

then multiplying by 1, yz, zx, xy, xw, yw, zw, $xyzw$, reducing and eliminating the quantities in the outside row,

$$x,\ y,\ z,\ w,\ yzw,\ zwx,\ wxy,\ xyz,$$

we have the result

$$\begin{vmatrix} 1 & 1 & 1 & 1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & c & b & \cdot & \cdot & 1 & \cdot & 1 \\ c & \cdot & a & \cdot & 1 & \cdot & \cdot & 1 \\ b & a & \cdot & \cdot & \cdot & \cdot & 1 & 1 \\ d & \cdot & \cdot & a & \cdot & 1 & 1 & \cdot \\ \cdot & d & \cdot & b & 1 & \cdot & 1 & \cdot \\ \cdot & \cdot & d & c & 1 & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & a & b & c & d \end{vmatrix} = 0;$$

so if we multiply the equations by x, y, z, w, yzw, zwx, wxy, and xyz, reduce and eliminate the quantities in the outside row,

$$1,\ yz,\ zx,\ xy,\ xw,\ yw,\ zw,\ xyzw,$$

we have the result

$$\begin{vmatrix} a & \cdot & \cdot & \cdot & 1 & 1 & 1 & \cdot \\ b & \cdot & 1 & 1 & \cdot & 1 & \cdot & \cdot \\ c & 1 & \cdot & 1 & 1 & \cdot & \cdot & \cdot \\ d & \cdot & \cdot & \cdot & 1 & 1 & 1 & \cdot \\ \cdot & d & \cdot & \cdot & \cdot & c & b & 1 \\ \cdot & \cdot & d & \cdot & c & \cdot & a & 1 \\ \cdot & \cdot & \cdot & d & b & a & \cdot & 1 \\ \cdot & a & b & c & \cdot & \cdot & \cdot & 1 \end{vmatrix} = 0,$$

which however is not essentially distinct from the form before obtained, but may be derived from it by an interchange of lines and columns.

And in general for any even number of quadratic radicals the two forms are not essentially distinct, but may be derived from each other by interchanging lines and columns, while for an odd number of quadratic radicals the two forms cannot be so derived from each other, but are essentially distinct.

I was indebted to Mr Sylvester for the remark that the above process applies to radicals of a higher order than the second. To take the simplest case, suppose

$$x + y = 0, \qquad x^3 = a, \qquad y^3 = b;$$

and multiply first by 1, x^2y, xy^2; this gives

$$x + y = 0, \qquad x^2y + xy^2 = 0, \qquad ay + x^2y^2 = 0,$$

or, eliminating,

$$\begin{vmatrix} 1 & 1 & \cdot \\ \cdot & \cdot & 1 \\ \cdot & a & 1 \\ b & \cdot & 1 \end{vmatrix} = 0;$$

[*the original prints a 3×3 array of dots and units corresponding to the elimination scheme; the determinant condition is what is significant.*]

next multiply by x, y, x^2y^2; this gives

$$x^2 + xy = 0, \qquad y^2 + xy = 0, \qquad bx^2 + ay^2 = 0,$$

or, eliminating,

$$\begin{vmatrix} 1 & \cdot & 1 \\ \cdot & 1 & 1 \\ b & a & \cdot \end{vmatrix} = 0\,;$$

and lastly, multiply by x^2, y^2, xy; this gives

$$a + x^2y = 0, \qquad b + xy^2 = 0, \qquad x^2y + xy^2 = 0,$$

or, eliminating,

$$\begin{vmatrix} a & 1 & \cdot \\ b & \cdot & 1 \\ \cdot & 1 & 1 \end{vmatrix} = 0,$$

where it is to be remarked that the second and third forms are not essentially distinct, since the one may be derived from the other by the interchange of lines and columns.

Applying the preceding process to the system

$$x + y + z = 0, \qquad x^3 = a, \qquad y^3 = b, \qquad z^3 = c,$$

multiply first by 1, xyz, $x^2y^2z^2$, x^2z, y^2x, z^2y, x^2y, y^2z, z^2x, reduce and eliminate the quantities in the outside row,

$$x,\ y,\ z,\ y^2z^2,\ z^2x^2,\ x^2y^2,\ x^2yz,\ y^2zx,\ z^2xy,\ z^2x^2,\ x^2y^2,$$

the result is a determinantal equation

$$\Delta_1 = 0$$

[whose schematic form, written out by Cayley as a 9×9 array of 1's, $\cdot$'s, and a, b, c's, is — for legibility — abridged here; see facsimile, p. 43].

next multiply by x, y, z, y^2z^2, z^2x^2, x^2y^2, x^2yz, y^2zx, z^2xy, reduce and eliminate the quantities in the outside row,

$$x^2,\ y^2,\ z^2,\ yz,\ zx,\ xy,\ xy^2z^2,\ yz^2x^2,\ zx^2y^2,$$

the result is a second determinantal equation $\Delta_2 = 0$ of the same size;

lastly, multiply by x^2, y^2, z^2, yz, zx, xy, $x^2y^2z^2$, $y^2z^2x^2$, $x^2y^2z^2$, reduce and eliminate the quantities in the outside row,

$$1,\ x^2y^2,\ y^2z^2,\ z^2x^2,\ x^2yz,\ y^2zx,\ z^2xy,\ xy^2z,\ yz^2x,$$

the result is a third determinantal equation $\Delta_3 = 0$:

where, as in the case of two cubic radicals, two forms, viz. the first and third forms of the rational equation, are not essentially distinct, but may be derived from each other by interchanging lines and columns.

And in general, whatever be the number of cubic radicals, two of the three forms are not essentially distinct, but may be derived from each other by interchanging lines and columns.

[The three large determinantal arrays printed at pp. 43–44 of the original are partially illegible in the present scan and have been replaced by their schematic descriptions; the substantive content — the existence and equivalence relations — is preserved.]

110.

NOTE ON THE TRANSFORMATION OF A TRIGONOMETRICAL EXPRESSION.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 61–62.]

THE differential equation

$$\frac{dx}{(a+x)\sqrt{c+x}} + \frac{dy}{(a+y)\sqrt{c+y}} + \frac{dz}{(a+z)\sqrt{c+z}} = 0,$$

integrated so as to be satisfied when the variables are simultaneously infinite, gives by direct integration

$$\tan^{-1}\sqrt{\frac{a-c}{c+x}} + \tan^{-1}\sqrt{\frac{a-c}{c+y}} + \tan^{-1}\sqrt{\frac{a-c}{c+z}} = 0\,;$$

and, by Abel's theorem,

$$\begin{vmatrix} 1 & x & (a+x)\sqrt{c+x} \\ 1 & y & (a+y)\sqrt{c+y} \\ 1 & z & (a+z)\sqrt{c+z} \end{vmatrix} = 0.$$

To show *à posteriori* the equivalence of these two equations, I represent the determinant by the symbol $\square$, and expressing it in the form

$$\square = \begin{vmatrix} 1, & a+x, & (a+x)\sqrt{c+x} \\ 1, & a+y, & (a+y)\sqrt{c+y} \\ 1, & a+z, & (a+z)\sqrt{c+z} \end{vmatrix},$$

I write for the moment $\xi = \sqrt{\dfrac{a-c}{c+x}}$, $\eta = \sqrt{\dfrac{a-c}{c+y}}$, $\zeta = \sqrt{\dfrac{a-c}{c+z}}$, so that

$$(a+x) = (a-c)\left(1+\tfrac{1}{\xi^2}\right), \qquad (a+x)\sqrt{c+x} = (a-c)^{\frac{3}{2}}\left(\tfrac{1}{\xi^3}+\tfrac{1}{\xi}\right),$$

and similarly for $(a+y)$, $(a+z)$, etc. Then

$$\square = \frac{(a-c)^{\frac{5}{2}}}{\xi^3\eta^3\zeta^3}\begin{vmatrix} \xi^2, & \xi^2+1, & \xi^2+1 \\ \eta^2, & \eta^2+1, & \eta^2+1 \\ \zeta^2, & \zeta^2+1, & \zeta^2+1 \end{vmatrix}$$

$$= \frac{(a-c)^{\frac{5}{2}}}{\xi^3\eta^3\zeta^3}\left(\xi+\eta+\zeta-\xi\eta\zeta\right)\begin{vmatrix} 1, & \xi, & \xi^2 \\ 1, & \eta, & \eta^2 \\ 1, & \zeta, & \zeta^2 \end{vmatrix};$$

or, replacing ξ, η, ζ by their values, we have identically

$$\begin{vmatrix} 1, & x, & (a+x)\sqrt{c+x} \\ 1, & y, & (a+y)\sqrt{c+y} \\ 1, & z, & (a+z)\sqrt{c+z} \end{vmatrix} = K(x,y,z;a,c)\left[\sqrt{\tfrac{a-c}{c+x}} + \sqrt{\tfrac{a-c}{c+y}} + \sqrt{\tfrac{a-c}{c+z}} - \sqrt{\tfrac{a-c}{c+x}}\sqrt{\tfrac{a-c}{c+y}}\sqrt{\tfrac{a-c}{c+z}}\right],$$

where $K(x,y,z;a,c)$ is the symmetric prefactor displayed above; and the equation

$$\sqrt{\tfrac{a-c}{c+x}} + \sqrt{\tfrac{a-c}{c+y}} + \sqrt{\tfrac{a-c}{c+z}} - \sqrt{\tfrac{a-c}{c+x}}\sqrt{\tfrac{a-c}{c+y}}\sqrt{\tfrac{a-c}{c+z}} = 0$$

is of course equivalent to the trigonometrical equation

$$\tan^{-1}\sqrt{\tfrac{a-c}{c+x}} + \tan^{-1}\sqrt{\tfrac{a-c}{c+y}} + \tan^{-1}\sqrt{\tfrac{a-c}{c+z}} = 0,$$

which shows the equivalence of the two equations in question.

111.

ON A THEOREM OF M. LEJEUNE-DIRICHLET'S.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 163–165.]

THE following formula,

$$\Sigma\, q^{ax^2+2bxy+cy^2} \cdot \Sigma\, q^{a'x'^2+2b'x'y'+c'y'^2} \cdots \;=\; 2\,\Sigma\left[S^{n-c}\,\epsilon^{\frac{1}{2}(n-1)}\right]\left(\tfrac{n}{P}\right)\, q^{nn'\cdots}, \qquad (3)$$

is given in Lejeune-Dirichlet's well-known memoir "Recherches sur diverses applications &c." (*Crelle*, t. xxi. [1840] p. 8). The notation is as follows: — On the left-hand side (a, b, c), (a', b', c'), ... are a system of properly primitive forms to the negative determinant $-D$ (i.e. a system of positive forms); x, y are positive or negative integers including zero, such that in the sum $\Sigma\, q^{ax^2+2bxy+cy^2}$, $ax^2 + 2bxy + cy^2$ is prime to $2D$, and similarly in the other sums; q is indeterminate and the summations extend to the values first mentioned, of x and y. On the right-hand side we have to consider the form of D, viz. we have $D = PS^2$ or else $D = 2PS^2$, where S^2 is the greatest square factor in D and where P is odd: this obviously defines P, and the values of S, ϵ, which are always ± 1 (or, as I prefer to express it, are always $\pm$) are given as follows, viz.

$D = PS^2$,	$P \equiv 1 \pmod 4$,	$S, \epsilon = +, +,$
$D = PS^2$,	$P \equiv 3 \pmod 4$,	$S, \epsilon = -, +,$
$D = 2PS^2$,	$P \equiv 1 \pmod 4$,	$S, \epsilon = +, -,$
$D = 2PS^2$,	$P \equiv 3 \pmod 4$,	$S, \epsilon = -, -;$

n, n' are any positive numbers prime to $2D$, $\left(\frac{D}{n}\right)$ is Legendre's symbol as generalized by Jacobi, viz. in general if p be a positive or negative prime not a factor of n, then $\left(\frac{n}{p}\right) = +$ or $-$ according as n is or is not a quadratic residue of p (or, what is the same thing, p being positive, $\left(\frac{n}{p}\right) \equiv n^{\frac{1}{2}(p-1)} \pmod p$), and for $P = pp'p'' \ldots$,

$$\left(\tfrac{n}{P}\right) \;=\; \left(\tfrac{n}{p}\right)\left(\tfrac{n}{p'}\right)\left(\tfrac{n}{p''}\right)\cdots$$

and the summation extends to all the values of n, n' of the form above mentioned. In the particular case $D = -1$, it is necessary that the second side should be doubled.

The method of reducing the equation is indicated in the memoir. The following are a few particular cases.

$D = -1$, $\quad \Sigma\, q^{x^2+y^2} \;=\; 4\,\Sigma\left(\frac{-1}{n}\right) q^n$,

$$\text{or}\qquad 2\left(1+2q+2q^4+2q^9+\cdots\right)\left(q+q^4+q^9+q^{16}+\cdots\right) \;=\; \frac{q}{1-q} - \frac{q^3}{1-q^3} + \frac{q^5}{1-q^5} - \frac{q^7}{1-q^7} + \&\text{c.}$$

$D = -2$, $\quad \Sigma\, q^{x^2+2y^2} \;=\; 2\,\Sigma\left(\frac{-2}{n}\right) q^n$,

$$\text{or}\qquad \left(1 + 2q^2 + 2q^8 + 2q^{18}\cdots\right)\left(q + q^3 + q^9 + q^{17}\cdots\right) \;=\; \frac{q}{1-q^2} + \frac{q^3}{1-q^6} - \frac{q^5}{1-q^{10}} - \&\text{c.}$$

an example given in the memoir.

$D = -3$,

$$(q+q^4+q^9+q^{16}+q^{25}\cdots)\left(1+2q^2+2q^6+2q^{12}\cdots\right) \;+\; 2\,(q^2+q^3+q^6+q^{12}+\cdots)(q^4+q^9+q^{16}+q^{25}\cdots)$$

$$= \frac{q}{1-q^2} - \frac{q+q^2}{1-q^3} + \frac{q^3-q^4}{1-q^6} + \frac{q^7-q^8}{1-q^9} + \&c.$$

I am not aware that the above theorem is quoted or referred to in any subsequent memoir on Elliptic Functions, or on the class of series to which it relates; and the theorem is so distinct in its origin and form from all other theorems relating to the same class of series, and, independently of the researches in which it originates, so remarkable as a result, that I have thought it desirable to give a detached statement of it in this paper.

112.

DEMONSTRATION OF A THEOREM RELATING TO THE PRODUCTS OF SUMS OF SQUARES.

[From the Philosophical Magazine, vol. iv. (1852), pp. 515–519.]

Mr Kirkman, in his paper "On Pluquaternions and Homoid Products of Sums of n Squares" (*Phil. Mag.* vol. xxxiii. [1848] pp. 447–459 and 494–509), quotes from a note of mine the following passage: — "The complete test of the possibility of the product of 2^m squares by 2^m squares reducing itself to a sum of 2^m squares is the following — forming the complete systems of triplets for (2^m-1) things, if eab, ecd, fac, fdb be any four of them, we must have, paying attention to the signs alone,

$$(\pm\, eab)(\pm\, ecd) \;=\; (\pm\, fac)(\pm\, fdb)\,;$$

i.e. if the first two are of the same sign, the last two must be so also, and *vice versa*; I believe that, for a system of seven, two conditions of this kind being satisfied would imply the satisfaction of all the others: it remains to be shown that the complete system of conditions cannot be satisfied for fifteen things." I propose to explain the meaning of the theorem, and to establish the truth of it, without in any way assuming the existence of imaginary units.

The identity to be established is

$$(w^2+a^2+b^2+\cdots)(w_1^2+a_1^2+b_1^2+\cdots) \;=\; w_{\prime\prime}^2+a_{\prime\prime}^2+b_{\prime\prime}^2+\cdots,$$

where the 2^m quantities $w, a, b, c, \ldots$ and the 2^m quantities $w_1, a_1, b_1, c_1, \ldots$ are given quantities in terms of which the 2^m quantities $w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$ have to be determined.

Without attaching any meaning whatever to the symbols $a_0, b_0, c_0, \ldots$ I write down the expressions

$$ww_1+aa_0+bb_0+cc_0+\cdots, \qquad w_1a_0+a_1b_0+b_1c_0+\cdots,$$

and I multiply as if $a_0, b_0, c_0, \ldots$ really existed, taking care to multiply without making any transposition in the order *inter se* of two symbols a_0, b_0 combined in the way of multiplication. This gives a quasi-product

$$\begin{aligned} ww_1 &+ (aw_1+a_1w)\,a_0+(bw_1+b_1w)\,b_0+\cdots \\ &+ aa_1a_0^2+bb_1b_0^2+\cdots \\ &+ ab_1a_0b_0+a_1b\,b_0a_0+\cdots. \end{aligned}$$

Suppose, now, that a quasi-equation, such as

$$a_0b_0c_0 \;=\; +,$$

means that in the expression of the quasi-product

$$a_0b_0,\; b_0c_0,\; c_0a_0,\; c_0b_0,\; a_0c_0,\; b_0a_0$$

are to be replaced by

$$c_0,\; a_0,\; b_0,\; -a_0,\; -b_0,\; -c_0\,;$$

and that a quasi-equation, such as $a_0b_0c_0=-$, means that in the expression of the quasi-product

$$b_0c_0,\; c_0a_0,\; a_0b_0,\; c_0b_0,\; a_0c_0,\; b_0a_0$$

are to be replaced by

$$-a_0,\; -b_0,\; -c_0,\; a_0,\; b_0,\; c_0\,.$$

It is in the first place clear that the quasi-equation $a_0b_0c_0 = +$ may be written in any one of the six forms

$$a_0b_0c_0 \;=\; b_0c_0a_0 \;=\; c_0a_0b_0 \;=\; -a_0c_0b_0 \;=\; -c_0b_0a_0 \;=\; -b_0a_0c_0\,,$$

and so for the quasi-equation $a_0b_0c_0 = -$. This being premised, if we form a system of quasi-equations, such as

$$a_0b_0c_0, \quad e_0a_0b_0, \quad \ldots,$$

where the system of triplets contains each duad once, and once only, and the arbitrary signs are chosen at pleasure; if, moreover, in the expression of the quasi-product we replace $a_0^2, b_0^2, \ldots$ each by -1, it is clear that the quasi-product will assume the form

$$w_{\prime\prime}w_1 + a_{\prime\prime}a_0 + b_{\prime\prime}b_0 + c_{\prime\prime}c_0 + \cdots\,,$$

$w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$ being determinate functions of $w, a, b, c, \ldots$; $w_1, a_1, b_1, c_1, \ldots$, homogeneous of the first order in the quantities of each set; the value of $w_{\prime\prime}$ being obviously in every case

$$w_{\prime\prime} \;=\; ww_1 - aa_1 - bb_1 - cc_1 - \cdots\,,$$

and $a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$ containing in every case the terms $aw_1 + a_1w$, $bw_1 + b_1w$, $cw_1 + c_1w$, $\ldots$ but the form of the remaining terms depending as well on the triplets entering into the system of quasi-equations as on the values given to the signs $\pm$; the quasi-equations serving, in fact, to prescribe a rule for the formation of certain functions $w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}, \ldots$, the properties of which functions may afterwards be investigated.

Suppose, now, that the system of quasi-equations is such that

$$e_0a_0b_0, \quad e_0c_0d_0$$

being any two of its triplets, with a common symbol e_0, there occur also in the system the triplets

$$f_0a_0c_0, \quad f_0d_0b_0, \quad g_0a_0d_0, \quad g_0b_0c_0\,;$$

and suppose that the corresponding portion of the system is

$$e_0a_0b_0 = \epsilon, \quad e_0c_0d_0 = \epsilon', \quad f_0a_0c_0 = \zeta, \quad f_0d_0b_0 = \zeta', \quad g_0a_0d_0 = \iota, \quad g_0b_0c_0 = \iota',$$

where $\epsilon, \zeta, \iota, \epsilon', \zeta', \iota'$ each of them denote one of the signs $+$ or $-$; then $e_{\prime\prime}, f_{\prime\prime}, g_{\prime\prime}$ will contain respectively the terms

$$\begin{aligned}
&\epsilon\,(ab_1 - a_1b) - \epsilon'\,(cd_1 - c_1d),\\
&\zeta\,(ac_1 - a_1c) + \zeta'\,(db_1 - d_1b),\\
&\iota\,(ad_1 - a_1d) + \iota'\,(bc_1 - b_1c)\,;
\end{aligned}$$

and $e_{\prime\prime}^2 + f_{\prime\prime}^2 + g_{\prime\prime}^2$ contains the terms

$$\begin{aligned}
(a^2 + b^2 + c^2 + d^2)&(a_1^2 + b_1^2 + c_1^2 + d_1^2) - a^2a_1^2 - b^2b_1^2 - c^2c_1^2 - d^2d_1^2\\
&+ 2\big[\,\epsilon\epsilon'\,(ab_1 - a_1b)(cd_1 - c_1d)\\
&\quad + \zeta\zeta'\,(ac_1 - a_1c)(db_1 - d_1b)\\
&\quad + \iota\iota'\,(ad_1 - a_1d)(bc_1 - b_1c)\big]\,;
\end{aligned}$$

and by taking account of the terms $ew_1 + e_1w$, $fw_1 + f_1w$, $gw_1 + g_1w$ in $e_{\prime\prime}, f_{\prime\prime}, g_{\prime\prime}$ respectively, we should have had besides in $e_{\prime\prime}^2 + f_{\prime\prime}^2 + g_{\prime\prime}^2$ the terms

$$(e^2 + f^2 + g^2)\,w_1^2 + (e_1^2 + f_1^2 + g_1^2)\,w^2 + 2(ee_1 + ff_1 + gg_1)\,ww_1.$$

Also $w_{\prime\prime}^2$ contains the terms

$$a^2a_1^2 + b^2b_1^2 + c^2c_1^2 + d^2d_1^2 - 2(ee_1 + ff_1 + gg_1)\, ww_1\,;$$

whence it is easy to see that

$$\begin{aligned} & w_{\prime\prime}^2 + a_{\prime\prime}^2 + b_{\prime\prime}^2 + c_{\prime\prime}^2 + \cdots \\ &\quad = (w^2 + a^2 + b^2 + c^2 + \cdots)(w_1^2 + a_1^2 + b_1^2 + c_1^2 + \cdots) \\ &\qquad + 2\Sigma\big[\, \epsilon\epsilon'\,(ab_1 - a_1b)(cd_1 - c_1d) \\ &\qquad\qquad + \zeta\zeta'\,(ac_1 - a_1c)(db_1 - d_1b) \\ &\qquad\qquad + \iota\iota'\,(ad_1 - a_1d)(bc_1 - b_1c)\big], \end{aligned}$$

where the summation extends to all the quadruplets formed each by the combination of two duads such as ab and cd, or ac and db, or ad and bc, i.e. two duads, which, combined with the same common letter (in the instances just mentioned e, or f, or g), enter as triplets into the system of quasi-equations; so that if $\nu = 2^m - 1$, the number of quadruplets is

$$\tfrac{1}{2}\cdot\tfrac{1}{2}(\nu-1)\cdot\tfrac{1}{2}(\nu-3)\cdot\nu\cdot 1 \;=\; \tfrac{1}{8}\nu(\nu-1)(\nu-3),$$

and the terms under the sign Σ will vanish identically if only

$$\epsilon\epsilon' \;=\; \zeta\zeta' \;=\; \iota\iota'\,;$$

but the relation $\epsilon\epsilon' = \iota\iota'$ is of the same form as the equation $\epsilon\epsilon' = \zeta\zeta'$; hence if all the relations

$$\epsilon\epsilon' \;=\; \zeta\zeta'$$

are satisfied, the terms under the sign Σ vanish, and we have

$$(w_{\prime\prime}^2 + a_{\prime\prime}^2 + b_{\prime\prime}^2 + c_{\prime\prime}^2 + \cdots) \;=\; (w^2 + a^2 + b^2 + c^2 + \cdots)(w_1^2 + a_1^2 + b_1^2 + c_1^2 + \cdots),$$

which is thus shown to be true, upon the suppositions

1. That the system of quasi-equations is such that

   $$e_0a_0b_0, \quad e_0c_0d_0,$$

   being any two of its triplets with a common symbol e_0, there occur also in the system the triplets

   $$f_0a_0c_0, \quad f_0d_0b_0, \quad g_0a_0d_0, \quad g_0b_0c_0\,.$$

2. That for any two pairs of triplets, such as

   $$e\,ab, \;\; e\,cd \quad\text{and}\quad f\,ac, \;\; f\,db\,,$$

   the product of the signs of the triplets of the first pair is equal to the product of the signs of the triplets of the second pair.

In the case of fifteen things a, b, c, ... the triplets may, as appears from Mr Kirkman's paper, be chosen so as to satisfy the first condition; but the second condition involves, as Mr Kirkman has shown, a contradiction; and therefore the product of two sums, each of them of sixteen squares, is not a sum of sixteen squares. It is proper to remark, that this demonstration, although I think rendered clearer by the introduction of the idea of the system of triplets furnishing the rule for the formation of the expressions $w_{\prime\prime}, a_{\prime\prime}, b_{\prime\prime}, c_{\prime\prime}$, &c., is not in principle different from that contained in Prof. Young's paper "On an Extension of a Theorem of Euler, &c.", *Irish Transactions*, vol. xxi. [1848, pp. 311–341].

113.

NOTE ON THE GEOMETRICAL REPRESENTATION OF THE INTEGRAL $\int dx \div \sqrt{(x+a)(x+b)(x+c)}$.

[From the Philosophical Magazine, vol. v. (1853), pp. 281–284.]

THE equation of a conic passing through the points of intersection of the conics

$$x^2+y^2+z^2 = 0, \qquad ax^2+by^2+cz^2 = 0,$$

is of the form

$$w\,(x^2+y^2+z^2)+ax^2+by^2+cz^2 = 0,$$

where w is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of w a quadratic equation, the roots of which may be considered as parameters for determining the line in question. Let one of the values of w be considered as equal to a constant quantity k, the line is always a tangent to the conic

$$k\,(x^2+y^2+z^2)+ax^2+by^2+cz^2 = 0,$$

and taking $w=p$ for the other value of w, p is a parameter determining the particular tangent, or, what is the same thing, determining the point of contact of this tangent.

The equation of the tangent is easily seen to be

$$x\sqrt{b-c}\,\sqrt{a+k}\,\sqrt{a+p}+y\sqrt{c-a}\,\sqrt{b+k}\,\sqrt{b+p}+z\sqrt{a-b}\,\sqrt{c+k}\,\sqrt{c+p} = 0;$$

suppose that the tangent meets the conic $x^2+y^2+z^2=0$ (which is of course the conic corresponding to $w=\infty$) in the points P, P', and let θ, ∞ be the parameters of the point P, and θ', ∞ the parameters of the point P', i.e. (repeating the definition of the terms) let the tangent at P of the conic $x^2+y^2+z^2=0$ be also touched by the conic $\theta\,(x^2+y^2+z^2)+ax^2+by^2+cz^2=0$, and similarly for θ'. The coordinates of the point P are given by the equations

$$x:y:z = \sqrt{b-c}\,\sqrt{a+\theta} : \sqrt{c-a}\,\sqrt{b+\theta} : \sqrt{a-b}\,\sqrt{c+\theta};$$

and substituting these values in the equation of the line PP', we have

$$(b-c)\sqrt{a+k}\,\sqrt{a+p}\,\sqrt{a+\theta}+(c-a)\sqrt{b+k}\,\sqrt{b+p}\,\sqrt{b+\theta}+(a-b)\sqrt{c+k}\,\sqrt{c+p}\,\sqrt{c+\theta} = 0, \quad \ldots (*)$$

an equation connecting the quantities p, θ. To rationalize this equation, write

$$\begin{aligned}\sqrt{(a+k)(a+p)(a+\theta)} &= \lambda+\mu a,\\ \sqrt{(b+k)(b+p)(b+\theta)} &= \lambda+\mu b,\\ \sqrt{(c+k)(c+p)(c+\theta)} &= \lambda+\mu c,\end{aligned}$$

values which evidently satisfy the equation in question. Squaring these equations, we have equations from which λ^2, $\lambda\mu$, μ^2 may be linearly determined; and making the necessary reductions, we find

$$\begin{aligned}\lambda^2 &= abc+kp\theta,\\ -2\lambda\mu &= bc+ca+ab-(p\theta+kp+k\theta),\\ \mu^2 &= a+b+c+k+p+\theta,\end{aligned}$$

or, eliminating λ, μ,

$$\big[bc+ca+ab-(p\theta+kp+k\theta)\big]^2-4\,(a+b+c+k+p+\theta)\,(abc+kp\theta) \;=\; 0, \qquad \ldots(*)$$

which is the rational form of the former equation marked $(*)$. It is clear from the symmetry of the formula, that the same equation would have been obtained by the elimination of L, M from the equations

$$\begin{aligned}\sqrt{(k+a)(k+b)(k+c)} \;&=\; L+Mk,\\ \sqrt{(p+a)(p+b)(p+c)} \;&=\; L+Mp,\\ \sqrt{(\theta+a)(\theta+b)(\theta+c)} \;&=\; L+M\theta,\end{aligned}$$

and it follows from Abel's theorem (but the result may be verified by means of Euler's fundamental integral in the theory of elliptic functions), that if

$$\Pi x \;=\; \int_\infty^x \frac{dx}{\sqrt{(x+a)(x+b)(x+c)}}\,,$$

then the algebraical equations $(*)$ are equivalent to the transcendental equation

$$\pm\Pi k\pm\Pi p\pm\Pi\theta \;=\; 0\,;$$

the arbitrary constant which should have formed the second side of the equation having been determined by observing that the algebraical equation gives for $p=\theta$, $k=\infty$, a system of values, which, when the signs are properly chosen, satisfy the transcendental equation. In fact, arranging the rational algebraical equation according to the powers of k, it becomes

$$\begin{aligned}&k^2\,(p-\theta)^2-2k\,\big[\,p\theta\,(p+\theta)+2(a+b+c)\,p\theta+(bc+ca+ab)(p+\theta)+2abc\,\big]\\ &+p^2\theta^2-2(bc+ca+ab)\,p\theta-4abc(p+\theta)+b^2c^2+c^2a^2+a^2b^2-2a^2bc-2b^2ca-2c^2ab \;=\; 0, \qquad (*)\end{aligned}$$

which proves the property in question, and is besides a very convenient form of the algebraical integral. The ambiguous signs in the transcendental integral are not of course arbitrary (indeed it has just been assumed that for $p=\theta$, Πp and $\Pi\theta$ are to be taken with opposite signs), but the discussion of the proper values to be given to the ambiguous signs would be at all events tedious, and must be passed over for the present.

It is proper to remark, that $p=\theta$ gives not only, as above supposed, $k=\infty$, but another value of k, which, however, corresponds to the transcendental equation

$$\pm\Pi k\pm 2\Pi p \;=\; 0\,;$$

the value in question is obviously

$$k \;=\; \frac{p^2-2(bc+ca+ab)\,p^2-8abc\,p+b^2c^2+c^2a^2+a^2b^2-2a^2bc-2b^2ca-2c^2ab}{(p+a)(p+b)(p+c)}\,.$$

Consider, in general, a cubic function $ax^3+3bx^2y+3cxy^2+dy^3$, or, as I now write it in the theory of invariants, $(a,b,c,d)\,(x,y)^3$, the Hessian of this function is

$$\big(ac-b^2,\ \tfrac12(ad-bc),\ bd-c^2\big)\,(x,y)^2,$$

and applying this formula to the function $(p+a)(p+b)(p+c)$, it is easy to write the equation last preceding in the form

$$\tfrac14\,k \;=\; p-(a+b+c)-\frac{9\ \text{Hessian}\,\{(p+a)(p+b)(p+c)\}}{(p+a)(p+b)(p+c)}\,,$$

which is a formula for the duplication of the transcendent Πx.

Reverting now to the general transcendental equation

$$\pm\Pi k \pm \Pi p \pm \Pi\theta \;=\; 0,$$

we have in like manner

$$\pm\Pi k \pm \Pi p \pm \Pi\theta' \;=\; 0\,;$$

and assuming a proper correspondence of the signs, the elimination of Πp gives

$$\Pi\theta' - \Pi\theta \;=\; \pm 2\,\Pi k,$$

i.e. if the points P, P' upon the conic $x^2 + y^2 + z^2 = 0$ are such that their parameters θ, θ' satisfy this equation, the line PP' will be constantly a tangent to the conic

$$k\,(x^2 + y^2 + z^2) + (ax^2 + by^2 + cz^2) \;=\; 0.$$

Hence also, if the parameters k, k', k'' of the conics

$$\begin{aligned}
k\,(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 \;&=\; 0,\\
k'(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 \;&=\; 0,\\
k''(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 \;&=\; 0,
\end{aligned}$$

satisfy the equation

$$\Pi k + \Pi k' + \Pi k'' \;=\; 0,$$

there are an infinity of triangles inscribed in the conic $x^2 + y^2 + z^2 = 0$, and the sides of which touch the last-mentioned three conics respectively.

Suppose $2\Pi k = \Pi\kappa$ (an equation the algebraic form of which has already been discussed), then

$$\Pi\theta' - \Pi\theta \;=\; \Pi\kappa,$$

$\theta = \infty$ gives $\theta' = \kappa$; or, observing that $\theta = \infty$ corresponds to a point of intersection of the conics $x^2 + y^2 + z^2 = 0$, $ax^2 + by^2 + cz^2 = 0$, κ is the parameter of the point in which a tangent to the conic $k\,(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0$ at any one of its intersections with the conic $x^2 + y^2 + z^2 = 0$ meets the last-mentioned conic. Moreover, the algebraical relation between θ, θ' and k (where, as before remarked, κ is a given function of k) is given by a preceding formula, and is simpler than that between θ, θ' and k.

The preceding investigations were, it is hardly necessary to remark, suggested by a well-known memoir of the late illustrious Jacobi, and contain, I think, the extension which he remarks it would be interesting to make of the principles in such memoir to a system of two conics. I propose reverting to the subject in a memoir to be entitled "Researches on the Porism of the in- and circumscribed triangle." *[This was, I think, never written.]*

114.

ANALYTICAL RESEARCHES CONNECTED WITH STEINER'S EXTENSION OF MALFATTI'S PROBLEM.

[From the Philosophical Transactions of the Royal Society of London, vol. cxlii. for the year 1852, pp. 253–278: Received April 12, — Read May 27, 1852.]

The problem, in a triangle to describe three circles each of them touching the two others and also two sides of the triangle, has been termed after the Italian geometer by whom it was proposed and solved, Malfatti's problem. The problem which I refer to as Steiner's extension of Malfatti's problem is as follows: — "To determine three sections of a surface of the second order, each of them touching the two others, and also two of three given sections of the surface of the second order," a problem proposed in Steiner's memoir, "Einige geometrische Betrachtungen," *Crelle*, t. i. [1826, pp. 161–184]. The geometrical construction of the problem in question is readily deduced from that given in the memoir just mentioned for a somewhat less general problem, viz. that in which the surface of the second order is replaced by a sphere; it is for the sake of the analytical developments to which the problem gives rise, that I propose to resume here the discussion of the problem. The following is an analysis of the present memoir:

§ 1. Contains a lemma which appears to me to constitute the foundation of the analytical theory of the sections of a surface of the second order.

§ 2. Contains a statement of the geometrical construction of Steiner's extension of Malfatti's problem.

§ 3. Is a verification, founded on a particular choice of coordinates, of the construction in question.

§ 4. In this section, referring the surface of the second order to absolutely general coordinates, and after an incidental solution of the problem to determine a section touching three given sections, I obtain the equations for the solution of Steiner's extension of Malfatti's problem.

§ 5. Contains a separate discussion of a system of equations, including as a particular case the equations obtained in the preceding section.

§§ 6 and 7. Contain the application of the formulæ for the general system to the equations in § 4, and the development and completion of the solution.

§ 8. Is an extension of some preceding formulæ to quadratic functions of any number of variables.

§ 1. Lemma relating to the sections of a surface of the second order.

If

$$ax^2 + by^2 + cz^2 + dw^2 + 2fyz + 2gzx + 2hxy + 2lxw + 2myw + 2nzw \;=\; 0$$

be the equation of a surface of the second order, and

$$\mathfrak{A}x^2 + \mathfrak{B}y^2 + \mathfrak{C}z^2 + \mathfrak{D}w^2 + 2\mathfrak{F}yz + 2\mathfrak{G}zx + 2\mathfrak{H}xy + 2\mathfrak{L}xw + 2\mathfrak{M}yw + 2\mathfrak{N}zw \;=\; 0$$

the reciprocal equation, the condition that the two sections

$$\lambda x + \mu y + \nu z + \rho w \;=\; 0, \qquad \lambda' x + \mu' y + \nu' z + \rho' w \;=\; 0,$$

may touch, is

$$\begin{aligned}&\left(\mathfrak{A}\lambda^2+\mathfrak{B}\mu^2+\mathfrak{C}\nu^2+\mathfrak{D}\rho^2+2\mathfrak{F}\mu\nu+2\mathfrak{G}\nu\lambda+2\mathfrak{H}\lambda\mu+2\mathfrak{L}\lambda\rho+2\mathfrak{M}\mu\rho+2\mathfrak{N}\nu\rho\right)^{\frac{1}{2}}\\&\quad\times\left(\mathfrak{A}\lambda'^2+\mathfrak{B}\mu'^2+\mathfrak{C}\nu'^2+\mathfrak{D}\rho'^2+2\mathfrak{F}\mu'\nu'+2\mathfrak{G}\nu'\lambda'+2\mathfrak{H}\lambda'\mu'+2\mathfrak{L}\lambda'\rho'+2\mathfrak{M}\mu'\rho'+2\mathfrak{N}\nu'\rho'\right)^{\frac{1}{2}}\\&\quad=\ \mathfrak{A}\lambda\lambda'+\mathfrak{B}\mu\mu'+\mathfrak{C}\nu\nu'+\mathfrak{D}\rho\rho'+\mathfrak{F}(\mu\nu'+\mu'\nu)+\mathfrak{G}(\nu\lambda'+\nu'\lambda)+\mathfrak{H}(\lambda\mu'+\lambda'\mu)\\&\qquad+\mathfrak{L}(\lambda\rho'+\lambda'\rho)+\mathfrak{M}(\mu\rho'+\mu'\rho)+\mathfrak{N}(\nu\rho'+\nu'\rho)\,;\end{aligned}$$

and in particular if the equation of the surface be

$$ax^2+by^2+cz^2+2fyz+2gzx+2hxy+pw^2\ =\ 0,$$

the condition of contact is

$$\begin{aligned}&\left(\mathfrak{A}\lambda^2+\mathfrak{B}\mu^2+\mathfrak{C}\nu^2+2\mathfrak{F}\mu\nu+2\mathfrak{G}\nu\lambda+2\mathfrak{H}\lambda\mu+K\,\tfrac{\rho^2}{p}\right)^{\frac{1}{2}}\\&\quad\times\left(\mathfrak{A}\lambda'^2+\mathfrak{B}\mu'^2+\mathfrak{C}\nu'^2+2\mathfrak{F}\mu'\nu'+2\mathfrak{G}\nu'\lambda'+2\mathfrak{H}\lambda'\mu'+K\,\tfrac{\rho'^2}{p}\right)^{\frac{1}{2}}\\&\quad=\ \mathfrak{A}\lambda\lambda'+\mathfrak{B}\mu\mu'+\mathfrak{C}\nu\nu'+\mathfrak{F}(\mu\nu'+\mu'\nu)+\mathfrak{G}(\nu\lambda'+\nu'\lambda)+\mathfrak{H}(\lambda\mu'+\lambda'\mu)+K\,\tfrac{\rho\rho'}{p}\,,\end{aligned}$$

in which last formula

$$\mathfrak{F}=gh-af,\quad \mathfrak{G}=hf-bg,\quad \mathfrak{H}=fg-ch,$$
$$K\ =\ abc-af^2-bg^2-ch^2+2fgh.$$

§ 2.

In order to state in the most simple form the geometrical construction for the solution of Steiner's extension of Malfatti's problem, let the given sections be called for conciseness the *determinators* [1]; any two of these sections lie in two different cones, the vertices of which determine with the line of intersection of the planes of the determinators, two planes which may be termed *bisectors*; the six bisectors pass three and three through four straight lines; and it will be convenient to use the term bisectors to denote, not the entire system, but any three bisectors passing through the same line. Consider three sections, which may be termed *tactors*, each of them touching a determinator and two bisectors, and three other sections (which may be termed *separators*) each of them passing through the point of contact of a determinator and tactor and touching the other two tactors; the separators will intersect in a line which passes through the point of intersection of the determinators. The three required sections, or as I shall term them the *resultors*, are determined by the conditions that each resultor touches two determinators and two separators, the possibility of the construction being implied as a theorem. The *à posteriori* verification may be obtained as follows:—

§ 3.

Let $x=0$, $y=0$, $z=0$ be the equations of the resultors, $w=0$ the equation of the polar of the point of intersection of the resultors. Since the resultors touch two and two, the equation of the surface is easily seen to be of the form

$$2yz+2zx+2xy+w^2\ =\ 0.\qquad (=)$$

[1] I use the words "determinators," &c. to denote indifferently the sections or the planes of the sections; the context is always sufficient to prevent ambiguity.

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The determinators are sections each of them touching two resultors, but otherwise arbitrary; their equations are

$$-\alpha x + \tfrac{1}{2\alpha}\,y + \tfrac{1}{2\alpha}\,z + w \ = \ 0, \qquad \tfrac{1}{2\beta}\,x - \beta y + \tfrac{1}{2\beta}\,z + w \ = \ 0, \qquad \tfrac{1}{2\gamma}\,x + \tfrac{1}{2\gamma}\,y - \gamma z + w \ = \ 0.$$

The separators are sections each of them touching two resultors at their point of contact (or what is the same thing, passing through the line of intersection of two resultors), and all of them having a line in common. Their equations may be taken to be

$$cy - bz \ = \ 0, \qquad az - cx \ = \ 0, \qquad bx - ay \ = \ 0.$$

[The memoir continues from § 3 onward at p. 60 et seq., outside the present PDF pp. 51–75 slice.]

[2]The reciprocal form is, it should be noted,

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy - 2w^2 \ = \ 0.$$

CAYLEY, Collected Mathematical Papers, Vol. II.

Scan-verified retype: source PDF pp. 76–104 (book pp. 60–88).

Paper 114 (concluded), Paper 115 (opening).

114 (continued).

ANALYTICAL RESEARCHES CONNECTED WITH STEINER'S EXTENSION OF MALFATTI'S PROBLEM.

[Continuation of § 3, beginning at book p. 60.]

[The opening of § 3, with the surface $2yz + 2zx + 2xy + w^2 = 0$ and the three determinators $-\alpha x + \frac{1}{2\alpha}y + \frac{1}{2\alpha}z + w = 0$, $\frac{1}{2\beta}x - \beta y + \frac{1}{2\beta}z + w = 0$, $\frac{1}{2\gamma}x + \frac{1}{2\gamma}y - \gamma z + w = 0$, was given in the preceding slice. We resume at p. 60, where the tactors are to be found.]

The values of a, b, c remaining to be determined. Now before fixing the values of these quantities, we may find three sections each of them touching a determinator at a point of intersection with the section which corresponds to it of the sections

$$cy - bz = 0, \qquad az - cx = 0, \qquad bx - ay = 0,$$

and touching the other two of the last-mentioned sections; and when a, b, c have their proper values the sections so found are the *tactors*. For, let $\lambda x + \mu y + \nu z + \rho w = 0$ be the equation of a section touching the determinator $-\alpha x + \frac{1}{2\alpha}y + \frac{1}{2\alpha}z + w = 0$, and the two sections $bx - ay = 0$, $az - cx = 0$; and suppose

$$\Delta^2 \;=\; \lambda^2 + \mu^2 + \nu^2 - 2\mu\nu - 2\nu\lambda - 2\lambda\mu - 2\rho^2\,;$$

the conditions of contact with the sections $bx - ay = 0$, $az - cx = 0$ are found to be

$$\begin{aligned}(b+a)\,\Delta \;&=\; (b+a)\lambda - (b+a)\mu - (b-a)\nu,\\ (c+a)\,\Delta \;&=\; (c+a)\lambda - (c-a)\mu - (c+a)\nu,\end{aligned}$$

values, however, which suppose a correspondence in the signs of the radicals. Thence $(b+a)\mu = (c+a)\nu$; or since the ratios only of the quantities λ, μ, ν, ρ are material, $\mu = c + a$, $\nu = b + a$, and therefore

$$\Delta^2 \;=\; \lambda^2 - 2\,(2a+b+c)\,\lambda + (b-c)^2 - 2\rho^2 \;=\; (\lambda - b - c)^2,$$

or

$$\rho^2 \;=\; -2\,(a\lambda + bc).$$

Hence the equation to a section touching $bx - ay = 0$, $az - cx = 0$ is

$$\lambda x + (c+a)\,y + (b+a)\,z + \sqrt{-2\,(a\lambda + bc)}\;w \;=\; 0,$$

and to express that this touches the determinator in question, we have

$$\pm\alpha\,(\lambda - b - c) \;=\; -\alpha\lambda - \tfrac{1}{\alpha}(2a+b+c) + 2\sqrt{-2\,(a\lambda + bc)},$$

and selecting the upper sign,

$$-\lambda - 2a\alpha \;=\; -2\sqrt{-2\,(a\lambda + bc)}\,;$$

whence

$$\lambda = -2a\,(a\alpha - \sqrt{-2bc}), \qquad \sqrt{-2\,(a\lambda + bc)} = (2a\alpha - \sqrt{-2bc}).$$

Therefore the section touching the determinator and the sections $bx - ay = 0$, $az - cx = 0$ is

$$-2a\,(a\alpha - \sqrt{-2bc})\,x + (c+a)\,y + (b+a)\,z + (2a\alpha - \sqrt{-2bc})\,w = 0\,;$$

and at the point of contact with the determinator

$$2yz + 2zx + 2xy + w^2 = 0.$$

[Book p. 61.] Eliminating w between the first and second equations, and between the second and third equations,

$$\sqrt{-2bc}\left(ax + \tfrac{1}{2\alpha}y + \tfrac{1}{2\alpha}z\right) + cy + bz = 0,$$

and from these equations $(cy - bz) = 0$, or the point of contact lies in the section $cy - bz = 0$. It follows that the equations of the *tactors* are

$$\begin{aligned}
-2a\,(a\alpha - \sqrt{-2bc})\,x + (c+a)\,y + (b+a)\,z + (2a\alpha - \sqrt{-2bc})\,w &= 0,\\
(c+b)\,x - 2\beta\,(b\beta - \sqrt{-2ca})\,y + (a+b)\,z + (2b\beta - \sqrt{-2ca})\,w &= 0,\\
(b+c)\,x + (a+c)\,y - 2\gamma\,(c\gamma - \sqrt{-2ab})\,z + (2c\gamma - \sqrt{-2ab})\,w &= 0,
\end{aligned}$$

where a, b, c still remain to be determined.

Now the separators pass through the point of intersection of the determinators; the equations of these give for the point in question,

$$\begin{aligned}
x : y : z : w = {} & (2\beta\gamma + 1)(-\alpha + \beta + \gamma + 2\alpha\beta\gamma)\\
& : (2\gamma\alpha + 1)(\alpha - \beta + \gamma + 2\alpha\beta\gamma)\\
& : (2\alpha\beta + 1)(\alpha + \beta - \gamma + 2\alpha\beta\gamma)\\
& : -1 + \alpha^2 + \beta^2 + \gamma^2 + 4\alpha^2\beta^2\gamma^2\,;
\end{aligned}$$

and the values of a, b, c are therefore

$$\begin{aligned}
a : b : c = {} & (2\beta\gamma + 1)(-\alpha + \beta + \gamma + 2\alpha\beta\gamma)\\
& : (2\gamma\alpha + 1)(\alpha - \beta + \gamma + 2\alpha\beta\gamma)\\
& : (2\alpha\beta + 1)(\alpha + \beta - \gamma + 2\alpha\beta\gamma),
\end{aligned}$$

which are to be substituted for a, b, c in the equations of the separators and tactors respectively.

Now proceeding to find the *bisectors*, let $\lambda x + \mu y + \nu z + \rho w = 0$ be the equation of a section touching the determinators, and suppose as before $\Delta^2 = \lambda^2 + \mu^2 + \nu^2 - 2\mu\nu - 2\nu\lambda - 2\lambda\mu - 2\rho^2$; the conditions of contact are

$$\begin{aligned}
\pm\beta\,\Delta &= \beta\lambda - (\beta + \tfrac{1}{2\beta})\mu + \beta\nu - 2\rho,\\
\mp\gamma\,\Delta &= \gamma\lambda + \gamma\mu - (\gamma + \tfrac{1}{2\gamma})\nu - 2\rho,
\end{aligned}$$

[Book p. 62.] where it is necessary, for the present purpose, to give opposite signs to the radicals. For if the radicals had the same sign, it would follow that

$$\tfrac{1}{\beta}\Big[\beta\lambda - (\beta + \tfrac{1}{2\beta})\mu + \beta\nu - 2\rho\Big] - \tfrac{1}{\gamma}\Big[\gamma\lambda + \gamma\mu - (\gamma + \tfrac{1}{2\gamma})\nu - 2\rho\Big] = 0,$$

hence the section $\lambda x + \mu y + \nu z + \rho w = 0$ would pass through the point

$$x : y : z : w \; = \; 0 : \tfrac{1}{2\beta^2} : -\tfrac{1}{2\gamma^2} : -\tfrac{1}{\beta} + \tfrac{1}{\gamma}\,,$$

or the section would be a tangent section of the two determinators of the same class with the resultor $x = 0$, which ought not to be the case. The proper formula is

$$\tfrac{1}{\beta}\Big[\beta\lambda - (\beta + \tfrac{1}{2\beta})\mu + \beta\nu - 2\rho\Big] + \tfrac{1}{\gamma}\Big[\gamma\lambda + \gamma\mu - (\gamma + \tfrac{1}{2\gamma})\nu - 2\rho\Big] \; = \; 0,$$

and this equation being satisfied, the section

$$\lambda x + \mu y + \nu z + \rho w \; = \; 0$$

passes through a point

$$x : y : z : w \; = \; -\tfrac{1}{\beta^2} - \tfrac{1}{\gamma^2} : \tfrac{1}{\gamma^2} : \tfrac{1}{\beta^2} : -\tfrac{2}{\beta} - \tfrac{2}{\gamma}\,.$$

The bisector passes through this point and the line of intersection of the determinators; its equation is

$$\tfrac{1}{\beta}\Big(\tfrac{1}{2\beta}x - \beta y + \tfrac{1}{2\beta}z + w\Big) - \tfrac{1}{\gamma}\Big(\tfrac{1}{2\gamma}x + \tfrac{1}{2\gamma}y - \gamma z + w\Big) \; = \; 0,$$

or, reducing and completing the system, the equations of the bisectors are

$$\begin{aligned}
\Big(\tfrac{1}{2\beta^2} - \tfrac{1}{2\gamma^2}\Big)x \; + \; \Big(\gamma + \tfrac{1}{2\beta^2}\Big)y \; + \; \Big(\tfrac{1}{2\gamma^2} - \beta\Big)z \; + \; \Big(\tfrac{1}{\beta} - \tfrac{1}{\gamma}\Big)w \; &= \; 0,\\
\Big(\tfrac{1}{2\gamma^2} - \alpha\Big)x \; + \; \Big(\tfrac{1}{2\gamma^2} - \tfrac{1}{2\alpha^2}\Big)y \; + \; \Big(\tfrac{1}{2\alpha^2} - \gamma\Big)z \; + \; \Big(\tfrac{1}{\gamma} - \tfrac{1}{\alpha}\Big)w \; &= \; 0,\\
\Big(\alpha - \tfrac{1}{2\beta^2}\Big)x \; + \; \Big(\tfrac{1}{2\alpha^2} - \beta\Big)y \; + \; \Big(\tfrac{1}{2\alpha^2} - \tfrac{1}{2\beta^2}\Big)z \; + \; \Big(\tfrac{1}{\alpha} - \tfrac{1}{\beta}\Big)w \; &= \; 0.
\end{aligned}$$

In order to verify the geometrical construction, it only remains to show that each bisector touches two tactors. Consider the bisector and tactor

$$-\Big(1 + \tfrac{c}{2\beta^2}\Big)x + \Big(1 + \tfrac{a}{2\gamma^2}\Big)y + \Big(\tfrac{a}{2\beta^2} - \tfrac{b}{2\gamma^2}\Big)z - \tfrac{1}{a}\Big(\tfrac{1}{\gamma} - \tfrac{1}{\beta}\Big)w \; = \; 0,$$

$$-2a\,(a\alpha - \sqrt{-2bc})\,x + (c + a)\,y + (b + a)\,z + (2a\alpha - \sqrt{-2bc})\,w \; = \; 0;$$

and represent these for a moment by

$$\lambda x + \mu y + \nu z + \rho w \; = \; 0, \qquad \lambda' x + \mu' y + \nu' z + \rho' w \; = \; 0\,;$$

[Book p. 63.] if Δ be the same as before, and Δ' the like function of λ', μ', ν', ρ'; also if

$$\Phi \; = \; \lambda\lambda' + \mu\mu' + \nu\nu' - (\mu\nu' + \mu'\nu) - (\nu\lambda' + \nu'\lambda) - (\lambda\mu' + \lambda'\mu) - 2\rho\rho',$$

then

$$\Delta^2 \; = \; \Big(2 + \tfrac{1}{a\beta}\Big)^2,$$

$$\Delta'^2 \; = \; (2a\alpha^2 - 2\alpha\sqrt{-2bc} + b + c)^2,$$

$$\Phi \; = \; a\alpha^2\Big(2 + \tfrac{1}{a\beta}\Big)^2 - 2\alpha\sqrt{-2bc}\Big(2 + \tfrac{1}{a\beta}\Big) + c\Big(2 + \tfrac{1}{a\beta}\Big),$$

and the condition of contact $\Delta\Delta' = \Phi$ (taking the proper sign for the radicals) becomes

$$\Big(2 + \tfrac{1}{a\beta}\Big)\,(2a\alpha^2 - 2\alpha\sqrt{-2bc} + b + c) \; = \; a\alpha^2\Big(2 + \tfrac{1}{a\beta}\Big) - 2\alpha\sqrt{-2bc}\Big(2 + \tfrac{1}{a\beta}\Big) + c\Big(2 + \tfrac{1}{a\beta}\Big),$$

or reducing,

$$a\alpha - b\beta + c\,\frac{\alpha-\beta}{2\alpha\beta+1} \;=\; 0,$$

an equation which is evidently not altered by the interchange of a, α and b, β. The conditions, in order that each bisector may touch two tactors, reduce themselves to the three equations,

$$\begin{aligned}
a\alpha - b\beta + c\,\frac{\alpha-\beta}{2\alpha\beta+1} &= 0,\\
b\beta - c\gamma + a\,\frac{\beta-\gamma}{2\beta\gamma+1} &= 0,\\
c\gamma - a\alpha + b\,\frac{\gamma-\alpha}{2\gamma\alpha+1} &= 0,
\end{aligned}$$

which are satisfied by the values above found for the quantities a, b, c. The possibility and truth of the geometrical construction are thus demonstrated.

§ 4.

Let it be in the first instance proposed to find the equation of a section touching all or any of the sections $x=0$, $y=0$, $z=0$ of the surface of the second order,

$$ax^2+by^2+cz^2+2fyz+2gzx+2hxy+pw^2 \;=\; 0.$$

Any section whatever of this surface may be written in the form

$$\sqrt{(a\lambda^2+b\mu^2+c\nu^2+2f\mu\nu+2g\nu\lambda+2h\lambda\mu-K)}\;\rho\,/\sqrt{-p}\cdot w$$
$$+(a\lambda+h\mu+g\nu)\,x+(h\lambda+b\mu+f\nu)\,y+(g\lambda+f\mu+c\nu)\,z \;=\; 0,$$

where

$$\nabla^2 \;=\; a\lambda^2+b\mu^2+c\nu^2+2f\mu\nu+2g\nu\lambda+2h\lambda\mu-K,$$

[Book p. 64.] and λ, μ, ν are indeterminate. And considering any other section represented by a like equation,

$$\sqrt{(a\lambda'^2+b\mu'^2+c\nu'^2+2f\mu'\nu'+2g\nu'\lambda'+2h\lambda'\mu'-K)}\;\rho'\,/\sqrt{-p}\cdot w$$
$$+(a\lambda'+h\mu'+g\nu')\,x+(h\lambda'+b\mu'+f\nu')\,y+(g\lambda'+f\mu'+c\nu')\,z \;=\; 0,$$

where

$$\nabla'^2 \;=\; a\lambda'^2+b\mu'^2+c\nu'^2+2f\mu'\nu'+2g\nu'\lambda'+2h\lambda'\mu'-K,$$

it may be shown, by means of the lemma previously given, that the condition of contact is

$$a\lambda\lambda'+b\mu\mu'+c\nu\nu'+f(\mu\nu'+\mu'\nu)+g(\nu\lambda'+\nu'\lambda)+h(\lambda\mu'+\lambda'\mu)+K\,\tfrac{\rho\rho'}{p} \;=\; \nabla\nabla'.$$

Suppose that λ', μ', ν' satisfy the equations

$$\begin{aligned}
a\lambda'+h\mu'+g\nu' &= 0,\\
h\lambda'+b\mu'+f\nu' &= 0,\\
g\lambda'+f\mu'+c\nu' &= 0,
\end{aligned}$$

so that the last-mentioned section becomes $x = 0$; and observing that the first of the above equations may be transformed into

$$a\lambda'^2 + b\mu'^2 + c\nu'^2 \;=\; \frac{K}{a},$$

it is easy to obtain $\lambda' = \sqrt{\mathfrak{A}}$, $\mu' = \frac{\mathfrak{H}}{\sqrt{\mathfrak{A}}}$, $\nu' = \frac{\mathfrak{G}}{\sqrt{\mathfrak{A}}}$. The condition of contact thus becomes

$$\frac{K}{\sqrt{\mathfrak{A}}}\,\rho^2 + K \;=\; 0\,;$$

and taking the under sign, $\lambda = \sqrt{\mathfrak{A}}$, so that if in the above-written equation we establish all or any of the equations $\lambda = \sqrt{\mathfrak{A}}$, $\mu = \sqrt{\mathfrak{B}}$, $\nu = \sqrt{\mathfrak{C}}$, we have the equation of a section touching all or the corresponding sections of the sections

$$x = 0, \qquad y = 0, \qquad z = 0.$$

In particular we have for a solution of the problem of tactions, the following equation of the section touching $x = 0$, $y = 0$, $z = 0$:

$$\begin{aligned}(a\sqrt{\mathfrak{A}} + h\sqrt{\mathfrak{B}} + g\sqrt{\mathfrak{C}})\,x + (h\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + f\sqrt{\mathfrak{C}})\,y + (g\sqrt{\mathfrak{A}} + f\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}})\,z \\ + \frac{\rho}{\sqrt{-p}}\sqrt{2\,(\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})(\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})}\;\cdot w \;=\; 0.\end{aligned}$$

Anticipating the use of a notation the value of which will subsequently appear, or putting

[Book p. 65.]

$$K^2 \;=\; -\mathrm{f}^4 - \mathrm{g}^4 - \mathrm{h}^4 + 2\mathrm{g}^2\mathrm{h}^2 + 2\mathrm{h}^2\mathrm{f}^2 + 2\mathrm{f}^2\mathrm{g}^2 - \frac{4\mathrm{f}^2\mathrm{g}^2\mathrm{h}^2}{J^2},$$

the equation of the section in question is

$$\frac{\mathrm{f}^2}{\sqrt{\mathfrak{A}}}\,(-\mathrm{f}^2 + \mathrm{g}^2 + \mathrm{h}^2)\,x + \frac{\mathrm{g}^2}{\sqrt{\mathfrak{B}}}\,(\mathrm{f}^2 - \mathrm{g}^2 + \mathrm{h}^2)\,y + \frac{\mathrm{h}^2}{\sqrt{\mathfrak{C}}}\,(\mathrm{f}^2 + \mathrm{g}^2 - \mathrm{h}^2)\,z + \frac{4\mathrm{f}^2\mathrm{g}^2\mathrm{h}^2}{J^2}\,\frac{\rho\sqrt{-pK}}{\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}}}\,w \;=\; 0.$$

I proceed to investigate a transformation of the equation for the section with an indeterminate parameter λ, which touches the two sections $y = 0$, $z = 0$. We have

$$a\nabla^2 \;=\; (a\lambda + h\mu + g\nu)^2 + (b\mu^2 + c\nu^2 - 2f\mu\nu) - \mathfrak{B} - K,$$

or, putting for μ^2 and ν^2 their values $\mathfrak{B}$, $\mathfrak{C}$ in the second term,

$$a\nabla^2 \;=\; (a\lambda + h\mu + g\nu)^2 + (\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})^2\,;$$

and introducing instead of λ an indeterminate quantity X, such that

$$a\lambda + h\mu + g\nu \;=\; (\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})\,X,$$

we have, also introducing throughout X instead of λ, and completing the substitution of $\sqrt{\mathfrak{B}}$, $\sqrt{\mathfrak{C}}$ for μ, ν, the equation of the section touching $y = 0$, $z = 0$, becomes

$$(ax + hy + gz)\,X + y\sqrt{\mathfrak{C}} + z\sqrt{\mathfrak{B}} + w\sqrt{-ap}\sqrt{1 + X^2} \;=\; 0.$$

It may be remarked here in passing, that this is a very convenient form for the demonstration of the theorem: "If two sections of a surface of the second order touch each other, and are also tangent sections (of the same class) to two fixed sections, then considering the planes through the axis of the fixed sections and the poles of the tangent sections, and also the tangent planes through this axis, the anharmonic ratio of the four planes is independent of the position of the moveable tangent sections;" where by the axis of the fixed sections is to be understood the line joining their poles.

The sections touching $z = 0$, $x = 0$ and $x = 0$, $y = 0$ are of course

$$\begin{aligned} z\sqrt{\mathfrak{A}} + x\sqrt{\mathfrak{C}} + (hx + by + fz)\,Y + w\sqrt{-bp}\sqrt{1+Y^2} &= 0, \\ x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)\,Z + w\sqrt{-cp}\sqrt{1+Z^2} &= 0. \end{aligned}$$

The conditions of contact of the sections represented by the above-written equations would be perhaps most simply obtained directly from the lemma, but it is proper to deduce it from the formula for contact used in the present memoir. If for shortness

$$\Phi(\pm) \;=\; a\lambda'\lambda'' + b\mu'\mu'' + c\nu'\nu'' + f(\mu'\nu'' + \mu''\nu') + g(\nu'\lambda'' + \nu''\lambda') + h(\lambda'\mu'' + \lambda''\mu') \pm K\,\tfrac{\rho'\rho''}{p}\,,$$

[Book p. 66.] where the symbol $\Phi(\pm)$ is used in order to mark the essentially different character of the results corresponding to the different values of the ambiguous sign, then

$$\begin{aligned} bc\,\Phi(-) \;&=\; f(h\lambda' + b\mu' + f\nu')(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + (\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(h\lambda'' + b\mu'' + f\nu'') \\ &\quad + f(-a\sqrt{\mathfrak{B}\mathfrak{C}} + h\sqrt{\mathfrak{C}\mathfrak{A}} + g\sqrt{\mathfrak{A}\mathfrak{B}} - aF - (\mathfrak{G}\mathfrak{H} - aF)) \\ &= \cdots \\ &= (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{f\,YZ + \sqrt{\mathfrak{A}}(Y+Z) + f\}. \end{aligned}$$

That is, $bc\,\Phi(-) = (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{f\,YZ + \sqrt{\mathfrak{A}}(Y+Z) + f\}$.

What, however, is really required[1], is the value of $\Phi(+)$; to find this, we have

$$bc\,\Phi(+) \;=\; bc\,\Phi(-) + 2bcK$$

$$= \;(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{f\,YZ + \sqrt{\mathfrak{A}}(Y+Z) + f\} + 2bcK - 2f(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}),$$

the second line of which is

$$2\,(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\left\{\frac{bcK - f(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})}{(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})}\right\} \;=\; 2\,(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\,\sqrt{\mathfrak{A}}\,\theta,$$

[Book p. 67.] where

$$\theta \;=\; \frac{1}{K}\left(\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}} + \mathfrak{F}\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{B}} + \mathfrak{H}\sqrt{\mathfrak{C}}\right),$$

and consequently

$$bc\,\Phi(+) \;=\; (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\big[f\,YZ + \sqrt{\mathfrak{A}}\,(Y+Z) + f + 2\theta\sqrt{\mathfrak{A}}\big],$$

[1] It may be shown without difficulty that the $(-)$ sign would imply that the sections touching $z = 0$, $x = 0$, and $x = 0$, $y = 0$ were sections touching $x = 0$ at the same point. By taking the $(-)$ sign in each equation we should have the solution of the problem "to determine three sections of a surface of the second order, the two sections of each pair touching one of three given sections at the same point," which is not without interest; the solution may be completed without any difficulty.

a reduction, which on account of its peculiarity, I have thought right to work out in full.

The condition of contact is

$$\Phi(+) \;=\; \nabla'\nabla'' \;=\; \tfrac{1}{\sqrt{bc}}\,(\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})\,\sqrt{1+Y^2}\,\sqrt{1+Z^2}.$$

Hence finally, the condition in order that the sections

$$\begin{aligned} z\sqrt{\mathfrak{A}}+(hx+by+fz)\,Y+x\sqrt{\mathfrak{C}}+w\sqrt{-bp}\sqrt{1+Y^2} &= 0,\\ x\sqrt{\mathfrak{B}}+y\sqrt{\mathfrak{A}}+(gx+fy+cz)\,Z+w\sqrt{-cp}\sqrt{1+Z^2} &= 0,\end{aligned}$$

(the former of which is a section touching $z=0$, $x=0$, and the latter a section touching $x=0$, $y=0$) may touch, is

$$fYZ+\sqrt{\mathfrak{A}}\,(Y+Z)+(f+2\theta\sqrt{\mathfrak{A}})-\sqrt{bc}\sqrt{1+Y^2}\sqrt{1+Z^2} \;=\; 0.$$

The preceding researches show that the solution of Steiner's extension of Malfatti's problem depends on a system of equations, such as the system mentioned at the commencement of the following section.

§ 5.

Consider the system of equations

$$\begin{aligned} \alpha+\beta(Y+Z)+\gamma YZ+\delta\sqrt{1+Y^2}\sqrt{1+Z^2} &= 0,\\ \alpha'+\beta'(Z+X)+\gamma' ZX+\delta'\sqrt{1+Z^2}\sqrt{1+X^2} &= 0,\\ \alpha''+\beta''(X+Y)+\gamma'' XY+\delta''\sqrt{1+X^2}\sqrt{1+Y^2} &= 0\,;\end{aligned}$$

these equations may, it will be seen, be solved by quadratics only, when the coefficients satisfy the relations

$$\frac{\gamma-\alpha}{\gamma^2-\alpha^2} \;=\; \frac{\gamma'-\alpha'}{\gamma'^2-\alpha'^2} \;=\; \frac{\gamma''-\alpha''}{\gamma''^2-\alpha''^2}\,.$$

[Book p. 68.] It may be remarked that these equations are satisfied by

$$\beta=0,\quad \beta'=0,\quad \beta''=0,\quad \gamma=\delta,\quad \gamma'=\delta',\quad \gamma''=\delta'',$$

or if we write

$$\frac{\alpha}{\gamma}=-l,\quad \frac{\alpha'}{\gamma'}=-m,\quad \frac{\alpha''}{\gamma''}=-n,$$

the equations become, by a simple reduction,

$$\begin{aligned} Y^2+Z^2+2l\,YZ &= l^2-1,\\ Z^2+X^2+2m\,ZX &= m^2-1,\\ X^2+Y^2+2n\,XY &= n^2-1,\end{aligned}$$

which are equivalent to the equations discussed in my paper "On a system of Equations connected with Malfatti's Problem and on another Algebraical System," *Cambridge and Dublin Mathematical Journal*, t. iv. [1849] pp. 270–275, [79]; the solution might have been effected by the direct method, which I shall here adopt, of eliminating any one of the variables between the two equations into which it enters, and combining the result with the third equation.

Writing the second and third equations under the form

$$A' + B'X + C'\sqrt{1+X^2} \;=\; 0,$$
$$A'' + B''X + C''\sqrt{1+X^2} \;=\; 0,$$

the result of the elimination may be presented in the form

$$A'A'' + B'B'' - C'C'' \;=\; \sqrt{A'^2 + B'^2 - C'^2} \cdot \sqrt{A''^2 + B''^2 - C''^2},$$

which is most easily obtained by writing $X = \tan\phi$ and operating with the symbol $\cos^{-1}$; but if the rationalized equations be represented by

$$\lambda' + 2\mu' X + \nu' X^2 \;=\; 0, \qquad \lambda'' + 2\mu'' X + \nu'' X^2 \;=\; 0,$$

the form

$$(\lambda'\nu'' - \lambda''\nu')^2 \;=\; 4(\lambda'\nu' - \mu'^2)(\lambda''\nu'' - \mu''^2) \;-\; (\lambda'\nu'' - 2\mu'\mu'')^2 \cdots$$

leads easily to the result in question. The values which enter are

$$A' = \alpha' + \beta' Z, \qquad A'' = \alpha'' + \beta'' Y,$$
$$B' = \beta' + \gamma' Z, \qquad B'' = \beta'' + \gamma'' Y,$$
$$C' = \delta'\sqrt{1+Z^2}, \qquad C'' = \delta''\sqrt{1+Y^2}\,;$$

whence, in the first place, by the equation connecting Y, Z,

$$C'C'' \;=\; -\frac{\delta'\delta''}{\delta}\,\{\alpha + \beta(Y+Z) + \gamma YZ\}.$$

[Book p. 69.] It is obviously convenient that $A'A'' + B'B''$ should be symmetrical with respect to Y and Z, and this will be the case if

$$\alpha'\beta'' + \beta'\gamma'' \;=\; \alpha''\beta' + \beta''\gamma',$$

that is, if

$$\beta'\,(\gamma'' - \alpha'') \;=\; \beta''\,(\gamma' - \alpha'),$$

or, assuming that the equations are symmetrically related to the system, we have the first set of relations between the coefficients, relations which are satisfied by

$$\alpha = \gamma + 2\phi\beta, \qquad \alpha' = \gamma' + 2\phi\beta', \qquad \alpha'' = \gamma'' + 2\phi\beta'',$$

and the values of α, α', α'' will be considered henceforth as given by these conditions.

We have

$$A'A''+B'B''-C'C'' \;=\; \alpha'\alpha''+\beta'\beta''+(\gamma'\beta''+\gamma''\beta'+2\phi\beta'\beta'')(Y+Z)+(\beta'\beta''+\gamma'\gamma'')\,YZ+\frac{\delta'\delta''}{\delta}\,\{\alpha+\beta(Y+Z)+\gamma YZ\}$$

The quantities $A'^2+B'^2-C'^2$, $A''^2+B''^2-C''^2$ are quadratic functions of Z and Y respectively, and with proper relations between the coefficients, we may assume

$$(A'^2 + B'^2 - C'^2)(A''^2 + B''^2 - C''^2) \;=\; U^2 + k\big[(\alpha + \beta(Y+Z) + \gamma YZ)^2 - \delta^2(1+Y^2)(1+Z^2)\big],$$

in which U is a linear function of $Y+Z$ and YZ, and k and δ^2 are constants. The first side must, in the first place, be symmetrical with respect to Y and Z, or

$$\alpha'^2 + \beta'^2 - \delta'^2, \quad (\alpha' + \gamma')\,\beta', \quad \beta'^2 + \gamma'^2 - \delta'^2$$

must be proportional to

$$\alpha''^2 + \beta''^2 - \delta''^2, \quad (\alpha'' + \gamma'')\,\beta'', \quad \beta''^2 + \gamma''^2 - \delta''^2.$$

But since $(\alpha' + \gamma')\beta'$, $(\alpha'' + \gamma'')\beta''$ are proportional to $\gamma'^2 - \alpha'^2$, $\gamma''^2 - \alpha''^2$, it is only necessary that $\beta'^2 + \gamma'^2 - \delta'^2$ should be proportional to $\beta''^2 + \gamma''^2 - \delta''^2$, that is, that

$$\frac{\gamma'^2 - \alpha'^2}{\beta'^2 + \gamma'^2 - \delta'^2} = \frac{\gamma''^2 - \alpha''^2}{\beta''^2 + \gamma''^2 - \delta''^2},$$

or since the equations are supposed symmetrically related to the system, we must have the second set of relations between the coefficients. Suppose

$$\frac{\gamma - \alpha}{\beta^2 + \gamma^2 - \delta^2} = \frac{\gamma' - \alpha'}{\beta'^2 + \gamma'^2 - \delta'^2} = \frac{\gamma'' - \alpha''}{\beta''^2 + \gamma''^2 - \delta''^2} = -\frac{s}{\phi'},$$

then since

$$\gamma^2 - \alpha^2 = -4\,(\gamma + \phi\beta)\,\phi\beta, \quad \&\text{c.},$$

[Book p. 70.] we have

$$\begin{aligned}
\delta^2 &= \beta^2 + \gamma^2 - 4s\,(\gamma + \phi\beta)\,\beta, \\
\delta'^2 &= \beta'^2 + \gamma'^2 - 4s\,(\gamma' + \phi\beta')\,\beta', \\
\delta''^2 &= \beta''^2 + \gamma''^2 - 4s\,(\gamma'' + \phi\beta'')\,\beta'',
\end{aligned}$$

and δ, δ', δ'' will be supposed henceforth to satisfy these equations.

We have next

$$\begin{aligned}
A'^2 + B'^2 - C'^2 &= 4\,(\gamma' + \phi\beta')\,\beta'\,s\,(Z + \tfrac{\beta'}{\phi} + sZ^2), \\
A''^2 + B''^2 - C''^2 &= 4\,(\gamma'' + \phi\beta'')\,\beta''\,s\,(Y + \tfrac{\beta''}{\phi} + sY^2),
\end{aligned}$$

which may be simplified by writing

$$s = \frac{\mu - \phi}{1 + \mu^2}, \qquad \tfrac{1+\mu^2}{\mu-\phi} = \frac{1 + \nu^2}{\mu - \phi},$$

where μ, ν are to be considered as given functions of s and ϕ. These values give

$$\begin{aligned}
A'^2 + B'^2 - C'^2 &= 4\,(\gamma' + \phi\beta')\,\beta'\,s\,(Z + \mu)\,(Z + \nu), \\
A''^2 + B''^2 - C''^2 &= 4\,(\gamma'' + \phi\beta'')\,\beta''\,s\,(Y + \mu)\,(Y + \nu).
\end{aligned}$$

Hence, putting for simplicity

$$b^2 = 4\,(\gamma' + \phi\beta')\,(\gamma'' + \phi\beta'')\,\beta'\beta'',$$

we have

$$b^2 s^2\,(Z + \mu)(Z + \nu)(Y + \mu)(Y + \nu) = U^2 + k\big[(\alpha + \beta(Y + Z) + \gamma YZ)^2 - \delta^2(1 + Y^2)(1 + Z^2)\big];$$

and the two sides have next to be expressed in terms of $Y + Z$ and YZ.

If for symmetry we write

$$\xi = 1, \qquad \eta = Y + Z, \qquad \zeta = YZ,$$

then

$$b^2s^2\,(\mu^2\xi+\mu\eta+\zeta)(\nu^2\xi+\nu\eta+\zeta) \;=\; U^2+k\big[(\alpha\xi+\beta\eta+\gamma\zeta)^2-\delta^2\{(\zeta-\xi)^2+\eta^2\}\big],$$

and U is now to be considered as a linear function of ξ, η, ζ.

The condition that the first side of the equation may divide into factors, gives an equation for determining k; and since the condition is satisfied for $k=\infty$, the equation will be linear, and it is easily seen that the value is $k=\frac{b^2s^2}{\delta^2}\,(\mu-\nu)^2$. In fact

$$4\,(\mu^2\xi+\mu\eta+\zeta)(\nu^2\xi+\nu\eta+\zeta) \;=\; (\mu^2\xi+\mu\eta+\nu^2\xi+\nu\eta+2\zeta)^2+(\mu-\nu)^2\big[(\zeta-\xi)^2+\eta^2\big],$$

hence

[Book p. 71.] and we may assume

$$U \;=\; \frac{\mu-\nu}{\delta}\Big\{\Big(\Lambda+\frac{1}{\Lambda}\Big)(\alpha\xi+\beta\eta+\gamma\zeta)-\delta\,\Big(\Lambda-\frac{1}{\Lambda}\Big)(\xi+\zeta)\Big\},$$

subject to its being shown that the value of Λ given by the comparison of coefficients is constant. The comparison of coefficients gives

$$2\mu+2\nu \;=\; \frac{\mu-\nu}{\delta}\Big\{\Big(\Lambda+\tfrac{1}{\Lambda}\Big)\alpha-\Big(\Lambda-\tfrac{1}{\Lambda}\Big)\delta\Big\},$$

$$4\,\xi \;=\; \frac{\mu-\nu}{\delta}\Big\{\Big(\Lambda+\tfrac{1}{\Lambda}\Big)\gamma-\Big(\Lambda-\tfrac{1}{\Lambda}\Big)\delta\Big\},$$

$$4\,(1-\mu^2) \;=\; \frac{\mu-\nu}{\delta}\,\Big(\Lambda+\tfrac{1}{\Lambda}\Big)(\gamma-\alpha),$$

the first and third of which give

$$4\,(1-\mu^2) \;=\; \frac{\mu-\nu}{\delta}\,\Big(\Lambda+\tfrac{1}{\Lambda}\Big)(\gamma-\alpha),$$

which will be identical with the second, if

$$\frac{2(1-\mu\nu)}{\mu+\nu} \;=\; \frac{\beta}{\gamma-\alpha} \;=\; -\,2\phi,$$

which follows at once from the equation $\nu=\frac{1+\mu\phi}{\mu-\phi}$.

Forming next the two equations

$$\ldots$$

these will be equivalent to a single equation, if

$$\tfrac{1}{4}\,(\mu+\nu)^2\,\delta^2 \;=\; (\mu+\nu)^2(\beta^2+\gamma^2)-4\,(\mu+\nu)\beta\gamma-4\,(\mu^2-1)\beta^2,$$

[Book p. 72.] or, finally, if

$$\delta^2+\delta^2 \;=\; \beta^2+\gamma^2-4s\beta\big[\gamma+\tfrac{\mu\nu-1}{\mu+\nu}\beta\big] \;=\; \beta^2+\gamma^2-4s\,(\gamma+\phi\beta)\,\beta,$$

which is in fact the case.

Writing the equations for

$$\Lambda \;+\; \tfrac{1}{\Lambda} \;=\; \tfrac{2\delta}{(\mu-\nu)\beta s}\,(\gamma-2s\beta),$$

$$\Lambda \;-\; \tfrac{1}{\Lambda} \;=\; \tfrac{2}{(\mu-\nu)\beta s\delta}\,(\gamma-2s\beta-2\phi\beta),$$

in the form

$$\ldots$$

and substituting in

$$U \;=\; \frac{\mu-\nu}{\delta}\left\{\left(\Lambda+\tfrac{1}{\Lambda}\right)(\alpha\xi+\beta\eta+\gamma\zeta)-\delta\left(\Lambda-\tfrac{1}{\Lambda}\right)(\xi+\zeta)\right\},$$

we have

$$U \;=\; \frac{2}{\beta s}\left\{(-\beta+2s\gamma+2\phi\gamma)\,\xi+(\gamma-2s\beta)\,\eta+(-\beta+2s\gamma+4\phi\gamma)\,\zeta\right\}\frac{1}{\delta},$$

and consequently, multiplying by

$$b \;=\; 2\sqrt{(\gamma'+\phi\beta')(\gamma''+\phi\beta'')\beta'\beta''},$$

we have

$$\tfrac{1}{\delta}\sqrt{(\gamma'+\phi\beta')(\gamma''+\phi\beta'')\beta'\beta''}\left\{(-\beta+2s\gamma+2\phi\gamma)\,\xi+(\gamma-2s\beta)\,\eta+(-\beta+2s\gamma+4\phi\gamma)\,\zeta\right\},$$

or collecting the different terms which enter into the equation

$$A'A''+B'B''-C'C'' \;=\; \sqrt{A'^2+B'^2-C'^2}\cdot\sqrt{A''^2+B''^2-C''^2},$$

the result is

$$\begin{aligned}&\left\{(\alpha'\alpha''+\beta'\beta'')\,\xi+(\gamma'\beta''+\gamma''\beta'+2\phi\beta'\beta'')\,\eta+(\beta'\beta''+\gamma'\gamma'')\,\zeta\right\}+\tfrac{\delta'\delta''}{\delta}(\alpha\xi+\beta\eta+\gamma\zeta)\\&\quad-\tfrac{1}{\delta}\sqrt{(\gamma'+2\phi\beta')(\gamma''+2\phi\beta'')\beta'\beta''}\left\{(-\beta+2s\gamma+2\phi\gamma)\,\xi+(\gamma-2s\beta)\,\eta+(-\beta+2s\gamma+4\phi\gamma)\,\zeta\right\} \;=\; 0,\end{aligned}$$

which, combined with the first equation written under the form

$$\alpha\xi+\beta\eta+\gamma\zeta+\delta\sqrt{(\zeta-\xi)^2+\eta^2} \;=\; 0,$$

determines the ratios of ξ, η, ζ, that is, the values of $Y+Z$ and YZ.

§ 6.

The system of equations

$$\begin{aligned}(f+2\theta\sqrt{\mathfrak{A}})+\sqrt{\mathfrak{A}}\,(Y+Z)+f\,YZ-\sqrt{bc}\,\sqrt{1+Y^2}\sqrt{1+Z^2} &\;=\; 0,\\(g+2\theta\sqrt{\mathfrak{B}})+\sqrt{\mathfrak{B}}\,(Z+X)+g\,ZX-\sqrt{ca}\,\sqrt{1+Z^2}\sqrt{1+X^2} &\;=\; 0,\\(h+2\theta\sqrt{\mathfrak{C}})+\sqrt{\mathfrak{C}}\,(X+Y)+h\,XY-\sqrt{ab}\,\sqrt{1+X^2}\sqrt{1+Y^2} &\;=\; 0,\end{aligned}$$

where

$$\theta \;=\; \tfrac{1}{K}\left(\sqrt{\mathfrak{ABC}}+\mathfrak{F}\sqrt{\mathfrak{A}}+\mathfrak{G}\sqrt{\mathfrak{B}}+\mathfrak{H}\sqrt{\mathfrak{C}}\right),$$

on which depends the solution of Steiner's extension of Malfatti's problem, is at once seen to belong to the class of equations treated of in the preceding section, and we have $\phi=0$, $s=0$. The equations at the conclusion of the preceding section become

$$\begin{aligned}&\left\{\sqrt{\mathfrak{BC}}+gh+2\theta\,(g\sqrt{\mathfrak{C}}+h\sqrt{\mathfrak{B}})+4\theta^2\sqrt{\mathfrak{BC}}\right\}\xi+\left\{g\sqrt{\mathfrak{C}}+h\sqrt{\mathfrak{B}}+2\theta\sqrt{\mathfrak{BC}}\right\}\eta+\left\{\sqrt{\mathfrak{BC}}+gh\right\}\zeta\\&\quad-a\left[(f+2\theta\sqrt{\mathfrak{A}})\,\xi+\sqrt{\mathfrak{A}}\,\eta+f\zeta\right]-\tfrac{1}{\sqrt{bc}}\sqrt{(g+\theta\sqrt{\mathfrak{B}})(h+\theta\sqrt{\mathfrak{C}})\sqrt{\mathfrak{BC}}}\left[(\sqrt{\mathfrak{A}}-2\theta f)\,\xi-f\eta+\sqrt{\mathfrak{A}}\zeta\right] \;=\; 0,\\&\left\{(f+2\theta\sqrt{\mathfrak{A}})\,\xi+\sqrt{\mathfrak{A}}\,\eta+f\zeta\right\}^2-bc\left\{(\xi-\zeta)^2+\eta^2\right\} \;=\; 0,\end{aligned}$$

which may also be written

$$\begin{aligned}&(\sqrt{\mathfrak{BC}}+\mathfrak{F})(\xi+\zeta)+(-a\sqrt{\mathfrak{A}}+g\sqrt{\mathfrak{B}}+h\sqrt{\mathfrak{C}}+2\theta\sqrt{\mathfrak{BC}})(\eta+2\theta\xi)\\ &\quad-\tfrac{1}{\sqrt{bc}}\sqrt{(g+\theta\sqrt{\mathfrak{B}})(h+\theta\sqrt{\mathfrak{C}})\sqrt{\mathfrak{BC}}}\,\left[(\sqrt{\mathfrak{A}}-2\theta f)\,\xi-f\eta+\sqrt{\mathfrak{A}}\zeta\right] \;=\; 0,\\ &\sqrt{\mathfrak{A}}\left\{f(\xi+\zeta)+(\eta+2\theta\xi)\right\}^2-bc\left\{(\xi-\zeta)^2+\eta^2\right\} \;=\; 0.\end{aligned}$$

Hence, observing that

$$\begin{aligned}g+\theta\sqrt{\mathfrak{B}} \;&=\; \tfrac{1}{K}\,(\sqrt{\mathfrak{BC}}+\mathfrak{F})(\sqrt{\mathfrak{AB}}+\mathfrak{H}),\\ h+\theta\sqrt{\mathfrak{C}} \;&=\; \tfrac{1}{K}\,(\sqrt{\mathfrak{BC}}+\mathfrak{F})(\sqrt{\mathfrak{AC}}+\mathfrak{G}),\\ -a\sqrt{\mathfrak{A}}+h\sqrt{\mathfrak{B}}+g\sqrt{\mathfrak{C}}+2\theta\sqrt{\mathfrak{BC}} \;&=\; (\sqrt{\mathfrak{BC}}+\mathfrak{F}),\end{aligned}$$

and putting for a moment

$$4\,\lambda \;=\; \sqrt{(\sqrt{\mathfrak{AB}}+\mathfrak{H})(\sqrt{\mathfrak{AC}}+\mathfrak{G})\sqrt{\mathfrak{BC}}},$$

and therefore

$$\sqrt{(g+\theta\sqrt{\mathfrak{B}})(h+\theta\sqrt{\mathfrak{C}})\sqrt{\mathfrak{BC}}} \;=\; (\sqrt{\mathfrak{BC}}+\mathfrak{F})\,\lambda,$$

[Book p. 74.] the first equation divides by $(\sqrt{\mathfrak{BC}}+\mathfrak{F})$, and the result is

$$(\xi+\zeta)+\theta\,(\eta+2\theta\xi)-\tfrac{2\lambda}{\sqrt{bc}}\left[\sqrt{\mathfrak{A}}\,(\xi+\zeta)-f\,(\eta+2\theta\xi)+\sqrt{\mathfrak{A}}\,(\eta+2\theta\xi)-(1+\theta^2)\,\xi\right] \;=\; 0.$$

Also, by an easy transformation, the second equation becomes

$$\sqrt{\mathfrak{A}}\left[(\xi+\zeta)-\left\{\sqrt{\mathfrak{A}}\,(\xi+\zeta)-f\,(\eta+2\theta\xi)\right\}\right]^2-bc\left\{(\xi+\zeta)-(1+\theta^2)\,\xi\right\}^2 \;=\; 0,$$

or putting

$$\xi+\zeta+\theta\,(\eta+2\theta\xi) \;=\; \Theta,\qquad \tfrac{1}{\sqrt{bc}}\left[\sqrt{\mathfrak{A}}\,(\xi+\zeta)-f\,(\eta+2\theta\xi)\right] \;=\; \Phi,\qquad \zeta \;=\; \Psi,$$

the equations become

$$\Theta-2\lambda\Phi \;=\; 0,\qquad -\Phi^2+4\lambda\left(\Theta-(1+\theta^2)\,\Psi\right) \;=\; 0\,;$$

hence eliminating Φ,

$$\left(\tfrac{\Theta}{2\lambda}\right)^2 \;=\; \tfrac{\Theta^2}{(1+\theta^2)\,\Psi}\left(1-\tfrac{1+\theta^2}{4\lambda^2}\right),$$

or, observing that

$$1+\theta^2 \;=\; \tfrac{1}{K^2}\,(\sqrt{\mathfrak{BC}}+\mathfrak{F})(\sqrt{\mathfrak{CA}}+\mathfrak{G})(\sqrt{\mathfrak{AB}}+\mathfrak{H}),$$

and reducing, we obtain a determination of Ψ; also $\Theta=2\lambda\Phi$ gives

$$\Phi \;=\; \frac{K\Theta}{2\sqrt{(\sqrt{\mathfrak{AB}}+\mathfrak{H})(\sqrt{\mathfrak{AC}}+\mathfrak{G})\sqrt{\mathfrak{BC}}}}\,.$$

Suppose

$$\sqrt{\mathfrak{BC}}+\mathfrak{F} \;=\; \mathfrak{a},\quad \sqrt{\mathfrak{BC}}-\mathfrak{F} \;=\; \mathfrak{a}_1,\quad \tfrac{\sqrt{\mathfrak{BC}}}{\beta\gamma} \;=\; \beta_2\gamma_2,$$

then substituting

$$\xi:\eta:\zeta \;=\; \left(\tfrac{\mathfrak{a}}{\mathfrak{a}_1}\right)^{1/2}\left(1+\tfrac{\sqrt{\mathfrak{a}}}{\sqrt{\beta_2\gamma_2}}\right)\;:\;\cdots$$

[Book p. 75.] that is,

$$\begin{aligned}\xi+\zeta+\theta\,(\eta+2\theta\xi)-\tfrac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}}\left[\sqrt{\mathfrak{A}}\,(\xi+\zeta)-f\,(\eta+2\theta\xi)\right] &= 0,\\ \xi+\zeta+\theta\,(\eta+2\theta\xi)-\tfrac{4bc\,(\mathfrak{a}+\mathfrak{a}_1)}{\beta_2\gamma_2}\left(1-\tfrac{\sqrt{\mathfrak{a}}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}}\right)\zeta &= 0;\end{aligned}$$

these may be written

$$L\xi+M\eta+N\zeta = 0, \qquad L'\xi+M'\eta+N'\zeta = 0,$$

where

$$\begin{aligned}L &= 1+2\theta^2-\tfrac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}}\,(\sqrt{\mathfrak{A}}-2\theta f),\\ M &= \theta+\tfrac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}}\,f,\\ N &= 1-\tfrac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}}\,\sqrt{\mathfrak{A}},\\ L' &= 1+2\theta^2-\tfrac{4bc\,(\mathfrak{a}+\mathfrak{a}_1)}{\beta_2\gamma_2}\left(1-\tfrac{\sqrt{\mathfrak{a}}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}}\right), \quad M' = \theta, \quad N' = 1,\end{aligned}$$

or since ξ, η, ζ are equal to 1, $Y+Z$, YZ respectively,

$$1:Y+Z:YZ = MN'-M'N:NL'-N'L:LM'-L'M.$$

Also

...

whence

$$\begin{aligned}Y+Z &= -\frac{2\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}\,\sqrt{\beta_2\gamma_2}}{K}\left(1-\tfrac{\sqrt{\mathfrak{a}}}{\sqrt{\beta_2\gamma_2}}\right)\left(1-\tfrac{\sqrt{\mathfrak{a}_1}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}}\right)-2\theta,\\ YZ &= \frac{2\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}\,\sqrt{\beta_2\gamma_2}}{K}\left(\theta+\tfrac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}}\,f\right)\left(1-\tfrac{\sqrt{\mathfrak{a}}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}}\right)-1;\end{aligned}$$

and by forming the analogous expressions for $Z+X$ and ZX, and $X+Y$ and XY, the values of X, Y, Z may be determined. But the equations in question simplify themselves in a remarkable manner by the notation before alluded to.

[Book p. 76.] Suppose

$$\mathrm{f} = \sqrt[4]{\mathfrak{A}}\sqrt{\sqrt{\mathfrak{B}\mathfrak{C}}-\mathfrak{F}}, \quad \mathrm{g} = \sqrt[4]{\mathfrak{B}}\sqrt{\sqrt{\mathfrak{C}\mathfrak{A}}-\mathfrak{G}}, \quad \mathrm{h} = \sqrt[4]{\mathfrak{C}}\sqrt{\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H}}, \quad J = \sqrt{2}\,\sqrt[4]{\mathfrak{A}\mathfrak{B}\mathfrak{C}},$$

these values give

$$\begin{aligned}\tfrac{K\sqrt{\mathfrak{A}}}{\sqrt{\mathfrak{B}\mathfrak{C}}}\,\mathfrak{a} &= 2\,\mathrm{f}^2\left(1-\tfrac{\mathrm{f}^2}{J^2}\right),\\ \tfrac{K\sqrt{bc}}{\sqrt{\mathfrak{A}}} &= 2\,\mathrm{g}\,\mathrm{h}\sqrt{1-\tfrac{\mathrm{g}^2}{J^2}}\sqrt{1-\tfrac{\mathrm{h}^2}{J^2}}\,,\\ \tfrac{Kf}{\sqrt{\mathfrak{A}}} &= \mathrm{f}^2-\mathrm{g}^2-\mathrm{h}^2+\tfrac{2\mathrm{g}^2\mathrm{h}^2}{J^2}\,,\\ K\theta &= -\mathrm{f}^2-\mathrm{g}^2-\mathrm{h}^2+2J^2\,,\\ K^2 &= -\mathrm{f}^4-\mathrm{g}^4-\mathrm{h}^4+2\mathrm{g}^2\mathrm{h}^2+2\mathrm{h}^2\mathrm{f}^2+2\mathrm{f}^2\mathrm{g}^2-\tfrac{4\mathrm{f}^2\mathrm{g}^2\mathrm{h}^2}{J^2}\,.\end{aligned}$$

Applying these results to the preceding formulae and forming for that purpose the equations

$$\tfrac{2\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}\sqrt{\beta_2\gamma_2}}{K}\cdot 4\,\mathrm{g}\,\mathrm{h}\,\lambda = \tfrac{J^2}{\mathrm{f}}\,, \qquad \tfrac{\sqrt{\mathfrak{a}_1}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}} = \tfrac{\mathrm{f}}{J}\,,$$

$$\mathrm{g\,h}\,K\theta + \frac{J^2}{\sqrt{\mathfrak{A}}}\,Kf \;=\; (J^2 - \mathrm{g\,h})(\mathrm{f}^2 - (\mathrm{g}-\mathrm{h})^2) - 2\,\mathrm{g\,h}\,(\mathrm{g}-\mathrm{h})^2\,,$$

we have

$$\begin{aligned} K(Y+Z) + 2K\theta \;&=\; 4\,(J^2 - \mathrm{g\,h})\left(1 - \tfrac{\mathrm{f}}{J}\right), \\ K^2 YZ + K^2 \;&=\; \left\{(J^2 - \mathrm{g\,h})(\mathrm{f}^2 - (\mathrm{g}-\mathrm{h})^2) - 2\,\mathrm{g\,h}\,(\mathrm{g}-\mathrm{h})^2\right\}\left(1 - \tfrac{\mathrm{f}}{J}\right); \end{aligned}$$

the former of which, combined with the similar equations for $Z+X$ and $X+Y$, gives for X, Y, Z the values to be presently stated, and these values will of course verify the second equation and the corresponding equations for ZX and XY.

Recapitulating the preceding notation, if $x=0$, $y=0$, $z=0$ are the equations of the given sections, $w=0$ the equation of the polar plane of their point of intersection with respect to the surface,

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 \;=\; 0,$$

the equation of the surface, $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$, $\mathfrak{F}$, $\mathfrak{G}$, $\mathfrak{H}$ as usual, and

[Book p. 77.] putting

$$\mathrm{f} \;=\; \sqrt[4]{\mathfrak{A}}\sqrt{\sqrt{\mathfrak{BC}} - \mathfrak{F}}\,,\quad \mathrm{g} \;=\; \sqrt[4]{\mathfrak{B}}\sqrt{\sqrt{\mathfrak{CA}} - \mathfrak{G}}\,,\quad \mathrm{h} \;=\; \sqrt[4]{\mathfrak{C}}\sqrt{\sqrt{\mathfrak{AB}} - \mathfrak{H}}\,,\quad J \;=\; \sqrt{2}\,\sqrt[4]{\mathfrak{ABC}}\,,$$

then the equations of the required sections are

$$\begin{aligned} (ax + hy + gz)\,X + y\sqrt{\mathfrak{C}} + z\sqrt{\mathfrak{B}} + w\sqrt{-ap}\sqrt{1+X^2} \;&=\; 0, \\ x\sqrt{\mathfrak{C}} + (hx + by + fz)\,Y + z\sqrt{\mathfrak{A}} + w\sqrt{-bp}\sqrt{1+Y^2} \;&=\; 0, \\ x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)\,Z + w\sqrt{-cp}\sqrt{1+Z^2} \;&=\; 0, \end{aligned}$$

where X, Y, Z are to be determined by the following equations,

$$\begin{aligned} (f + 2\theta\sqrt{\mathfrak{A}}) + \sqrt{\mathfrak{A}}\,(Y+Z) + f\,YZ - \sqrt{bc}\,\sqrt{1+Y^2}\sqrt{1+Z^2} \;&=\; 0, \\ (g + 2\theta\sqrt{\mathfrak{B}}) + \sqrt{\mathfrak{B}}\,(Z+X) + g\,ZX - \sqrt{ca}\,\sqrt{1+Z^2}\sqrt{1+X^2} \;&=\; 0, \\ (h + 2\theta\sqrt{\mathfrak{C}}) + \sqrt{\mathfrak{C}}\,(X+Y) + h\,XY - \sqrt{ab}\,\sqrt{1+X^2}\sqrt{1+Y^2} \;&=\; 0; \end{aligned}$$

and the solution of which, putting

$$\mathrm{f} \;=\; \sqrt[4]{\mathfrak{A}}\sqrt{\sqrt{\mathfrak{BC}} - \mathfrak{F}}\,,\quad \mathrm{g} \;=\; \sqrt[4]{\mathfrak{B}}\sqrt{\sqrt{\mathfrak{CA}} - \mathfrak{G}}\,,\quad \mathrm{h} \;=\; \sqrt[4]{\mathfrak{C}}\sqrt{\sqrt{\mathfrak{AB}} - \mathfrak{H}}\,,\quad J \;=\; \sqrt{2}\,\sqrt[4]{\mathfrak{ABC}}\,,$$

is given by the equations

$$\begin{aligned} KX \;&=\; \tfrac{2\mathrm{f\,g\,h}}{J} + (-\mathrm{f} + \mathrm{g} + \mathrm{h})^2 - 2(-\mathrm{f} + \mathrm{g} + \mathrm{h})\,J, \\ KY \;&=\; \tfrac{2\mathrm{f\,g\,h}}{J} + (\mathrm{f} - \mathrm{g} + \mathrm{h})^2 - 2(\mathrm{f} - \mathrm{g} + \mathrm{h})\,J, \\ KZ \;&=\; \tfrac{2\mathrm{f\,g\,h}}{J} + (\mathrm{f} + \mathrm{g} - \mathrm{h})^2 - 2(\mathrm{f} + \mathrm{g} - \mathrm{h})\,J. \end{aligned}$$

Instead of the direct but very tedious process by which these values of X, Y, Z have been obtained, we may substitute the following *à posteriori verification.*[2]

[2]It is perhaps worth noticing that the value of the quantity λ previously made use of is $\lambda^2 \;=\; -\frac{1}{K^2}\big(\mathrm{f}^2 - \mathrm{g}^2 - \mathrm{h}^2 + (J + \mathrm{f} - \mathrm{g} - \mathrm{h})\,J\big)\cdots\cdot$

We have

$$\begin{aligned}
K^2(1+X^2) &= 4\,(-\mathrm{f}+\mathrm{g}+\mathrm{h})^2 J^2\big(1+\tfrac{\mathrm{f}}{J}\big)\big(1-\tfrac{\mathrm{g}}{J}\big)\big(1-\tfrac{\mathrm{h}}{J}\big),\\
K^2\sqrt{1+Y^2}\sqrt{1+Z^2} &= 4\,(\mathrm{f}^2-(\mathrm{g}-\mathrm{h})^2)J^2\big(1-\tfrac{\mathrm{f}}{J}\big)\sqrt{1-\tfrac{\mathrm{g}^2}{J^2}}\sqrt{1-\tfrac{\mathrm{h}^2}{J^2}},\\
K^2(1+YZ) &= 4\,\big(1-\tfrac{\mathrm{f}}{J}\big)\big\{(J^2-\mathrm{g}\,\mathrm{h})(\mathrm{f}^2-(\mathrm{g}-\mathrm{h})^2)-2\,\mathrm{g}\,\mathrm{h}\,(\mathrm{g}-\mathrm{h})^2\big\},\\
K^2(Y+Z)-2\mathrm{f}^2-2\mathrm{g}^2-2\mathrm{h}^2+4J^2 &= 4\,\big(1-\tfrac{\mathrm{f}}{J}\big)(J^2-\mathrm{g}\,\mathrm{h}).
\end{aligned}$$

[Book p. 78.] Putting also

$$\mathrm{f}^2-\mathrm{g}^2-\mathrm{h}^2+\tfrac{2\mathrm{g}^2\mathrm{h}^2}{J^2}\,K^2(1+YZ) \;=\; 4\,\big(1-\tfrac{\mathrm{f}}{J}\big)\big\{(\mathrm{f}^2-(\mathrm{g}-\mathrm{h})^2)\big[(J^2-\mathrm{g}\,\mathrm{h})(\mathrm{f}^2-(\mathrm{g}-\mathrm{h})^2)-2\,\mathrm{g}\,\mathrm{h}\,(\mathrm{g}-\mathrm{h})^2\big]-4\,\mathrm{g}^2\mathrm{h}^2\,(\mathrm{g}-\mathrm{h})^2(J^2$$

we have, after reductions in which the symmetry of the expressions plays an essential part,

$$4\,\big(1-\tfrac{\mathrm{f}}{J}\big)\big(\mathrm{f}^2-(\mathrm{g}-\mathrm{h})^2\big)\,2\,\mathrm{g}\,\mathrm{h}\,J^2\,\big(1-\tfrac{\mathrm{g}}{J}\big)\big(1-\tfrac{\mathrm{h}}{J}\big),$$

and the values obtained above give also

$$2\,\mathrm{g}\,\mathrm{h}\,\sqrt{1+Y^2}\sqrt{1+Z^2}\,\sqrt{1+X^2}\,K^2,$$

which shows that the relation between Y and Z is verified by the assumed values of these quantities, and the other two equations are of course also verified. The solution of the problem will be rendered more complete if the equations of the required sections and of the auxiliary sections made use of in the geometrical construction are expressed in terms of f, g, h, J.

§ 7.

First, to substitute in the equations of the required sections or resultors. Writing the first equation in the form

$$\tfrac{K^2}{2\sqrt{\mathfrak{BC}}}\,\big[\sqrt{\mathfrak{BC}}\,ax+(hX+\sqrt{\mathfrak{C}})\,y+(gX+\sqrt{\mathfrak{B}})\,z+\sqrt{-ap}\sqrt{1+X^2}\,w\big] \;=\; 0,$$

the coefficient of x will be

$$\tfrac{a}{\sqrt{\mathfrak{BC}}}\,\big\{-\tfrac{\mathrm{f}^2}{J^2}(-\mathrm{f}^2+\mathrm{g}^2+\mathrm{h}^2)\,K^2\,\big(1-\tfrac{\mathrm{f}}{J}\big)-\mathrm{g}\,\mathrm{h}\,K^2\,\big(1-\tfrac{\mathrm{f}}{J}\big)\big\}\,\cdots\,,$$

[Book p. 79.] or, as it is convenient to write it,

$$\mathrm{f}\,(-\mathrm{f}+\mathrm{g}+\mathrm{h})\,\tfrac{\mathrm{f}}{\sqrt{\mathfrak{A}}}\,\Big(1-\tfrac{\mathrm{f}}{J}\Big)\,x.$$

The coefficient of y is

$$\tfrac{1}{2\sqrt{\mathfrak{B}}}\,\Big\{\big[-\mathrm{f}^4-\mathrm{g}^4-\mathrm{h}^4+2\mathrm{g}^2\mathrm{h}^2+2\mathrm{h}^2\mathrm{f}^2+2\mathrm{f}^2\mathrm{g}^2-\tfrac{4\mathrm{f}^2\mathrm{g}^2\mathrm{h}^2}{J^2}\big]\cdots\Big\},$$

or, after all reductions,

$$\big(\mathrm{f}^2-\mathrm{g}^2+\mathrm{h}^2\big)\,\tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\,\Big(1-\tfrac{\mathrm{f}}{J}\Big)\,y,$$

and similarly the coefficient of z is

$$\big(\mathrm{f}^2+\mathrm{g}^2-\mathrm{h}^2\big)\,\tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\,\Big(1-\tfrac{\mathrm{f}}{J}\Big)\,z,$$

and the coefficient of w is

$$\tfrac{1}{J}\Big(1+\tfrac{\mathrm{f}}{J}\Big)(-\mathrm{f}+\mathrm{g}+\mathrm{h})\cdot 2\sqrt{J^2-\mathrm{f}^2}\sqrt{1-\tfrac{\mathrm{g}^2}{J^2}}\sqrt{1-\tfrac{\mathrm{h}^2}{J^2}}\,\sqrt{-ap}\,w.$$

Hence, forming the equation of the resultor in question, and by means of it those of the other resultors, the equations of the resultors are

$$\begin{aligned}
&\mathrm{f}\,(-\mathrm{f}+\mathrm{g}+\mathrm{h})\,\tfrac{\mathrm{f}}{\sqrt{\mathfrak{A}}}\,\big(1-\tfrac{\mathrm{f}}{J}\big)\,x+\mathrm{g}\,(\mathrm{f}^2-\mathrm{g}^2+\mathrm{h}^2)\,\tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\,\big(1-\tfrac{\mathrm{f}}{J}\big)\,y\\
&\qquad+\mathrm{h}\,(\mathrm{f}^2+\mathrm{g}^2-\mathrm{h}^2)\,\tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\,\big(1-\tfrac{\mathrm{f}}{J}\big)\,z+2\sqrt{J^2-\mathrm{f}^2}\sqrt{1-\tfrac{\mathrm{g}^2}{J^2}}\sqrt{1-\tfrac{\mathrm{h}^2}{J^2}}\,\sqrt{-ap}\,w \;=\; 0,\\
&\mathrm{f}\,(-\mathrm{f}^2+\mathrm{g}^2+\mathrm{h}^2)\,\tfrac{\mathrm{f}}{\sqrt{\mathfrak{A}}}\,\big(1-\tfrac{\mathrm{g}}{J}\big)\,x+\mathrm{g}\,(\mathrm{f}-\mathrm{g}+\mathrm{h})\,\tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\,\big(1-\tfrac{\mathrm{g}}{J}\big)\,y\\
&\qquad+\mathrm{h}\,(\mathrm{f}^2+\mathrm{g}^2-\mathrm{h}^2)\,\tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\,\big(1-\tfrac{\mathrm{g}}{J}\big)\,z+2\sqrt{J^2-\mathrm{g}^2}\sqrt{1-\tfrac{\mathrm{f}^2}{J^2}}\sqrt{1-\tfrac{\mathrm{h}^2}{J^2}}\,\sqrt{-bp}\,w \;=\; 0,\\
&\mathrm{f}\,(\mathrm{f}^2-\mathrm{g}^2+\mathrm{h}^2)\,\tfrac{\mathrm{f}}{\sqrt{\mathfrak{A}}}\,\big(1-\tfrac{\mathrm{h}}{J}\big)\,x+\mathrm{g}\,(\mathrm{f}^2+\mathrm{g}^2-\mathrm{h}^2)\,\tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\,\big(1-\tfrac{\mathrm{h}}{J}\big)\,y\\
&\qquad+\mathrm{h}\,(\mathrm{f}+\mathrm{g}-\mathrm{h})\,\tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\,\big(1-\tfrac{\mathrm{h}}{J}\big)\,z+2\sqrt{J^2-\mathrm{h}^2}\sqrt{1-\tfrac{\mathrm{f}^2}{J^2}}\sqrt{1-\tfrac{\mathrm{g}^2}{J^2}}\,\sqrt{-cp}\,w \;=\; 0,
\end{aligned}$$

values which might be somewhat simplified by writing $\mathfrak{x}$, $\mathfrak{y}$, $\mathfrak{z}$, $\mathfrak{w}$ instead of

$$\tfrac{\mathrm{f}}{\sqrt{\mathfrak{A}}}\big(1-\tfrac{\mathrm{f}}{J}\big)x,\quad \tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\big(1-\tfrac{\mathrm{g}}{J}\big)y,\quad \tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\big(1-\tfrac{\mathrm{h}}{J}\big)z,\quad \sqrt{-p}\sqrt{1-\tfrac{\mathrm{f}^2}{J^2}}\sqrt{1-\tfrac{\mathrm{g}^2}{J^2}}\sqrt{1-\tfrac{\mathrm{h}^2}{J^2}}\,w\,;$$

and it may be also remarked that the coefficients as well of these formulae as of those which follow may be elegantly expressed in terms of the parts of a triangle having f, g, h for its sides.

The equations of the separators are found by taking the differences two and two of the equations of the resultors (this requires to be verified *à posteriori*); thus subtracting the third equation from the second, the result contains a constant factor

$$\tfrac{2}{J}\,\big[4\,\mathrm{f}^2\mathrm{g}^2\mathrm{h}^2-J\,(\mathrm{f}^2-(\mathrm{g}-\mathrm{h})^2)\,((\mathrm{g}+\mathrm{h})^2-\mathrm{f}^2)\big],$$

[Book p. 80.] equivalent to (after rejection of factor)

. . .

Rejecting the factor in question and forming the analogous two equations, the equations of the separators are

$$\begin{aligned}
-\;\tfrac{\mathrm{g}-\mathrm{h}}{\mathrm{f}}\cdot\tfrac{\mathrm{f}}{\sqrt{\mathfrak{A}}}\,\big(1-\tfrac{\mathrm{f}}{J}\big)\,x\;+\;\tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\,\big(1-\tfrac{\mathrm{g}}{J}\big)\,y\;-\;\tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\,\big(1-\tfrac{\mathrm{h}}{J}\big)\,z\;&=\;0,\\
\tfrac{\mathrm{f}}{\sqrt{\mathfrak{A}}}\,\big(1-\tfrac{\mathrm{f}}{J}\big)\,x\;-\;\tfrac{\mathrm{h}-\mathrm{f}}{\mathrm{g}}\cdot\tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\,\big(1-\tfrac{\mathrm{g}}{J}\big)\,y\;+\;\tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\,\big(1-\tfrac{\mathrm{h}}{J}\big)\,z\;&=\;0,\\
-\;\tfrac{\mathrm{f}}{\sqrt{\mathfrak{A}}}\,\big(1-\tfrac{\mathrm{f}}{J}\big)\,x\;+\;\tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\,\big(1-\tfrac{\mathrm{g}}{J}\big)\,y\;-\;\tfrac{\mathrm{f}-\mathrm{g}}{\mathrm{h}}\cdot\tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\,\big(1-\tfrac{\mathrm{h}}{J}\big)\,z\;&=\;0\,;
\end{aligned}$$

[Book p. 81.] and from the mode of formation of these equations it is obvious that the separators have a line in common.

The equations of the determinators being $x=0$, $y=0$, $z=0$, the equations of the *tactors* are

$$\sqrt{\mathfrak{B}}\,z-\sqrt{\mathfrak{C}}\,y\;=\;0,\quad \sqrt{\mathfrak{C}}\,x-\sqrt{\mathfrak{A}}\,z\;=\;0,\quad \sqrt{\mathfrak{A}}\,y-\sqrt{\mathfrak{B}}\,x\;=\;0\,;$$

and if $\alpha x+\beta y+\gamma z+\delta w=0$ be the equation of the tactor touching

$$x=0,\quad \sqrt{\mathfrak{B}}\,x-\sqrt{\mathfrak{C}}\,z=0 \text{ and } \sqrt{\mathfrak{A}}\,y-\sqrt{\mathfrak{B}}\,x=0,$$

the conditions of contact are

$$\mathfrak{A}\left(\mathfrak{A}\alpha^2+\cdots+\tfrac{K}{p}\,\delta^2\right) \;=\; (\mathfrak{A}\alpha+\mathfrak{H}\beta+\mathfrak{G}\gamma)^2,$$

$$2\sqrt{\mathfrak{A}\mathfrak{B}}\,(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})\left(\mathfrak{A}\alpha^2+\cdots+\tfrac{K}{p}\,\delta^2\right) \;=\; \left\{(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})(\alpha\sqrt{\mathfrak{A}}-\beta\sqrt{\mathfrak{B}})+\gamma\,(\mathfrak{G}\sqrt{\mathfrak{B}}-\mathfrak{H}\sqrt{\mathfrak{C}})\right\}^2,$$

$$2\sqrt{\mathfrak{A}\mathfrak{C}}\,(\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{G})\left(\mathfrak{A}\alpha^2+\cdots+\tfrac{K}{p}\,\delta^2\right) \;=\; \left\{(\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{G})(\alpha\sqrt{\mathfrak{A}}-\gamma\sqrt{\mathfrak{C}})+\beta\,(\mathfrak{H}\sqrt{\mathfrak{C}}-\mathfrak{G}\sqrt{\mathfrak{B}})\right\}^2;$$

whence

$$\tfrac{1}{\sqrt{\mathfrak{B}}}\sqrt{2\sqrt{\mathfrak{A}\mathfrak{B}}\,(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})}\,(\mathfrak{A}\alpha+\mathfrak{H}\beta+\mathfrak{G}\gamma) \;=\; (\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})\sqrt{\mathfrak{A}}\alpha-(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})\sqrt{\mathfrak{B}}\beta+(\mathfrak{G}\sqrt{\mathfrak{B}}-\mathfrak{H}\sqrt{\mathfrak{C}})\gamma,$$

$$\tfrac{1}{\sqrt{\mathfrak{C}}}\sqrt{2\sqrt{\mathfrak{A}\mathfrak{C}}\,(\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{G})}\,(\mathfrak{A}\alpha+\mathfrak{H}\beta+\mathfrak{G}\gamma) \;=\; (\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{G})\sqrt{\mathfrak{A}}\alpha+(\mathfrak{H}\sqrt{\mathfrak{C}}-\mathfrak{G}\sqrt{\mathfrak{B}})\beta-(\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{G})\sqrt{\mathfrak{C}}\gamma,$$

and

$$a\alpha^2+b\beta^2+c\gamma^2-2f\beta\gamma-2g\gamma\alpha-2h\alpha\beta \;=\; 0,$$

and putting for a moment

$$\sqrt{\mathfrak{A}}\,\mu \;=\; \sqrt{\mathfrak{B}\mathfrak{C}}-\mathfrak{F}-\sqrt{2}\,(\sqrt{\mathfrak{B}\mathfrak{C}}-\mathfrak{F}),$$

$$\sqrt{\mathfrak{A}}\,\nu \;=\; \sqrt{\mathfrak{B}\mathfrak{C}}-\mathfrak{F}-\sqrt{2}\,\sqrt{\mathfrak{A}\mathfrak{B}}\,(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})$$

after some reductions, and observing that the ratios only of the quantities α, β, γ, δ are material, we obtain

$$\alpha \;=\; \tfrac{K}{\sqrt{\mathfrak{A}}}\,(K+b\nu+g\mu), \quad \beta \;=\; \tfrac{K}{\sqrt{\mathfrak{B}}}\,(\cdots), \quad \gamma \;=\; \tfrac{K}{\sqrt{\mathfrak{C}}}\,(\cdots), \quad \delta \;=\; \tfrac{K}{\sqrt{\mathfrak{A}}}\,\sqrt{-(b\nu^2+c\mu^2+2f\mu\nu)\cdot p};$$

[Book p. 82.] and it is easily seen also that the coordinates of the point of contact are

$$x=0, \quad y=\nu, \quad z=\mu, \quad w \;=\; -\tfrac{\delta\sqrt{\mathfrak{A}}}{pK};$$

also

$$\mu \;=\; -\tfrac{\mathrm{f\,g}}{\sqrt{\mathfrak{B}}}\left(1-\tfrac{\mathrm{g}}{J}\right), \qquad \nu \;=\; -\tfrac{\mathrm{f\,h}}{\sqrt{\mathfrak{C}}}\left(1-\tfrac{\mathrm{h}}{J}\right).$$

Hence substituting and introducing throughout the quantities f, g, h, J, also forming the analogous equations, the equations of the tactors are

$$\begin{aligned}&\left\{\mathrm{f}^2(-\mathrm{f}^2+\mathrm{g}^2+\mathrm{h}^2)+(\mathrm{g}+\mathrm{h})\,J\left(\mathrm{f}^2-(\mathrm{g}-\mathrm{h})^2-\tfrac{2\mathrm{g}^2\mathrm{h}^2}{J^2}\right)\right\}\tfrac{\mathrm{f}}{\sqrt{\mathfrak{A}}}\,x\\&\quad+2\sqrt{X}\,\mathrm{g\,h}\left(1-\tfrac{\mathrm{g}}{J}\right)\left(1-\tfrac{\mathrm{h}}{J}\right)(\mathrm{f}^2-(\mathrm{g}-\mathrm{h})^2)\sqrt{-p}\,w \;=\; 0,\end{aligned}$$

$$\begin{aligned}&\left\{\mathrm{g}^2(\mathrm{f}^2-\mathrm{g}^2+\mathrm{h}^2)+(\mathrm{h}+\mathrm{f})\,J\left(\mathrm{g}^2-(\mathrm{h}-\mathrm{f})^2-\tfrac{2\mathrm{h}^2\mathrm{f}^2}{J^2}\right)\right\}\tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\,y\\&\quad+2\sqrt{Y}\,\mathrm{h\,f}\left(1-\tfrac{\mathrm{h}}{J}\right)\left(1-\tfrac{\mathrm{f}}{J}\right)(\mathrm{g}^2-(\mathrm{h}-\mathrm{f})^2)\sqrt{-p}\,w \;=\; 0,\end{aligned}$$

$$\begin{aligned}&\left\{\mathrm{h}^2(\mathrm{f}^2+\mathrm{g}^2-\mathrm{h}^2)+(\mathrm{f}+\mathrm{g})\,J\left(\mathrm{h}^2-(\mathrm{f}-\mathrm{g})^2-\tfrac{2\mathrm{f}^2\mathrm{g}^2}{J^2}\right)\right\}\tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\,z\\&\quad+2\sqrt{Z}\,\mathrm{f\,g}\left(1-\tfrac{\mathrm{f}}{J}\right)\left(1-\tfrac{\mathrm{g}}{J}\right)(\mathrm{h}^2-(\mathrm{f}-\mathrm{g})^2)\sqrt{-p}\,w \;=\; 0.\end{aligned}$$

[Book p. 83.] It is obvious, from the equations, that each separator passes through the point of contact of a tactor and determinator; it consequently only remains to be shown that each separator touches two tactors. Consider the tactor which has been represented by $\alpha x+\beta y+\gamma z+\delta w=0$; the unreduced values of the coefficients give

$$\mathfrak{A}\alpha+\mathfrak{H}\beta+\mathfrak{G}\gamma \;=\; \tfrac{K^2}{\sqrt{\mathfrak{A}}}, \quad \mathfrak{H}\alpha+\mathfrak{B}\beta+\mathfrak{F}\gamma \;=\; \tfrac{K^2}{\sqrt{\mathfrak{A}}}\,(\mathfrak{H}+\nu), \quad \mathfrak{G}\alpha+\mathfrak{F}\beta+\mathfrak{C}\gamma \;=\; \tfrac{K^2}{\sqrt{\mathfrak{A}}}\,(\mathfrak{G}+\mu),$$

$$\sqrt{\mathfrak{A}\alpha^2 + \cdots + \tfrac{K}{p}\,\delta^2} \;=\; \tfrac{1}{\sqrt{\mathfrak{A}}}\,(\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma) \;=\; K^2.$$

Represent for a moment the separator $\ell x + my + nz + sw = 0$. Then putting $\mathfrak{A}\ell^2 + \cdots + \frac{K}{p}\,s^2 = \Box^2$, since

$$\mathfrak{A}\ell^2 + \cdots + \tfrac{K}{p}\,s^2 \;=\; K^2\Big\{\ell\,\big(1-\tfrac{\mathrm{f}}{J}\big) - \tfrac{\mathrm{g}}{\sqrt{\mathfrak{B}}}\,(\mathfrak{H}+\nu)\,\big(1-\tfrac{\mathrm{g}}{J}\big)\big(1-\tfrac{\mathrm{h}}{J}\big) - (\mathrm{f}-\mathrm{g})\tfrac{\mathrm{h}}{\sqrt{\mathfrak{C}}}\,\big(1-\tfrac{2\mathrm{g}}{J}\big)\big(1-\tfrac{\mathrm{h}}{J}\big)\Big\} \;=\; \tfrac{K^2}{J}\,\Big(-(\mathrm{f}-\mathrm{g})^2 + \mathrm{h}(\mathrm{f}+\mathrm{g})$$

the condition of contact becomes

$$\Box \;=\; \tfrac{1}{J}\,\Big\{-(\mathrm{f}-\mathrm{g})^2 + \mathrm{h}(\mathrm{f}+\mathrm{g}) - \tfrac{2\mathrm{fgh}}{J}\Big\}\,;$$

or, forming the value of $\Box^2$ and substituting,

$$\big(\mathrm{f}^2 - (\mathrm{g}-\mathrm{h})^2 - \tfrac{2\mathrm{g}^2\mathrm{h}^2}{J^2}\big)\big(1-\tfrac{\mathrm{h}}{J}\big)\Big(1-\tfrac{\mathrm{f}}{J}\Big)\Big(1-\tfrac{\mathrm{g}}{J}\Big) \;=\; \tfrac{1}{J}\,\Big(-(\mathrm{f}-\mathrm{g})^2 + \mathrm{h}(\mathrm{f}+\mathrm{g}) - \tfrac{2\mathrm{fgh}}{J}\Big)^2,$$

which may be verified without difficulty; and thus the construction for the resultors is shown to be true.

§ 8.

Several of the formulae of the preceding sections of this memoir apply to any number of variables. Consider the surface (i.e. hypersurface)

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + \cdots + pw^2 \;=\; 0,$$

and the section (i.e. hypersection)

$$(a\lambda + h\mu + g\nu + \cdots)\,x + (h\lambda + b\mu + f\nu + \cdots)\,y + (g\lambda + f\mu + c\nu + \cdots)\,z + \cdots + \sqrt{-p}\,\nabla\,w \;=\; 0,$$

where

$$\nabla^2 \;=\; a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu + \cdots - K,$$

the condition of contact with any other section represented by a similar equation is

$$a\lambda\lambda' + b\mu\mu' + c\nu\nu' + \cdots + f(\mu\nu' + \mu'\nu) + g(\nu\lambda' + \nu'\lambda) + h(\lambda\mu' + \lambda'\mu) + \cdots + K\,\tfrac{\rho\rho'}{p} \;=\; \nabla\nabla',$$

where K is the determinant formed with the coefficients a, b, c, f, g, h, $\ldots$. And consequently, by establishing all or any of the equations $\lambda = \sqrt{\mathfrak{A}}$, $\mu = \sqrt{\mathfrak{B}}$, $\nu = \sqrt{\mathfrak{C}}$, $\ldots$, we have the condition in order that the section in question may touch all or the corresponding sections of the sections $x = 0$, $y = 0$, $z = 0$, $\ldots$.

Let n be the number of the variables x, y, z, $\ldots$; then $K^{n-1} = \begin{vmatrix} \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \cdots \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \cdots \\ \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \cdots \\ \vdots & & & \end{vmatrix}$,

[Book p. 84.] also

$$K^{n-1}\big[(a\lambda + h\mu + g\nu + \cdots)\,x + (h\lambda + b\mu + f\nu + \cdots)\,y + (g\lambda + f\mu + c\nu + \cdots)\,z + \cdots\big]$$

$$= -\begin{vmatrix} x & y & z & \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix},$$

whence also

$$K^{n-2}(\nabla^2 + K) = -\begin{vmatrix} \lambda & \mu & \nu & \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix} \quad \text{or} \quad K^{n-2}\nabla^2 = -\begin{vmatrix} 1 & \lambda & \mu & \nu\cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix},$$

[Book p. 85.] and the equation of the section in question becomes

$$-\begin{vmatrix} x & y & z & \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)}\sqrt{-p}\sqrt{-\begin{vmatrix} 1 & \lambda & \mu & \nu\cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix}}\cdot t = 0,$$

also the condition of contact with the corresponding section is

$$-\begin{vmatrix} \pm 1 & \lambda & \mu & \nu\cdots \\ \lambda' & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu' & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \nu' & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix} = \sqrt{-\begin{vmatrix} 1 & \lambda & \mu & \nu\cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix}}\sqrt{-\begin{vmatrix} 1 & \lambda' & \mu' & \nu'\cdots \\ \lambda' & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu' & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \nu' & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix}}.$$

In particular the equation of the sections which touches all the sections $x = 0$, $y = 0$, $z = 0$, ..., is

$$-\begin{vmatrix} x & y & z & \cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)}\sqrt{-p}\sqrt{-\begin{vmatrix} 1 & \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}}\cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix}}\cdot t = 0.$$

Again, the equations of the section touching $y = 0$, $z = 0$, ... and the section touching $x = 0$, $z = 0$, ... are

$$-\begin{vmatrix} x & y & z & \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)}\sqrt{-p}\sqrt{-\begin{vmatrix} 1 & \lambda & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}}\cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix}}\cdot t = 0,$$

$$-\begin{vmatrix} x & y & z & \cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)}\sqrt{-p}\sqrt{-\begin{vmatrix} 1 & \sqrt{\mathfrak{A}} & \mu & \sqrt{\mathfrak{C}}\cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G}\cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F}\cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C}\cdots \\ \vdots & & & \end{vmatrix}}\cdot t = 0,$$

[Book p. 86.] and the condition of contact of these two sections is

$$-\begin{vmatrix} 1 & \lambda & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} \cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix} = \sqrt{-\begin{vmatrix} 1 & \lambda & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix}} \sqrt{-\begin{vmatrix} 1 & \sqrt{\mathfrak{A}} & \mu & \sqrt{\mathfrak{C}} \cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix}}.$$

It would seem from the appearance of these equations that there should be some simpler method of obtaining the solution than the method employed in the previous part of this memoir.

115.

NOTE ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED POLYGON.

[From the Philosophical Magazine, vol. vi. (1853), pp. 99–102.]

The equation of a conic passing through the points of intersection of the conics

$$U = 0, \qquad V = 0$$

is of the form

$$wU + V \;=\; 0,$$

where w is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of w a quadratic equation, the roots of which may be considered as parameters for determining the line in question. Let one of the values of w be considered as equal to a given constant k; the line is always a tangent to the conic

$$kU + V \;=\; 0\,;$$

and taking $w = p$ for the other value of w, p is a parameter determining the particular tangent, or, what is the same thing, the point of contact of this tangent.

Suppose the tangent meets the conic $U = 0$ (which is of course the conic corresponding to $w = \infty$) in the points P, P', and let θ, ∞ be the parameters of the point P, and θ', ∞ the parameters of the point P'. It follows from my "Note on the Geometrical representation of the Integral $\int dx \div \sqrt{(x+a)(x+b)(x+c)}$," [113] [3] and from the theory of invariants, that if $\square\xi$ represent the "Discriminant" of $\xi U + V$

[Book p. 88.] (I now use the term discriminant in the same sense in which determinant is sometimes used, viz. the discriminant of a quadratic function $ax^2+by^2+cz^2+2fyz+2gzx+2hxy$ or $(a,b,c,f,g,h)(x,y,z)^2$, is the determinant $k = abc - af^2 - bg^2 - ch^2 + 2fgh$), and if

$$\Pi\,\xi \;=\; \int_\infty^\xi \frac{d\xi}{\sqrt{\square\xi}}\,,$$

then the following theorem is true, viz.

"If (θ, ∞), (θ', ∞) are the parameters of the points P, P' in which the conic $U = 0$ is intersected by the tangent, the parameter of which is p, of the conic $kU + V = 0$, then the equations

$$\begin{aligned} \Pi\,\theta \;&=\; \Pi\,p - \Pi\,k, \\ \Pi\,\theta' \;&=\; \Pi\,p + \Pi\,k, \end{aligned}$$

determine the parameters θ, θ' of the points in question." And again,

"If the variable parameters θ, θ' are connected by the equation

$$\Pi\,\theta' - \Pi\,\theta \;=\; 2\,\Pi\,k,$$

then the line PP' will be a tangent to the conic $kU + V = 0$." Whence, also,

[3] I take the opportunity of correcting an obvious error in the note in question, viz. $a^2+b^2+c^2-2bc-2ca-2ab$ is throughout written instead of (what the expression should be) $a^2+b^2+c^2-2a^2bc-2b^2ca-2c^2ab$. [This correction is made, *ante* p. 55.]

"If the sides of a triangle inscribed in the conic $U = 0$ touch the conics

$$k\,U + V = 0, \qquad k'\,U + V = 0, \qquad k''\,U + V = 0,$$

then the equation

$$\Pi\,k + \Pi\,k' + \Pi\,k'' \;=\; 0$$

must hold good between the parameters k, k', k''."

And, conversely, when this equation holds good, there are an infinite number of triangles inscribed in the conic $U = 0$, and the sides of which touch the three conics; and similarly for a polygon of any number of sides.

The algebraical equivalent of the transcendental equation last written down is

$$\begin{vmatrix} 1, & k, & \sqrt{\square k} \\ 1, & k', & \sqrt{\square k'} \\ 1, & k'', & \sqrt{\square k''} \end{vmatrix} \;=\; 0\,;$$

let it be required to find what this becomes when $k = k' = k'' = 0$; we have

$$\sqrt{\square k} \;=\; A + B\,k + C\,k^2 + \ldots,$$

[Continuation of Paper 115 lies beyond the present slice; the body extends from book p. 88 onward into the next PDF range.]

122.

On the Homographic Transformation of a Surface of the Second Order into Itself.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 326–333.]

The following theorems in plane geometry, relating to polygons of any number (odd or even) of sides, are, well known.

"If there be a polygon of $(\alpha + 1)$ sides inscribed in a conic, and so if the sides pass through given points, the $(\alpha+1)$th side will envelope a conic having double contact with the given conic." And "If there be a polygon of $(\alpha + 1)$ sides inscribed in a conic, and so of the sides touch conics having double contact with the given conic, the $(\alpha+1)$th side will envelope a conic having double contact with the given conic." The second theorem of course includes the first, but it is the two separately that I propose to extend to a surface of the second order.

As regards the corresponding theory in geometry of three dimensions, Sir W. Hamilton has given a theorem relating to polygons of an odd number of sides, which may be thus stated: "If there be a polygon of $(2\alpha + 1)$ sides inscribed in a surface of the second order, and 2α of the sides pass through given points, the $(2\alpha + 1)$th side will constantly touch two surfaces of the second order, each of them intersecting the given surface of the second order in the same four lines[1]."

The entire theory depends upon what may be termed the transformation of a surface of the second order into itself, or analytically, upon the transformation of a quadratic form of four indeterminates into itself. I use for shortness the term trans- formation simply; but this is to be understood as meaning a homographic transformation, or in analytic language, a transformation by means of linear substitutions. It will be convenient to remark at the outset, that if two points of a surface of the second order have the relation contemplated in the data of Sir W. Hamilton's theorem (viz. if the line joining the two points pass through a fixed point), the transformation is, using the language of the *Recherches Arithmétiques*, an *improper* one, but that the relation contemplated in the conclusion of the theorem (viz. that of two points of a surface of the second order, connected by a line touching two surfaces of the second order each of them intersecting the given surface of the second order in the same four lines) depends upon a *proper* transformation; and that the circumstance that an even number of improper transformations is required in order to make a proper trans- formation (that this circumstance, I say), is the reason why the theorem applies to polygons in which an even number of sides pass through fixed points, that is, to polygons of an *odd* number of sides.

Consider, in the first place, two points of a surface of the second order such that the line joining them passes through a given point. Let x, y, z, w be current coordinates[2], and let the

[1] See *Phil. Mag.* vol. xxxix. (1851), p. 396. The form in which the theorem is exhibited by Sir W. Hamilton is somewhat different; the surface considered as being not ellipsoid, and the two surfaces touched by the last or $(2\alpha + 1)$th side of the polygon are spoken of as having focal connection with two confocal hyperboloids of two sheets, having respectively double contact with the given ellipsoid; the student is, in fact, a quadric conic which is taken as an analytical premise. (The conclusion is necessarily of the kind mentioned; but when this is once understood, the language of the text may be properly employed. It may be proper to explain that hyperboloids of one sheet inscribed in an ellipsoid is, by analogy with the case in plano of the hyperbola of imaginary transformation, the ulterior expression of the doubled conic of a surface of the second order than is implied.

[2] Strictly speaking, it is the *ratios* of these quantities, x, y, z, w, which are the coordinates, and consequently, even when the point is given, the values x, y, z, w are merely indeterminate to a factor *près*. So that in assuming that a point is given, we should write $x : y : z : w = a : b : c : d$ and so on, but when this is once understood, the language of the text may be properly employed. It may be proper to explain that the equation $(a, b, c, d \rangle\ x, y, z, w)^2 = \mathrm{Re} xx + yy^2$, where "$\rangle$" means $a(x + b(y + c)z + dw)$. The system of coefficients may frequently be indicated by a single coefficient only ; thus in the text $(a, \ldots)(x, y, z, w)^2$ stands for the most general quadratic function of four variables.

equation of the surface be

$$(a, \ldots \between x,\ y,\ z,\ w)^2 = 0,$$

and take for the coordinates of the two points on the surface $x_1,\ y_1,\ z_1,\ w_1$ and $x_2,\ y_2,\ z_2,\ w_2$, and for the coordinates of the fixed point $\alpha,\ \beta,\ \gamma,\ \delta$. Write for shortness

$$(a, \ldots \between \alpha, \beta, \gamma, \delta)^2 = p,$$

$$(a, \ldots \between \alpha, \beta, \gamma, \delta \,\rangle\, x, y, z, w) = q_1,$$

$$(a, \ldots \between \alpha, \beta, \gamma, \delta \,\rangle\, x_2, y_2, z_2, w_2) = q_2,$$

then the coordinates $x_2,\ y_2,\ z_2,\ w_2$ are determined by the very simple formula

$$\begin{aligned} x_2 &= x_1 - \frac{2q_1}{p}\, y_1, \\ y_2 &= y_1 - \frac{2q_1}{p}\, y_1, \\ z_2 &= z_1 - \frac{2q_1}{p}\, y_1, \\ w_2 &= w_1 - \frac{2q_1}{p}\, y_1. \end{aligned}$$

In fact, the point $(\xi, \eta, \zeta, \omega)$ lies in the line joining the points (x_1, y_1, z_1, w_1) and (x_2, y_2, z_2, w_2); and to show that this line touches the surface $V = 0$, it is only necessary to form the equation of the tangent plane at the point $(\xi, \eta, \zeta, \omega)$ to the surface in question; this is

$$(\alpha, \ldots \between (x, y, z, w \,\rangle\, \xi, \eta, \zeta, \omega)^2 = 0,$$

that is, the point (x_1, y_1, z_1, w_1) and (x_2, y_2, z_2, w_2) will correspond to each other by means of the above formulæ, and that they are determined by means of the equations $\xi,\ \eta,\ \zeta,\ \omega$ are given by the $x_1 + x_2,\ y_1 + y_2,\ z_1 + z_2,\ w_1 + w_2$.

Suppose now that we consider three points 1, 2, 3 of the surface, the lines joining 1, 2 and 2, 3 passing respectively through given points; we have first

$$x_2 = x_1 - \tfrac{2q_1}{p}\alpha, \quad y_2 = y_1 - \tfrac{2q_1}{p}\beta,$$

$$z_2 = z_1 - \tfrac{2q_1}{p}\gamma, \quad w_2 = w_1 - \tfrac{2q_1}{p}\delta,$$

where $q_1,\ \beta,\ \gamma,\ \delta$ are the parameters which may subject to the condition

$$\alpha^2 + \beta^2 - c^2 = 0,$$

the equation of the variable conic may be written

$$[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2) - 2(xx - \xi) - 2(yy - \eta)\beta = 0,$$

Or, what is the same thing, the system of equations is

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 0, \quad \xi + \eta = c\xi + c\eta$$

$$(2 - c)\alpha^2 - p(\alpha + c\eta)\alpha + c\eta\xi + (-c\eta + p + c\eta)\eta + (-c\eta - \mu c + c\eta)\eta = 0.$$

or, it may also be written,

$$(a^2 + p + a^2 c\eta + (a\eta - \mu)c + c\eta a^2)\alpha + (b\eta - \beta c)\eta + c(-c\eta + \mu_0)\eta + (-c\eta - p - c\eta + a\eta)\eta = 0.$$

Represent for a moment the form $\eta\eta - \mu c + \mu c\eta$ by V, and consider the surface $V = 0$. If we may also be written,

$$(\alpha^2 + \beta^2 + c\eta + (\eta\eta - c\eta + b\eta)c)\eta + p\alpha\eta = 0,$$

this discriminant is

$$(\eta + V\eta + \alpha\eta + V\eta + \theta^2,$$

which shows that the surface intersect in four lines. Suppose the discriminant vanishes ; we have for the determination of η a quadratic equation, which may be written

$$q'^2 + (2 + s)\, q' + R = 0\,;$$

but the roots of this equation by q', q'', then each of the functions $\eta + Vc'$, $\eta + Vc'$ will break up into linear factors, and we may write

$$\eta + V c'_r = R_r R_r,$$

$$\eta + V c''_r = R'_r R_r,$$

(V' and V'' are of course linear functions of R_r, and R_r, R'_r.) These which lead in evidence the fact of the two surfaces intersecting in four lines.

The equations

$$q_r + q_r = Rq_r, \quad q_r = q_r = 2q_r, \quad \alpha_r + \alpha_r = Rq_r, \text{ \&c.}$$

generating lines and in like manner in the point Q. Thus P, Q are tangent planes at the points A, B respectively, and the system of these polygons of generating lines is the system of common chords. The chord in question is a tangent to the two surfaces involved; it is therefore the line of intersection of the tangent planes through P to the same PA as belonging determinately to one or the other of the two systems of generating lines, then when P is given, the corresponding point Q is, it is clear, completely determined. What precedes may be recapitulated in the statement, that in the improper transformation of a surface of the second order into itself, we have, as corresponding points, the opposite angles of a skew quadrangle lying upon the surface, and having the other two opposite angles upon given lines of intersection of θ, ϕ, and then the proper transformation is obtained is not the general proper transformation.

It is necessary to remark, that, whenever the coordinates of the points Q are connected with the coordinates of the points B by means of the equations which belong to an improper transformation, the points P, Q have to each other the geometrical relation above mentioned, viz. there exist two plane sections θ, ϕ such that P, Q are the opposite angles of a skew quadrangle upon the surface, and having the other two angles upon θ and ϕ. Q is the opposite angles of a skew quadrangle upon the surface, having two of its opposite angles upon θ and ϕ respectively, and the two adjacent angles upon θ and ϕ.

It is, perhaps, hardly possible to embrace by a single intelligible synthesis the remaining side will satisfy the geometrical condition in virtue of which its extremities will be a pair of corresponding points, the pair being proper or improper according to the rule just explained.

I conclude with the remark, that we may by means of two plane sections of a surface of the second order obtain a proper transformation. For, if the generating lines through P meet the sections θ, ϕ in the points A, B respectively, and the remaining generating lines through A, B respectively meet in the points ϕ, θ respectively in B', A', and the remaining generating lines through B', A' respectively meet in a point P'; then will P, P' be a pair of corresponding points in a proper trans- formation.

2 Stone Buildings, January 11, 1854.

123.

ON THE GEOMETRICAL REPRESENTATION OF AN ABELIAN INTEGRAL.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 414–418.]

The equation of a surface passing through the curve of intersection of the surfaces

$$ax^2 + by^2 + cz^2 + dw^2 = 0,$$

$$ax^2 + b'y^2 + c'z^2 + d'w^2 = 0,$$

is of the form

$$k(ax^2 + by^2 + cz^2 + dw^2) + l(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

where k is an arbitrary parameter. Suppose that the surface touches a given plane, we have for the determination of k a certain equation the root of which may be considered as parameters defining the plane in question. Let one of the values of k be considered equal to a given quantity k, the plane touches the surface

$$k(ax^2 + by^2 + cz^2 + dw^2) + l(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

and the other two values of k may be considered as parameters defining the particular tangent plane, or what is the same thing, determining the point of contact with the surface.

Or more clearly, then — in order to determine the position of a point on the surface

$$k(ax^2 + by^2 + cz^2 + dw^2) + l(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

the tangent plane at the point in question is touched by two other surfaces

$$p(ax^2 + by^2 + cz^2 + dw^2) + q(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

$$p'(ax^2 + by^2 + cz^2 + dw^2) + q'(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

and similarly for the other lines.

I consider now a curve of double curvature (k) touching the surface (k); for the points of the curve (k), the parameter of one of the surfaces touched at the point in question is constant, that is, p, q are functions of the parameters p', q' of the other; or what is the same thing, writing $k = p \div q$, $k' = p' \div q'$, the coordinates of the point (k) of the curve (k) are

$$x : y : z : w = \sqrt{(\alpha + k)(\alpha + k')} : \sqrt{(\beta + k)(\beta + k')} : \sqrt{(\gamma + k)(\gamma + k')} : \sqrt{(\delta + k)(\delta + k')},$$

or under a somewhat more convenient form we have

$$x : y : z : w = \sqrt{(a + k)(a + k')} : \sqrt{(b + k)(b + k')} : \sqrt{(c + k)(c + k')} : \sqrt{(d + k)(d + k')},$$

where a, b, c, d are inversely as a, b', c', d'. Taking for a moment $k = k'$, this is a thing thus, viz. it is

$$x : y : z : w = (a + k) : (b + k) : (c + k) : (d + k),$$

which equation is satisfied by writing $(a + k)x : (b + k)y : (c + k)z : (d + k)w$, where α, β, γ, δ are inversely as a, b, c, d. Similarly, we have

$$x : y : z : w = \sqrt{(a + k)(a + k')} : \sqrt{(b + k)(b + k')} : \sqrt{(c + k)(c + k')} : \sqrt{(d + k)(d + k')},$$

for the conditions in order that the point $(k,\, k')$ may lie in the tangent plane at $(k'),\, p,\, q$ to the surface (k'). Similarly, we have

$$\sum (b+k)\sqrt{(c+k)(c+k')} + (c+k)\sqrt{(b+k)(b+k')} = 0,$$

for the condition in order that the point (k,p) may lie in the tangent plane at $(k',\, p,\, q)$ to the surface (k). Similarly,

$$\sum (a+k')(a+k)\sqrt{(a+k)(a+k') + (b+k')(b+k)\sqrt{(b+k)(b+k')}} = 0,$$

or making the same reductions give

$$\sum (a+a')\sqrt{(a+k)(a+k')} \cdot \sqrt{\frac{a+a}{a+a'}} = h \cdot \mu + a + b'',$$

and the like for the system of equations

$$\sqrt{(a+a')(a+\rho)(a+\sigma)(a+\tau)} \cdot \sqrt{\frac{a+a}{a+a'}} = h \cdot \mu + a + b''.$$

When in each system a is to be successively replaced by a, b, c, d, and ρ, σ, τ and μ, ν are indeterminate. Now dividing each equation of the one system by the corresponding equation in the other system, we see that the equation

$$\frac{a+b}{a+a'}\, h + \frac{1}{\mu+a+b''} = h + a,$$

is satisfied by the values a, b, c, d of a, and therefore, since the equation is in a only of the third order, that the equation in question must be identically true. We may therefore write

$$h + \mu + a + a'' = (a+a)(\lambda+\mu),$$

and the two systems of equations become identical with

$$\sqrt{(a+k)(a+k')(a+p)(a+q)(a+r)(a+s)} = h(a+a)(\rho+\mu).$$

A set of remarks which may be represented by the single equation

$$\Phi(x+y)\Phi(x+y) = \mu(x+y) + (\nu x + \nu y)(x+y),$$

where x is arbitrary; or in the same thing, writing $\theta = a+x,\, a+b,\, b+b$,

$$x(x+a)(x+b)(x+c)(x+d)(x+\alpha)(x+\beta) = h(\mu+a)(\rho+\mu).$$

Hence, putting the transcendental equation

$$\Pi p + \Pi q + \Pi p' + \Pi q' = \text{const.}$$

$$\Pi_1 p = \int \frac{p\, dx}{\sqrt{(a+x)(a+y)(a+z)(a+w)(a+\alpha)}},$$

we see that the algebraical equations between p, q, p', q' are equivalent to the transcendental equation

$$\Pi p + \Pi q + \Pi p' + \Pi q' = \text{const.}$$

The algebraical equations which connect θ, ϕ with p, q, p', q', may be exhibited under several different forms; thus, for instance, considering the point (k, k') as a point in the line joining (k, p), and (k', q'), the equation is

$$\begin{vmatrix} \sqrt{(a+p)(a+q)} & \sqrt{(a+k)(a+k')} & \sqrt{(a+p')(a+q')} \\ \sqrt{(b+p)(b+q)} & \sqrt{(b+k)(b+k')} & \sqrt{(b+p')(b+q')} \\ \sqrt{(c+p)(c+q)} & \sqrt{(c+k)(c+k')} & \sqrt{(c+p')(c+q')} \\ \sqrt{(d+p)(d+q)} & \sqrt{(d+k)(d+k')} & \sqrt{(d+p')(d+q')} \end{vmatrix} = 0,$$

i.e. the determinants formed by selecting any three of the four columns must vanish; the equations so obtained are equivalent (as they should be) to two independent equations.

Or, again, by considering (p, q, ϕ) first as a point in the tangent plane at (k, k') to the surface (k), and then as a point in the tangent plane at (k', p', q') to the surface (k'), we obtain

$$\sum (a+k)\sqrt{(a+p)(a+q)}\sqrt{(a+k)(a+k')} = 0,$$

$$\sum (a+k')\sqrt{(a+p)(a+q)}\sqrt{(a+k)(a+k')} = 0.$$

Or again, we may consider the line joining (k, k') and (p, q), the formulæ will satisfy the geometrical condition in virtue of which its extremities will be a pair of corresponding points, the pair being proper or improper according to the rule just explained.

The condition in order that the line joining the points (k, k', p, q) and (k', k'', p', q') may meet a curve which is represented by the single equation

$$\sum_a a x_a x_b x_c \cdots = 0,$$

may also be expressed in the form

$$2k\zeta + (\Pi p + \Pi q + \Pi p' + \Pi q') = \text{const.},$$

the equation entails

$$\Pi p + \Pi q + \Pi p' + \Pi q' = \text{const.}$$

or

$$\sum \frac{dx}{\sqrt{(a+x)(a+y)(a+z)(a+w)(a+\alpha)}} = \text{const.}$$

considering only the first integral, the second one being analogously.

The preceding formulæ are obviously very incomplete; but the connection which they point out between the geometrical question and the Abelian integral involving the root of a function of the sixth order, may I think be of service in the theory of these integrals.

124.

On a Property of the Caustic by Refraction of the Circle.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 427–431.]

M. St Laurent has shown (*Gergonne*, vol. xviii. [1827] p. 1), that in certain cases the caustic by refraction of a circle is identical with the caustic of refraction of a circle (the refracting circle and radiant point being, of course, properly chosen), and a very elegant demonstration of M. St Laurent's theorems is given by M. Gergonne in the same volume, p. 48. A similar method may be employed to demonstrate the more general theorem, that the same caustic by refraction of a circle may be considered as arising from six different systems of a radiant point, circle, and index of refraction. The demonstration is obtained by means of the secondary caustic, which (as is well known) an oval of Descartes. Such oval has three rectangular axes, and two corresponding circles and indices of refraction, as follows:—

Let x be the radius of the refracting circle, ξ, η be the coordinates of the radiant point, and the secondary caustic in the envelope of the circle

$$x^2 + \beta^2 = c^2,$$

where α, β are parameters which vary subject to the condition

$$\alpha^2 + \beta^2 - c^2 = 0,$$

the equation of the variable circle may be written

$$[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2) - 2(xx - \xi) - 2(yy - \eta)]\beta = 0,$$

which is of the form

$$U + \lambda + B(\lambda^2 + \delta) = 0,$$

the envelope is therefore

$$U^2 = A^2(\lambda + B),$$

Hence substituting, we have for the equation of the envelope, i.e. for the secondary caustic,

$$[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2)]^2 = 4c^2[(xx - \xi)^2 + (yy - \eta)^2],$$

and this may perhaps be considered as the standard form.

To show that this equation belongs to a Descartes' oval, suppose for greater con- venience $\eta = 0$, and write

$$\rho^2(x^2 + y^2) = \rho^2 - \xi^2 + 2c\eta\sqrt{kx - \xi}\xi + a^2,$$

multiplying this equation by $1 - \frac{1}{\alpha}$, and adding to each side $a^2\left(x - \frac{\xi}{a}\right) + (x - \xi)^2 + y^2$, we have

$$\left(1 - \frac{1}{\alpha}\right)[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2)]^2 + \left(x - \frac{\xi}{a}\right)^2$$

$$= (x - \xi)^2 + y^2 + 2c\left(x - \frac{1}{a}\right)\sqrt{(x - \xi)^2 + y^2} + c^2\left(x - \frac{1}{a}\right)^2,$$

or reducing

$$\alpha^2\left[\left(x - \frac{\xi}{a}\right)^2 + y^2\right] = \left[\sqrt{(x - \xi)^2 + y^2} + c\left(x - \frac{1}{a}\right)\right]^2;$$

again, multiplying the same equation by $\frac{1}{a^2}\left(1 - \frac{1}{a^2}\right)$, and adding to each side $\frac{1}{a^2}\left(1 - \frac{\xi}{a}\right)^2 + \frac{1}{a^2}y^2 + \xi^2 + y^2$, we have

$$\frac{1}{a^2}\left(1 - \frac{\xi}{a}\right)^2 + \frac{1}{a^2}y^2 + (x - \xi)^2 + y^2 + 2\frac{c}{a}\left(1 - \frac{\xi}{a}\right)\sqrt{(x - \xi)^2 + y^2} + \frac{c^2}{a^2}\left(1 - \frac{\xi}{a}\right)^2,$$

or reducing,

$$\left(x - \frac{\xi}{a}\right)^2 + y^2 = \frac{1}{a^2}\left[\sqrt{(x - \xi)^2 + y^2} + \frac{c}{a}\left(1 - \frac{\xi}{a}\right)\right]^2.$$

Hence, extracting the square roots of each side of the equations thus found, we have the equation of the secondary caustic in either of the forms

$$\sqrt{\left(x-\frac{\xi}{a}\right)^2+y^2}=\frac{1}{a}\sqrt{(x-\xi)^2+y^2}+\frac{c}{a}\left(x-\frac{1}{a}\right),$$
$$\sqrt{\left(x-\frac{\xi}{a}\right)^2+y^2}=\frac{1}{a}\sqrt{(x-\xi)^2+y^2}+\frac{c}{a}\left(1-\frac{\xi}{a}\right),$$

to which are to be joined

$$\sqrt{\left(x-\frac{\xi}{a}\right)^2+y^2}\sqrt{\left(x-\frac{\xi}{a'}\right)^2+y^2}+\tfrac{c}{a}\left(x-\frac{1}{a'}\right)+\tfrac{c}{a}\left(1-\frac{\xi}{a}\right)\sqrt{\left(x-\frac{\xi}{a'}\right)^2+y^2}+\tfrac{c}{a'}\left(1-\frac{\xi}{a}\right)\sqrt{1-a\xi^2+y^2}=$$

Write successively,

$$\begin{array}{llll}
\xi'=\xi, & c'=c, & \mu'=\mu\,; & (1)\\
\xi'=\dfrac{\xi}{a}, & c'=\dfrac{c}{a}, & \mu'=\dfrac{1}{\mu}\,; & (\alpha)\\
\xi'=\dfrac{1}{\xi}, & c'=\dfrac{c}{\xi}, & \mu'=\dfrac{\mu}{\xi}\,; & (\beta)\\
\xi'=\xi, & c'=c, & \mu'=\dfrac{1}{\mu}\,; & (\gamma)\\
\xi'=\dfrac{\xi}{a}, & c'=\dfrac{c}{a}, & \mu'=\mu\,; & (\delta)\\
\xi'=\dfrac{1}{\xi}, & c'=\dfrac{c}{\xi}, & \mu'=\dfrac{1}{\mu}\,; & (\epsilon)
\end{array}$$

or, what is the same thing,

$$\begin{array}{llll}
\xi'=\xi, & c'=c, & \mu'=\mu\,; & (1)\\
\xi'=\xi, & c'=\xi, & \mu'=\xi/\mu\,; & (\alpha)\\
\xi'=\xi, & c'=\xi/\mu, & \mu'=\mu\,; & (\beta)\\
\xi'=\xi, & c'=c, & \mu'=1/\mu\,; & (\gamma)\\
\xi'=c/\mu, & c'=c, & \mu'=\mu\,/c\,; & (\delta)\\
\xi'=c/\mu, & c'=c, & \mu'=1/\mu\,; & (\epsilon)
\end{array}$$

then, whichever system of values of ξ, c, μ be substituted for ξ, c, μ, we have in each case identically the same secondary caustic, the effect of the substitutions being simply to interchange the different forms of the equation; and we have therefore identically the same caustic. By writing

$$(\xi,c,\mu)\to(\xi',c',\mu')\;=\;*(\xi,c,\mu),$$

where α, β, γ, δ, ϵ will be functional symbols, such as are treated of in my paper "On the Theory of Groups as depending on the symbolic equation $\theta^n=1$," [125], and it is easy to verify the equations

$$\begin{aligned}
1&=\alpha\beta=\beta\alpha=\gamma^2=\delta\epsilon,\\
\alpha&=\beta\gamma=\gamma\beta=\delta^2=\epsilon\delta,\\
\beta&=\alpha\gamma=\gamma\alpha=\epsilon\delta=\delta\epsilon,\\
\gamma&=\alpha\beta=\beta\alpha=\delta\epsilon=\epsilon\delta,\\
\delta&=\alpha\beta=\beta\alpha=\gamma\delta=\delta\gamma,\\
\epsilon&=\alpha\delta=\delta\alpha=\gamma\epsilon=\epsilon\gamma.
\end{aligned}$$

Suppose, for example, $\xi = c$, i.e. let the radiant point be in the circumference; then in the fourth system $\xi' = c$, $c' = c$, $\mu' = \frac{1}{\mu}$, i.e. since the radius of the refracting circle is the radius of the circle and the index of refraction reciprocal, this is one of M. St Laurent's theorems, viz.

THEOREM. The caustic by refraction of a circle when the radiant point is in the circumference, is the same caustic by refraction for the same radiant circle and a reciprocal index of refraction. The first of these is a group of eighteen, viz.

Again, let $\xi = c\mu$, the fifth system gives $\xi' = \frac{c}{\mu}$, $c' = c$, $\mu' = -1$, or the new system is in this case also a reflecting system. This is the other of M. St Laurent's theorems, viz.

THEOREM. The caustic by reflection of a circle when the distance of the radiant point from the centre is equal to the radius of the circle multiplied by the index of refraction, is the caustic by reflection of the same circle for a radiant point which is the image of the first radiant point.

Of course it is to be understood that the image of a point means a point whose distance from the centre is equal to square of radius $\div$ distance.

2 Stone Buildings, Nov. 2, 1853.

125.

ON THE THEORY OF GROUPS, AS DEPENDING ON THE SYMBOLIC EQUATION $\theta^n = 1$.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 40–47.]

Let θ be a symbol of operation, which may, if we please, have for its operand, not a single quantity x, but a system $(x, y, \ldots)$, so that

$$\theta(x, y, \ldots) = (x', y', \ldots),$$

where x', y', ... are any functions whatever of x, y, ..., it is in not even necessary that $x', y', \ldots$ should be the same in number with x, y, ... In particular $x', y', \ldots$ may represent a permutation of x, y, ... θ is in this case what is termed a substitution; and if, instead of a set x, y, ..., we consider points or magnitudes (or, even, opera- tions) any how related, then θ may be an operator giving rise to a substitution, or it may be in particular a substitution, in like manner. The general question of the analytical representation of a substitution does not enter into the present plan, but the symbols θ, ϕ, etc., θ, θ', etc. are interpreted in such manner that the symbols $\theta\phi$, $\phi\theta$, etc. have a definite signification, but this need not always be the case; for instance, in the theory of matrices, the symbols θ, ϕ, etc. do not in general satisfy the commutative law $\theta\phi = \phi\theta$, so that, $\theta\phi$, $\phi\theta$, etc. are to be considered as in general different symbols. In other respects also the analogy is by no means perfect, and the words "operation" and "operand" are used as the only ones really suitable; the theory of operators, or as it may be called, the theory of operations, in the theory of analytical operations functional process which gives rise to the symbols in question, may be considered analogous to or having the same meaning as multi- plied together so as to form a table, thus:—

Further Symbols

	1	α	β	$\ldots$
1	1	α	β	$\ldots$
α	α	α^2	$\alpha\beta$	$\ldots$
β	β	$\beta\alpha$	β^2	$\ldots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	

that as well rows as columns of the square will contain all the symbols 1, α, β, ... It follows that when there is no entry of any symbol of the group without repetition, and by such order whatever, is a symbol of the group. Suppose that this group

$$1,\ \alpha,\ \beta,\ \gamma,\ \ldots,$$

contains a symbol β which may be such that there exists in the same group $\theta^n = 1$.[3]

satisfies the equation $\theta^n = 1$, where n is the smallest number for which this is the case, then we may speak of θ as a composite symbol of any order n less than n a composite number. It follows then by a like reasoning as for β, γ, the system 1, α, β, γ is a subgroup of θ. It follows from this, in particular, that there can be no system in which any of the others, but are associative, $\theta(\phi\chi) = (\theta\phi)\chi$. The expression of the symbol of an operation θ is given by the symbol $\theta = a + b\alpha + c\beta + \ldots$, the form of $\theta = a + b\alpha + c\beta + \ldots$, the symbols at the same time being known are commutative or $\theta\phi = \phi\theta$, the system is said to be Abelian. There may be involved THEOREM.— If the symbols of the group be multiplied together two and two in any order, the entire system being known are not necessarily distinct. It may be done by means of the substitution equation $\theta^n = 1$, where n is some positive integer; this is the case of a system of all the roots of the equation $x^n = 1$, in the same notation, the symbols of the equation are

$$1,\ \alpha,\ \alpha^2,\ \ldots,\ \alpha^{n-1} \quad (\alpha^n = 1).$$

This definition explains the meaning of the equation $\theta^n = 1$; viz. this means therefore represent the group symbols, viz.

$$1,\ \alpha,\ \beta,\ \gamma,\ \ldots, \qquad (\beta^n = 1),$$

thus multiplying each term by a further factor, we have for the group 1, α, $\alpha\beta$, $a\alpha\beta$, but we must be equal either to 1 or to one of the roots. In the former case our group is—

$$1,\ \alpha,\ \beta \quad (\alpha^n = 1).$$

which is analogous to the system of roots of the ordinary equation $\theta^n - 1 = 0$. For the sake of comparison with what follows, I remark, that representing the last- mentioned group by

$$1,\ \alpha,\ \beta,\ \gamma,$$

we have the table

	1	α	β	γ
1	1	α	β	γ
α	α	β	γ	1
β	β	γ	1	α
γ	γ	1	α	β

If, on the other hand, $\alpha^2 = 1$, then it is easy by similar reasoning to show that we must have $\beta = \beta\alpha$, so that the group is of the form

$$1,\ \alpha,\ \beta,\ \alpha\beta, \quad (\alpha^2 = 1,\ \beta^2 = 1,\ \alpha\beta = \beta\alpha),$$

or if expressed under the group by

$$1,\ \alpha,\ \beta,\ \gamma,$$

we have the table

[3] The idea of a group as applied to permutations or substitutions is due to Galois, and the introduction of it may be considered as marking an epoch in the progress of the theory of algebraical equations.

	1	α	β	γ
1	1	α	β	γ
α	α	1	γ	β
β	β	γ	1	α
γ	γ	β	α	1

or, if we please, the symbols are such that

$$\alpha^2 = \beta^2 = \gamma^2 = 1,$$

$$\alpha = \beta\gamma = \gamma\beta,\ \beta = \alpha\gamma = \gamma\alpha,\ \gamma = \alpha\beta = \beta\alpha,$$

[and we have thus a group essentially distinct from that of the system of roots of the ordinary equation $x^n - 1 = 0$.]

Systems of this form are of frequent occurrence in analysis, and it is only on account of their extreme simplicity that they have not been expressly remarked. For instance, in the theory of elliptic functions, if u be the parameter, and

$$\alpha(u) = \frac{u}{u}, \quad \beta(u) = (-u), \quad \gamma(u) = -\frac{1}{u},$$

although, indeed, in this case (treating forms which take properly equivalent as identical) we have $\alpha = \beta$, and therefore $\gamma = 1$, in which point of view this group is simply a group of two symbols 1, α ($\alpha^2 = 1$).

Again, in the theory of matrices, if I denote the operation of inversion, and tr that of transposition, (I do not stop to explain in terms as the example may be passed over), we may write

$$\alpha = I, \quad \beta = \mathrm{tr}, \quad \gamma = I \cdot \mathrm{tr} = \mathrm{tr} \cdot I.$$

I proceed to the case of a group of six symbols,

$$1,\ \alpha,\ \beta,\ \gamma,\ \delta,\ \epsilon,$$

which may be considered as representing a system of roots of the symbolic equation

$$\theta^6 = 1.$$

It is in the first place to be shown that there is at least one root which is a prime root of $\theta^6 = 1$, or (to use a simpler expression) a root having the index 3. If it clear that if there were a simpler expression a root having the index 2. It is true that if the index 3, θ must be done by means of this root would have the index 2; then in the square of θ would also have this index, that is, it would have the index 6 which is the cube root of the symbol. The symbol $\theta = 1$ may always be considered as having the index 6, considered in the symbol $\theta = 1$, the index 3 of a root having the index 3. It is impossible that *all* the roots should have the index 3. This may be done by means of the simpler observation: if so, then, since their product has also the index 2 in the symbol $\theta = 1$, this is to have the index 6.

Assume that the symbols α, β, γ have the index 3. The square of any one of this root, e.g. α, is then α^2, and these are contained in the group, must have either the index 6 or the index 2 (or to be a simpler expression) a root having the index 3, having each of the order 4. It is easy to see that any of the roots of θ, and that the group of the order 4. It is easy to see that any of the roots of θ, and that the group is, in fact, 1, $\gamma\delta$, $\delta\gamma$, $\delta\gamma^2$, which is, in fact, the group of six having the index 3, having obtained ; I prefer commencing with the assumption of a root having the index 3. Suppose that α is such a root, the group must clearly be of the form

$$1,\ \alpha,\ \alpha^2,\ \beta,\ \alpha\beta,\ \alpha^2\beta \quad (\alpha^3 = 1);$$

and multiplying the entire group by γ as a nearer factor, it becomes γ, $\alpha\gamma$, $\alpha^2\gamma$, γ', $\alpha\gamma'$, $\alpha^2\gamma'$; we must therefore have $\beta\gamma = \alpha\beta$ or $\gamma\beta = \alpha\beta$. But the latter sup- position $\gamma' = \alpha\beta$ leads also to the same name in which case we may assume $\gamma' = 1$; then we have β^2 also equal to 1, α or α^2, but the first two cases lead to the suppo- sition $\gamma^2 = \alpha$ or $\beta\gamma = 1$, and to a system of six symbols 1, α, α^2, $\gamma\beta$, $\gamma\alpha\beta$, $\gamma\alpha^2\beta$, when we must have either $\gamma\alpha = \alpha\gamma$ or else $\gamma\alpha = \alpha^2\gamma$. The former assumption leads to the group

$$1,\ \alpha,\ \alpha^2,\ \gamma,\ \alpha\gamma,\ \alpha^2\gamma\ (\alpha^3 = 1,\ \gamma^2 = 1,\ \alpha\gamma = \gamma\alpha),$$

which is, in fact, analogous to the system of roots of the ordinary equation $x^n - 1 = 0$. For the sake of comparison with what follows, I remark, that representing the last- mentioned group by the general symbols

$$1,\ \alpha,\ \beta,\ \gamma,\ \delta,\ \epsilon,$$

we have the table

	1	α	β	γ	δ	ϵ
1	1	α	β	γ	δ	ϵ
α	α	β	γ	δ	ϵ	1
β	β	γ	δ	ϵ	1	α
γ	γ	δ	ϵ	1	α	β
δ	δ	ϵ	1	α	β	γ
ϵ	ϵ	1	α	β	γ	δ

while if we represent the second of these two forms, viz. the group

$$1,\ \alpha,\ \alpha^2,\ \gamma\beta,\ \gamma\alpha\beta,\ \gamma\alpha^2\beta,\quad (\alpha^3 = 1,\ \gamma^2 = 1,\ \gamma\alpha = \alpha^2\gamma),$$

by the same general symbols

$$1,\ \alpha,\ \beta,\ \gamma,\ \delta,\ \epsilon,$$

we have the table

	1	α	β	γ	δ	ϵ
1	1	α	β	γ	δ	ϵ
α	α	β	1	δ	ϵ	γ
β	β	1	α	ϵ	γ	δ
γ	γ	δ	ϵ	1	α	β
δ	δ	ϵ	γ	α	β	1
ϵ	ϵ	γ	δ	β	1	α

or, what is the same thing, the symbols are such that

$$\alpha^3 = \beta^3 = \gamma^2 = 1,$$

$$\alpha = \beta\gamma = \gamma\beta\,;\quad \beta = \alpha\gamma = \gamma\alpha,$$

$$\gamma = \delta\beta = \alpha\beta,$$

[and we have thus a group essentially distinct from that of the system of roots of the ordinary equation $x^n - 1 = 0$.]

Systems of this form are of frequent occurrence in analysis, and it is only on account of their extreme simplicity that they have not been expressly remarked. For instance, in the theory of elliptic functions, if u be the parameter, α a fixed root, and tr denotes the operation of transposition, we have the group symbols which may be expressed in the form

$$\alpha(u) = \frac{u - c}{u - d},\quad \beta(u) = (-u),\quad \gamma(u) = -\frac{1}{u},$$

although, indeed, in this case (treating forms which take properly equivalent as identical) we have $\alpha\beta = \gamma$, and although the small group $\{1, \alpha, \beta, \gamma\}$ is simply a group of two symbols 1, α, $(\alpha^2 = 1)$.

If we represent the second of two, and only two, essentially distinct forms of a group of six.

If we represent the second of these two forms, viz. the group

$$1,\ \alpha,\ \beta,\ \alpha^2,\ \beta\alpha,\ \beta\alpha^2,$$

$$\alpha = \beta,\ \gamma = \beta\alpha,\ \delta = \beta\alpha^2,$$

and we have thus two, and only two, essentially distinct forms of a group of six.

126.

On the Theory of Groups, as depending on the symbolic equation $\theta^n = 1$. Second Part.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 408–409.]

Imagine the symbols

$$L,\ M,\ N,\ \ldots$$

such that (L being any symbol of the system),

$$L^{-1}L = LL^{-1} = N,\ LN = N, \ldots$$

is the group

$$1,\ \alpha,\ \beta,\ \ldots,$$

then, in the first place, M being any other symbol of the system, $M^{-1}LM$, $M^{-1}M$, ... will be the same group 1, α, β, ... In fact, the system L, M, N, ... may be written L, $L\alpha$, $L\beta$, ..., and so the system M, βM, γM, ... will be $M^{-1}LM$, $M^{-1}\alpha LM = (M^{-1}LM)\alpha M$, ... which belongs to the group 1, α, β, ...

Next it may be shown that

$$LL',\ ML',\ NL',\ \ldots$$

is a group, although LL' may not in general the same group as 1, α, β, ... In fact, writing $M = L\alpha$, $N = L\beta$, ..., the symbols just written down are

$$LL',\ L\alpha L',\ L\beta L',\ \ldots,$$

and we have e.g. $L\alpha L' = L\alpha L' = L\sigma L', = L\sigma L'$, where σ belongs to the group 1, α, β, ... ;

[Pages 130: continuation in similar group-theory style omitted in full transcription.]

127.

On the Homographic Transformation of a Surface of the Second Order into Itself.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 208–212; continuation of **122**.]

I pass to the original transformation. Sir W. R. Hamilton has given (in the note, p. 723 of his *Lectures on Quaternions* [Dublin, 1853]) the following theorem:—If there be a polygon of 2α sides inscribed in a surface of the second order, and $(2\alpha - 1)$ of the sides pass through given points, the side or $(2\alpha-1)$th side will constantly touch two cones circumscribed about the surface of the second order. The relation between the extremities of the $(2\alpha-1)$th side is that of two points connected by the general improper transformation; in other words, if there be a surface of the second order into itself, then have, as corresponding points, the opposite angles of a skew quadrangle lying upon the surface, and having the other two opposite angles upon given lines.

The tangent planes through P to the surface in question form an involution; the two sections which are the silicongugates of the involution pass through P, and having to consider the corresponding points θ, ϕ in the same theorems, viz. that the two points which are the involution of θ and ϕ, the surface in the manner described in the result of PA as belonging determinately to one of the two systems of generating lines, then P is given, the corresponding point Q is, it is clear, completely determined. What precedes may be recapitulated in the statement, that in the improper transformation of a surface of the second order into itself, we have, as corresponding points, the opposite angles of a skew quadrangle lying upon the surface, and having the other two opposite angles upon given lines.

I proceed to establish (by M. Hermite's method) the formulæ for the particular case in question. The thing required is to find x_2, y_2, z_2, w_2 linear functions of x_1, y_1, z_1, w_1, such that

$$ax_2^2+by_2^2+cz_2^2+dw_2^2=ax_1^2+by_1^2+cz_1^2+dw_1^2,$$

$$(ay_1y_2+bz_1z_2+cz_1z_2+dw_1w_2)^2-(ax_1^2+by_1^2+cz_1^2+dw_1^2)(ax_2^2+by_2^2+cz_2^2+dw_2^2)$$
$$=Lx_1x_2+My_1y_2+Nz_1z_2+Sw_1w_2,$$

where L, M, N, S are constants of course, in the case in question functions of the constants a, b, c, d of the surface, and the linear functions of x_1, y_1, z_1, w_1 which are to be the values of x_2, y_2, z_2, w_2. Following the analysis, in which $L=4$, $M=4$, the following system of equations, in which (a,b,c,d) of the points minimums, and $M=ab+cd=2bc+da$, are obtained.

Following M. Hermite's method, the formulæ are obtained.

128.

Developments on the Porism of the In-and-Circumscribed Polygon.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 339–345.]

I propose to develope some particular cases of the theorems given in my paper, "Correction of two Theorems relating to the Porism of the in-and-circumscribed Polygon" (*Phil. Mag.* vol. vi. [1853], **114**]). The two theorems are as follows:–

Theorem. The condition that there may be inscribed in the conic $U=0$ an infinity of n-gons circumscribed about the conic $V=0$, depends upon the developement in ascending powers of ξ of the square root of the discriminant of $\xi U+V$, viz. if this square root be

$$A+B\xi+C\xi^2+D\xi^3+E\xi^4+F\xi^5+G\xi^6+H\xi^7+\dots,$$

then for $n=3$, 5, 7, &c. respectively, the conditions are

$$|\,C\,|=0,\quad \begin{vmatrix} C & D \\ D & E \end{vmatrix}=0,\quad \begin{vmatrix} C & D & E \\ D & E & F \\ E & F & G \end{vmatrix}=0,\ \text{\&c.};$$

and for $n=4$, 6, 8, &c. respectively, the conditions are

$$|\,D\,|=0,\quad \begin{vmatrix} D & E \\ E & F \end{vmatrix}=0,\quad \begin{vmatrix} D & E & F \\ E & F & G \\ F & G & H \end{vmatrix}=0,\ \text{\&c.}$$

THEOREM. In the case where the conics are replaced by the two circles which (taken in their respective conjugates) are equivalent to a polygon inscribed in the one and circumscribed about the other, the values of x which give the position of ξ on which the dimensions are replaced by the

$$\alpha^2+\beta^2-R^2=0, \quad (x-c)^2+y^2-r^2=0,$$

then the discriminant, the square root of which appears as before, is

$$A+B\xi+C\xi^2+D\xi^3+E\xi^4+\dots.$$

Write for a moment

$$A=R\xi+(\xi^2+D\xi^3+E\xi^4+\dots=\sqrt{(1+\alpha\xi)(1+\beta\xi)(1+\gamma\xi)},$$

then the discriminant, the square root of which appears as before, is

$$(1+\xi)\,r^2=\alpha+\beta+\gamma-3\alpha\beta,$$

then

$$\begin{aligned} A&=1,\\ 2B&=a+b+c,\\ -3C&=a^2+b^2+c^2-3bc,\\ 16D&=a^3+b^3+c^3-3a(b^2+c^2+bc),\\ -128E^2&=a^4+b^4+c^4-4(b^3+c^3+abc^2+a^2bc)+6a^2bc. \end{aligned}$$

To adapt these to the case of the two circles, we have to write

$$\alpha^2(1+\alpha\xi)(1+\beta\xi)(1+\gamma\xi)+4r^2=\beta+r^2(1+\beta)\xi^2+\frac{1}{4}r^2\xi^4,$$

and therefore

$$\alpha^2=1,$$
$$\alpha^2(\alpha+\beta+c)=-\xi^2,$$

values which satisfy identically $\alpha^2+y^2+z^2+w^2=c^2+(a^2+\alpha)-\beta+3\alpha\beta$, belong to an improper transformation. We have thus obtained the proper trans- formation of the surface of the second order $ax^2+by^2+cz^2+dw^2=0$ into itself.

Returning for a moment to the equations which lead to the surface $\Sigma y-\Sigma\omega=0$, it is easy to see that they without loss of generality, in the points equations (x_1,y_1,z_1,w_1) and (x_2,y_2,z_2,w_2), then,

The resulting system of equations is, in fact, identical with

$$ax_2^2+by_2^2+cz_2^2+dw_2^2=ax_1^2+by_1^2+cz_1^2+dw_1^2,$$

and forming the values, the determined values of x_2, y_2, z_2, w_2 are linear functions of x_1, y_1, z_1, w_1 of the form

$$MM'z_2=(aa'-bb'-cc'-dd')\,z_1+(\,ab'+a'b+cd'-c'd\,)\,y_1+(\,ac'+a'c-bd'+b'd\,)\,z_1-(\,b'c-bc'+a'd+ad'\,)\,w_1+(\,bd'-b$$

values which satisfy identically $x_2^2+y_2^2+z_2^2+w_2^2=x_1^2+y_1^2+z_1^2+w_1^2$, and which belong, as is supposed, to an improper transformation. We have thus obtained the proper trans- formation of the surface of the second order $ax^2+by^2+cz^2+dw^2=0$ into itself.

2 Stone Buildings, January 11, 1854.

129.

On the Porism of the In-and-Circumscribed Triangle, and on an Irrational Transformation of Two Ternary Quadratic Forms each into Itself.

[From the *Philosophical Magazine*, vol. vii. (1855), pp. 115–121.]

There is an irrational transformation of two ternary quadratic forms each into itself, based upon the solution of the following geometrical problem.

Given that the conic

$$lx + my + nz = 0$$

meets the conic

$$(a, b, c, f, g, h)(x, y, z)^2 = 0$$

in the points (x_0, y_0, z_0), (x_1, y_1, z_1), to find the other point of intersection.

The solution is exceedingly simple. Take $(a,\ b,\ c)$ for the coordinates of the other point of intersection, we must have identically with respect to x, y, z,

$$(a, \ldots \rangle\, x, y, z)^2 - 2(\alpha, \ldots \rangle\, x_0, y_0, z_0 \rangle\, x, y, z) = (lx + my + nz)\big(\beta_0(x - x_0) + \beta_1(y - y_0) + \beta_2(z - z_0)\big),$$

to a constant factor *près*. Assume successively $y = 0$, $z = 0$; $x = 0$, $z = 0$; $x = 0$, $y = 0$, then we have

$$\begin{aligned} l : m : n = {} & y_0z_0(\alpha x_0 + gz_0 + hy_0) - x_0(ax_0 + hy_0 + gz_0) : \\ & z_0x_0(hx_0 + by_0 + fz_0) - y_0(hx_0 + by_0 + fz_0) : \\ & x_0y_0(gx_0 + fy_0 + cz_0) - z_0(gx_0 + fy_0 + cz_0), \end{aligned}$$

or as they may be more simply written,

$$l : m : n = y_0z_0(ax_0^2 - by_0^2 - cz_0^2 - 2fy_0z_0) : z_0x_0(\ldots) : x_0y_0(\ldots).$$

[Pages 146–147: development of the transformation theory.]

We have, then, identically

$$lx_0(lx + my + nz) + xx_0(my - nz)(xz - yx_0) + y_0(mz - ny)(xy - zx_0) + lz_0(\ldots) - hx_0z_0(\ldots) = 0,$$

which shows that if θ_2, η_2 may be as well to give the corresponding solution of the problem.

[Pages 147–148: further reductions; the irrational transformation

$$\sqrt{lx + my + nz} \quad \text{and} \quad \sqrt{(a, \ldots)(x, y, z)^2}$$

provide an irrational transformation of the two forms each into itself.]

I shall once consider l, m, n as given functions of x_0, y_0, z_0, satisfying identically the equations

$$lx_0 + my_0 + nz_0 = 0, \quad Nc + na = 0, \quad xx_0 + yy_0 + zz_0 = 0,$$

equations which express that the line $lx + my + nz = 0$ is the tangent at the point (x_0, y_0, z_0) to the conic $(a, b, c, \ldots)(x, y, z)^2 = 0$. And I shall take for β, γ the following constant values, viz. in the case of the point (x, y, z) we shall have for the values:

$$\theta_1 = \theta_2 - q_0,\ (2qa - bx), \quad \eta_1 = \eta_2 - q_0,\ (2q\beta - ay), \quad \zeta_1 = \zeta_2 - q_0,\ (2q\gamma - cz),$$

and similarly for the values θ_2, η_2, ζ_2 in the same point. We obtain immediately

$$\theta_2^2 + \eta_2^2 + \zeta_2^2 = \theta_1^2 + \eta_1^2 + \zeta_1^2.$$

[Pages 149–150: continuation through the proof.]

The preceding investigations have been in my possession for about eighteen months.

2 Stone Buildings, April 18, 1855.

130.

DEUXIÈME MÉMOIRE SUR LES FONCTIONS DOUBLEMENT PÉRIODIQUES.

[From the *Journal de Mathématiques Pures et Appliquées* (Liouville), tom. xix. (1854), pp. 193–208; Sequel to Mémoire t. x. (1845), **25**.]

Je vais essayer de développer ici les propriétés qui se rapportent aux transformations linéaires des polynômes des fonctions p, g, Θ, Z, dont je me suis occupé dans le Mémoire sur les fonctions doublement périodiques que j'ai donné dans ce Recueil en 1845. Avant d'entrer en matière, je remarque que partant des expressions

$$\Omega = u + u', \qquad \Upsilon = v + v',$$

des deux périodes, on a $i = \sqrt{-1}$, on obtient, en définissant

$$\Omega' = u\nu - u'\nu', \qquad \Upsilon' = u'\nu' - u\nu',$$

les équations

$$\begin{aligned}\Omega\,\Upsilon' &= u\nu + u'\nu' = -(\nu\Omega - \nu'\Omega'),\\ \Upsilon\,\Omega' &= u\nu' + u'\nu = (\nu\Omega + \nu'\Omega'),\end{aligned}$$

et au moyen desquelles

$$B = \frac{\nu(\nu\Omega - \nu'\Omega')}{i\Upsilon\,\mathrm{mod.}\,(\nu\nu - \nu'\nu')}, \qquad B' = \frac{\nu'(\nu\Omega + \nu'\Omega')}{i\Upsilon\,\mathrm{mod.}\,(\nu\nu' - \nu'\nu)},$$

des quantités B, B' on déduit les formules

$$B + \beta = \frac{\nu\Upsilon}{i\,\mathrm{mod.}\,(\nu\nu' - \nu'\nu)}\Omega, \qquad B - B' = \frac{\nu\,\nu'}{i\,\mathrm{mod.}\,(\nu\nu' - \nu'\nu)}.$$

Je me mets attention aux transformations qui correspondent à des entiers impairs et premiers, et je suppose, de plus, que les transformations soient toujours propres, et régulières ; c'est-à-dire qu'on doit avoir

$$(2k+1)\,\Omega_r = \lambda\Omega + \mu\Upsilon, \quad \mathrm{mod.}\,(\mu,\nu) < T,$$

$$(2k+1)\,\Upsilon_r = \nu\Omega + \rho\Upsilon, \quad \mathrm{mod.}\,(\nu,\rho) < T,$$

où $2k+1$ est un nombre impair et premier, λ, μ, ν, ρ sont des entiers impairs ou pairs, et cômeptes de tel signe près, $\lambda\rho - \mu\nu = 2k+1$, et que les coefficients sont, savoir, soit signe près,

$$\lambda\rho - \mu\nu = 2k+1.$$

(condition pour que la transformation soit propre), et de même que les fonctions

$$\lambda \equiv 0, \quad \mu \equiv 0, \qquad \mathrm{mod}\,.\,2,$$

$$\nu \equiv 0, \quad \rho \equiv 1.$$

(condition pour que la transformation soit régulière).

On trouve tout de suite

$$\Omega = \rho\Omega_r - \mu\Upsilon_r, \qquad \Upsilon = -\nu\Omega_r + \lambda\Upsilon_r,$$

j'écris aussi

$$\Omega_r = u_r + u_r', \qquad \Upsilon_r = v_r + v_r',$$

et je suppose que R_r, β_r soient des fonctions de u_r, u'_r, v_r, v'_r analogues aux fonctions B, β, de u, u', v, v'.

Cela étant, je tire d'abord de l'équation

$$(2k+1)(B+\beta)\,\Omega = \frac{\nu}{i\,\text{mod.}\,(\nu\nu-\nu'\nu')}\Omega^2,$$

au moyen de laquelle

$$(B+\beta)\,\Omega = \frac{\nu}{i\,\text{mod.}\,(\nu\nu'-\nu'\nu)}\Omega^2,$$

au transformant en

$$\begin{aligned}[(2k+1)(B_r+\beta_r)\,\Omega &= \frac{\nu_r}{i\,\text{mod.}\,(\nu_r\nu'-\nu'_r\nu)}\Omega_r^2, \\ &= \frac{\nu_r}{i\,\text{mod.}\,(\nu_r\nu'-\nu'_r\nu)}\big(\rho\,\Omega_r-\mu\,\Upsilon_r\big)^2, \\ &= \frac{\rho^2\nu-2\rho\mu(\nu+\nu')+\mu^2\nu'}{i\,\text{mod.}\,(\nu_r\nu'-\nu'_r\nu)}\,\big(\rho\Omega_r-\mu\Upsilon_r\big).\end{aligned}$$

[Pages 152–156: continuation; the second Mémoire develops the expression of $J_r x$, $g_r x$, $Z_r x$ in terms of the original Jx, gx, Zx, the products

$$J_r x = e^{-\frac{1}{2k+1}(B_r-2k+1\cdot B)\,x\,\pi i}\prod\frac{J(x+s\psi_r)}{J(s\psi_r)},$$

the cases “Premier cas” ($p_r=1$, $q_r=2q_\theta$) and “Deuxième cas” ($p_r=2k+1$, $q_r=2$), and the corresponding Ω_r, Υ_r formulæ. The exposition continues in **[157–208]** of this same Mémoire 130.]

Cela fait voir que

$$e^{-i\pi x} = e^{-i\pi x},$$

le coefficient A étant donné au moyen de l'équation

$$A = \sum\frac{1}{(m,n)}\,,$$

où la somme est prise, comme auparavant, entre les limites

$$\text{mod}\,.\,(m,n) < T, \qquad T=\infty.$$

Mais il n'est pas permis d'écrire

$$A = \iint\frac{dx\,dy}{(m,n)^2}.$$

En effet, cette intégrale n'est pas le premier terme d'une suite dont il faudrait, pour obtenir un résultat exact, prendre deux termes ; le second terme de la suite serait une intégrale prise le long de tout le contour, et le second terme ne serait pas, comme dans le premier cas, égale à zéro. Pour trouver la valeur de A, je remarque que les fonctions doublement périodiques p, g, Θ, Z, qui sont contenues dans les formules ordinaires, peuvent satisfaire aux conditions

$$\mathit{mod.}\,(m,n) < T, \quad T=\infty,$$

puisqu'on a l'analyse $T=x$, et puis l'on obtient que la fonction $Ly+My^2$. Cela étant, on substituera pour u la valeur, je forme l'équation

$$\frac{y(x+s\psi)}{y(x)} = e^{-bx-cx^2}\Pi\big(1+\tfrac{x}{(m,n)+s\psi}\big),$$

et j'écris successivement

$$y = \tfrac{1}{2}\Pi, \quad y = \tfrac{1}{2}T,$$

et qui donne pour les équations correspondante du produit infini double doublement $e^{i\pi x}$, Gx ; en comparant les valeurs ainsi obtenues avec les équations qui donnent les valeurs de $y(x+\frac{1}{2}\Pi)$, $y(x+\frac{1}{2}T)$, on trouve

$$L = 0, \qquad M = \frac{\pi}{\text{mod.}\,(\nu\nu' - \nu'\nu)},$$

et enfin,

$$\frac{y(x+s\psi)}{y(x)} = e^{-\frac{\pi}{\text{mod.}\,(\nu\nu'-\nu'\nu)}}\Pi\big(1+\tfrac{x}{(m,n)+s\psi}\big),$$

$$\text{mod}\ .\,(m,n) < T, \quad T = \infty.$$

Cela donne pour l'équation qu'il s'agissait d'établir ; il ne faut pas remarquer qu'il y ait remarque que pour $y = 0$, on doit considérer à part le facteur $1 + \frac{x}{(m,n)}$, lequel multiplié par $y(x)$ devient tout simplement x ; l'équation réduite donc dans ce cas.

Cayley's Collected Mathematical Papers, Vol. II, pp. 141–184

Reconstruction with OCR skeleton and PNG verification

128. (continued)

Developments on the Porism of the In-and-Circumscribed Polygon (continued).

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 339–345.]

(Continuation from p. 140.)

which, he remarks, is satisfied by $r = -p$ and $r = -\dfrac{pq}{p+q}$, and that consequently the rationalized equation will divide by $p + r$ and $pq - r(p + q)$; and he finds, after the division,

$$p^2q^2 + p^2q^2(p+q)\,r - pq(p+q)^2\,r^2 - (p+q)(p-q)^2\,r^3 = 0,$$

which, restoring for p, q their values $R + a$, $R - a$, is the very equation above found.

The form given by Steiner (*Crelle*, t. ii. p. 289) is

$$r(R-a)^2 = (R+a)\sqrt{(R-r+a)(R-r-a)} + (R+a)\sqrt{(R-r-a)\cdot 2R},$$

which, putting p, q instead of $R + a$, $R - a$, is

$$qr = p\sqrt{(p-r)(q-r)} + p\sqrt{(q-r)(q+p)};$$

and Jacobi has shown in his memoir, "Anwendung der elliptischen Transcendenten u. s. w.," *Crelle*, t. iii. [1828] p. 376, that the rationalized equation divides (like that of Fuss) by the factor $pq - (p + q)r$, and becomes by that means identical with the rational equation given by Fuss.

In the case of two concentric circles $a = 0$, and putting $\dfrac{r}{R} = M$, we have

$$\sqrt{1-M} = \sqrt{1-M}.$$

This is, in fact, the very formula which corresponds to the general case of two conics having double contact. For suppose that the polygon is inscribed in the conic $U = 0$, and circumscribed about the conic $U + P^2 = 0$, we have then to find the discriminant of $\xi U + U + P^2$, i.e. of $(1 + \xi)U + P^2$. Let K be the discriminant of U, and let F be what the polar reciprocal of U becomes when the variables are replaced by the coefficients of P, or, what is the same thing, let $-F$ be the determinant obtained by bordering K (considered as a matrix) with the coefficients of P. The discriminant of $(1 + \xi)U + P^2$ is $(1 + \xi)^3K + (1 + \xi)^2F$, i.e. it is

$$(1+\xi)^2K\big[(1+\xi) + \tfrac{F}{K}\big], \quad \text{i.e. } (1+\xi)^2K(1+M\xi),$$

where $M = 1 + \dfrac{F}{K}$; or, what is the same thing, M is the discriminant of U divided by the discriminant of $U + P^2$. And M having this meaning, the condition of there being inscribed in the conic U an infinity of n-gons circumscribed about the conic $U + P^2 = 0$, is found by means of the series

$$A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + \&\text{c.} = (1+\xi)\sqrt{1+M\xi}.$$

We have, therefore,

$$\begin{aligned}
A &= 1,\\
2B &= M+2,\\
8C &= -M^2+4M,\\
16D &= M^3-2M^2,\\
-128E &= 5M^4-8M^3,\\
&\text{\&c.},\\
1024\,(CE-D^2) &= M^4(M^2-12M+16),\\
&\text{\&c.}
\end{aligned}$$

Hence for the triangle, quadrangle and pentagon, the conditions are

I. For the triangle,

$$M+2=0.$$

II. For the quadrangle,

$$M-4=0.$$

III. For the pentagon,

$$M^2-12M+16=0;$$

and so on.

It is worth noticing, that, in the case of two conics having a 4-point contact, we have $F=0$, and consequently $M=1$. The discriminant is therefore $(1+\xi)^3$, and as this does not contain any variable parameter, the conics cannot be determined so that there may be for a given value of n (nor, indeed, for any value whatever of n) an infinity of n-gons inscribed in the one conic, and circumscribed about the other conic.

The geometrical properties of a triangle, &c. inscribed in a conic and circumscribed about another conic, these two conics having double contact with each other, are at once obtained from those of the system in which the two conics are replaced by concentric circles. Thus, in the case of a triangle, if ABC be the triangle, and α, β, γ be the points of contact of the sides with the inscribed conic, then the tangents to the circumscribed conic at A, B, C meet the opposite sides BC, CA, AB in points lying in the chord of contact, the lines $A\alpha$, $B\beta$, $C\gamma$ meet in the pole of contact, and so on.

In the case of a quadrangle, if $ACEG$ be the quadrangle, and b, d, f, h the points of contact with the inscribed conic, then the tangents to the circumscribed conic at the pair of opposite angles A, E and the corresponding diagonal CG, and in like manner the tangents at the pair of opposite angles C, G and the corresponding diagonal AE, meet in the chord of contact. Again, the pairs of opposite sides AC, EG, and the line dh joining the points of contact of the other two sides with the inscribed conic, and the pairs of opposite sides AG, CE, and the line hf joining the points of contact of the other two sides with the inscribed conic, meet in the chord of contact. The diagonals AE, CG, and the lines bf, dh through the points of contact of pairs of opposite sides with the inscribed conic, meet in the pole of contact, &c.

The beautiful systems of 'focal relations' for regular polygons (in particular for the pentagon and the hexagon), given in Sir W. R. Hamilton's *Lectures on Quaternions*, [Dublin, 1853] Nos. 379–393, belong, it is clear, to polygons which are inscribed in and circumscribed about two conics having double contact with each other. In fact, the focus of a conic is a point such that the lines joining such point with the circular points at infinity (i.e. the points in which a circle is intersected by the line infinity) are tangents to the conic. In the case of two concentric circles, these are to be considered as touching in the circular points at infinity; and consequently, when the concentric circles are replaced by two conics having double contact, the circular points at infinity are replaced by the points of contact of the two conics.

Thus, in the figure (which is simply Sir W. R. Hamilton's figure 81 put into perspective), the system of relations

$$\begin{gathered} F, G(\,..)\, ABCI, \\ G, H(\,..)\, BCDK, \\ H, I(\,..)\, CDEF, \\ I, K(\,..)\, DEAG, \\ K, F(\,..)\, EABH, \end{gathered}$$

will mean, $F, G(\,..)\, ABCI$, that there is a conic inscribed in the quadrilateral $ABCI$ such that the tangents to this conic through the points F and G pass two and two through the points of contact of the circumscribed and the inscribed conics, and similarly for the other relations of the system. As the figure is drawn, the tangents in question are of course (as the tangents through the foci in the case of the two concentric circles) imaginary.

2 Stone Buildings, March 7, 1854.

129.

On the Porism of the In-and-Circumscribed Triangle, and on an Irrational Transformation of Two Ternary Quadratic Forms each into itself.

[From the *Philosophical Magazine*, vol. ix. (1855), pp. 513–517.]

There is an irrational transformation of two ternary quadratic forms each into itself, based upon the solution of the following geometrical problem,

Given that the line

$$lx + my + nz = 0$$

meets the conic

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0$$

in the point (x_1, y_1, z_1)*; to find the other point of intersection.*

The solution is exceedingly simple. Take (x_2, y_2, z_2) for the coordinates of the other point of intersection, we must have identically with respect to x, y, z,

$$\begin{aligned} &(a, \dots \mid x, y, z)^2 \cdot (\mathfrak{A}, \dots \mid l, m, n)^2 - k(lx + my + nz)^2 \\ &= (a, \dots \mid x_1, y_1, z_1 \mid x, y, z) \cdot (a, \dots \mid x_2, y_2, z_2 \mid x, y, z) \end{aligned}$$

to a constant factor près.

Assume successively $x, y, z = \mathfrak{A}, \mathfrak{H}, \mathfrak{G};\ \mathfrak{H}, \mathfrak{B}, \mathfrak{F};\ \mathfrak{G}, \mathfrak{F}, \mathfrak{C}$; it follows that

$$\begin{aligned} x_2 : y_2 : z_2 = y_1 z_1 &\big[\mathfrak{A}(a, \dots \mid l, m, n)^2 - (\mathfrak{A}l + \mathfrak{H}m + \mathfrak{G}n)^2\big] \\ : z_1 x_1 &\big[\mathfrak{B}(a, \dots \mid l, m, n)^2 - (\mathfrak{H}l + \mathfrak{B}m + \mathfrak{F}n)^2\big] \\ : x_1 y_1 &\big[\mathfrak{C}(a, \dots \mid l, m, n)^2 - (\mathfrak{G}l + \mathfrak{F}m + \mathfrak{C}n)^2\big]; \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} x_2 : y_2 : z_2 = {} & y_1 z_1 (bn^2 + cm^2 - 2fmn) \\ & : z_1 x_1 (cl^2 + an^2 - 2gnl) \\ & : x_1 y_1 (am^2 + bl^2 - 2hlm). \end{aligned}$$

It is not necessary for the present purpose, but it may be as well to give the corresponding solution of the problem

Given that one of the tangents through the point (ξ, η, ζ) *to the conic*

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0$$

is the line $l_1 x + m_1 y + n_1 z = 0$*; to find the equation to the other tangent.*

Let $l_2 x + m_2 y + n_2 z = 0$ be the other tangent, then

$$\begin{aligned} (a, \dots \mid \xi, \eta, \zeta)^2 \cdot (a, \dots \mid x, y, z)^2 - \big[(a, \dots \mid \xi, \eta, \zeta \mid x, y, z)\big]^2 & \\ = (l_1 x + m_1 y + n_1 z)(l_2 x + m_2 y + n_2 z) & \end{aligned}$$

to a constant factor près. Assume successively $y = 0, z = 0$; $z = 0, x = 0$; $x = 0, y = 0$; then we have

$$\begin{aligned} l_2 : m_2 : n_2 = {} & m_1 n_1 \big[a(a, \dots \mid \xi, \eta, \zeta)^2 - (a\xi + h\eta + g\zeta)^2\big] \\ & : n_1 l_1 \big[b(a, \dots \mid \xi, \eta, \zeta)^2 - (h\xi + b\eta + f\zeta)^2\big] \\ & : l_1 m_1 \big[c(a, \dots \mid \xi, \eta, \zeta)^2 - (g\xi + f\eta + c\zeta)^2\big]; \end{aligned}$$

or, as they may be more simply written,

$$\begin{aligned} l_2 : m_2 : n_2 = {} & m_1 n_1 (\mathfrak{B}\zeta^2 + \mathfrak{C}\eta^2 - 2\mathfrak{F}\eta\zeta) \\ & : n_1 l_1 (\mathfrak{C}\xi^2 + \mathfrak{A}\zeta^2 - 2\mathfrak{G}\zeta\xi) \\ & : l_1 m_1 (\mathfrak{A}\eta^2 + \mathfrak{B}\xi^2 - 2\mathfrak{H}\xi\eta). \end{aligned}$$

Returning now to the solution of the first problem, I shall for the sake of simplicity consider the formulæ obtained by taking for the equation of the conic,

$$\alpha x^2 + \beta y^2 + \gamma z^2 = 0.$$

We see, therefore, that if this conic be intersected by the line $lx + my + nz = 0$ in the points (x_1, y_1, z_1) and (x_2, y_2, z_2), then

$$\begin{aligned} x_2 : y_2 : z_2 = {} & y_1 z_1 (\gamma m^2 + \beta n^2) \\ & : z_1 x_1 (\alpha n^2 + \gamma l^2) \\ & : x_1 y_1 (\beta l^2 + \alpha m^2). \end{aligned}$$

We have, in fact, identically

$$\begin{aligned} & l y_1 z_1 (\beta n^2 + \gamma m^2) + m z_1 x_1 (\gamma l^2 + \alpha n^2) + n x_1 y_1 (\alpha m^2 + \beta l^2) \\ & \quad = (\alpha mn x_1 + \beta nl y_1 + \gamma lm z_1)(l x_1 + m y_1 + n z_1) - lmn(\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2). \end{aligned}$$

Also,

$$\begin{aligned}&\alpha y_1^2z_1^2(\beta n^2+\gamma m^2)^2+\beta z_1^2x_1^2(\gamma l^2+\alpha n^2)^2+\gamma x_1^2y_1^2(\alpha m^2+\beta l^2)^2\\ &\quad=\alpha\beta\gamma\big[-l^2x_1^2-m^2y_1^2-n^2z_1^2\\ &\qquad+(my_1+nz_1)^2l^2x_1^2+(nz_1+lx_1)^2m^2y_1^2+(lx_1+my_1)^2n^2z_1^2-2lmnx_1y_1z_1(lx_1+my_1+nz_1)\big]\\ &\qquad-(l^2\beta\gamma x_1^2+m^2\gamma\alpha y_1^2+n^2\alpha\beta z_1^2)(\alpha x_1^2+\beta y_1^2+\gamma z_1^2);\end{aligned}$$

which show that if $lx_1+my_1+nz_1=0$ and $\alpha x_1^2+\beta y_1^2+\gamma z_1^2=0$, then also $lx_2+my_2+nz_2=0$ and $\alpha x_2^2+\beta y_2^2+\gamma z_2^2=0$: this is, of course, as it should be.

I shall now consider l, m, n as given functions of x_1, y_1, z_1 satisfying identically the equations

$$\begin{aligned}lx_1+my_1+nz_1&=0,\\ l^2bc+m^2ca+n^2ab&=0,\end{aligned}$$

equations which express that $lx+my+nz=0$ is the tangent from the point (x_1,y_1,z_1) to the conic $ax^2+by^2+cz^2=0$. And I shall take for α, β, γ the following values, viz.

$$\begin{aligned}\alpha&=ax_1^2+by_1^2+cz_1^2-a(x_1^2+y_1^2+z_1^2),\\ \beta&=ax_1^2+by_1^2+cz_1^2-b(x_1^2+y_1^2+z_1^2),\\ \gamma&=ax_1^2+by_1^2+cz_1^2-c(x_1^2+y_1^2+z_1^2);\end{aligned}$$

so that x_1, y_1, z_1 continuing absolutely indeterminate, we have identically $\alpha x_1^2+\beta y_1^2+\gamma z_1^2=0$. Also taking Θ as a function of x_1, y_1, z_1, the value of which will be subsequently given, I write

$$\begin{aligned}x_2&=\Theta y_1z_1(\beta n^2+\gamma m^2),\\ y_2&=\Theta z_1x_1(\gamma l^2+\alpha n^2),\\ z_2&=\Theta x_1y_1(\alpha m^2+\beta l^2);\end{aligned}$$

so that x_1, y_1, z_1 are arbitrary, and x_2, y_2, z_2 are taken to be determinate functions of x_1, y_1, z_1. The point (x_2,y_2,z_2) is geometrically connected with the point (x_1,y_1,z_1) as follows, viz. (x_2,y_2,z_2) is the point in which the tangent through (x_1,y_1,z_1) to the conic $ax^2+by^2+cz^2=0$ meets the conic passing through the point (x_1,y_1,z_1) and the points of intersection of the conics $ax^2+by^2+cz^2=0$ and $x^2+y^2+z^2=0$. Consequently, in the particular case in which (x_1,y_1,z_1) is a point on the conic $x^2+y^2+z^2=0$, the point (x_2,y_2,z_2) is the point in which this conic is met by the tangent through (x_1,y_1,z_1) to the conic $ax^2+by^2+cz^2=0$.

It has already been seen that $lx_1+my_1+nz_1=0$ and $\alpha x_1^2+\beta y_1^2+\gamma z_1^2=0$ identically; consequently we have identically $lx_2+my_2+nz_2=0$ and $\alpha x_2^2+\beta y_2^2+\gamma z_2^2=0$. The latter equation, written under the form

$$(ax_1^2+by_1^2+cz_1^2)(x_2^2+y_2^2+z_2^2)-(x_1^2+y_1^2+z_1^2)(ax_2^2+by_2^2+cz_2^2)=0,$$

shows that if x_2, y_2, z_2 are such that $x_2^2+y_2^2+z_2^2=x_1^2+y_1^2+z_1^2$, then that also $ax_2^2+by_2^2+cz_2^2=ax_1^2+by_1^2+cz_1^2$. I proceed to determine Θ so that we may have $x_2^2+y_2^2+z_2^2=x_1^2+y_1^2+z_1^2$. We obtain immediately

$$\begin{aligned}\tfrac{1}{\Theta^2}(x_2^2+y_2^2+z_2^2)&=(l^2x_1^2+m^2y_1^2+n^2z_1^2)(\alpha^2x_1^2+\beta^2y_1^2+\gamma^2z_1^2)\\ &\quad-(\alpha^2l^2x_1^4+\beta^2m^2y_1^4+\gamma^2n^2z_1^4-2\beta\gamma mnl^2y_1^2z_1^2-2\gamma\alpha nlm^2z_1^2x_1^2-2\alpha\beta lmn^2x_1^2y_1^2);\end{aligned}$$

write for a moment

$$ax_1^2+by_1^2+cz_1^2=p,\quad x_1^2+y_1^2+z_1^2=q,\quad\text{so that}\quad\alpha=p-aq,\ \beta=p-bq,\ \gamma=p-cq,$$

then

$$\begin{aligned}
\alpha^2x_1^2+\beta^2y_1^2+\gamma^2z_1^2 &= qp^2-2p\cdot pq+(a^2x_1^2+b^2y_1^2+c^2z_1^2)\,q^2\\
&= q\bigl\{(a^2x_1^2+b^2y_1^2+c^2z_1^2)\,q-p^2\bigr\}\\
&= q\bigl\{(b-c)^2y_1^2z_1^2+(c-a)^2z_1^2x_1^2+(a-b)^2x_1^2y_1^2\bigr\}.
\end{aligned}$$

Moreover,

$$\begin{aligned}
&\alpha^2l^2x_1^4+\beta^2m^2y_1^4+\gamma^2n^2z_1^4-2\beta\gamma mnl^2y_1^2z_1^2-2\gamma\alpha nlm^2z_1^2x_1^2-2\alpha\beta lmn^2x_1^2y_1^2\\
&\quad= p^2\bigl\{l^2x_1^4+m^2y_1^4+n^2z_1^4-2mny_1^2z_1^2-2nlz_1^2x_1^2-2lmx_1^2y_1^2\bigr\}\\
&\qquad-2pq\bigl\{al^2x_1^4+bm^2y_1^4+cn^2z_1^4-(b+c)mny_1^2z_1^2-(c+a)nlz_1^2x_1^2-(a+b)lmx_1^2y_1^2\bigr\}\\
&\qquad+q^2\bigl\{a^2l^2x_1^4+b^2m^2y_1^4+c^2n^2z_1^4-2bcmny_1^2z_1^2-2canlz_1^2x_1^2-2ablmx_1^2y_1^2\bigr\},
\end{aligned}$$

the first line of which vanishes in virtue of the equation $lx_1+my_1+nz_1=0$; we have therefore

$$\begin{aligned}
\tfrac{1}{\Theta^2}(x_2^2+y_2^2+z_2^2)-(x_1^2+y_1^2+z_1^2)=(l^2x_1^2+m^2y_1^2+n^2z_1^2)\bigl\{(b-c)^2y_1^2z_1^2+(c-a)^2z_1^2x_1^2+(a-b)^2x_1^2y_1^2\bigr\}\\
+2pq\{\cdots\}-q^2\{\cdots\}.
\end{aligned}$$

Hence reducing the function on the right-hand side, and putting

$$\tfrac{1}{\Theta^2}(x_2^2+y_2^2+z_2^2)\div(x_1^2+y_1^2+z_1^2)=1,$$

we have

$$\begin{aligned}
\tfrac{1}{\Theta^2}&=a^2l^4x_1^4+b^2m^4y_1^4+c^2n^4z_1^4\\
&\quad+(c^2m^4-2b^2m^2n^2)\,y_1^2z_1^2+(a^2n^4-2c^2n^2l^2)\,z_1^2x_1^2+(b^2l^4-2a^2l^2m^2)\,x_1^2y_1^2\\
&\quad+(b^2n^4-2c^2m^2n^2)\,y_1^2z_1^2+(c^2l^4-2a^2n^2l^2)\,z_1^2x_1^2+(a^2m^4-2b^2l^2m^2)\,x_1^2y_1^2\\
&\quad+\bigl\{l^4(b-c)^2+m^4(c-a)^2+n^4(a-b)^2\\
&\qquad+2m^2n^2(bc-ca-ab)+2n^2l^2(-bc+ca-ab)+2l^2m^2(-bc-ca+ab)\bigr\}\,x_1^2y_1^2z_1^2.
\end{aligned}$$

The value of Θ might probably be expressed in a more simple form by means of the equations $lx_1+my_1+nz_1=0$ and $l^2bc+m^2ca+n^2ab=0$, even without solving these equations; but this I shall not at present inquire into.

Recapitulating, l, m, n are considered as functions of x_1, y_1, z_1 determined (to a common factor près) by the equations

$$\begin{aligned}
lx_1+my_1+nz_1&=0,\\
l^2bc+m^2ca+n^2ab&=0;
\end{aligned}$$

Θ is determined as above, and then writing

$$\begin{aligned}
\alpha&=ax_1^2+by_1^2+cz_1^2-a(x_1^2+y_1^2+z_1^2),\\
\beta&=ax_1^2+by_1^2+cz_1^2-b(x_1^2+y_1^2+z_1^2),\\
\gamma&=ax_1^2+by_1^2+cz_1^2-c(x_1^2+y_1^2+z_1^2),
\end{aligned}$$

we have

$$\begin{aligned}
x_2&=\Theta y_1z_1(\beta n^2+\gamma m^2),\\
y_2&=\Theta z_1x_1(\gamma l^2+\alpha n^2),\\
z_2&=\Theta x_1y_1(\alpha m^2+\beta l^2);
\end{aligned}$$

and these values give

$$\begin{gathered}
lx_2 + my_2 + nz_2 = 0, \\
x_2^2 + y_2^2 + z_2^2 = x_1^2 + y_1^2 + z_1^2, \\
ax_2^2 + by_2^2 + cz_2^2 = ax_1^2 + by_1^2 + cz_1^2.
\end{gathered}$$

In connexion with the subject I may add the following transformation, viz. if

$$3\sqrt{\alpha}\, x' = \sqrt{3\alpha}\,(y - z) + \sqrt{(3\alpha - 2\beta)(x^2 + y^2 + z^2) + 2\beta(yz + zx + xy)},$$

then reciprocally

$$3\sqrt{\beta}\, x = -\sqrt{3\beta}\,(y' - z') + \sqrt{(3\beta - 2\alpha)(x'^2 + y'^2 + z'^2) + 2\alpha(y'z' + z'x' + x'y')},$$

and

$$\begin{gathered}
x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2, \\
\beta\,(x^2 + y^2 + z^2 - yz - zx - xy) = \alpha\,(x'^2 + y'^2 + z'^2 - y'z' - z'x' - x'y').
\end{gathered}$$

Suppose $1 + \rho + \rho^2 = 0$, then

$$x^2 + y^2 + z^2 - yz - zx - xy = (x + \rho y + \rho^2 z)(x + \rho^2 y + \rho z);$$

and in fact

$$\begin{aligned}
3\sqrt{\alpha}\,(x' + \rho^2 y' + \rho z') &= -\sqrt{3\alpha}\,(1 + 2\rho)(x + \rho y + \rho^2 z), \\
3\sqrt{\alpha}\,(x' + \rho y' + \rho^2 z') &= \sqrt{3\beta}\,(1 + 2\rho)(x + \rho^2 y + \rho z).
\end{aligned}$$

The preceding investigations have been in my possession for about eighteen months.

2 Stone Buildings, April 18, 1855.

130.

Deuxième Mémoire sur les Fonctions Doublement Périodiques.

[From the *Journal de Mathématiques Pures et Appliquées* (Liouville), tom. xix. (1854), pp. 193–208 : Sequel to Memoir t. x. (1845), **25**.]

Je vais essayer de développer ici les propriétés qui se rapportent aux transformations linéaires des périodes des fonctions yx, gx, Ωx, Zx, dont je me suis occupé dans le Mémoire sur les fonctions doublement périodiques que j'ai donné dans ce Recueil en 1845. Avant d'entrer en matière, je remarque que, partant des expressions

$$\begin{aligned}
\Omega &= \omega + \omega' i, \\
\mathrm{T} &= v + v' i,
\end{aligned}$$

des deux périodes, où $i = \sqrt{-1}$, on obtient, en écrivant

$$\begin{aligned}
\Omega^* &= \omega - \omega' i, \\
\mathrm{T}^* &= v - v' i,
\end{aligned}$$

les équations

$$\begin{aligned}
\Omega\,\mathrm{T}^* &= \omega v + \omega' v' - i(\omega v' - \omega' v), \\
\Omega^*\mathrm{T} &= \omega v + \omega' v' + i(\omega v' - \omega' v);
\end{aligned}$$

au moyen desquelles et des valeurs

$$B = \frac{\pi(\omega v' - \omega' v)}{\Omega\,\mathrm{T}\ \mathrm{mod.}\,(\omega v' - \omega' v)}, \quad \beta = \frac{\pi(\omega v + \omega' v')}{\Omega\,\mathrm{T}\ \mathrm{mod.}\,(\omega v' - \omega' v)}$$

des quantités B, β, on déduit les formules

$$B + \beta = \frac{\pi}{\Omega\ \mathrm{mod.}\,(\omega v' - \omega' v)}\,\Omega^*, \qquad B - \beta = \frac{\pi}{\mathrm{T}\ \mathrm{mod.}\,(\omega v' - \omega' v)}\,\mathrm{T}^*.$$

Je ne fais attention qu'aux transformations qui correspondent à des entiers impairs et premiers, et je suppose, de plus, que la transformation soit toujours propre et régulière, c'est-à-dire qu'en écrivant

$$(2k+1)\,\Omega_, = \lambda\Omega + \mu\mathrm{T} = (\lambda, \mu),$$
$$(2k+1)\,\mathrm{T}_, = \nu\Omega + \rho\mathrm{T} = (\nu, \rho),$$

où $2k+1$ est un entier positif, impair et premier, et où λ, μ, ν, ρ sont des entiers tels, qu'au signe près, $\lambda\rho - \mu\nu$ soit égal à $2k+1$, je suppose

$$\lambda\rho - \mu\nu = 2k+1,$$

(condition pour que la transformation soit propre), et, en outre,

$$\begin{aligned} &\lambda \equiv 1, \quad \mu \equiv 0, \quad (\mathrm{mod}\ 2) \\ &\nu \equiv 0, \quad \rho \equiv 1, \end{aligned}$$

(condition pour que la transformation soit régulière).

On trouve tout de suite

$$\Omega = \rho\Omega_, - \mu\mathrm{T}_, = (\rho, -\mu)_,,$$
$$\mathrm{T} = -\nu\Omega_, + \lambda\mathrm{T}_, = (-\nu, \lambda)_,,$$

j'écris aussi

$$\Omega^*_, = \omega_, - \omega'_, i, \quad \mathrm{T}^*_, = v_, - v'_, i,$$

et je suppose que $B_,$, $\beta_,$ soient des fonctions de $\omega_,$, $v_,$ telles que les fonctions B, β le sont de ω, v.

Cela étant, je forme d'abord l'équation

$$(2k+1)(\omega_, v'_, - \omega'_, v_,) = \omega v' - \omega' v,$$

au moyen de laquelle l'équation

$$(B + \beta)\,\Omega = \frac{\pi}{\mathrm{mod.}\,(\omega v' - \omega' v)}\,\Omega^*$$

se transforme en

$$(2k+1)(B_, + \beta_,)\,\Omega = \frac{\pi}{\mathrm{mod.}\,(\omega v' - \omega' v)}\,\Omega^*_,.$$

De là

$$\big[(2k+1)(B_, + \beta_,) - B\big]\,\Omega = \frac{\pi}{\Omega}\,\xi(\omega, \lambda),$$

ou enfin

$$\big[(2k+1)(B_, + \beta_,) - B\big]\,\Omega = -\frac{\pi}{\Omega}\,\beta_,(\omega, \mu),$$

et de même

$$\big[(2k+1)(B_, - \beta_,) - B_,\big]\,\Omega = -\frac{\pi}{\Omega}\,\xi_,(\omega,\lambda);$$

équations qui seront bientôt utiles.

Je suppose d'abord que $2k+1$ soit égal à l'unité, transformation que l'on peut nommer triviale. La fonction yx est définie par l'équation

$$yx = e^{-\frac{1}{2}Bx^2}x\prod\Big[1+\frac{x}{(m,n)}\Big], \quad \text{mod.}\,(m,n) < T,\ T=\infty;$$

dans $(m,n) = m\Omega + n\mathrm{T}$, les entiers m, n doivent prendre toutes les valeurs positives ou négatives (le seul système $m=0$, $n=0$ excepté) qui satisfont à l'inégalité

$$\text{mod.}\,(m,n) < T,$$

dont le second membre T sera ensuite supposé infini. Soit $y_,x$ la fonction correspondante pour les périodes $\Omega_,$, $\mathrm{T}_,$, on aura

$$y_,x = e^{-\frac{1}{2}B_,x^2}x\prod\Big[1+\frac{x}{(m,n)_,}\Big], \quad \text{mod.}\,(m,n)_, < T,\ T=\infty.$$

Or

$$\begin{aligned}(m,n)_, &= m\Omega_, + n\mathrm{T}_,\\ &= m(\lambda\Omega+\mu\mathrm{T}) + n(\nu\Omega+\rho\mathrm{T})\\ &= (\lambda m+\nu n)\,\Omega + (\mu m+\rho n)\,\mathrm{T}\\ &= m_,\Omega + n_,\mathrm{T}\\ &= (m_,,n_,).\end{aligned}$$

En écrivant, comme nous venons de le faire,

$$\begin{aligned}m_, &= \lambda m + \nu n,\\ n_, &= \mu m + \rho n,\end{aligned}$$

on voit tout de suite qu'à chaque système de valeurs entières de m, n, correspond un système, et un seul système, de valeurs entières de $m_,$, $n_,$; et que de chaque système de valeurs entières de $m_,$, $n_,$, correspond un système, et un seul système, de valeurs entières de m, n ; de plus, les systèmes $m=0$, $n=0$ et $m_,=0$, $n_,=0$, correspondent l'un à l'autre. Il est donc permis d'écrire

$$\prod\Big[1+\frac{x}{(m,n)}\Big] = \prod\Big[1+\frac{x}{(m,n)_,}\Big],$$

les limites comme auparavant ; car, à cause de

$$(m,n)_, = (m_,,n_,),$$

la condition pour les limites, savoir

$$\text{mod.}\,(m,n)_, < T,\ T=\infty,$$

devient

$$\text{mod.}\,(m,n) < T,\ T=\infty.$$

Cela donne enfin l'équation

$$y_,x = e^{-\frac{1}{2}(B_,-B)x^2}\,yx,$$

et, au moyen de cette équation, on obtient une équation correspondante pour la transformation de l'une quelconque des fonctions yx, gx, Ωx, Zx, définies par les équations

$$\begin{aligned}
yx &= e^{-\frac{1}{2}Bx^2} x \prod \Big[1 + \frac{x}{(m,n)}\Big], \quad \text{mod.}\,(m,n) < T,\\
gx &= e^{-\frac{1}{2}Bx^2} \prod \Big[1 + \frac{x}{(m,n)}\Big], \quad \text{mod.}\,(m,n) < T, \quad T = \infty,\\
\Omega x &= e^{-\frac{1}{2}Bx^2} \prod \Big[1 + \frac{x}{(m,n)}\Big], \quad \text{mod.}\,(m,n) < T,\\
Zx &= e^{-\frac{1}{2}Bx^2} \prod \Big[1 + \frac{x}{(m,n)}\Big], \quad \text{mod.}\,(m,n) < T,
\end{aligned}$$

(équations dans lesquelles $m = m + \frac{1}{2}$, $n = n + \frac{1}{2}$). Je prends par exemple la fonction gx, et j'écris dans l'équation entre $y_,x$ et yx, $x + \frac{1}{2}\Omega$, au lieu de x. Soit pour un moment $\rho = 2\rho' + 1$, $\mu = 2\mu'$; cela donne

$$x + \tfrac{1}{2}\Omega = x + \tfrac{1}{2}(\rho\Omega_, - \mu \mathrm{T}_,) = x + (\rho', -\mu')_,.$$

Donc

$$y_,(x + \tfrac{1}{2}\Omega) = e^{\frac{1}{2}B_,(\rho',-\mu')^2_,}\, M_,\, g_,x,$$

c'est-à-dire

$$y_,(x + \tfrac{1}{2}\Omega_,) = e^{\frac{1}{2}B_,(\rho',-\mu')^2_,}\, M_,\, g_,x;$$

de plus,

$$y(x + \tfrac{1}{2}\Omega) = e^{\frac{1}{2}B\Omega^2}\, M\, gx.$$

Ces substitutions étant effectuées, les coefficients M, $M_,$ doivent être éliminés en écrivant $x = 0$; cela donne

$$g_,x = e^{-\frac{1}{2}(B_,-B)x^2 \cdot \text{etc.}}\, gx,$$

ou enfin, au moyen d'une équation déjà trouvée,

$$g_,x = e^{-\frac{1}{2}(B_,-B)x^2}\, gx,$$

et de même pour les fonctions Ωx, Zx.

Donc enfin, en représentant par Jx l'une quelconque des fonctions yx, gx, Ωx, Zx, on aura

$$J_,x = e^{-\frac{1}{2}(B_,-B)x^2}\, Jx,$$

où $J_,x$ est ce que devient Jx au moyen d'une transformation triviale (propre et régulière) des périodes.

Je passe à présent à la transformation pour un nombre impair et premier $(2k+1)$ quelconque ; mais pour cela on a besoin de connaitre la valeur de la fonction

$$u' = \prod \Big[1 + \frac{x}{(m,n) + y}\Big], \quad \text{mod.}\,\{(m,n) + y\} < T,\ T = \infty,$$

où $y = a + bi$ est une quantité réelle ou imaginaire quelconque.

Soit u ce que devient u' en prenant pour la condition par rapport aux limites

$$\text{mod.}\,(m,n) < T,\ T = \infty;$$

on trouve sans peine

$$u = \frac{x+y}{y} \cdot \frac{y(x+y)}{y(y)}.$$

Pour trouver u', je forme l'équation

$$u : u' = \prod \Big[1 + \frac{x}{(m,n)}\Big],$$

la limite inférieure du produit infini double étant

$$\text{mod.}\,\{(m,n) + y\} > T,$$

et la limite supérieure

$$\text{mod.}\,(m,n) < T,\ \ T = \infty;$$

cela donne

$$\log u - \log u' = x \sum \frac{1}{(m,n)^2} - \tfrac{1}{2} x^2 \sum \frac{1}{(m,n)^3} + \cdots,$$

car on peut démontrer que

$$\sum \frac{1}{(m,n)^k} = 0, \quad k \geq 2.$$

Pour cela, observons que m et n étant infinis puisque T l'est, la première des sommes dont il s'agit peut se remplacer par l'intégrale double

$$I = \iint \frac{dm\, dn}{(m,n)^2},$$

laquelle (en écrivant $m = r\cos\theta$, $n = r\sin\theta$, ce qui donne, comme on sait, $dm\, dn = r\, dr\, d\theta$) devient

$$I = \iint \frac{dr\, d\theta}{r(\Omega\cos\theta + \mathrm{T}\sin\theta)^2},$$

d'où

$$I = \iint \frac{(\log r)\, d\theta}{(\Omega\cos\theta + \mathrm{T}\sin\theta)^2},$$

en prenant $(\log r)$ entre les limites convenables. Pour trouver ces limites, j'écris

$$(m,n) + y = r(\Omega\cos\theta + \mathrm{T}\sin\theta) + y,$$

ce qui donne

$$\text{mod.}^2\{(m,n) + y\} = \big\{r(\Omega\cos\theta + \mathrm{T}\sin\theta) + y\big\}\big\{r(\Omega^*\cos\theta + \mathrm{T}^*\sin\theta) + y^*\big\},$$

savoir, à l'une des limites

$$\begin{aligned} &r^2(\Omega\cos\theta + \mathrm{T}\sin\theta)(\Omega^*\cos\theta + \mathrm{T}^*\sin\theta) \\ &\quad + r\big\{y^*(\Omega\cos\theta + \mathrm{T}\sin\theta) + y(\Omega^*\cos\theta + \mathrm{T}^*\sin\theta)\big\} + yy^* = 0; \end{aligned}$$

ou, en négligeant les puissances négatives de T,

$$r = \frac{T}{\sqrt{(\Omega\cos\theta + \mathrm{T}\sin\theta)(\Omega^*\cos\theta + \mathrm{T}^*\sin\theta)}} - \frac{1}{2}\Big[\frac{y}{\Omega\cos\theta + \mathrm{T}\sin\theta} + \frac{y^*}{\Omega^*\cos\theta + \mathrm{T}^*\sin\theta}\Big],$$

et à l'autre limite,

$$r = \frac{T}{\sqrt{(\Omega\cos\theta + \mathrm{T}\sin\theta)(\Omega^*\cos\theta + \mathrm{T}^*\sin\theta)}}.$$

Or, en représentant ces deux équations par

$$r = R - \phi, \quad r = R,$$

on trouve, pour la valeur de $(\log r)$ entre les deux limites,

$$\log R - \log(R-\phi) = -\log\Big(1-\frac{\phi}{R}\Big) = 0,$$

à cause de la valeur infinie de R. Ainsi la somme cherchée est nulle ; et il est tout clair que les sommes suivantes $\sum \frac{1}{(m,n)^k}$, &c., se réduisent de même à zéro.

Donc enfin,

$$\log u - \log u' = x\sum\frac{1}{(m,n)^2}.$$

Cela fait voir que

$$u' = e^{-kx}u,$$

le coefficient k étant donné au moyen de l'équation

$$k = \sum\frac{1}{(m,n)^2},$$

où la somme est prise, comme auparavant, entre les limites

$$\text{mod.}\,\{(m,n)+y\} > T,\quad \text{mod.}\,(m,n) < T,\ T=\infty.$$

Mais il n'est pas permis d'écrire

$$k = \iint\frac{dm\,dn}{(m,n)^2}.$$

En effet, cette intégrale n'est que le premier terme d'une suite dont il faudrait, pour obtenir un résultat exact, prendre deux termes ; le second terme de la suite serait une intégrale prise le long d'un contour, et il serait, ce me semble, très-difficile d'en trouver la valeur. Pour trouver la valeur de k, je remarque que k sera fonction linéaire des quantités T, y, y^*, $\frac{1}{T}$, &c., qui entrent dans les valeurs de r ; donc, puisqu'en dernière analyse $T=\infty$, k ne peut être que de la forme $Ly+My^*$. Cela étant, en substituant pour u' sa valeur, je forme l'équation

$$\frac{y(x+y)}{y(y)} = e^{-Bxy}e^{(-Bxy+Lxy+Mxy^*)x^2}\prod\Big[1+\frac{x}{(m,n)+y}\Big]^{-1},\quad \text{mod.}\,\{(m,n)+y\}<T,\ T=\infty,$$

et j'écris successivement

$$y=\tfrac12\Omega,\quad y=\tfrac12\mathrm{T},$$

ce qui donne pour les valeurs correspondantes du produit infini double $e^{-\frac12 Bx^2}gx$ et $e^{-\frac12 Bx^2}\Omega x$. En comparant les valeurs ainsi obtenues avec les équations qui donnent les valeurs de $y(x+\frac12\Omega)$, $y(x+\frac12\mathrm{T})$, on trouve

$$L=0,\quad M=\frac{\pi}{\text{mod.}\,(\omega v'-\omega' v)};$$

ou enfin,

$$\frac{y(x+y)}{y(y)} = e^{-Bxy}e^{\beta xy^*}\prod\Big[1+\frac{x}{(m,n)+y}\Big]^{-1},\quad \text{mod.}\,\{(m,n)+y\}<T,\ T=\infty,$$

laquelle est l'équation qu'il s'agissait d'établir. Il est à peine nécessaire de faire la remarque que pour $y=0$, on doit considérer à part le facteur $1+\frac{x}{y}$, lequel multiplié par $y(y)$ devient tout simplement x ; l'équation subsiste donc dans ce cas.

En revenant au problème des transformations linéaires, partant des équations

$$(2k+1)\,\Omega_{,} = \lambda\Omega+\mu\mathrm{T},$$
$$(2k+1)\,\mathrm{T}_{,} = \nu\Omega+\rho\mathrm{T},$$

je suppose d'abord que les coefficients λ, ν ne satisfassent pas à la fois aux deux conditions

$$\lambda \equiv 0, \quad \nu \equiv 0 \pmod{2k+1},$$

et je prends p, q des entiers quelconques tels, que $\lambda p + \nu q$ ne soit pas $\equiv 0$, mod. $(2k+1)$.

Cela étant, soient

$$\begin{aligned} \lambda p + \nu q &= p_{,}, \\ \mu p + \rho q &= q_{,}, \\ (2k+1)\,\psi &= p\Omega + q\mathrm{T}, \end{aligned}$$

et, par conséquent,

$$\psi = p\Omega_{,} + q\mathrm{T}_{,}.$$

Je forme l'équation

$$(m_{,}, n_{,}) + s\psi = (m, n)_{,},$$

savoir

$$m_{,}\Omega + n_{,}\mathrm{T} + s\psi = m\Omega_{,} + n\mathrm{T}_{,},$$

c'est-à-dire

$$\begin{aligned} \lambda m + \nu n - sp_{,} &= (2k+1)\, m_{,}, \\ \mu m + \nu n - sq_{,} &= (2k+1)\, n_{,}, \end{aligned}$$

ou, ce qui est la même chose,

$$\begin{aligned} m - sp &= m_{,}\rho - n_{,}\nu, \\ n - sq &= -m_{,}\mu + n_{,}\lambda. \end{aligned}$$

Or, $m_{,}$, $n_{,}$, s étant des entiers donnés, m, n seront aussi des entiers ; de même, m, n étant des entiers donnés, on trouve de $-k$ à $-k$ un entier s qui donne $m_{,}$ un entier. Mais cela étant, $n_{,}$ sera aussi un entier ; car autrement $n_{,}$ serait une fraction ayant pour dénominateur, lequel on voudrait, des nombres $2k+1$, λ, ν, ce qui est impossible à moins que

$$\lambda \equiv 0, \quad \nu \equiv 0 \pmod{2k+1}.$$

Mais si ces équations avaient lieu, on trouverait d'abord s de manière à avoir $m_{,}$ entier, et alors, puisqu'on n'a pas aussi

$$\mu \equiv 0, \quad \rho \equiv 0 \pmod{2k+1}$$

(en effet, cela est impossible à cause de l'équation $\lambda\rho - \mu\nu = 2k+1$), on démontrerait, comme auparavant, pour $n_{,}$, que $m_{,}$ est entier. Donc, enfin, m, n étant des entiers donnés, on trouve pour $m_{,}$, $n_{,}$, s un système d'entiers tel que s soit compris de k à $-k$, et l'on voit sans peine qu'il n'y a qu'un seul système de cette espèce.

A présent, partant de l'équation

$$\frac{y(x+y)}{y(y)} = e^{-Bxy+\beta xy^*} \prod \Big[1 + \frac{x}{(m,n)+y}\Big]^{-1},$$

(et faisant attention à la particularité que présente le cas de $y = 0$), j'écris successivement

$$y = 0, \quad y = \pm\psi, \ \ldots, \ y = \pm k\psi,$$

et je forme le produit des équations ainsi trouvées. Cela donne, à cause de $(m_,,n_,)+s\psi=(m,n)_,$,

$$\prod\frac{y(x+s\psi)}{y(s\psi)}=e^{-Bxy\,\text{etc.}}\prod\Big[1+\frac{x}{(m_,,n_,)+s\psi}\Big]^{-1},$$

la condition, par rapport aux limites, étant

$$\text{mod.}\,(m,n)_,<T,\ \ T=\infty.$$

Or

$$y_,x=e^{-\frac{1}{2}B_,x^2}x\prod\Big[1+\frac{x}{(m_,,n_,)}\Big],$$

avec la même condition, par rapport aux limites ; donc, enfin,

$$y_,x=\frac{e^{-\frac{1}{2}B_,x^2+\cdots}}{e^{\cdots}}\prod\frac{y(x+s\psi)}{y(s\psi)},$$

où, dans le numérateur, s doit avoir toutes les valeurs entières depuis $s=-k$ jusqu'à $s=+k$, y compris $s=0$, et dans le dénominateur ces mêmes valeurs, hormis la valeur $s=0$.

Il est, à présent, facile de faire voir que cette propriété subsiste pour l'une quelconque des fonctions yx, gx, Ωx, Zx ; en effet, pour la démontrer pour gx, j'écris $x+\frac{1}{2}\Omega$ au lieu de x ; en prenant, pour un moment, $\rho=2\rho'+1$, $\mu=2\mu'$; cela donne

$$x+\tfrac{1}{2}\Omega=x+(\rho',-\mu')_,.$$

Donc

$$y_,(x+\tfrac{1}{2}\Omega)=e^{\frac{1}{2}B_,(\rho',-\mu')^2_,}\,M\,g_,x.$$

A présent, partant de l'équation

$$\frac{y(x+y)}{y(y)}=e^{-Bxy+\beta xy^*}\prod\Big[1+\frac{x}{(m,n)+y}\Big]^{-1}$$

(en faisant attention à la particularité que présente le cas de $y=0$), j'écris successivement

$$y=0,\quad y=\pm\psi,\ \ldots,\ y=\pm k\psi,$$

et je forme le produit des équations ainsi trouvées. Cela donne, à cause de $(m_,,n_,)+s\psi=(m,n)_,$,

$$\prod\frac{y(x+s\psi)}{y(s\psi)}=e^{-Bxy+\cdots}\prod\Big[1+\frac{x}{(m_,,n_,)+s\psi}\Big]^{-1},$$

la condition, par rapport aux limites, étant

$$\text{mod.}\,(m,n)_,<T,\ \ T=\infty.$$

(*D'autres manipulations algébriques analogues conduisent aux formules concrètes pour la transformation des fonctions yx, gx, Ωx, Zx, exprimées par produits sur les indices $s=-k,\ldots,k$ et $s=-k,\ldots,k$ avec $s\neq0$ au dénominateur. Les formules définitives s'écrivent comme suit :*)

$$\Omega_,=(\Omega-\psi)\,\mathrm{T}_,=(\mathrm{T}-\psi),\ \text{etc.},$$

et, en résumant, pour les fonctions transformées, on obtient des systèmes du type

$$\begin{aligned}\Omega_,&=\Omega_{,1}(2\Omega+2\mathrm{T}_,),\\ \mathrm{T}_,&=\Omega_{,1}\Omega+\mathrm{T}_{,1}\mathrm{T}=\psi,\\ \mathrm{T}_{,2}&=\mathrm{T}.\end{aligned}$$

Les paramètres particuliers correspondent au cas général en fonctions de Ω, T ; on obtient également, par un moyen d'une transformation triviale, on obtiendrait

$$B' = \Omega + 2B_{,}\,\mathrm{T}, \quad \mathrm{T}_{,} = \mathrm{T},$$

et puis

$$\psi = \frac{1}{2k+1}\,\Omega, \quad \mathrm{T}_{,} = \mathrm{T},$$

et, de là,

$$\frac{\Omega_{,}}{\mathrm{T}_{,}} = \frac{1}{2k+1} \cdot \frac{\Omega}{\mathrm{T}}.$$

Les équations correspondantes pour la deuxième sont :

$$\begin{aligned}
&p = 2k+1, \quad q = 0,\\
&\lambda = 2k+1, \quad \nu = 0,\\
&\mu = 0, \quad \rho = 1,\\
&\rho_{,} = 1, \quad q_{,} = 0,\\
&\psi_{,} = \frac{1}{2k+1}\left[(2k+1)\,\Omega + (2k+1)\,\mathrm{T}\right],\\
&\Omega_{,1} = \Omega,\\
&\mathrm{T}_{,1} = (2k+1)\,\Omega + \mathrm{T},
\end{aligned}$$

ce qui donne

$$\frac{\Omega_{,}}{\mathrm{T}_{,}} = (2k+1)\,\frac{\Omega}{\mathrm{T}}.$$

J'ajoute, sans m'arrêter pour les démontrer, quelques formules de transformation pour le nombre 3 ; je trouve d'abord

$$\begin{aligned}
&\Omega_{,} = \tfrac{1}{3}\Omega, \quad \mathrm{T}_{,} = \mathrm{T},\\
&B_{,} = e^{-\frac{1}{2}B\Omega^2 - \beta\Omega\Omega^*}\,\frac{y(x+\frac{1}{3}\Omega)\,y(x-\frac{1}{3}\Omega)}{y^2(\frac{1}{3}\Omega)},\\
&G_{,} = e^{-\frac{1}{2}B\Omega^2 - \beta\Omega\Omega^*}\,\frac{y(x+\frac{1}{3}\Omega)\,y(x-\frac{1}{3}\Omega)}{B^2(\frac{1}{3}\Omega)},\\
&Z_{,} = e^{-\frac{1}{2}B\Omega^2 - \beta\Omega\Omega^*}\,\frac{y(x+\frac{1}{3}\Omega)\,y(x-\frac{1}{3}\Omega)}{B^2(\frac{1}{3}\Omega)}.
\end{aligned}$$

Ces équations donnent, en introduisant les fonctions elliptiques ϕx, fx, Fx données au moyen de

$$\phi x = \frac{Yx}{Bx}, \quad fx = \frac{gx}{Bx}, \quad Fx = \frac{Gx}{Bx},$$

les équations

$$\phi_{,}x = \phi x \cdot \frac{fx \cdot f(x+\frac{1}{3}\Omega)\,f(x-\frac{1}{3}\Omega)}{f^2(\frac{1}{3}\Omega)},$$

dont la seconde peut s'écrire sous la forme

$$f_{,}x = \frac{1 - x^2\,\dfrac{P}{Q}(\frac{1}{3}\Omega)\,\phi x}{1 + x^2\dfrac{P}{Q}(\frac{1}{3}\Omega)\,\phi x}, \quad \text{etc.}$$

On trouve en effet, en mettant comme à l'ordinaire $\Phi = \phi^2 x$,

$$c'^2 = b\phi c' \,, \quad e'^2 = (b - c')\, c' \,,$$

et puis

$$\begin{aligned} \phi_, x &= \frac{1 - c(1-b)\,\phi x}{1 - c(c-b)\,\phi x}, \\ f_, x &= \frac{1 - (c+b)\,\phi x}{1 - c(c-b)\,\phi x}, \\ F_, x &= \frac{1 - c\phi x}{1 - c(c-b)\,\phi x}, \end{aligned}$$

formules qui correspondent à celles de la transformation de Lagrange. Les équations (β), (γ) donnent en mettant ces valeurs de ϕ, f, F, laquelle, égalée à la valeur qui vient d'être trouvée, donne

$$\frac{xc\,(\overline{\phi}\,P(\frac{1}{3}\Omega))}{B(x - \frac{1}{3}\Omega)\,B(x + \frac{1}{3}\Omega)} = \frac{\phi x\, fx}{1 - c(c-b)\,\phi x}.$$

(*Suit la dérivation par formules analogues du système correspondant pour la transformation* $\Omega_, = \Omega$, $\mathrm{T}_, = \frac{1}{3}\mathrm{T}$. *Le texte expose ensuite le système correspondant, puis poursuit avec les formules définitives.*)

On obtient tout de suite, pour le premier cas, le système d'équations

$$\begin{aligned} \rho_, &= 1, \quad q_, = 2q_,, \\ \lambda &= 1, \quad \nu = 2, \\ \mu &= 0, \quad \rho = (2k+1), \\ p &= 1, \quad q = 1, \\ \psi &= \tfrac{1}{2k+1}\left[(2k+1) + 2\,\mathrm{T}_,\right] \mathrm{T}, \\ \Omega_, &= \Omega, \\ \mathrm{T}_, &= (2k+1)\,\mathrm{T} + \psi, \\ \mathrm{T}_{,2} &= \mathrm{T}. \end{aligned}$$

Le tout particulier le plus simple est celui de $q = 0$, cela donne

$$\psi = \Omega_,, \quad \mathrm{T}_, = \mathrm{T};$$

et, de là,

$$\frac{\Omega_,}{\mathrm{T}_,} = \frac{1}{2k+1} \cdot \frac{\Omega}{\mathrm{T}},$$

et même le cas général se réduit à celui-ci, car, au moyen d'une transformation triviale, on obtiendrait

$$B' = \Omega + 2B_,\mathrm{T}, \quad \mathrm{T}_, = \mathrm{T},$$

et puis

$$\psi = \frac{1}{2k+1}\Omega', \quad \mathrm{T}_, = \mathrm{T},$$

et, de là,

$$\frac{\Omega_,}{\mathrm{T}_,} = \frac{1}{2k+1} \cdot \frac{\Omega}{\mathrm{T}}.$$

Les équations correspondantes pour la deuxième sont :

$$\begin{aligned}
p &= 2k+1, \quad q = 0, \\
\lambda &= 2k+1, \quad \nu = 0, \\
\mu &= 0, \quad \rho = 1, \\
p_, &= 1, \quad q_, = 0, \\
\psi &= \tfrac{1}{2k+1}\big[(2k+1)\,\Omega + (2k+1)\,\mathrm{T}\big], \\
\Omega_{,1} &= \Omega, \\
\mathrm{T}_{,1} &= (2k+1)\,\Omega + \mathrm{T},
\end{aligned}$$

ce qui donne

$$\frac{\Omega_,}{\mathrm{T}_,} = (2k+1)\,\frac{\Omega}{\mathrm{T}}.$$

J'ajoute, sans m'arrêter pour les démontrer, quelques formules de transformation pour le nombre 3 ; je trouve d'abord

$$\begin{aligned}
\Omega_, &= \tfrac{1}{3}\Omega, \quad \mathrm{T}_, = \mathrm{T}, \\
y_,x &= e^{-\frac{1}{2}\cdots x^2}\, gx\, Ex, \\
g_,x &= e^{-\frac{1}{2}\cdots x^2}\, \frac{E(x+\frac{1}{3}\Omega)\,E(x-\frac{1}{3}\Omega)}{E^2(\frac{1}{3}\Omega)}, \\
\Omega_,x &= e^{-\frac{1}{2}\cdots x^2}\, \frac{E(x+\frac{1}{3}\Omega-\frac{1}{2}\mathrm{T})\,E(x-\frac{1}{3}\Omega-\frac{1}{2}\mathrm{T})}{E^2(\frac{1}{3}\Omega-\frac{1}{2}\mathrm{T})}, \\
Z_,x &= e^{-\frac{1}{2}\cdots x^2}\, \frac{E(x+\frac{1}{3}\Omega-\frac{1}{2}\mathrm{T})\,E(x-\frac{1}{3}\Omega-\frac{1}{2}\mathrm{T})}{E^2(\frac{1}{3}\Omega-\frac{1}{2}\mathrm{T})}.
\end{aligned}$$

Ces équations donnent, en introduisant les fonctions elliptiques ϕx, fx, Fx données ci-dessus, les équations

$$\begin{aligned}
\phi_,x &= \phi x \cdot \frac{f(x+\frac{1}{3}\Omega)\,f(x-\frac{1}{3}\Omega)}{f^2(\frac{1}{3}\Omega)}, \\
f_,x &= \frac{1-c\phi x}{1-c(c-b)\,\phi x}, \\
F_,x &= \frac{1-c\phi x}{1-c(c-b)\,\phi x};
\end{aligned}$$

équations qui sont, sauf la notation dont je me suis déjà servi dans la mémoire cité, les mêmes que celles de Jacobi.

Londres, Stone Buildings, 23 Fév. 1852.

131.

Nouvelles Recherches sur les Covariants.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. xlvii. (1854), pp. 109–125.]

Je me sers de la notation

$$(a_0, \ldots a_n \mid x, y)^n$$

pour représenter la fonction

$$a_0x^n + \tfrac{n}{1}\, a_1x^{n-1}y + \cdots + a_ny^n;$$

en supposant que les coefficients $a_0', a_1', \dots a_n'$ soient donnés par l'équation

$$(a_0', \dots a_n' \mid \lambda x + \mu y, \lambda' x + \mu' y)^n = (a_0, a_1, \dots a_n \mid x, y)^n,$$

supposé identique par rapport à x, y, soit $\phi(a_0, a_1, \dots a_n; x, y)$ une fonction des coefficients et des variables, telle que

$$\phi(a_0', a_1', \dots a_n'; x, y) = (\lambda\mu' - \lambda'\mu)^a\, \phi(a_0, a_1, \dots a_n; x, y),$$

cette fonction ϕ est nommée dans M. Cauvarian, et dans le cas particulier de ϕ une fonction des seuls coefficients, un *Invariant* de la fonction donnée.

Je suppose d'abord que les nouveaux coefficients soient donnés par l'équation

$$(a_0', \dots a_n' \mid x, y)^n = \phi(a_0, a_1, \dots a_n; x, y);$$

cela suppose les relations

$$\begin{aligned} a_0' &= a_0, \\ a_1' &= a_1 + \lambda a_0, \\ a_2' &= a_2 + 2\lambda a_1 + \lambda^2 a_0, \\ &\&\text{c.} \end{aligned}$$

Il faut donc que le covariant ϕ satisfasse à l'équation

$$\phi(a_0', a_1', \dots a_n'; x, y) = \phi(a_0, a_1, \dots a_n; x, y + \lambda x),$$

ce qui donne

$$(a_0, a_1, \dots a_n; x + py + p^2z) = \phi(a_0, a_1, \dots a_n; x, y),$$

où en posant

$$\begin{aligned} a_0'' &= a_0, \\ a_1'' &= a_1 + \lambda a_0, \\ a_2'' &= a_2 + 2\lambda a_1 + \lambda^2 a_0, \\ &\&\text{c.}, \end{aligned}$$

le covariant ϕ doit satisfaire aux équations

$$\phi(a_0'', a_1'', \dots a_n''; x, y) = \phi(a_0, a_1, \dots a_n; x, y).$$

(*La suite du mémoire développe la théorie des covariants en partant de cette définition : étude des opérateurs $\Box$, ∇ et de leurs propriétés algébriques, énoncé du théorème fondamental sur la représentation des covariants par produits de racines, et applications aux invariants des formes binaires de degrés bas. Le texte n'étant constitué que de formules denses et d'identités algébriques avec de nombreux indices, nous reproduisons ci-dessous les principales équations et énoncés retenus du texte original.*)

Et réciproquement, toute fonction homogène par rapport aux coefficients et issue par rapport aux variables, qui satisfait à ces équations, est un covariant de la fonction donnée.

Par exemple, l'invariant $\phi = ac - b^2$ de la fonction $ax^2 + 2bxy + cy^2$ satisfait aux équations

$$(ab + 2bk)\,k - (bk + ck)^2 = (2bk - a^2k)\,k - 0,$$

et l'invariant $\phi = ace - bd^2 + abe^2 - 3bcd - b^3 + 2ad^3$ de la fonction $ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4$ satisfait aux équations

$$(ak_0 + 3bk)\,k_1 = (ak_0, k_1)\,k_0.$$

Il est clair qu'on ne considérait que les fonctions qui restent les mêmes en échange dans les deux indices la coefficient a_0, $a_1, \ldots a_n$ et les variables x, y, respectivement. Le nombre des coefficients a_0, $a_1, \ldots a_n$ est supérieur, le nombre des coefficients indéterminés ; il est exactement égal au nombre des coefficients dans la fonction donnée. Mais il y a quelque autre indication à faire avant de pouvoir considérer comme arbitraires les indices β, &c., dont la série indique le caractère du covariant cherché.

Cherchons par exemple pour la fonction $ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4$ un covariant ϕ de la forme

$$\phi = Ax^4y + Bbx^3y^2 + Cx^2y^3 + Dy^4,$$

contenant les quatre coefficients indéterminés A, B, C, D. En substituant dans l'équation les mêmes après substitutions convenables, on obtient

$$3(aA)x^2dy^4 + (4bB + 4cC)x^2y^3 + (4cB + 4dD)xy^3 + (8e)y^4 = 0,$$

ou bien, puisque cette équation doit avoir lieu indépendamment de la quantité k (qui est arbitraire), il faut que les coefficients soient séparément nuls. On a donc les quatre équations

$$aA = 0, \quad 4bB = 0, \quad 4cB = 0, \quad 8e = 0,$$

ce qui donne le système $A : B : C : D$, et l'on voit que cette série est complètement déterminée, et l'on obtient en effet

$$\phi = ac + bd - 6acd - 4eb^2 + 3bc^2.$$

La circonstance suivante réclame néanmoins de l'attention. Il s'agit d'énoncer que dans le calcul, le problème de trouver le nombre des coefficients d'un cota donné ; problème qui n'a aucun lieu de l'autre.

(*La suite développe les notations symboliques Ω, $\Box$, etc., et culmine avec l'énoncé du théorème central de l'article :*)

Théorème. Tout covariant ϕ de la fonction

$$(a_0, \ldots a_n \mid x, y)^n$$

satisfait aux deux équations à différences partielles

$$(\Box - p)\,\phi = 0, \quad (\triangle - q)\,\phi = 0,$$

où

$$\Box = na_1\partial_{a_0} + (n-1)\,a_2\partial_{a_1} + \cdots + a_n\partial_{a_{n-1}} + y\partial_x,$$
$$\triangle = na_0\partial_{a_1} + (n-1)\,a_1\partial_{a_2} + \cdots + a_{n-1}\partial_{a_n} + x\partial_y,$$

et réciproquement.

Inversement on a, en notant d'abord $a^n = b$, puis en considérant des trois quantités indépendantes a, b, c, la solution générale de l'équation à différences partielles

$$ap\,\partial_a + 2bp\,\partial_b + 3cp\,\partial_c = 0,$$

en supposant un poids de p, est indépendante de a, b, c, et l'on considère ainsi cette propriété fondamentale : *tout covariant est composé de termes de la première espèce*; et résultat ainsi obtenu général des covariants forment, lorsque cela est permis, par cette même observation, une suite indiquant la longueur des polynômes à poids p.

Pour les polynômes binaires, l'énoncé du théorème prend la forme suivante :

Théorème. Pour qu'une fonction ϕ des coefficients $a_0, a_1, \dots a_n$ d'une forme binaire de degré n, soit un *invariant*, il faut et il suffit qu'elle satisfasse simultanément aux deux équations

$$\Box\,\phi = 0, \quad \triangle\,\phi = 0,$$

où $\Box$, $\triangle$ sont les opérateurs ci-dessus, restreints à leur partie agissant sur les seuls coefficients.

(*Le texte poursuit avec des démonstrations détaillées, des applications aux discriminants des formes du degré* 2, 3, 4, *et des exemples numériques.*)

Pour le discriminant

$$\nabla = ax^2 + 2bxy + cy^2$$

de la fonction de degré 2, le discriminant est $4ac - b^2$. Nous savons alors

$$\nabla = -b^2(4ac - b^2) + B^2 + Cb^2,$$

en faisant $F = 4ab + 2b^3$, B, C étant donnés par

$$\eta\,B = F(4xy^2 - by^3), \quad 2yC = F(B),$$

c'est-à-dire $B = 12acb - b^3$, $C = -27a^2b$, et de là

$$\nabla = -27a^2b + 18acb^2 - 4ac^3 + b^2c^2 - 4a^3c^2,$$

valeur qui correspond en effet au discriminant ordinaire. En changeant d'une manière convenable les coefficients.

Londres, Stone Buildings, 25 Févr. 1852.

132.

Réponse à une Question Proposée par M. Steiner.

(Aufgabe 4, *Crelle* t. xxxi. (1846) p. 90.)

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. x. (1855), pp. 277–278.]

En partant des deux théorèmes :

I. Qu'il existe un nombre infini de surfaces du second ordre qui touchent une surface du second ordre aux six arêtes d'un quelconque hexaèdre quadrilatéral ;

II. Que le lieu d'intersection de trois plans rectangles qui touchent une surface du second ordre est une sphère concentrique avec la surface, tandis que pour les hyperboloïdes un seul nappe le lieu d'intersection est réduit à un plan,

M. Steiner suppose le cas d'un parallélépipède rectangle, en raisonnant d'une côte P de ce parallélépipède P ; par lequel passent trois plans principaux. Les trois plans du parallélépipède P rencontrent deux surfaces ses faces A, A', B, B', C, C'. Avec ces six plans principaux P, &c. on aurait six points de contact, qui pourraient former une figure du second ordre. Mais c'est dans ce cas, on doit avoir un point de l'espace considérée, P, dont les trois plans principaux à une surface dont les plans tangents A, A', B, B', C, C' contiennent les six points et faces du parallélépipède, et celles trois faces, sphériques, et en outre, lorsque le parallélépipède P est aux trois plans principaux d'un point d'une surface dont les trois faces sont parallèles aux faces de A', B', C'. La contradiction qui apparaitra dans le cas général est par conséquent admissible que pour les six points.

Aussi d'attention le solution des équations (A), je vais démontrer quelque propriété générale qui appartient à ce théorème. Pour abréger, je considére Δ le système principal de la surface; ce dernier que les coefficients a_0, $a_1, \ldots a_n$ et les coefficients de la même forme donnent en effet le même nom et méritera mériter que les nouvelles indices ne sont pas r, mais s. Donc, tout autre que sont a, b, c, d, …; on considère ensuite quelques points jusqu'au système principal P, et l'on cherche les valeurs qu'elles donnent dans la fonction a, b, c, d, …, f, qui en a l'expression. On considère alors les équations symétriques de la fonction a et de la fonction f, et l'on considère la quantité $(\Phi - P^2 = 0)$, en effet, ce théorème est bien déterminé, et le théorème β et le II sont tous deux légitimes.

133.

Sur un Théorème de M. Schläfli.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. l. (1855), pp. 278–282.]

On lit dans le § 13 d'un mémoire très intéressant de M. Schläffli intitulé "Über die Resultanten eines Systems mehrerer algebraischer Gleichungen," (*Mém. de l'Acad. de Vienne*, t. iv. [1852]) un très beau théorème sur les Zésultantes.

Pour faire voir plus clairement, en quoi consiste ce théorème, je prends un cas particulier. Soit

$$\begin{aligned} U &= ax^2 + 3bx^2y + 3cxy^2 + dy^3 + (a, b, c, d \mid x, y)^3, \\ V &= ax^3 + 2bxy + cy^2 = (a, b, c \mid x, y)^2. \end{aligned}$$

Je fais $p = a^3$, $q = ay$, $r = pq$, et je forme les opérateurs

$$\begin{aligned} \nabla &= \tfrac{1}{2}\partial_a + p\partial_b + q\partial_c, \\ \triangle &= \tfrac{1}{2}\partial_a + p\partial_b + q\partial_c, \end{aligned}$$

lesquels, opérant sur U, donnent

$$r(p\xi + qy - C), \quad p(p\xi + qy + C),$$

opérant sur V, donnent

$$p\xi + qy + r^2.$$

Cela étant, soit $\phi = 0$ le résultant des équations $U = 0$, $V = 0$, c'est-à-dire l'équation que l'on obtient en éliminant x, y entre les deux équations $U = 0$, $V = 0$, on a

$$\phi = \det \begin{pmatrix} a & 3b & 3c & d & \cdot \\ \cdot & a & 3b & 3c & d \\ a & 2b & c & \cdot & \cdot \\ \cdot & a & 2b & c & \cdot \\ \cdot & \cdot & a & 2b & c \end{pmatrix}.$$

Je suppose que les opérations ∇, $\triangle$ opèrent sur le résultant ϕ, et que cela donne les fonctions $\nabla\phi$, $\triangle\phi$, on trouve sans peine

$$\triangle\phi = (p\xi + q\eta - C)\phi,$$
$$\nabla\phi = (p\xi + q\eta + C)\phi,$$

ou en črivant (ξ, η, ζ) pour les résultats de ϕ et (ξ) pour les mêmes résultats de ϕ, on aurait à considérer ce résultant comme une fonction symétrique de la fonction ; ce résultant est ce qui est nommé dans le théorème de M. Schläffli.

Généralement, on suppose que l'on ait autant de fonctions U, V, W, … que d'indéterminés x, y, z, …, ces fonctions étant des degrés respectifs μ, μ', …, des nombres indéterminés, et l'on pose pour chacune des fonctions du système l'opérateur ∇ correspônd ce que ce que devient l'opérateur ∇ pour la fonction U, dans laquelle on substitue à la place des indéterminées x, y, z, W, …, des facteurs p, q, r, &c. ; avec les indices respectifs des fonctions U, V, W, …, et de la fonction dène faite, on a un théorème dont je ne pourrais indiquer la démonstration de M. Schläffli, mais dont la valeur du résultant ϕ est telle que cela est en outre une fonction symétrique.

Cela étant, soit ϕ le résultant des fonctions U, V, W, …, c'est-à-dire l'équation que l'on obtient en éliminant x, y, z, … ; et soit r le théorème de l'opération ∇ comme l'on a vu pour les hyperboloïdes. Cela donne en effet pour les résultants à fonction de ∇.

Théorème. La démonstration donnée dans la mémoire cité est, on ne peut plus, simple et élégante. Elle expose d'abord sur un théorème connu (démontré au reste § 6) qui peut être énoncé ainsi ; savoir, en supposant que les équations $U = 0$, $V = 0$, … soient satisfaites, on aura (près un facteur indépendant de ξ, η, …),

$$\mathfrak{V}\phi = \xi\,(p\xi + q\eta + \cdots)^{\mu-\mu'}(\xi, \eta, \ldots), \quad \mathfrak{V}\phi = \xi\,(p\xi + q\eta + \cdots)^{\mu-\mu''}(\xi, \eta, \ldots), \quad \text{\&c.}$$

Puis, cela fait, est fondée sur la démonstration démontrée (§ 12), savoir : le résultant des fonctions

$$\begin{gathered} t\,(p\xi + q\eta + \cdots)^{\mu} + f'(\xi, \eta, \ldots), \\ t'(p\xi + q\eta + \cdots)^{\mu'} + f''(\xi, \eta, \ldots), \\ \vdots \end{gathered}$$

(où f, f', … sont des polynômes de degrés μ, μ', … en ξ, η, &c., des constantes quelconques, en supposant que μ ne soit plus petit qu'aucun autre des indices μ, μ', …, et en posant $\pi = \dfrac{\pi}{\mu} + \dfrac{\pi}{\mu'} + \cdots$, le résultant Φ du système sera tout au plus du degré $\dfrac{\pi}{\mu}$ par rapport à t et de même par rapport aux coefficients de f, f', … Le degré par rapport à les coefficients est ai ; le degré par rapport aux coefficients de f, f', … sera donc au moins un infiniment petit de l'ordre k contenu à $\phi^{\pi-\mu}$ comme *facteur*. Dans cette supposition les équations $\mathfrak{V}\phi = 0$, $\mathfrak{V}\phi = 0$, &c. deviendront :

$$\begin{gathered} t\,(p\xi + q\eta + \cdots)^{\mu} + f(\xi, \eta, \ldots) = 0, \\ t'\,(p\xi + q\eta + \cdots)^{\mu'} + f'(\xi, \eta, \ldots) = 0, \\ \vdots \end{gathered}$$

où f, f', … sont des polynômes de degrés μ, μ', … dont les coefficients sont infiniment petits par rapport à tous les autres plus du degré $\dfrac{\pi}{\mu}$ par rapport à t et de même par rapport aux coefficients de f, f', … Le degré par rapport à ces coefficients est ai (assujettis à la seule condition ci-dessus mentionnée) étant d'ailleurs arbitraires, on suit que le résultant Φ contiendra en général cette même puissance $\phi^{\pi-\mu}$ *comme facteur* ; ce qu'il s'agissait de démontrer.

(*Le texte poursuit avec le développement de calculs explicites et la démonstration complète pour des systèmes binaires et ternaires. La dernière portion du texte (p. 184) contient quelques détails sur les hautes puissances de* $\phi^{\pi-\mu}$ *qui apparaissent comme facteurs dans le résultant complément, le tout encadré par des observations sur l'importance du théorème dans le contexte des résultants multidimensionnels.*)

Cayley's Collected Mathematical Papers, Vol. II, pp. 185–234

Reconstruction with OCR skeleton and PNG verification

134. Remarques sur la notation des fonctions algébriques.

REMARQUES SUR LA NOTATION DES FONCTIONS ALGÉBRIQUES.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 282–285.]

Je me sers de la notation

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix}$$

pour représenter ce que j'appelle une *matrice*; savoir un *système* de quantités rangées en forme de *carré*, mais d'ailleurs tout à fait indépendantes (je ne parle pas ici des matrices *rectangulaires*). Cette notation me paraît très commode pour la théorie des équations linéaires; j'écris par exemple

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix} (x, y, z, \dots)$$

pour représenter le système des équations

$$\begin{aligned} \xi &= \alpha x + \beta y + \gamma z \dots, \\ \eta &= \alpha' x + \beta' y + \gamma' z \dots, \\ \zeta &= \alpha'' x + \beta'' y + \gamma'' z \dots. \end{aligned}$$

On obtient par là l'équation

$$(x, y, z, \dots) = \begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix}^{-1} (\xi, \eta, \zeta, \dots),$$

qui représente le système d'équations qui donne x, y, z, … en termes de ξ, η, ζ, …, et on se trouve ainsi conduit à la notation

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix}^{-1}$$

de la matrice *inverse*. Les termes de cette matrice sont des fractions, ayant pour *dénominateur* commun le *déterminant* formé avec les termes de la matrice originale; les *numérateurs* sont les déterminants mineurs formés avec les termes de cette même matrice en supprimant l'une quelconque des lignes et l'une quelconque des colonnes.

Soit encore

$$(x, y, z, \dots) = \begin{vmatrix} a, & b, & c, & \dots \\ a', & b', & c', & \dots \\ a'', & b'', & c'', & \dots \\ & \vdots & & \end{vmatrix} (x, y, z, \dots),$$

on peut écrire

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix} \begin{vmatrix} a, & b, & c, & \dots \\ a', & b', & c', & \dots \\ a'', & b'', & c'', & \dots \\ & \vdots & & \end{vmatrix} (x, y, z, \dots),$$

et l'on parvient ainsi à l'idée d'une *matrice composée*, par ex.

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix} \begin{vmatrix} a, & b, & c, & \dots \\ a', & b', & c', & \dots \\ a'', & b'', & c'', & \dots \\ & \vdots & & \end{vmatrix}.$$

On voit d'abord que la valeur de cette matrice composée est

$$\begin{vmatrix} (\alpha, \beta, \gamma, \dots)(a, a', a'', \dots), & (\alpha, \beta, \gamma, \dots)(b, b', b'', \dots), \dots \\ (\alpha', \beta', \gamma', \dots)(a, a', a'', \dots), & (\alpha', \beta', \gamma', \dots)(b, b', b'', \dots) \dots \\ \vdots & \end{vmatrix}$$

où $(\alpha, \beta, \gamma, \dots)(a, a', a'', \dots) = \alpha a + \beta a' + \gamma a'' + \dots$. Il faut faire attention, dans la composition des matrices, de combiner les *lignes* de la matrice à gauche avec les *colonnes* de la matrice à droite, pour former les *lignes* de la matrice composée. Il y aurait bien des choses à dire sur cette théorie de matrices, laquelle doit, il me semble, précéder la théorie des Déterminants.

Une notation semblable peut être employée dans la théorie des fonctions *quadratiques*. En effet, on peut dénoter par

$$\left(\begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix} \right) (\xi, \eta, \zeta)(x, y, z)$$

la fonction *linéo-linéaire*

$$\begin{aligned} &(\alpha\xi + \beta\eta + \gamma\zeta \dots)x \\ &+ (\alpha'\xi + \beta'\eta + \gamma'\zeta \dots)y \\ &+ (\alpha''\xi + \beta''\eta + \gamma''\zeta \dots)z \\ &\;\vdots \end{aligned}$$

et de là par

$$\left(\begin{vmatrix} a, & h, & g, & \dots \\ h, & b, & f, & \dots \\ g, & f, & c, & \dots \\ & \vdots & & \end{vmatrix} \right) (x, y, z, \dots)^2$$

la fonction *quadratique*

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy \ldots$$

que je représente aussi par

$$(a, b, c, \ldots\ f, g, h, \ldots)(x, y, z, \ldots)^2.$$

Je remarque qu'en général je représente une fonction rationnelle et intégrale, homogène et des *degrés* m, m', &c., par rapport aux indéterminées x, y, &c., x', y', &c., de la manière suivante:

$$(\diamond)(x, y, \ldots)^m (x', y', \ldots)^{m'} \ldots.$$

Une fonction rationnelle et intégrale, homogène et du degré m par rapport aux deux indéterminées x, y sera donc représentée par

$$(\diamond)(x, y)^m.$$

En introduisant dans cette notation les *coefficients*, j'écris par exemple

$$(a, b, c, d \diamond x, y)^3,$$

pour représenter la fonction

$$ax^3 + 3bx^2y + 3cxy^2 + dy^3,$$

tandis que je me sers de la notation

$$(a, b, c, d \diamond x, y)^3,$$

pour représenter la fonction

$$ax^3 + bx^2y + cxy^2 + dy^3,$$

et de même pour les fonctions d'un degré quelconque. J'ai trouvé cette distinction très commode.

[In the foregoing Paper as here printed, except in the expression in the second line of this page,)(is used instead of)(); it appears by a remark (*Crelle*, t. li, errata) that the manuscript had the interlaced parentheses)(). Moreover in the manuscript ( ) was used for a Matrix, which was thus distinguished from a Determinant, but in the absence of any real ambiguity, no alteration has been made in this respect. In the reprint of subsequent papers from Crelle, the arrowhead)(or)() is used instead of ( ).]

135. Note sur les covariants d'une fonction quadratique, cubique, ou biquadratique à deux indéterminées.

NOTE SUR LES COVARIANTS D'UNE FONCTION QUADRATIQUE, CUBIQUE, OU BIQUADRATIQUE À DEUX INDÉTERMINÉES.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 285–287.]

La théorie d'une fonction à deux indéterminées d'un degré quelconque, par exemple

$$(\diamond)(x, y)^m,$$

dépend du système des covariants de la fonction, lequel est censé contenir la fonction elle-même.

Pour une fonction *quadratique* le système de covariants est

$$\begin{aligned}&(a,b,c)(x,y)^2,\\&ac-b^2.\end{aligned}$$

Pour la fonction *cubique*, le système est

$$\begin{aligned}&(a,b,c,d)(x,y)^3,\\&(ac-b^2,\ ad-bc,\ bd-c^2)(x,y)^2,\\&(-a^2d+3abc-2b^3,\ -abd+2ac^2-b^2c,\ acd-2b^2d+bc^2,\ ad^2-3bcd+2c^3)(x,y)^3,\\&-a^2d^2+6abcd-4ac^3-4b^3d+3b^2c^2,\end{aligned}$$

fonctions lesquelles, en supposant qu'on les représente par U, H, Φ, $\square$, satisfont identiquement à l'équation

$$\Phi^2+\square U^3=-4H^3.$$

Pour la fonction *biquadratique*, le système est

$$\begin{aligned}&(a,b,c,d,e)(x,y)^4,\\&ae-4bd+3c^2,\\&(ac-b^2,\ 2ad-2bc,\ ae+2bd-3c^2,\ 2be-2cd,\ ce-d^2)(x,y)^4,\\&ace+2bcd-ad^2-b^2e-c^3,\end{aligned}$$

$$\left\{\begin{array}{l}-a^2d+3abc-2b^3,\\-a^2e-2abd+9ac^2-6b^2c,\\-5abe+15acd-10b^2d,\\+10a^2d-10b^2e,\\+5ade+10bd^2-15bce,\\+ae^2+2bde-9c^2e+6cd^2,\\+be^2-3cde+2d^3,\end{array}\right\}(x,y)^6,$$

et ces fonctions, en supposant qu'on les représente par U, I, H, J, Φ, satisfont identiquement à l'équation

$$JU^3-IU^2H+4H^3=-\Phi^2.$$

J'ajoute à ce système la fonction

$$\begin{aligned}I^3-27J^2=a^3e^3&-12a^2bde^2-18a^2c^2e^2+54a^2cd^2e-27a^2d^4\\&+54ab^2ce^2-6ab^2d^2e-180abc^2de+108abcd^3+81ac^4e\\&-54ac^3d^2-27b^4e^2-64b^3d^3+108b^2cde-54bcd^3\\&+36b^2c^2d^2,\end{aligned}$$

qui est le *discriminant* de la fonction biquadratique.

Pour donner une application de ces formules, soit proposé de résoudre une équation quadratique, cubique ou biquadratique, ou autrement dit: de trouver un *facteur linéaire* de la fonction quadratique, cubique, ou biquadratique.

Il est assez singulier que pour la fonction quadratique la solution est en quelque sorte plus compliquée que pour les deux autres. En effet, il n'existe pas de solution symétrique, à moins qu'on n'introduise des quantités arbitraires et superflues; savoir, on trouve pour facteur linéaire de $(a, b, c)(x, y)^2$ l'expression

$$(\alpha, b, c)(x, \beta)(x, y) \pm \sqrt{-\square}\,(\beta x - \alpha y),$$

où $(a, b, c)(\alpha, \beta)(x, y)$ dénote $a\alpha x + b(\alpha y + \beta x) + c\beta y$.

Pour la fonction *cubique*, l'équation $\Phi^2 + \square U^3 = -4H^3$ fait voir que les deux fonctions $\Phi + U\sqrt{-\square}$, $\Phi - U\sqrt{-\square}$ sont l'une et l'autre des cubes parfaits. L'expression

$$\sqrt[3]{\tfrac{1}{2}(\Phi + U\sqrt{-\square})} - \sqrt[3]{\tfrac{1}{2}(\Phi - U\sqrt{-\square})}$$

sera donc une fonction linéaire de x, y; et puisque cette fonction s'évanouit pour $U = 0$, elle ne sera autre chose que l'un des facteurs linéaires de $(a, b, c, d)(x, y)^3$.

Pour la fonction *biquadratique*, en partant de l'équation

$$JU^3 - IU^2H + 4H^3 = -\Phi^2,$$

j'écrirai

$$M = \frac{I^3}{4J^2},$$

et je mets l'équation sous la forme

$$(1, 0, -M, M)(H, JU)^3 = -\tfrac{1}{4}P^2\Phi^2.$$

Donc, en supposant que w_1, w_2, w_3 soient les racines de l'équation

$$(1, 0, -M, M)(w, 1)^3 = 0,$$

ou plus simplement de l'équation

$$w^2 - M(w - 1) = 0,$$

ces expressions $IH - w_1JU$, $IH - w_2JU$, $IH - w_3JU$ seront toutes trois des carrés de fonctions quadratiques. L'expression

$$(w_2 - w_3)\sqrt{(IH - w_1JU)} + (w_3 - w_1)\sqrt{(IH - w_2JU)} + (w_1 - w_2)\sqrt{(IH - w_3JU)}$$

sera donc une fonction quadratique, et on voit sans peine qu'elle sera le carré d'une fonction *linéaire*. Or cette expression s'évanouit pour $U = 0$; donc ce sera précisément le carré de l'un quelconque des facteurs linéaires de $(a, b, c, d, e)(x, y)^4$.

L'équation identique pour les covariants d'une fonction biquadratique donne lieu aussi (remarque que je dois à M. Hermite) à une transformation très élégante de l'intégrale elliptique $\int dx \div \sqrt{(a, b, c, d, e)(x, 1)^4}$.

136. Sur la transformation d'une fonction quadratique en elle-même par des substitutions linéaires.

Sur la transformation d'une fonction quadratique en elle-même par des substitutions linéaires.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 288–299.]

Il s'agit de trouver les transformations linéaires d'une fonction quadratique $(\diamond)(x, y, z, \ldots)^2$ en *elle-même*, c'est-à-dire de trouver pour $(x, y, z, \ldots)$ des fonctions linéaires de x, y, z, $\ldots$ telles que

$$(\diamond)(x, y, z, \ldots)^2 = (\diamond)(x, y, z, \ldots)^2.$$

En représentant la fonction quadratique par

$$(\diamond)(x, y, z, \ldots)^2 = \left|\begin{matrix} a, & h, & g, & \ldots \\ h, & b, & f, & \ldots \\ g, & f, & c, & \ldots \\ & \vdots & & \end{matrix}\right| (x, y, z, \ldots)^2,$$

la solution qu'a donnée M. Hermite de ce problème peut être résumée dans la seule équation

$$(x, y, z, \ldots) = \left|\begin{matrix} a, & h, & g, & \ldots \\ h, & b, & f, & \ldots \\ g, & f, & c, & \ldots \\ & \vdots & & \end{matrix}\right|^{-1} \left|\begin{matrix} a, & h+\nu, & g-\mu, & \ldots \\ h-\nu, & b, & f+\lambda, & \ldots \\ g+\mu, & f-\lambda, & c, & \ldots \\ & \vdots & & \end{matrix}\right| \left|\begin{matrix} a, & h-\nu, & g+\mu, & \ldots \\ h+\nu, & b, & f-\lambda, & \ldots \\ g-\mu, & f+\lambda, & c, & \ldots \\ & \vdots & & \end{matrix}\right|^{-1} (x, y, z, \ldots$$

où λ, μ, ν, $\ldots$ sont des quantités quelconques.

En effet, pour démontrer que cela est une solution, on n'a qu'à reproduire dans un ordre inverse le procédé de M. Hermite. En introduisant les quantités auxiliaires $(\xi, \eta, \zeta, \ldots)$, on peut remplacer l'équation par les deux équations

$$\left|\begin{matrix} a, & h, & g, & \ldots \\ h, & b, & f, & \ldots \\ g, & f, & c, & \ldots \\ & \vdots & & \end{matrix}\right| (x, y, z, \ldots) = \left|\begin{matrix} a, & h+\nu, & g-\mu, & \ldots \\ h-\nu, & b, & f+\lambda, & \ldots \\ g+\mu, & f-\lambda, & c, & \ldots \\ & \vdots & & \end{matrix}\right| (\xi, \eta, \zeta, \ldots),$$

$$\left|\begin{matrix} a, & h, & g, & \ldots \\ h, & b, & f, & \ldots \\ g, & f, & c, & \ldots \\ & \vdots & & \end{matrix}\right| (x, y, z, \ldots) = \left|\begin{matrix} a, & h-\nu, & g+\mu, & \ldots \\ h+\nu, & b, & f-\lambda, & \ldots \\ g-\mu, & f+\lambda, & c, & \ldots \\ & \vdots & & \end{matrix}\right| (\xi, \eta, \zeta, \ldots),$$

qui donnent tout de suite d'abord

$$(\diamond)(x, y, z, \ldots)(\xi, \eta, \zeta, \ldots) = (\diamond)(\xi, \eta, \zeta, \ldots)(y, \ldots),$$

et puis

$$x + x = 2\xi, \quad y + y = 2\eta, \quad z + z = 2\zeta, \quad \&c.$$

On obtient par là

$$\begin{aligned} (\diamond)(x, y, z, \ldots) = (\diamond)&(2\xi - x,\ 2\eta - y,\ 2\zeta - z, \ldots)^2 \\ &= 4(\diamond)(\xi, \eta, \zeta, \ldots) - 4(\diamond)(\xi, \eta, \zeta, \ldots)(x, y, z, \ldots) \\ &\quad + (\diamond)(x, y, z, \ldots)^2, \end{aligned}$$

c'est-à-dire l'équation

$$(\diamond)(x, y, z, \ldots)^2 = (\diamond)(x, y, z, \ldots)^2,$$

qu'il s'agissait de vérifier.

Je remarque que la transformation est toujours *propre*. En effet, le déterminant de transformation est

$$\begin{vmatrix} a, & h, & g, & \dots \\ h, & b, & f, & \dots \\ g, & f, & c, & \dots \\ & \vdots & & \end{vmatrix}^{-1} \begin{vmatrix} a, & h-\nu, & g+\mu, & \dots \\ h+\nu, & b, & f-\lambda, & \dots \\ g-\mu, & f+\lambda, & c, & \dots \\ & \vdots & & \end{vmatrix}^{-1} \begin{vmatrix} a, & h+\nu, & g-\mu, & \dots \\ h-\nu, & b, & f+\lambda, & \dots \\ g+\mu, & f-\lambda, & c, & \dots \\ & \vdots & & \end{vmatrix}.$$

Or les déterminants qui entrent dans les deux termes moyens, ne contiennent l'un ou l'autre que des puissances paires de λ, μ, ν, ... Donc ces deux déterminants sont égaux, et les quatre termes du produit sont *réciproques* deux à deux; le déterminant de transformation est donc $+1$, et la transformation est *propre*.

Pour obtenir une transformation *impropre*, il faut considérer une fonction quadratique qui contienne, outre les indéterminées x, y, z, ..., une indéterminée θ, et puis réduire à zéro la valeur des coefficients de cette indéterminée θ. Les valeurs de x, y, z, ... en contiendront pas θ, et en représentant par $\mathfrak{D}$ le déterminant de transformation pour la forme aux indéterminées x, y, z, ..., on aura par là $\mathfrak{D} = -\theta$; le déterminant de transformation pour la forme aux indéterminées x, y, z, ... multiplié par -1. Le déterminant de transformation pour la forme aux indéterminées x, y, z, ... sera donc -1, c'est-à-dire, la transformation sera *impropre*.

Au lieu de la formule de transformation ci-dessus, on peut se servir des formules

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} a, & h+\nu, & g-\mu, \dots \\ h-\nu, & b, & f+\lambda, \dots \\ g+\mu, & f-\lambda, & c \quad \dots \\ & \vdots & \end{vmatrix}^{-1} \begin{vmatrix} a, & h, & g, \dots \\ h, & b, & f, \dots \\ g, & f, & c, \dots \\ & \vdots & \end{vmatrix} (x, y, z, \dots),$$

$$x = 2\xi - x, \quad y = 2\eta - y, \quad z = 2\zeta - z, \quad \&c.$$

Par exemple, en supposant que la forme à transformer soit

$$ax^2 + by^2 + cz^2 + \&c.,$$

on aura

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} a, & \nu, & -\mu, \dots \\ -\nu, & b, & \lambda, \dots \\ \mu, & -\lambda, & c, \dots \\ & \vdots & \end{vmatrix}^{-1} (ax, by, cz, \dots),$$

$$x = 2\xi - x, \quad y = 2\eta - y, \quad z = 2\zeta - z, \quad \&c.$$

de manière qu'en posant

$$\begin{vmatrix} a, & \nu, & -\mu, \dots \\ -\nu, & b, & \lambda, \dots \\ \mu, & -\lambda, & c, \dots \\ & \vdots & \end{vmatrix} = k,$$

$$\begin{vmatrix} a, & \nu, & -\mu, \dots \\ -\nu, & b, & \lambda, \dots \\ \mu, & -\lambda, & c, \dots \\ & \vdots & \end{vmatrix}^{-1} = \frac{1}{k} \begin{vmatrix} A, & B, & C, \dots \\ A', & B', & C', \dots \\ A'', & B'', & C'', \dots \\ & \vdots & \end{vmatrix},$$

on aura

$$(x, y, z, \ldots) = \frac{1}{k} \begin{vmatrix} 2A - \dfrac{k}{a}, & 2B, & 2C, & \ldots \\ 2A', & 2B' - \dfrac{k}{b}, & 2C', & \ldots \\ 2A'', & 2B'', & 2C'' - \dfrac{k}{c}, & \ldots \\ & \vdots & & \end{vmatrix} (ax, by, cz, \ldots).$$

ce qui est l'équation pour la transformation *propre* en elle-même, de la fonction $ax^2 + by^2 + cz^2 + \&c$. On déduira, comme dans le cas général, la formule pour la transformation *impropre*. On trouvera des observations sur cette formule dans le mémoire "Recherches ultérieures sur les déterminants gauches" [137].

Je reviens à l'équation générale

$$(\diamond)(x, y, z, \ldots)^2 = (\diamond)(x, y, z, \ldots)^2,$$

et je suppose seulement que x, y, z, $\ldots$ soient des fonctions linéaires de x, y, z, $\ldots$ qui satisfont à cette équation sans supposer rien davantage par rapport à la forme de la solution. Cela étant, je forme les fonctions linéaires $x - ax$, $y - ay$, $z - az$, &c., où a est une quantité quelconque, et je considère la fonction

$$(\diamond)(x - ax,\ y - ay,\ z - az, \ldots)^2,$$

laquelle, en la développant, devient

$$(1 + a^2)(\diamond)(x, y, z, \ldots)^2 - 2a\,(\diamond)(x, y, z, \ldots\ x, y, z, \ldots);$$

et en développant de la même manière la fonction quadratique

$$(\diamond)\big(x - \tfrac{1}{a}x,\ y - \tfrac{1}{a}y,\ z - \tfrac{1}{a}z, \ldots\big)^2,$$

on obtient l'équation identique

$$(\diamond)(x - ax,\ y - ay,\ z - az, \ldots)^2 = (\diamond)\big(x - \tfrac{1}{a}x,\ y - \tfrac{1}{a}y,\ z - \tfrac{1}{a}z, \ldots\big)^2.$$

Soit $\Box$ le déterminant formé avec les coefficients de fonctions linéaires $x - ax$, $y - ay$, $z - az$, &c. En supposant que le nombre des indéterminées x, y, z, &c., est n, $\Box$ sera évidemment une fonction rationnelle et intégrale du degré n par rapport à a. Soit de même $\Box'$ le déterminant formé avec les coefficients de

$$x - \tfrac{1}{a}x, \quad y - \tfrac{1}{a}y, \quad z - \tfrac{1}{a}z, \quad \&c.;$$

l'équation qui vient d'être trouvée, donne $\Box' = a^n\Box$, c'est-à-dire $\Box = \pm a^n\Box'$. Cela fait voir que les coefficients du premier et du dernier terme de l'avant-dernier terme, &c., sont égaux, aux signes près. De plus, le coefficient de la plus haute puissance a^n est toujours ± 1, et on voit sans peine qu'en supposant d'abord que a soit *impair*, on a pour la transformation *propre*:

$$\Box = (1,\ P,\ \ldots P,\ 1)(a, 1)^n;$$

et pour la transformation *impropre*

$$\Box = (1,\ -P,\ \ldots - P,\ 1)(a, -1)^n;$$

équation qui peut être changée en celles-ci: $\Box = -(1,\ -P,\ \ldots P,\ 1)(a, 1)^n$. Puis, en supposant que n soit *pair*, on a pour la transformation *propre*:

$$\Box = (1,\ P,\ \ldots P,\ 1)(a, 1)^n;$$

et pour la transformation *impropre*

$$\Box = (1,\ -P,\ \ldots P,-1)(a,1)^n;$$

le coefficient moyen étant dans ce cas égal à zéro. Ces théorèmes pour la forme du déterminant des fonctions linéaires $x - ax$, $y - ay$, $z - az$, $\ldots$ sont dus à M. Hermite.

Il y a à remarquer que la forme $(\diamond)(x,y,z,\ldots)^2$ est indéterminée; c'est-à-dire, on suppose seulement que x, y, z, $\ldots$ soient des fonctions linéaires de x, y, z, $\ldots$, telles qu'il y ait une forme quadratique $(\diamond)(x,y,z,\ldots)^2$ pour laquelle l'équation $(\diamond)(x,y,z,\ldots)^2 = (\diamond)(x,y,z,\ldots)^2$ est satisfaite.

Je regarde d'un autre point de vue ce problème de la transformation en ellemême, d'une fonction quadratique par des substitutions linéaires. Je suppose que x, y, z, &c., sont des fonctions linéaires de x, y, z, $\ldots$, et je cherche les conditions pour que la fonction linéaire, il faut que $\beta = 1/\beta$, &c. Le symbole βa ne se trouve et dans le déterminant, ni au côté droit. Cependant il convient de poser $\beta a = -a\beta$; car cela étant, il serait permis de transformer les Pfaffians, en écrivant par exemple $\alpha\beta 12 = -\beta\alpha 12$.

Je prends en passant que, si avant de poser l'équation $\beta a = -\alpha\beta$, on suppose que les deux symboles a, β deviennent identiques (ce pour exemple on écrit $a = \beta = 5$), on aurait par exemple

$$a\beta.12 = a\beta.12 + a1.2\beta + a2.\beta 1 = 55.12 + 51.25 + 52.51 = 55.12,\ \text{\&c.},$$

et en retrouvant ainsi la formule pour le développement de $\overline{12345 \mid 12345}$.

La nouvelle formule peut être appliquée immédiatement au développement des *déterminants mineurs*. En effet, en se servant de la notation des matrices, on aura

$$\begin{vmatrix} 11, & 12, & 13 \\ 21, & 22, & 23 \\ 31, & 32, & 33 \end{vmatrix}^{-1} = \frac{1}{\overline{\overline{123 \mid 123}}} \begin{vmatrix} +23\,\overline{23}, & -13\,\overline{23}, & -12\,\overline{32} \\ -23\,\overline{13}, & +13\,\overline{13}, & -21\,\overline{31} \\ -32\,\overline{12}, & -31\,\overline{21}, & +12\,\overline{12} \end{vmatrix},$$

$$\begin{vmatrix} 11, & 12, & 13, & 14 \\ 21, & 22, & 23, & 24 \\ 31, & 32, & 33, & 34 \\ 41, & 42, & 43, & 44 \end{vmatrix}^{-1} = \frac{1}{\overline{\overline{1234 \mid 1234}}} \begin{vmatrix} +234\,\overline{234}, & -134\,\overline{234}, & -124\,\overline{324}, & -123\,\overline{324} \\ -234\,\overline{134}, & +134\,\overline{134}, & -214\,\overline{314}, & -213\,\overline{413} \\ -324\,\overline{124}, & -314\,\overline{214}, & +124\,\overline{124}, & -312\,\overline{412} \\ -423\,\overline{123}, & -413\,\overline{213}, & -412\,\overline{312}, & +123\,\overline{123} \end{vmatrix},$$

et ainsi de suite. On suppose toujours que chaque terme de la matrice à droite soit divisé par le dénominateur commun. On voit que les déterminants mineurs qui correspondent à des termes tels que xx sont des déterminants *ordinaires*, avec le signe +, tandis que les déterminants mineurs qui correspondent à des termes tels que $\alpha\beta$, sont des quatre termes du produit sont *réciproques* deux à deux; le déterminant de transformation est donc ± 1, et la transformation est *propre*.

137. Recherches ultérieures sur les déterminants gauches.

RECHERCHES ULTÉRIEURES SUR LES DÉTERMINANTS GAUCHES.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 299–313: Continuation of the Memoir t. XXXII. (1846) and t. XXXVIII. (1849); **52** and **69**.]

J'ai déjà donné une formule pour le développement d'un *déterminant gauche*. En prenant, pour fixer les idées, un cas particulier, soit

$$\overline{12345 \mid 12345} = \begin{vmatrix} 11, & 12, & 13, & 14, & 15 \\ 21, & 22, & 23, & 24, & 25 \\ 31, & 32, & 33, & 34, & 35 \\ 41, & 42, & 43, & 44, & 45 \\ 51, & 52, & 53, & 54, & 55 \end{vmatrix},$$

(où $12 = -21$, &c., tandis que les quantités 11, 22, &c. ne s'évanouissent pas). Cette formule peut être écrite comme suit:

$$\begin{aligned}
\overline{12345 \mid 12345} = {} & 11 \cdot 22 \cdot 33 \cdot 44 \cdot 55 \\
& + 11 \cdot 22 \cdot 33 \cdot (45)^2 \\
& + 11 \cdot 22 \cdot 44 \cdot (35)^2 \\
& + 11 \cdot 22 \cdot 55 \cdot (34)^2 \\
& + 11 \cdot 33 \cdot 44 \cdot (25)^2 \\
& + 11 \cdot 33 \cdot 55 \cdot (24)^2 \\
& + 11 \cdot 44 \cdot 55 \cdot (23)^2 \\
& + 22 \cdot 33 \cdot 44 \cdot (15)^2 \\
& + 22 \cdot 33 \cdot 55 \cdot (14)^2 \\
& + 22 \cdot 44 \cdot 55 \cdot (13)^2 \\
& + 33 \cdot 44 \cdot 55 \cdot (12)^2 \\
& + 11 \cdot (2345)^2 \\
& + 22 \cdot (1345)^2 \\
& + 33 \cdot (1245)^2 \\
& + 44 \cdot (1235)^2 \\
& + 55 \cdot (1234)^2.
\end{aligned}$$

Les expressions 12, 1234, &c. à droite sont ici des *Pfaffians.* On a

$$12 = 12,$$

$$1234 = 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23,$$

et en écrivant encore un terme, pour mieux présenter la loi:

$$\begin{aligned}
123456 = {} & 12 \cdot 34 \cdot 56 + 13 \cdot 45 \cdot 62 + 14 \cdot 56 \cdot 23 + 15 \cdot 62 \cdot 34 + 16 \cdot 23 \cdot 45 \\
& + 12 \cdot 35 \cdot 64 + 13 \cdot 46 \cdot 25 + 14 \cdot 52 \cdot 36 + 15 \cdot 63 \cdot 42 + 16 \cdot 24 \cdot 53 \\
& + 12 \cdot 36 \cdot 54 + 13 \cdot 42 \cdot 56 + 14 \cdot 53 \cdot 62 + 15 \cdot 64 \cdot 23 + 16 \cdot 25 \cdot 34.
\end{aligned}$$

J'ai trouvé récemment une formule analogue pour le développement d'un *déterminant gauche bordé*, tel que

$$\overline{\alpha 1234 \mid \beta 1234} = \alpha\beta \begin{vmatrix} \alpha\beta, & \alpha 1, & \alpha 2, & \alpha 3, & \alpha 4 \\ 1\beta, & 11, & 12, & 13, & 14 \\ 2\beta, & 21, & 22, & 23, & 24 \\ 3\beta, & 31, & 32, & 33, & 34 \\ 4\beta, & 41, & 42, & 43, & 44 \end{vmatrix};$$

cette formule est:

$$\begin{aligned}\overline{\alpha 1234 \mid \beta 1234} = \alpha\beta \cdot 11 \cdot 22 \cdot 33 \cdot 44 \\ + \alpha\beta \cdot 12 \cdot 12 \cdot 33 \cdot 44 \\ + \alpha\beta \cdot 13 \cdot 13 \cdot 22 \cdot 44 \\ + \alpha\beta \cdot 14 \cdot 14 \cdot 22 \cdot 33 \\ + \alpha\beta \cdot 23 \cdot 23 \cdot 11 \cdot 44 \\ + \alpha\beta \cdot 24 \cdot 24 \cdot 11 \cdot 33 \\ + \alpha\beta \cdot 34 \cdot 34 \cdot 11 \cdot 22 \\ + \alpha\beta \cdot 1234 \cdot 1234 \\ + \alpha 1 \cdot \beta 1 \cdot 22 \cdot 33 \cdot 44 \\ + \alpha 2 \cdot \beta 2 \cdot 11 \cdot 33 \cdot 44 \\ + \alpha 3 \cdot \beta 3 \cdot 11 \cdot 22 \cdot 44 \\ + \alpha 4 \cdot \beta 4 \cdot 11 \cdot 22 \cdot 33 \\ + \alpha 123 \cdot \beta 123 \cdot 44 \\ + \alpha 124 \cdot \beta 124 \cdot 33 \\ + \alpha 134 \cdot \beta 134 \cdot 22 \\ + \alpha 234 \cdot \beta 234 \cdot 11.\end{aligned}$$

Il est à peine nécessaire de remarquer que dans les Pfaffians à droite, qui entrent des symboles tels que 1α, $\beta 1$, &c., qui ne se trouvent pas dans le déterminant dont il s'agit, il faut entendre $1\alpha = -\alpha 1$, $\beta 1 = -1\beta$, &c. Le symbole $\beta\alpha$ ne se trouve ni dans le déterminant, ni au côté droit. Cependant il convient de poser $\beta\alpha = -\alpha\beta$; car cela étant, il serait permis de transformer les Pfaffians, en écrivant par exemple $\alpha\beta 12 = -\beta\alpha 12$.

Je remarque en passant que, si avant de poser l'équation $\beta\alpha = -\alpha\beta$, on suppose que les deux symboles α, β deviennent identiques (ce pour exemple on écrit $\alpha = \beta = 5$), on aurait par exemple

$$\alpha\beta \cdot 12 = \alpha\beta \cdot 12 + \alpha 1 \cdot 2\beta + \alpha 2 \cdot \beta 1 = 55 \cdot 12 + 51 \cdot 25 + 52 \cdot 51 = 55 \cdot 12, \text{ \&c.},$$

et en retrouvant ainsi la formule pour le développement de $\overline{12345 \mid 12345}$.

La nouvelle formule peut être appliquée immédiatement au développement des *déterminants mineurs*. En effet, en se servant de la notation des matrices, on aura les expressions pour les matrices inverses, comme on le voit dans le mémoire.

(En effet, en développant les deux côtés, on obtient

$$11{\cdot}22{\cdot}33{+}11{\cdot}(23)^2{+}22{\cdot}(13)^2{+}33{\cdot}(12)^2 = 11{\cdot}(22{\cdot}33{+}(23)^2){+}22{\cdot}(21{\cdot}33{+}23{\cdot}13){+}31{\cdot}(31{\cdot}22{+}32{\cdot}12).$$

On voit que les deux termes marqués par un † se détruisent et que l'équation est *identique*. On doit avoir de même

$$\begin{aligned}\overline{1234 \mid 1234} = 11 \cdot \overline{234 \mid 234} \\ - 12 \cdot \overline{234 \mid 134} \\ - 13 \cdot \overline{324 \mid 124} \\ - 14 \cdot \overline{423 \mid 123},\end{aligned}$$

ou, ce qui est la même chose:

$$\begin{aligned}\overline{1234 \mid 1234} = 11 \cdot \overline{234 \mid 234} \\ + 21 \cdot \overline{234 \mid 134} \\ + 31 \cdot \overline{324 \mid 124} \\ + 41 \cdot \overline{423 \mid 123};\end{aligned}$$

c'est-à-dire, en développant des deux côtés:

$$\begin{aligned}&11 \cdot 22 \cdot 33 \cdot 44 + 11 \cdot 22 \cdot 33 \cdot (34)^2 + 11 \cdot 44 \cdot (23)^2 \\ &+ 22 \cdot 33 \cdot (14)^2 + 33 \cdot 44 \cdot (12)^2 + (1234)^2 \\ &\quad = 11 \cdot \{22 \cdot 33 \cdot 44 + 22 \cdot (34)^2 + 33 \cdot (24)^2 + 44 \cdot (23)^2 + (234)^2\} \\ &\quad + 21 \cdot \{21 \cdot 33 \cdot 44 + 23 \cdot 13 \cdot 44 + 24 \cdot 14 \cdot 33 + 34 \cdot (2\dagger)^2\} \\ &\quad + 31 \cdot \{31 \cdot 22 \cdot 44 + 32 \cdot 12 \cdot 44 + 34 \cdot 14 \cdot 22 + 24 \cdot (3\dagger)^2\} \\ &\quad + 41 \cdot \{41 \cdot 22 \cdot 33 + 42 \cdot 12 \cdot 33 + 43 \cdot 13 \cdot 22 + 23 \cdot (4\dagger)^2\}.\end{aligned}$$

Cette expression est en effet identique, comme on le voit en observant que les six termes marqués par un † se détruisent deux à deux, et que les trois termes marqués par un $(*)$ sont l'ensemble équivalents à $(1234)^2$.)

Je remarque que le nombre des termes du développement du déterminant gauche est toujours une puissance de 2, et que pour cela, ce nombre se réduit à la moitié en réduisant à zéro un terme quelconque xx. Mais outre cela, le déterminant peut dans cette supposition s'exprimer comme une somme d'une certaine quantité de *carrés* de fonctions d'un ordre inférieur à l'unité. Je considère par exemple le déterminant gauche $\overline{123 \mid 123}$. En faisant $33 = 0$ et accentuant, pour y mettre plus de clarté, tous les symboles, on trouve

$$\overline{123 \mid 123} = 11' \cdot (23')^2 + 22' \cdot (13')^2.$$

De là, en dérivant

$$\begin{aligned}&11 = 13' \cdot 11', \quad 12 = 11' \cdot 23', \\ &22 = 13' \cdot 22',\end{aligned}$$

on obtient

$$\overline{12 \mid 12} = 11 \cdot 22 + (12)^2 = 13' \cdot \{22' \cdot (13')^2 + 11' \cdot (23')^2\},$$

c'est-à-dire

$$\overline{12 \mid 12} = 11' \cdot 14' \cdot \overline{123 \mid 123}.$$

On a de même

$$\overline{1234 \mid 1234} = 11' \cdot 22' \cdot (34')^2 + 11' \cdot 33' \cdot (24')^2 + 22' \cdot 33' \cdot (14')^2 + (1234')^2,$$

et de là, en dérivant

$$\begin{aligned}&11 = 14' \cdot 11', \quad 12 = 11' \cdot 24', \quad 23 = 1234', \\ &22 = 14' \cdot 22', \quad 13 = 11' \cdot 34', \\ &33 = 14' \cdot 33',\end{aligned}$$

on obtient

$$\begin{aligned}\overline{123 \mid 123} &= 11 \cdot 22 \cdot 33 + 11 \cdot (23)^2 + 22 \cdot (31)^2 + 33 \cdot (12)^2 \\ &= 11' \cdot 14'\{22' \cdot 33' \cdot (14')^2 + (1234')^2\} + 11' \cdot 22' \cdot (34')^2 + 11' \cdot 33' \cdot (24')^2,\end{aligned}$$

c'est-à-dire

$$\overline{123 \mid 123} = 11' \cdot 14' \cdot \overline{1234 \mid 1234}.$$

De même

$$\begin{aligned}\overline{12345 \mid 12345} = {} & 11' \cdot 22' \cdot 33' \cdot 44' \cdot (55')^2 \\ & + 11' \cdot 22' \cdot 33' \cdot (45')^2 \\ & + 11' \cdot 22' \cdot 44' \cdot (35')^2 \\ & + 11' \cdot 33' \cdot 44' \cdot (25')^2 \\ & + 22' \cdot 33' \cdot 44' \cdot (15')^2 \\ & + 11' \cdot (2345')^2 \\ & + 22' \cdot (1345')^2 \\ & + 33' \cdot (1245')^2 \\ & + 44' \cdot (1235')^2.\end{aligned}$$

Or il est permis d'écrire

$$\begin{array}{llll} 11 = 15' \cdot 11', & 12 = 11' \cdot 25', & 23 = 1235', & 1234 = 12345' \cdot 15', \\ 22 = 15' \cdot 22', & 13 = 11' \cdot 35', & 24 = 1245', & \\ 33 = 15' \cdot 33', & 14 = 11' \cdot 45', & 34 = 1345', & \\ 44 = 15' \cdot 44'. & & & \end{array}$$

En effet, les quantités à gauche ne sont liées entre elles que par la seule équation $1234 = 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23$, qui est satisfaite identiquement par les valeurs à substituer pour ces quantités qui y entrent. Cela étant, on trouve d'abord:

$$\overline{1234 \mid 1234} = 11'(15')^2\, \overline{12345 \mid 12345}.$$

Je prends encore un exemple. On a

$$\begin{aligned}\overline{123456 \mid 123456} = {} & 11' \cdot 22' \cdot 33' \cdot 44' \cdot (56')^2 \\ & + 11' \cdot 22' \cdot 44' \cdot 55' \cdot (46')^2 \\ & + 11' \cdot 22' \cdot 44' \cdot 55' \cdot (36')^2 \\ & + 11' \cdot 33' \cdot 44' \cdot 55' \cdot (26')^2 \\ & + 22' \cdot 33' \cdot 44' \cdot 55' \cdot (16')^2 \\ & + 11' \cdot 22' \cdot (3456')^2 \\ & + 11' \cdot 33' \cdot (2456')^2 \\ & + 11' \cdot 44' \cdot (2356')^2 \\ & + 11' \cdot 55' \cdot (2346')^2 \\ & + 22' \cdot 33' \cdot (1456')^2 \\ & + 22' \cdot 44' \cdot (1356')^2 \\ & + 22' \cdot 55' \cdot (1346')^2 \\ & + 33' \cdot 44' \cdot (1256')^2 \\ & + 33' \cdot 55' \cdot (1246')^2 \\ & + 44' \cdot 55' \cdot (1236')^2 \\ & + (123456')^2.\end{aligned}$$

Ici, il est permis d'écrire

$$
\begin{aligned}
&11 = 16' \cdot 11', \quad 12 = 16' \cdot 26', \quad 23 = 1236', \quad 34 = 1346', \quad 45 = 1456',\\
&22 = 16' \cdot 22', \quad 13 = 16' \cdot 36', \quad 24 = 1246', \quad 35 = 1356',\\
&33 = 16' \cdot 33', \quad 14 = 16' \cdot 46', \quad 25 = 1256',\\
&44 = 16' \cdot 44', \quad 15 = 16' \cdot 56',\\
&55 = 16' \cdot 55',\\
&1234 = 2346' \cdot 11' \cdot 16', \quad 2345 = 123456' \cdot 16',\\
&1235 = 2356' \cdot 11' \cdot 16',\\
&1245 = 2456' \cdot 11' \cdot 16',\\
&1345 = 3456' \cdot 11' \cdot 16';
\end{aligned}
$$

car les valeurs des quantités à gauche satisfont identiquement aux équations qui ont lieu entre ces mêmes quantités. Par exemple l'équation $1234 = 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23$ devient $2346' \cdot 16' = 26' \cdot 1346' + 63' \cdot 1246' + 46' \cdot 1236'$.

Or l'expression à droite devient, en développant:

$$
\begin{aligned}
&26'(13' \cdot 46' + 63' \cdot 14' + 16' \cdot 34')\\
&+ 63'(12' \cdot 46' + 14' \cdot 62' + 16' \cdot 24')\\
&+ 46'(12' \cdot 36' + 13' \cdot 62' + 16' \cdot 23').
\end{aligned}
$$

et les termes qui contiennent le facteur $16'$, donnent ensemble $16' \cdot 2346'$, les autres termes se détruisent deux à deux. On obtient enfin, en effectuant la substitution:

$$\overline{12345 \mid 12345} = 11' \cdot (16')^2\, \overline{123456 \mid 123456};$$

et ainsi de suite.

Je fais les mêmes substitutions dans les matrices inverses, en supprimant cependant la dernière ligne et la dernière colonne de chaque matrice. On trouve ainsi, en y ajoutant les équations ci-dessus trouvées par rapport aux déterminants:

$$11' \cdot \overline{123 \mid 123} = \overline{12 \mid 12},$$

$$\frac{1}{13' \cdot \overline{123 \mid 123}} \begin{vmatrix} +23 \cdot 23, & -13 \cdot 23 \\ -23 \cdot 13, & +13 \cdot 13 \end{vmatrix} = \frac{1}{\overline{12 \mid 12}} \begin{vmatrix} -2\,\bar{2}\,2 + \dfrac{12 \cdot 12}{11}, & -1\,\bar{2} \\ +2\,\bar{1}\,1, & +2\,\bar{1}\,1 - \dfrac{1\,\bar{2}}{1} \end{vmatrix},$$

$$11' \cdot 14' \cdot \overline{1234 \mid 1234} = \overline{123 \mid 123},$$

$$\frac{1}{14' \cdot \overline{1234 \mid 1234}} \begin{vmatrix} +234 \cdot 234, & -134 \cdot 234, & -124 \cdot 324 \\ -234 \cdot 134, & +134 \cdot 134, & -214 \cdot 314 \\ -324 \cdot 124, & -314 \cdot 214, & +124 \cdot 124 \end{vmatrix} = \frac{1}{\overline{123 \mid 123}} \begin{vmatrix} -23 \cdot 23 + \dfrac{123 \cdot 123}{11}, & -2 \cdot 1 \cdot 2 \\ +23 \cdot 13, & +13 \cdot 13 - \dfrac{123 \cdot}{22} \end{vmatrix}$$

$$11' \cdot 15' \cdot \overline{12345 \mid 12345} = \overline{1234 \mid 1234},$$

$$\frac{1}{15' \cdot \overline{12345 \mid 12345}} \begin{vmatrix} +2345 \cdot 2345, & -1345 \cdot 2345, & -1245 \cdot 3245, & -1235 \cdot 4235 \\ -2345 \cdot 1345, & +1345 \cdot 1345, & -2145 \cdot 3145, & -2135 \cdot 4135 \\ -3245 \cdot 1245, & -3145 \cdot 2145, & +1245 \cdot 1245, & -3125 \cdot 4125 \\ -4235 \cdot 1235, & -4135 \cdot 2135, & -4125 \cdot 3125, & +1235 \cdot 1235 \end{vmatrix} = \frac{1}{\overline{1234 \mid 1234}} \begin{vmatrix} -234 \cdot 234 + \\ +234 \\ +324 \end{vmatrix}$$

et ainsi de suite.

Il est bon de changer un peu la forme de ces équations. On en déduit sans peine:

$$\frac{1}{13'\cdot\overline{123\,|\,123}}\begin{vmatrix} 2\cdot 23\cdot\overline{23}-\dfrac{\overline{23\cdot 23}}{11}, & -2\cdot 13\cdot\overline{23} \\ +2\cdot 23\cdot 13', & +2\cdot 13\cdot 13'-\dfrac{\overline{123\cdot 123}}{22}\end{vmatrix} = \frac{1}{\overline{12\,|\,12}}\begin{vmatrix} -2\cdot 2\,\overline{2}+\dfrac{12\cdot 12}{11}, & -2\cdot 1\,\overline{2} \\ +2\cdot 2\,\overline{1}\,1, & +2\cdot 1\,\overline{1}-\dfrac{1\,\overline{2}}{1}\end{vmatrix}$$

$$\frac{1}{14'\cdot\overline{1234\,|\,1234}}\begin{vmatrix} 2\cdot 234\cdot\overline{234}-\dfrac{1234\cdot 1234}{11}, & -2\cdot 134\cdot\overline{234}, & -2\cdot 124\cdot\overline{324} \\ -2\cdot 234\cdot\overline{134}, & 2\cdot 134\cdot 134'-\dfrac{1234\cdot 1234}{22'}, & -2\cdot 214\cdot\overline{314} \\ -2\cdot 324\cdot\overline{124}, & -2\cdot 314\cdot\overline{214}, & 2\cdot 124\cdot 124'-\dfrac{1234\cdot 1234}{33}\end{vmatrix} = \frac{}{\overline{123}}$$

$$= \frac{1}{\overline{123\,|\,123}}\begin{vmatrix} -2\cdot 23\cdot\overline{23}+\dfrac{123\cdot 123}{11}, & -2\cdot 13\cdot 23 \\ +2\cdot 23\cdot 13, & +2\cdot 13\cdot 13-\dfrac{123\cdot 123}{22}\end{vmatrix},$$

et ainsi de suite.

Les formules que je viens de présenter, peuvent être appliquées aussitôt à la solution de la question suivante: "Trouver x_1, x_2, x_3, &c., fonctions linéaires de x_1, x_2, x_3, &c. telles que

$$11\,x_1^2+22\,x_2^2+33\,x_3^2+\text{\&c.}=11\,{x_1}^2+22\,{x_2}^2+33\,{x_3}^2+\text{\&c.,}''$$

c'est-à-dire: transformer une fonction quadratique $11\,x_1^2+22\,x_2^2+33\,x_3^2+$ &c. en *elle-même* par des substitutions linéaires. Il suffira d'écrire les solutions pour le cas de trois indéterminées: on satisfait identiquement à l'équation

$$11\,x_1^2+22\,x_2^2+33\,x_3^2=11\,{x_1}^2+22\,{x_2}^2+33\,{x_3}^2$$

en dérivant

$$(x_1,\ x_2,\ x_3)=\frac{1}{\overline{123\,|\,123}}\begin{vmatrix} +2\cdot 23\cdot\overline{23}-\dfrac{\overline{123\cdot 123}}{11}, & -2\cdot 13\cdot\overline{23}, & -2\cdot 12\cdot\overline{32} \\ -2\cdot 23\cdot\overline{13}, & +2\cdot 13\cdot\overline{13}-\dfrac{\overline{123\cdot 123}}{22}, & -2\cdot 21\cdot\overline{31} \\ -2\cdot 32\cdot\overline{12}, & -2\cdot 31\cdot\overline{21}, & +2\cdot 12\cdot 12-\dfrac{\overline{123\cdot 123}}{33}\end{vmatrix}(11\,x_1,\ 2$$

Voilà la transformation *propre*. On tire la transformation *impropre* de $11\,x_1^2+22\,x_2^2$ en elle-même en écrivant $33=0$; car, cela posé, les valeurs de x_1, x_2 ne contiennent pas x_3, et l'on n'a plus besoin de la valeur de x_3. On obtient ainsi la solution suivante; savoir, on satisfait identiquement à l'équation

$$11\,x_1^2+22\,x_2^2=11\,{x_1}^2+22\,{x_2}^2$$

en dérivant

$$(x_1,\ x_2)=\frac{1}{\overline{123\,|\,123}}\begin{vmatrix} 2\cdot 23\cdot\overline{23}-\dfrac{\overline{123\cdot 123}}{11}, & -2\cdot 13\cdot\overline{23} \\ -2\cdot 23\cdot\overline{13}, & 2\cdot 13\cdot\overline{13}-\dfrac{\overline{123\cdot 123}}{22}\end{vmatrix}(11\,x_1,\ 22\,x_2),$$

ce qui se réduit, en faisant la substitution $11=13'\cdot 11'$, $22=13'\cdot 22'$, $12=11'\cdot 23'$, à la solution à celle-ci, savoir de satisfaire identiquement à l'équation $11\,x_1^2+22\,x_2^2=11\,{x_1}^2+22\,{x_2}^2$:

$$(x_1,\ x_2)=\frac{1}{\overline{12\,|\,12}}\begin{vmatrix} -2\cdot 2\,\overline{2}+\dfrac{12\cdot 12}{11}, & -2\cdot 1\,\overline{2} \\ +2\cdot 2\,\overline{1}, & +2\cdot 1\,\overline{1}-\dfrac{12\cdot 12}{22}\end{vmatrix}(11\,x_1,\ 22\,x_2),$$

ce qui est encore une transformation *impropre*, qui correspond de plus près à la formule pour la transformation *propre*; la seule différence est que les signes des termes de la première colonne de la matrice ont été changés.

En introduisant des lettres simples a, b, &c. à la place des symboles 11, 22, &c., je considère d'abord la transformation *propre*

$$ax^2 + by^2 = ax^2 + by^2.$$

Ici, en écrivant

$$\begin{vmatrix} 11, & 12 \\ 21, & 22 \end{vmatrix} = \begin{vmatrix} a, & \nu \\ -\nu, & b \end{vmatrix},$$

la formule donne

$$(x, y) = \frac{1}{ab + \nu^2} \begin{vmatrix} ab - \nu^2, & -2b\nu \\ 2\nu a, & ab - \nu^2 \end{vmatrix} (x, y).$$

La transformation *impropre*

$$ax^2 + by^2 = ax^2 + by^2$$

s'obtient au moyen de la formule donnée plus haut pour la transformation *propre* de la fonction $ax^2 + by^2 + ex^2$ en *elle-même.* En y écrivant $a = b = c = \omega$. On a ainsi pour la transformation propre:

$$(x, y, z) = \frac{1}{\omega^2 + \lambda^2 + \mu^2 + \nu^2} \begin{vmatrix} \omega\lambda - \mu^2 - \nu^2, & 2\lambda\mu - 2\nu\omega, & 2\nu\lambda + 2\mu\omega \\ 2\lambda\mu + 2\nu\omega, & \omega^2 - \lambda^2 + \mu^2 - \nu^2, & 2\mu\nu - 2\lambda\omega \\ 2\nu\lambda - 2\mu\omega, & 2\mu\nu + 2\lambda\omega, & \omega^2 - \lambda^2 - \mu^2 + \nu^2 \end{vmatrix} (x, y, z).$$

et en dérivant dans la formule pour la transformation impropre, $a = b = c = 1$, $d = 0$, en écrivant λ, μ, ν, $-\omega$ au lieu de ρ, σ, τ, ϕ, on obtient pour la transformation impropre

$$x^2 + y^2 + z^2 = x^2 + y^2 + z^2,$$

l'équation

$$(x, y, z) = \frac{1}{\omega^2 + \lambda^2 + \mu^2 + \nu^2} \begin{vmatrix} -\omega^2 + \lambda^2 + \mu^2 + \nu^2, & -2\lambda\mu + 2\nu\omega, & -2\nu\lambda - 2\mu\omega \\ -2\lambda\mu - 2\nu\omega, & -\omega^2 + \lambda^2 - \mu^2 + \nu^2, & 2\mu\nu - 2\lambda\omega \\ -2\nu\lambda + 2\mu\omega, & -2\mu\nu - 2\lambda\omega, & -\omega^2 + \lambda^2 + \mu^2 - \nu^2 \end{vmatrix} (x, y, z).$$

Pour obtenir la transformation *propre*

$$ax^2 + by^2 + cz^2 + dw^2 = ax^2 + by^2 + cz^2 + dw^2,$$

j'écris

$$\begin{vmatrix} 11, & 12, & 13, & 14 \\ 21, & 22, & 23, & 24 \\ 31, & 32, & 33, & 34 \\ 41, & 42, & 43, & 44 \end{vmatrix} = \begin{vmatrix} a, & \nu, & -\mu, & \rho \\ -\nu, & b, & \lambda, & \sigma \\ \mu, & -\lambda, & c, & \tau \\ -\rho, & -\sigma, & -\tau, & d \end{vmatrix};$$

cela donne d'abord, en mettant pour abréger

$$\phi = \lambda\rho + \mu\sigma + \nu\tau,$$

la valeur du déterminant

$$\overline{1234 \mid 1234} = abcd + bcd\rho^2 + cad\sigma^2 + abd\tau^2 + ad\lambda^2 + bd\mu^2 + cd\nu^2 + \phi^2$$

(ce que je représente par k).

J'ajoute aussi la valeur de la *matrice inverse*

$$\begin{vmatrix} 11, & 12, & 13, & 14 \\ 21, & 22, & 23, & 24 \\ 31, & 32, & 33, & 34 \\ 41, & 42, & 43, & 44 \end{vmatrix}^{-1},$$

savoir:

$$\begin{aligned} \frac{1}{k}\{&bcd + b\tau^2 + c\sigma^2 + d\lambda^2, \quad -cdv - v\tau^2 + d\lambda\mu - \sigma\rho, \\ &cdu + v\rho\tau + d\lambda\nu - \mu\tau, \quad acd + b\tau^2 + c\rho^2 + d\mu^2, \\ &- bdp - \nu\sigma\tau + d\mu\nu - \lambda\rho, \quad ucd + c\nu\rho + d\nu\mu + d\nu\rho - \mu\sigma, \\ &bcp + \lambda\phi + c\rho\tau - b\mu\tau, \quad acu + \mu\phi + d\rho\sigma + a\rho\tau, \end{aligned}$$

On a pour la transformation propre, l'équation $(x, y, z, w) =$

$$\frac{1}{R}\begin{vmatrix} -\rho^2 + \sigma^2 + \tau^2 + \omega^2 + \lambda^2 - \mu^2 - \nu^2 - \psi^2, & -2\omega + 2\tau\psi + 2\lambda\mu - 2\rho\sigma, \\ 2\omega\nu - 2\tau\psi + 2\lambda\mu - 2\rho\sigma, & \rho^2 - \sigma^2 + \tau^2 + \omega^2 - \lambda^2 + \mu^2 - \nu^2 - \psi^2, \\ -2\omega\mu + 2\rho\psi + 2\lambda\nu - 2\rho\tau, & 2\omega\lambda + 2\mu\nu - 2\sigma\psi - 2\sigma\tau, \\ 2\omega\lambda - 2\nu\psi + 2\mu\nu - 2\sigma\tau, & 2\omega\mu - 2\rho\psi + 2\lambda\nu - 2\rho\tau, \\ 2\omega\mu - 2\nu\psi + 2\mu\rho - 2\lambda\sigma, & -\rho^2 - \sigma^2 - \tau^2 + \omega^2 + \lambda^2 + \mu^2 + \nu^2 - \psi^2 \end{vmatrix}(x, y, z, w).$$

On peut changer la forme de cette expression, en y écrivant

$$\lambda = \tfrac{1}{2}(\alpha - \alpha'), \quad \mu = \tfrac{1}{2}(\beta - \beta'), \quad \nu = \tfrac{1}{2}(\gamma - \gamma'), \quad \psi = \tfrac{1}{2}(\delta - \delta'),$$

$$\rho = \tfrac{1}{2}(\alpha + \alpha'), \quad \sigma = \tfrac{1}{2}(\beta + \beta'), \quad \tau = \tfrac{1}{2}(\gamma + \gamma'), \quad \omega = \tfrac{1}{2}(\delta + \delta');$$

cela donne

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 - \alpha'^2 - \beta'^2 - \gamma'^2 - \delta'^2 = 0,$$
$$R = \tfrac{1}{4}(\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + \alpha'^2 + \beta'^2 + \gamma'^2 + \delta'^2),$$

de manière qu'en écrivant

$$M = \alpha^2 + \beta^2 + \gamma^2 + \delta^2, \quad M' = \alpha'^2 + \beta'^2 + \gamma'^2 + \delta'^2,$$

on obtient

$$R = \tfrac{1}{4}(M + M') = \sqrt{(MM')},$$

et la formule pour la transformation devient

$$(x, y, z, w) = \frac{1}{\sqrt{(MM')}}\begin{vmatrix} -\alpha\alpha' + \beta\beta' + \gamma\gamma' + \delta\delta', & -\alpha\delta' - \beta\alpha' - \gamma\beta' + \delta\gamma', \\ -\alpha\delta - \beta\alpha + \gamma\delta - \delta\gamma, & \alpha\alpha' - \beta\beta' + \gamma\gamma' + \delta\delta', \\ -\alpha\gamma' + \beta\delta' - \gamma\alpha' + \delta\beta', & \alpha\delta' - \beta\gamma' - \gamma\beta' + \delta\alpha', \\ -\alpha\beta' + \beta\gamma' - \gamma\delta' - \delta\alpha', & -\alpha\gamma' - \beta\delta' + \gamma\alpha' + \delta\beta' \end{vmatrix}(x, y, z, w).$$

On voit donc que même sans supposer l'équation $M = M'$, cette formule donne la transformation *propre*

$$x^2 + y^2 + z^2 + w^2 = x^2 + y^2 + z^2 + w^2.$$

Cette solution est à peu près de la même forme que la solution *impropre* donnée par Euler dans son mémoire "Problema algebraicum ob affectiones prorsus singulares memorabile," *Nov. Comm. Petrop.*, t. XV 1770, p. 75, et *Comm. Arith. collectae*, [4to. Petrop. 1849], t. 1. p. 427. Je remarque aussi que cette même solution peut être déduite de la théorie des *Quaternions*. En effet, i, j, k étant des quantités imaginaires telles que $i^2 = j^2 = k^2 = -1$, $jk = -kj = i$, $ki = -ik = j$, $ij = -ji = k$, on obtient, en effectuant la multiplication:

$$(xi + yj + zk + w) = \tfrac{1}{\sqrt{(MM')}}(\alpha i + \beta j + \gamma k + \delta)(xi + yj + zk + w)(\alpha' i + \beta' j + \gamma' k + \delta'),$$

x, y, z, w ayant les mêmes valeurs que dans la dernière formule de transformation. En changeant les signes des termes de la quatrième colonne, on en tire pour la transformation *impropre*

$$x^2 + y^2 + z^2 + w^2 = x^2 + y^2 + z^2 + w^2,$$

la formule suivante plus symétrique:

$$(x, y, z, w) = \frac{1}{\sqrt{(MM')}} \begin{vmatrix} -\alpha\alpha' + \beta\beta' + \gamma\gamma' + \delta\delta', & -\alpha\beta' - \beta\alpha' - \gamma\delta' + \delta\gamma' \\ -\alpha\beta' - \beta\alpha' + \gamma\delta' + \delta\gamma', & \alpha\alpha' - \beta\beta' + \gamma\gamma' + \delta\delta' \\ -\alpha\gamma' + \beta\delta' - \gamma\alpha' + \delta\beta', & -\alpha\gamma' - \beta\delta' - \gamma\alpha' - \delta\beta' \\ -\alpha\delta' - \beta\gamma' - \gamma\beta' + \delta\alpha', & \alpha\delta' + \beta\gamma' + \gamma\beta' - \delta\alpha' \end{vmatrix} (x, y, z, w).$$

Ces formules pour la transformation, tant propre qu'impropre, de la fonction $x^2+y^2+z^2+w^2$ en *elle-même*, sont utiles dans la théorie des polygones inscrits dans une surface du second ordre.

138. Recherches sur les matrices dont les termes sont des fonctions linéaires d'une seule indéterminée.

Recherches sur les matrices dont les termes sont des fonctions linéaires d'une seule indéterminée.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 313–317.]

Je pose la matrice

$$\begin{vmatrix} A, & B, & C, & \dots \\ A', & B', & C', & \dots \\ A'', & B'', & C'', & \dots \\ & \vdots & & \end{vmatrix}$$

dont les termes (n^2 en nombre) sont des fonctions linéaires d'une quantité s, et je considère le déterminant formé avec cette matrice, et les déterminants mineurs formés en supprimant un nombre quelconque des lignes et un nombre égal de colonnes de la matrice. En supprimant une *seule* ligne et une *seule* colonne, on obtient les *premiers mineurs*; en supprimant deux lignes et deux colonnes, on obtient les *seconds mineurs*; et ainsi de suite. Cela étant, je suppose que la quantité s a été trouvée en égalant à zéro le déterminant formé avec la matrice donnée; ce déterminant sera une fonction de s du n-ième degré qui généralement ne contiendra pas comme une fonction de s *aucun* facteur multiple. On voit donc qu'un facteur simple du déterminant ne peut pas entrer comme facteur dans les premiers mineurs (c'est-à-dire dans *tous* les premiers mineurs); mais en supposant que le déterminant ait des facteurs multiples, un facteur multiple du déterminant peut entrer comme facteur (simple ou multiple) dans les premiers mineurs, ou dans les mineurs d'un ordre plus élevé. Il importe de trouver le degré selon lequel un facteur multiple du déterminant peut entrer comme facteur des premiers mineurs, ou des mineurs d'un ordre quelconque donné.

Cela se fait très facilement au moyen d'une propriété générale des déterminants; si les mineurs du $(r+1)$ième ordre contiennent le facteur $(s-a)^a$ (c'est-à-dire, si tous les mineurs du cet ordre identiquement zéro; et si même le r-ième ordre contiennent le facteur $(s-a)^{a-1}$; et si les mineurs du $(r-1)$ième ordre contiendront au moins le facteur $(s-a)^{a-2}$. Autrement dit: les mineurs du $(r-1)$ième ordre contiendront le facteur $(s-a)^a$ où $\gamma \geq 2\beta - \alpha$, ou, ce qui est la même chose, $\alpha - 2\beta + \gamma \leq 0$; c'est-à-dire: en formant la suite des indices des puissances selon lesquelles le facteur $(s-a)$ doit entrer dans les mineurs premiers, seconds, &c. (il va sans dire que cette suite finit par les mineurs $\alpha, \beta, \gamma, \ldots$, la différences secondes seront positives [c'est-à-dire non négatives]. Je représente par α, β, γ, $\ldots$ la suite dont il s'agit; je suppose, pour fixer les idées, que δ soit le dernier terme qui ne s'évanouisse pas, et j'écris

$$\begin{array}{cccc} \alpha, & \beta, & \gamma, & \delta,\ 0,\ 0, \ldots \\ \alpha - \beta, & \beta - \gamma, & \gamma - \delta, & \delta,\ 0, \ldots \\ \alpha - 2\beta + \gamma, & \beta - 2\gamma + \delta, & \gamma - 2\delta, & \delta,\ 0, \ldots \end{array}$$

ici, quel que soit le nombre des termes, tous les nombres de la troisième ligne seront positifs, et en représentant ces nombres par f, f', f'', &c., on obtient:

$$\begin{aligned} \alpha &= f + 2f' + 3f'' + 4f''' + \ldots, \\ \beta &= f' + 2f'' + 3f''' + \ldots, \\ \gamma &= f'' + 2f''' + \ldots, \\ \delta &= f''' + \ldots, \end{aligned}$$

Il y a ici à considérer que le nombre a, indice de la puissance selon laquelle le facteur $(s-a)$ entre dans le déterminant, est donné; il sera donc permis de prendre pour f, f', f'', $\ldots$ des valeurs entières et positives quelconques (zéro y compris) qui satisfont à la première équation; les autres équations donnent alors les valeurs de $\beta, \gamma, \&c.$ On forme de cette manière une table des particularités qui peut présenter le facteur multiple $(s-a)^a$ du déterminant; cette table est aussi la table des nombres $\alpha, \beta, \&c.$ ainsi de suite; alors, dans les différentes formules qui viennent d'être données, le coefficient d'un terme x_q^p à droite sera

$$\frac{-k}{\overline{(\dagger\alpha,\ \beta,\ \gamma,\ \ldots)}\,\overline{(\alpha',\ \beta',\ \gamma',\ \ldots)}}$$

et le coefficient d'un terme $x_q x_p$ à gauche sera

$$\frac{-k}{\overline{(\dagger\alpha,\ \beta,\ \gamma,\ \ldots)}\,\overline{(\alpha',\ \beta',\ \gamma',\ \ldots)}},$$

où k dénote le *discriminant* de la fonction quadratique à gauche, et où les coefficients des fonctions quadratiques des dénominateurs sont les coefficients inverses de cette même fonction quadratique à gauche.[1]

On suppose maintenant que la fonction $(a,b,c)(x,y)^2$ se transforme en elle-même par la substitution $\alpha x + \beta y$, $\gamma x + \delta y$ au lieu de x, y, on aura $\alpha\delta - \beta\gamma = \pm 1$ ou -1, qui sont les deux cas d'une fonction binaire.

En écrivant d'abord $(\diamond)(x,y)^2 = (A,B,C)(x,y)^2$ il faut obtenir cette substitution identiquement; $(A,B,C)(\alpha x + \beta y, \gamma x + \delta y)^2 = (A,B,C)(x,y)^2$, c'est-à-dire $A(\alpha^2 - 1) = 0$, $B(\beta^2 - 1) = 0$, $C(ab-1) = 0$, $F(\alpha\beta) = 0$, $G(\alpha\beta) = 0$, $H(\alpha b - 1) = 0$, $= 0$, et on voit que pour obtenir une

[1] Je profite de cette occasion pour remarquer concernant des recherches que les formules données dans le note sur les fonctions du second ordre (t. XXVIII. [1840] p. 193) [71] pour les cas des trois et de quatre indéterminées, sont exactes, mais que je m'étais trompé dans la forme générale du théorème. (This correction is included vol. i. p. 289.)

solution dans laquelle la forme quadratique ne se réduit pas à un carré, ou à une fonction de deux indéterminées, il faut supposer par exemple $\alpha^2 - 1 = 0$, $bc - 1 = 0$. On a donc dans le cas d'une fonction ternaire:

$$\alpha^2 = 1, \quad bc = 1, \quad (\diamond)(x, y, z)^2 = l{x_3}^2 + mx_2x_3.$$

La transformation sera *propre*, ou *impropre*, suivant que $\alpha = +1$ ou $\alpha = -1$.

Dans le cas d'une fonction *quaternaire*, on obtient pour la transformation *propre*:

$$ab = cd = 1, \quad (\diamond)(x, y, z, w)^2 = l{x_3}^2 + mx_3x_4,$$

et pour la transformation *impropre*:

$$\alpha = -1, \quad b = c = d = 1, \quad (\diamond)(x, y, z, w)^2 = l{x_3}^2 + mx_3x_4 + nx_2x_4.$$

Dans le cas d'une fonction *quinaire* on obtient

$$\alpha^2 = 1, \quad bc = de = 1, \quad (\diamond)(x, y, z, w, t)^2 = l{x_3}^2 + mx_2x_3,$$

et la transformation est *propre* ou *impropre*, selon que $\alpha = +1$ ou $\alpha = -1$; et ainsi de suite.

Cette méthode a des difficultés dans le cas où l'équation en s a des racines égales. Je n'entre pas ici dans ce sujet.

Dans les formules qu'on vient de trouver, on peut considérer les coefficients l, m, &c. comme des quantités *arbitraires*. Mais en supposant que la fonction quadratique soit *donnée*, ces coefficients deviennent *déterminés*. On les trouvera par la formule suivante que je ne m'arrête pas à démontrer.

Soient α, β, γ, &c. les coefficients de la fonction linéaire x_k, et les coefficients de la fonction linéaire x_k, et ainsi de suite; alors, dans les différentes formules qui viennent d'être données, le coefficient d'un terme ${x_q}^2$ à droite sera

$$\frac{-k}{(\dagger\overline{\alpha,\ \beta,\ \gamma,\ \ldots})\,(\overline{\alpha',\ \beta',\ \gamma',\ \ldots})}$$

et le coefficient d'un terme x_qx_p à gauche sera

$$\frac{-k}{(\dagger\overline{\alpha,\ \beta,\ \gamma,\ \ldots})\,(\overline{\alpha',\ \beta',\ \gamma',\ \ldots})},$$

où k dénote le *discriminant* de la fonction quadratique à gauche, et où les coefficients des fonctions quadratiques des dénominateurs sont les coefficients inverses de cette même fonction quadratique à gauche.

J'ai déjà fait voir de quelle manière cette formule se rattache à la formule pour la transformation *propre*; la différence entre les formes de ces transformations dans ces cas simple est assez frappante.

Pour obtenir la transformation *propre*

$$ax^2 + by^2 + cz^2 = ax^2 + by^2 + cz^2,$$

j'écris

$$\begin{vmatrix} 11, & 12, & 13 \\ 21, & 22, & 23 \\ 31, & 32, & 33 \end{vmatrix} = \begin{vmatrix} a, & \nu, & -\mu \\ -\nu, & b, & \lambda \\ \mu, & -\lambda, & c \end{vmatrix};$$

cette formule donne

$$(x, y, z) = \frac{1}{abc + a\lambda^2 + b\mu^2 + c\nu^2} \times$$

$$\left| \begin{array}{ccc} abc + a\lambda^2 - b\mu^2 - c\nu^2, & 2(\lambda\mu - c\nu), & 2(\nu\lambda + b\mu)c \\ 2(\lambda\mu + c\nu)\, a, & abc - a\lambda^2 + b\mu^2 - c\nu^2, & 2(\mu\nu - a\lambda)\, c \\ 2(\nu\lambda - b\mu)\, a, & 2(\mu\nu + a\lambda)\, b, & abc - a\lambda^2 - b\mu^2 + c\nu^2 \end{array} \right| (x, y, z).$$

La transformation *impropre*

$$ax^2 + by^2 + cz^2 = ax^2 + by^2 + cz^2$$

peut être tirée de la transformation *propre* en elle-même de la fonction donnée ci-après $ax^2 + by^2 + cz^2 + dw^2$; en y écrivant $d = 0$, on obtient

$$(x, y, z) = \frac{1}{bc\rho^2 + ca\sigma^2 + ab\tau^2 + \phi^2} \times$$

$$\left| \begin{array}{ccc} -bc\rho^2 + ca\sigma^2 + ab\tau^2 - \phi^2, & -2b\rho\sigma - 2bc\rho\tau, & 2bc\sigma\tau - 2bcrp \\ 2a\sigma\rho - 2acp\sigma, & bc\rho^2 - ac\sigma^2 + ab\tau^2 - \phi^2, & -2a\tau\rho - 2ab\sigma\tau \\ -2a\tau\sigma + 2ab\tau\rho, & 2b\rho\tau - 2ab\sigma\tau, & bc\rho^2 + ac\sigma^2 - ab\tau^2 - \phi^2 \end{array} \right| (x, y, z).$$

139. An Introductory Memoir upon Quantics.

AN INTRODUCTORY MEMOIR UPON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLIV. for the year 1854, pp. 244–258. Received April 20,—Read May 4, 1854.]

1. The term Quantics is used to denote the entire subject of rational and integral functions, and of the equations and loci to which these give rise; the word "quantic" is an adjective, meaning *of such a degree*, but may be used substantively, the noun understood being (unless the contrary appear by the context) function; so used the word admits of the plural "quantics."

The quantities or symbols to which the expression "degree" refers, or (what is the same thing) in regard to which a function is considered as a quantic, will be spoken of as "facients." A quantic may always be considered as being, in regard to its facients, homogeneous, since to render it so, it is only necessary to introduce as a facient unity, or some symbol which is to be ultimately replaced by unity; and in the cases in which the facients are considered as forming two or more distinct sets, the quantic may, in like manner, be considered as homogeneous in regard to each set separately.

The expression "an equation," used without explanation, is to be understood as meaning the equation obtained by putting any quantic equal to zero. I make no absolute distinction between the words "degree" and "order" as applied to an equation or system of equations, but I shall in general speak of the order rather than the degree. The equations of a system may be independent, or there may exist relations of connexion between the different equations of the system; the subject of a system of equations so connected together is one of extreme complexity and difficulty. It will be sufficient to notice here, that in any system whatever of equations, assuming only that the equations are not more than sufficient to determine the ratios of the facients, and joining to the system so many linear equations between the facients as will render the order of the system determinate (that is, the order of the equation which determines any one of these ratios), the order of the equation which determines any one of the facients (or, what is clear from the context, the order of the equation so determined) is nothing else than the order of the equation.

2. An equation or system of equations represents, or is represented by, a locus. This assumes that the facients depend upon quantities x, y, $\dots$ the coordinates of a point in space; the entire series of points, the coordinates of which satisfy the equation or system of equations, constitutes the locus. To avoid complexity, it is proper to take the facients themselves as coordinates, or at all events to consider these facients as linear functions of the coordinates; this being the case, the order of the locus is the order of the equation, or system of equations.

3. I have spoken of the *coordinates of a point in space.* I consider that there is an ideal space of three dimensions, but of course in the ordinary acceptation of the word, space is of three dimensions; however, in a plane (the space of two ordinary plane geometry) is a space of two dimensions, and we may consider the line as a space of one dimension, the line itself being the only locus of points in the space. To extend the notion of dimensionality, I do not introduce a space of an absolute number of dimensions, that is, the consideration of a space of four or of more dimensions, but I consider the symbols x, y, z, $\dots$ as variable, and obtain a system of points in the space of n dimensions; the number of equations, considered as a result, ought to render the locus indifferently of the elements, or of the facients themselves. In what follows, I shall not enter so as to render the result, indifferently of the elements; but only investigate whether a locus is or is not affected by the actual or any other generality consistent with its definition (and not, in this is a point upon which it is not at present necessary to dwell).

The theory includes, as a very particular case, the ordinary theory of reciprocity; we have only to say that to any line shall the word "line" shall mean "point," and the word "point" shall mean "line," and that lines respectively shall be considered as a purely descriptive notion of relating to a point and a line respectively shall be considered as a purely descriptive notion of relating to a point and a line, and similarly with regard to curves of a higher order, when the ideas of reciprocity applicable to these curves are properly developed.

5. A quantic of the degrees m, m', $\dots$ in the sets $(x, y, \dots)$, $(x', y', \dots)$ &c. will for the most part be represented by a notation such as

$$(* \diamond x, y, \dots)^m (x', y', \dots)^{m'} \dots,$$

where the mark $*$ may be considered as indicative of the absolute generality of the quantic; any such quantic may of course be considered as the sum of a series of terms $x^a y^b \dots x'^{a'} y'^{b'} \dots$, &c. of the proper degrees in the different sets respectively, each term being multiplied by a coefficient; these coefficients may be mere numerical multiples of single letters or elements such as a, b, c, $\dots$, or else functions (in general rational and integral functions) of such elements; this explains the meaning of the expression "the elements of a quantic" in the case where the coefficients are mere numerical multiples of single letters or elements such as $(a, b, \dots)$, by the appropriate numerical coefficient of a term $x^a y^b \dots x'^{a'} y'^{b'} \dots$, I mean the coefficient of this term in the expansion of

$$(x + y + \dots)^m (x' + y' + \dots)^{m'} \dots);$$

and I represent by the notation

$$(a, b, \dots)(x, y, \dots)^m (x', y', \dots)^{m'} \dots),$$

a quantic in which each term is multiplied as well by the appropriate numerical coefficient as by the literal coefficient or element. On the other hand, I represent by the notation

$$(a, b, \dots)(x, y, \dots)^m (x', y', \dots)^{m'} \dots),$$

a quantic in which each term is multiplied only by the literal coefficient or element which belongs to it in the set $(a, b, \dots)$ of literal coefficients or elements. And a like distinction applies to the cases where the coefficients are functions of the elements $(a, b, \dots)$.

6. I consider now the quantic

$$(* \between x, y, \ldots)^m (x', y', \ldots)^{m'} \ldots,$$

and selecting any two facients of the same set, e.g. the facients x, y, I remark that there is always an operation upon the elements, tantamount as regards the quantic to the operation $x\partial_y$; viz. if we differentiate with respect to each element, multiply by perpetual functions of the elements and add, we obtain the same result as by differentiating with ∂_y and multiplying by x. For instance, as regards $3ax^2 + bxy + 5cy^2$ (the numerical coefficients are taken hyphenated), we have $3b\partial_a + 10b\partial_c$ tantamount to $x\partial_y$; and as regards any other case, I represent by $[x\partial_y]$ the operation upon the elements tantamount to $x\partial_y$, and I write down here the series of operations

$$[x\partial_y] - x\partial_y, \quad [x'\partial_y] - x'\partial_y, \quad \ldots$$

where x, y are considered as being successively replaced by every permutation of two different facients of the set $(x, y, \ldots)$, x', y' successively replaced by every permutation of two different facients of the set $(x', y', \ldots)$, and so on; thus I call an entire system, and is to be considered as forming one such system.

7. The definition of a covariant may however be generalized in two directions: we may instead of a single quantic consider two or more quantics; the operations $[x\partial_y]$, although represented by means of the same symbols x, y, will as regards the different quantics, different meanings, but the conditions for the covariance of the entire system, we have only to consider in place of the system spoken of in the original definition, the system

$$\Sigma[x\partial_y] - x\partial_y, \quad \ldots, \quad \Sigma[x'\partial_y] - x'\partial_y, \quad \ldots, \quad \text{\&c. \&c.},$$

and we obtain the definition of a covariant of two or more quantics.

Again, we may consider in connexion with each set of facients any number of new sets being equal to or greater than the number of facients in the set. We have, in place of the original system $(x, y, \ldots)$, the systems

$$[x\partial_y] - S(x\partial_y), \quad \ldots, \quad [x'\partial_y] - S(x'\partial_y), \quad \ldots, \quad \text{\&c. \&c.},$$

and we obtain the generalized definition of a covariant.

8. A covariant has been defined simply as a function reduced to zero by each of the operations of the entire system. But in dealing with given quantics, we may without loss of generality consider the covariant as a function of the like form with the quantic, i.e. as being a rational and integral function homogeneous in regard to each of the different sets separately, and as being also a rational and integral function of the elements. In particular, the covariant may be the quantic itself, and when this is the case the term "covariant" includes, as already remarked, "invariant."

It is proper to remark, that if the same quantic be represented by means of different sets of elements, then the symbols $[x\partial_y]$ which correspond to these different forms of the same quantic are more transformations of each other, i.e. they become in virtue of the relations between the different sets of elements identical.

9. What precedes is a return to and generalization of the method employed in the first part of the memoir published in the *Camb. Math. Journ.*, t. IV. [1845], and *Camb. and Dubl. Math. Journ.*, t. I. [1846], under the title "On Linear Transformations," [13 and 14], and Crelle, t. XXX. [1846], under the title "Mémoire sur les Hyperdéterminants," [*16], and which I shall refer to as my original memoir. I there consider in fact the invariants of a quantic

$$(* \between x_1,\ x_2 \ldots x_n)(y_1,\ y_2 \ldots y_n) \ldots$$

linear in regard to n sets each of n facients, and I represent the coefficients of a term $x_1y_2\rho_1\ldots$ by ref...; there is no difficulty in the way of α, β being any two different numbers each of the series 1, 2, ... n, the operation $[x_p\partial_{x_q}]$ is identical with the operation

$$\Sigma\Sigma\ldots\Big(ref\ldots\frac{d}{d\rho\rho\ldots}\Big),$$

where the summations refer to s, t, ... which pass respectively from 1 to n, both inclusive; and the condition that a function (in the present memoir, assumed to be an invariant, i.e. to contain only the coefficients, but the condition may be reduced to zero by the operation $[x_p\partial_{x_q}] - x_p\partial_{x_q}$, is of course simply the condition that such function may be reduced to zero by the operation

$$\Sigma\Sigma\ldots\Big(ref\ldots\frac{d}{d\rho\rho\ldots}\Big)u = 0$$

of my original memoir.

10. But the definition in the present memoir includes also the method made use of in the second part of my original memoir. This method is substantially as follows: consider for simplicity a quantic $U =$

$$(*\diamond x,\ y,\ldots)^m$$

containing only the single set $(x, y, \ldots)$, and let U_1, U_2, ... be what the quantic becomes when the set $(x, y, \ldots)$ is successively replaced by the sets $(x_1, y_1, \ldots)$, $(x_2, y_2, \ldots)$, ...; the number of these new sets being equal to or greater than the number of facients in the set. Suppose that A, B, C, ... are any of the determinants

$$\begin{vmatrix} \partial_{x_1}, & \partial_{y_1}, & \partial_{z_1}, & \ldots \\ \partial_{x_2}, & \partial_{y_2}, & \partial_{z_2}, & \ldots \\ & \vdots & & \end{vmatrix},$$

then forming the derivative

$$A^pB^qC^r\ldots U_1U_2\ldots,$$

where p, q, r, ... are any positive integers, the function so obtained is a covariant involving the original sets $(x, y, \ldots)$, $(x_1, y_1, \ldots)$, $(x_2, y_2, \ldots)$ &c.; and if after the differentiations we replace $(x_1, y_1, \ldots)$, $(x_2, y_2, \ldots)$ &c. by $(x, y, \ldots)$, the function is a covariant according to the definition given in this memoir. But to do this some preliminary explanations are necessary.

11. I consider any two operations P, Q, involving each or either of them differentiations in respect of variables contained in the other of the operations. It is required to investigate the effect of the operation $P\cdot Q$, where the operation Q is performed upon some operand Ω, and the operation P is then performed upon the product $Q\cdot\Omega$. P is to be performed as usual as well in respect of variables ∂_x, ∂_y, ... contained in Q as in respect of variables a, b, ... contained in Ω. Suppose that variables a, b, ... contained in Q are not contained in Ω. But to render derivative is a covariant according to the definition in this Memoir. But to do this some preliminary explanations are necessary.

11. I consider any two operations P, Q, involving each or either of them differentiations in respect of variables contained in the other of the quantic. It is to investigate the effect of the operation $P\cdot Q$, where the operation Q is performed upon some operand Ω, and the operation P is then performed upon the product $Q\Omega$, where the operation P is to be performed as usual as well in respect of variables ∂_x, ∂_y, ... contained in Q as in respect of variables a, b, ... contained in Ω. Suppose that ∂_a, ∂_b, ..., contained in Q, may be also derived for some new operand Q, and they may be derived for some new operand Q, and ditto perform the operation Q as operand.

We can therefore in this same thing, the covariant operated upon by Φ, is reduced to zero by the operation $\mathfrak{D} = [a\partial_b] - 8a\partial_b$.

12. I return to the expression

$$A^p B^q C^r \ldots U_1 U_2 \ldots,$$

and I suppose that after the changes of the sets $(x_1, y_1, \ldots)$, $(x_2, y_2, \ldots)$, &c. into the original set $(x, y, \ldots)$ may be deferred until after the operation $\mathfrak{D}$, provided that the result is a covariant, we must prove that it is reduced to zero by the operation $\mathfrak{D} = [a\partial_b] - 8a\partial_b$.

It is easy to see that the change of the sets $(x_1, y_1, \ldots)$, $(x_2, y_2, \ldots)$, &c. into the original set $(x, y, \ldots)$ &c. into the original set, we should before the deferred change have an operation $\mathfrak{D}$, provided by $Sa\partial_b$, where we please write $Sa\partial_b$, instead therefore $\mathfrak{D} = [a\partial_b] - Sa\partial_b$. Now we have, as before, $A \cdot \mathfrak{D} = \mathfrak{D} \cdot A + \mathfrak{D} \cdot (A)$, where, as before, $A(\mathfrak{D})$ denotes the result of the operation A performed upon $\mathfrak{D}$ as operand, and similarly $\mathfrak{D}(A)$ the result of the operation $\mathfrak{D}$ performed upon A as operand; but in like manner $\mathfrak{D}$ is convertible with B, C, &c., since it must be noticed as regards B, C, &c., that the simple symbols ∂_x, ∂_y, … are not in fact functions tantamount to changes which we have hereafter to make of x, y, … into x, y, …; and we therefore have

$$\mathfrak{D} \cdot A^p B^q C^r \ldots U_1 U_2 \ldots = 0,$$

or $A^p B^q C^r \ldots U_1 U_2 \ldots$ is a covariant, the proposition which was to be proved.

13. I pass to a theorem which leads to another method of finding the covariants of a quantic. For this purpose I consider the quantic

$$(* \diamond x,\ y \ldots \diamond x',\ y' \ldots),$$

the coefficients of which are mere numerical multiples of the elements $(a, b, c, \ldots)$; and in connexion with this quantic I consider the linear functions $\xi x + \eta y \ldots$, $\xi' x' + \eta' y' \ldots$, &c. which I treat as coefficients, may be represented by means of the quantities $(\xi, \eta, \ldots)$, $(\xi', \eta', \ldots)$, $(\xi, y, \ldots)$, $(\xi', y', \ldots)$, &c.

we may from the quantic (which by convenience I call U) form an operative quantic, $\Theta U = (\xi, \eta, \ldots \diamond x, y, \ldots)(\xi', \eta', \ldots \diamond x', y', \ldots)^{m'} \ldots$

the coefficients of which are now numerical multiples of the corresponding derivatives, ∂_a, ∂_b, ∂_c, … of the original quantic. Consider, the new operative quantic

$$\Theta U = (\xi, \eta, \ldots \diamond x, y, \ldots)(\xi', \eta', \ldots \diamond x', y', \ldots)^{m'} \ldots$$

i.e. a product of powers of the linear functions. And it is to be remarked that as regards the quantic Θ in like manner the covariants in respect to x, y &c. are more numerical multiples, of the new elements ξ, η, …, ξ', η' …, and we can therefore consider this quantic in the same way as we did consider Θ as a product of powers of the linear functions. And it is to be remarked that as regards the quantic Θ in like manner the covariants in respect of x, y &c. as elements, &c. And the term "covariant" includes, as before, as the other coefficients can be deduced by mere differentiations.

14. The general demonstration of this equation presents no real difficulty, but to avoid the necessity of fixing upon a notation to distinguish the coefficients of the different terms and for the sake of simplicity, I shall merely confine the explanation by an example. Consider the quantic

$$U = ax^3 + 3bx^2y + 3cxy^2 + dy^3;$$

this gives

$$\Theta = \xi\partial_a + \xi^2\eta\partial_b + \xi\eta^2\partial_c + \eta^3\partial_d;$$

or, if for greater clearness, ∂_a, ∂_b, ∂_c, ∂_d are represented by α, β, γ, δ, then

$$\Theta = \xi^3\alpha + \xi^2\eta\beta + \xi\eta^2\gamma + \eta^3\delta,$$

and we have

$$[x\partial_y] = 3b\partial_a + 2c\partial_b + d\partial_c,$$

and

$$[\xi\partial_\eta] = 3a\partial_b + 2b\partial_c + c\partial_d.$$

Now considering Φ as a function of ∂_a, ∂_b, ∂_c, ∂_d, or, what is the same thing, of α, β, γ, δ, we have to prove that

$$\Phi \cdot \Theta = \Theta \cdot \Phi(\Theta);$$

and if in the expression of Φ we write $a + \partial_a$, $\beta + \partial_b$, $\gamma + \partial_c$, $\delta + \partial_d$ for a, b, c, d (where only the symbols ∂_a, ∂_b, ∂_c, ∂_d are to be considered as affecting a, b, c, d, or as contained in the operand of Φ, and reject the first term (or zero term independent of ∂_a, ∂_b, ∂_c, ∂_d, and reject the first term of the operand $3a\partial_b + 2c\partial_b + d\partial_c$, by the expression) we obtain the first term of the result. This will give the form which the same value of Φ takes by the substitution of $\Phi(\Theta)$. This value is

$$\Phi(\Theta) = (3a\beta + 2b\gamma + c\delta)\Phi;$$

performing the differentiations ∂_a, ∂_b, ∂_c, ∂_d, its value is

$$\Phi([\xi\partial_\eta]) = (3a\xi^2 + 2b\eta + \gamma\xi)\Phi,$$

i.e. we have

$$\Phi(\Theta) = [\xi\partial_\eta]\Phi.$$

15. Suppose now that Φ is a covariant of Θ, then the operation Φ performed upon any covariant of U gives rise to a covariant of the system

$$(* \diamond x, y, \ldots \diamond \xi, \eta, \ldots) \ldots, \quad (\xi', \eta', \ldots \diamond x', y', \ldots), \quad \&c.$$

To prove this is to be in the first instance noticed, that as regards $(\xi, \eta, \ldots)$, &c. we have $[\xi\partial_\eta] = \eta\partial_\xi$, &c. Hence considering $[\xi\partial_\eta]$, &c. as referring to the quantic U, the operation $\Sigma[\xi\partial_\eta] - x\partial_y$, &c. are referring to zero by every covariant of the system, and therefore every covariant of the system must be reduced to zero by the operations

$$\mathfrak{D} = [x\partial_y] + \eta\partial_\xi - x\partial_y,$$

$$\mathfrak{D} = [\xi\partial_\eta] + y\partial_x - \xi\partial_\eta,$$

equations which it is obvious may be replaced by

$$\mathfrak{D} \cdot \Phi = \mathfrak{D} \cdot \Phi + \mathfrak{D} \cdot (\Phi),$$

$$\Phi \cdot \mathfrak{D} = \mathfrak{D} \cdot \Phi + \Phi \cdot (\mathfrak{D}),$$

and consequently (in virtue of the theorem),

$$\mathfrak{D} \cdot \Phi = \mathfrak{D} \cdot \Phi + \mathfrak{D} \cdot (\Phi),$$

$$\Phi \cdot \mathfrak{D} = \mathfrak{D} \cdot \Phi + \Phi \cdot (\mathfrak{D}),$$

and we have therefore

$$\mathfrak{D} \cdot \Phi - \Phi \cdot \mathfrak{D} = -(\eta\partial_\xi - \eta\partial_\xi)(\Phi);$$

or, since Φ is a covariant of Θ, i.e. since $\Theta\cdot\Phi=\Phi\cdot\Theta$. And since every covariant of the system reduces to zero by the operation $\mathfrak{D}\cdot\Phi$, and therefore by the operation $\Phi\cdot\mathfrak{D}$, such covariant will also be reduced to zero by the operation $\mathfrak{D}\cdot\Phi$, or what is the same thing, the covariant operated upon by Φ, is reduced to zero by the operation $\mathfrak{D}$ operating upon a covariant gives a covariant.

16. In the case of a quantic such as $U=$

$$(*\diamond x,\ y\ldots)^m,\quad (*\diamond x',\ y'\ldots)^{m'},\quad \&\text{c.}$$

we may instead of the new sets (ξ,η), (ξ',η'), …employ the sets $(y,-x)$, $(y',-x')$, &c. The operative quantic Θ is in this case defined by the equation $\Theta U=0$, and the operations Φ performed upon a covariant of U will give the analogous equation

$$\Phi([x\partial_y])=[x\partial_y](\Phi).$$

where on the left-hand side $[x\partial_y]$ refers to U, but on the right-hand side $[x\partial_y]$ refers to Θ, and instead of $[x\partial_y]+y\partial_x-x\partial_y$, we have simply $\mathfrak{D}=[x\partial_y]-x\partial_y$.

17. I pass next to the quantic

$$(*\diamond x,\ y)^m,$$

which I shall in general consider under the form

$$(a,\ b,\ldots b',\ a'\diamond x,\ y)^m,$$

but sometimes under the form

$$(a,\ b,\ldots b',\ a'\diamond x,\ y)^m,$$

the former notation denoting, it will be remembered,

$$ax^m+\frac{m}{1}bx^{m-1}y\ldots+\frac{m}{1}b'xy^{m-1}+a'y^m,$$

and the latter notation

$$ax^m+bx^{m-1}y\ldots+b'xy^{m-1}+a'y^m.$$

But in particular cases the coefficients will be represented by means of unaccentuated letters, thus $(a,b,c,d\diamond x,y)^3$ will be used to denote $ax^3+3bx^2y+3cxy^2+dy^3$, and $(a,b,c,d\diamond x,y)^3$ will be used to denote $ax^3+bx^2y+cxy^2+dy^3$, and so in all similar cases.

Applying the general methods to this case, we see that

$$\begin{aligned}&(a,\ b,\ldots b',\ a'\diamond x,\ y)^m,\\ &\qquad[y\partial_x]=ab\partial_a+2bb\partial_b+\ldots+mb'\partial_{a'},\\ &\qquad[x\partial_y]=mb\partial_a+(m-1)c\partial_b+\ldots+a'\partial_{b'};\end{aligned}$$

in fact, with these meanings of the symbols the quantic is reduced to zero by each of the operations $[y\partial_x]-y\partial_x$, $[x\partial_y]-x\partial_y$, and we may consequently take this as the definition of the operations $[y\partial_x]$, $[x\partial_y]$, hence supposing that the function U is a covariant or invariant of the quantic, we have for the definition of a covariant or invariant

$$[y\partial_x](U)=ay\partial_x(U),\quad [x\partial_y](U)=-(\alpha+\beta\ldots)(U)+\alpha b\partial_x(U),$$

in fact, after the differentiations, the new sets $(x_1,y_1),(x_2,y_2),\ldots$ may be replaced by the original set (x,y), will be a covariant of the quantic U. And it is to be remarked that as regards the quantic

$$\overline{12'\ 13'\ 23'}\ldots U_1U_2U_3\ldots,$$

i.e. a product of any linear functions, the new elements being merely the coefficients of the original quantic. The operative quantic Θ becomes in this case under consideration

$$\Theta = (b, \ldots b', a \diamond x, y)^m,$$

the signs being alternately positive and negative; in fact it is easy to verify that this expression is identical with the expression ΘU operating on a covariant of U.

18. But the quantic

$$(a,\ b, \ldots b',\ a' \diamond x,\ y)^m,$$

considered as decomposable into linear factors, may be expressible in the form

$$a(x - \alpha y)(x - \beta y) \ldots,$$

gives rise to a fresh series of results. We have here also

$$[y\partial_x] = -(\alpha + \beta + \ldots)\alpha\partial_a + \partial_\alpha + \partial_\beta \ldots,$$

$$[x\partial_y] = -(\alpha + \beta \ldots)\alpha\partial_a + \alpha^2\partial_\alpha + \beta^2\partial_\beta + \ldots;$$

in fact with these meanings of the symbols the quantic is reduced to zero by each of the operations $[y\partial_x] - y\partial_x$, $[x\partial_y] - x\partial_y$, which expresses the definition of the operations $[y\partial_x]$, $[x\partial_y]$, and any expression of U as a covariant or invariant of the quantic. And the expressions $(\alpha - \beta)(x - \alpha y) - (\alpha - \beta)(x - \beta y)$, identical.

19. Consider now the expression

$$a^p\Sigma(x - \alpha y)(x - \beta y)^q \ldots (x - \beta)^r \ldots,$$

where the sum of the indices j, p, …of all the simple factors which contain α, the sum of the indices k, p, …of all the simple factors which contain β, are respectively θ of the coefficient α; the index θ is in particular if $\phi = 0$, i.e. if V be a quantic of the order θ, then in particular the order m in the coefficients, the order θ in the facients x, y, and of the degree θ in the coefficients.

20. In connexion with this covariant

$$a^p\Sigma(x - \alpha y)(x - \beta y)^q \ldots (x - \beta y)^r \ldots,$$

of the order μ and of the degree θ in the coefficients, of the quantic $U =$

$$a(x - \alpha y)(x - \beta y) \ldots,$$

consider the covariant

$$\Sigma(\overline{12'} \ldots)V_1V_2 \ldots V_m$$

of a quantic $V =$

$$(* \diamond x,\ y)^\mu,$$

in which, after the differentiations, the new sets (x_1, y_1), (x_2, y_2), …may be replaced by the original set (x, y). The last-mentioned covariant will be of the order m in the coefficients, and in particular if $\phi = 0$, i.e. if V be a quantic of the order θ of the coefficients, then in particular the order m in the coefficients of the degree m in the coefficients, and of the degree θ in the coefficients, and that the new in like manner contain only numerical multiples of these invariants. The precise meaning of the law, in the last-mentioned point of view, requires some explanation. Suppose that we know all the linearly independent invariants of a quantic of the order m and degree θ in the coefficients, considered as a function of the order m and the degree θ in the coefficients, it can in like manner be obtained as a covariant of degree $\theta + m$ in the coefficients, viz. of the variables the sets (x_1, y_1), (x_2, y_2), &c.; this is the precise meaning in which I assert independent, and by a covariant of the order m, considered as a function of the variables (x_1, y_1), (x_2, y_2), &c.; this is the precise meaning in which I assert that the invariants of the degree θ in question are mere rational and integral functions of the invariants of the degree θ in question may be considered as a function of the degree θ in question, where they may merely numerical multiples of these different invariants, but in general I will not consider these invariants different invariants, but in general I will not consider these invariants different invariants because, in particular cases of the degree θ in question to assume that $\theta = m$, viz. of the invariants of the degree θ which are necessary will compromise all the invariants of that degree (if any) which are rational and integral functions of the invariants of an inferior order and of the degree θ.

21. The number of the really independent covariants of a quantic $(*\diamond x, y)^m$ is precisely equal to the order of the quantic, i.e. any covariant Q is a function (generally an irrational function) of any other two covariants and of the coefficients; or in like manner the number of the really independent invariants is, as may be easily seen by a process analogous to the one just employed, $m-2$ invariants. We may therefore say that every covariant (generally an irrational function), of $m-2$ invariants, considered as a function of the coefficients.

22. Consider any covariant of the quantic

$$(a,\ b, \ldots b',\ a' \diamond x,\ y)^m,$$

and let this be of the order μ and of the degree θ in the coefficients. It is very easily shown that $m\theta - \mu$ is necessarily even. In particular in the case of an invariant (i.e. when $\mu = 0$) so that a quantic of an odd order admits only of an even degree θ; but if $\mu = 0$ and $\mu\theta$ is necessarily even. In the former case one of the two numerical distinction between the two cases must be drawn; but as an important distinction between the two cases must be drawn; this is in fact only of an odd order admits only of an even degree.

23. There is another very simple condition which is satisfied by every covariant of the quantic

$$(a,\ b, \ldots b',\ a' \diamond x,\ y)^m,$$

viz. if we consider the facients (x, y) as being respectively of the weights $\frac{1}{2}$, $-\frac{1}{2}$, the coefficients $(a, b, \ldots b', a')$ as being respectively of the weights $\frac{1}{2}m$, $\frac{1}{2}m - 1$, $\ldots$, $-\frac{1}{2}m + 1$, $-\frac{1}{2}m$, then the covariant is of the weight zero. This is the most natural mode of stating the theorem, since the symbols, ∂_a, ∂_b, $\ldots$, $\partial_{b'}$, $\partial_{a'}$ are respectively of the weights $-\frac{1}{2}m$, $-\frac{1}{2}m + 1$, $\ldots$; the conditions can without negative numbers be of course be modified so as to effect upon the coefficients $(a, b, \ldots b', a')$ as being respectively of the weights 0, 1, $\ldots$, $m-1$, m, respectively, and the conditions of every covariant or term of the covariant of the weights $\frac{1}{2}(m\theta + \mu)$.[2]

24. The preceding laws as to the form of a covariant have been stated here by way of anticipation, principally for the sake of the remark, that they so far define the form of a covariant as to render it in very many cases practicable with a moderate amount of labour to complete the investigations by means of one or the other of the operations of the entire system. Now, besides the quantic itself, there are a variety of other functions which are reduced to zero by each of the operations of the entire system; any such function is said to be a covariant of the proper order and degree in the coefficients, in any particular case may very easily be found, and the first term of the proper function and degree in the coefficients, may form the second term induced by means by more differentiations. There can be no very important term in the second term, and so on, until either the remaining terms can be found by means of any inferior order; but the very important terms are very important terms in the case in question is that of such order a, and

[2] I may remark that it was only M. Hermite's important discovery of an invariant of the degree 18 of a quantic of the order 5, which removed an erroneous impression which I had been under from the commencement of the subject, that out was of necessity *evenly* even.

the first term being known, the remaining ones can (i.e. in practice in a very convenient manner for the calculation given to the question, viz. that of the highest power of either of the facients) be deduced by mere differentiations. There can be very important terms (i.e. in practice in a very convenient manner for the calculation given to the question, viz. that of the highest power of either of the facients) be noticed as regards covariants, that when the first or the last term of a covariant is known, the other coefficients can be deduced by mere differentiations.

Postscript added October 7th, 1854.—I have, since the preceding memoir was written, found with respect to the covariants of a quantic $(*\diamond x, y)^m$ a function of any number and degree in the coefficients satisfying the prescribed conditions as to symmetry and weight; such function, viz. the function which is to be obtained by Aronhold's method, may, in this manner be modified as follows:—if the coefficients $(a, b, \ldots, b', a')$ are considered as denoting the relations a, b, $c = a$ b, c, d where $\alpha\beta - \beta\gamma$, $\alpha c - \beta b$, $\ldots$, $\beta\gamma - \beta\beta$ b, $\alpha\delta - \beta\gamma$, are linear functions of a, b, c, d, the form $\delta(\alpha\beta - \beta\gamma)\alpha(\alpha c - \beta b)$, will be a function of the elements satisfying the prescribed conditions as to symmetry and weight: but I do not believe that any covariant or invariant other than the discovery of any covariant or invariant other than the discoveries of the facients is known. I have, since the above memoir was written, hope to return to the subject in a subsequent memoir.